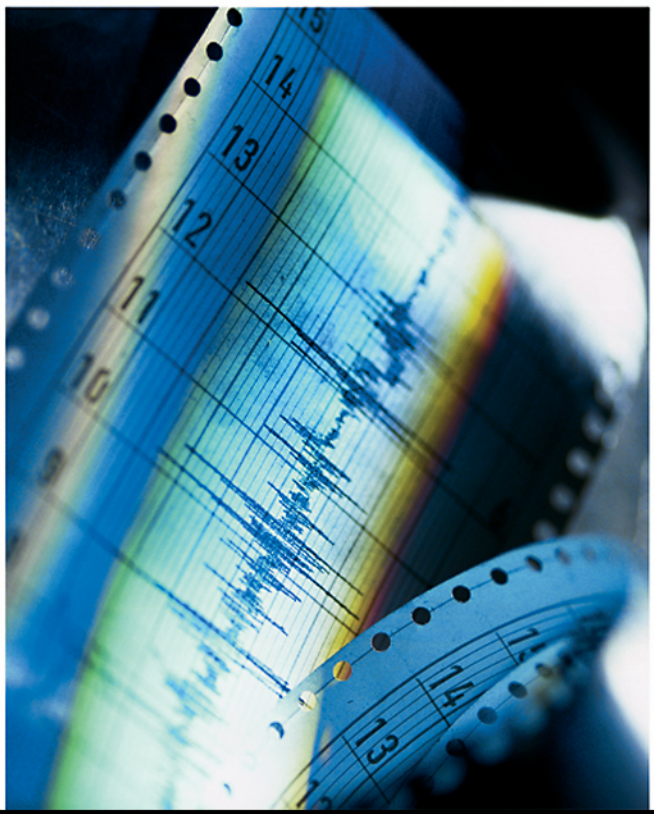


# SEISMIC DESIGN OF BUILDINGS TO EUROCODE 8



EDITED BY
**AHMED Y. ELGHAZOULI**

Spon Press

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# 1 Introduction: seismic design and Eurocode 8

*P. Bisch*

## 1.1 The Eurocodes

The European directive 'Construction Products' issued in 1989 comprises requirements relating to the strength, stability and fire resistance of construction. In this context, the structural Eurocodes are technical rules, unified at the European level, which aim to ensure the fulfilment of these requirements. They are a set of fifty-eight standards gathered into ten Eurocodes, providing the basis for the analysis and design of structures and of the constitutive materials. Complying with Eurocodes makes it possible to declare the conformity of structures and construction products and to apply CE (Conformité Européenne) marking to them (a requirement for many products, including most construction products, marketed within the European Union). Thus, Eurocodes constitute a set of standards of structural design, consistent in principle, which facilitates free distribution of products and services in the construction sector within the European Union.

Beyond the political goals pursued by the Union, the development of Eurocodes has also given rise to considerable technical progress, by taking into account the most recent knowledge in structural design, and producing technical standardisation across the European construction sector. The Eurocodes have been finalised in the light of extensive feedback from practitioners, since codes should reflect recognised practices current at the time of issue, without, however, preventing the progress of knowledge.

The methodology used to demonstrate the reliability (in particular, safety assessment) of structures is the approach referred to as 'semi-probabilistic', which makes use of partial coefficients applied to actions, material properties and covering the imperfections of analysis models and construction. The verification consists of analysing the failure modes of the structure, associated with limit states, in design situations with associated combinations of actions that can reasonably be expected to occur simultaneously.

Inevitably, the Eurocodes took many years to complete since, to reach general consensus, it was necessary to reconcile differing national experiences and requirements coming from both researchers and practising engineers.

## 1.2 Standardisation of seismic design

The first concepts for structural design in seismic areas, the subject of Eurocode 8 (EC8), were developed from experience gained in catastrophes such as the San Francisco earthquake in 1906 and the Messina earthquake in 1908.

At the very beginning, in the absence of experimental data, the method used was to design structures to withstand uniform horizontal accelerations of the order of 0.1*g*. After the Long Beach earthquake in 1933, the experimental data showed that the ground accelerations could be much higher, for instance 0.5*g*. Consequently, the resistance of certain structures could be explained only by the energy dissipation that occurred during the movement of the structure caused by the earthquake. The second generation of codes took into account on the one hand the amplification due to the dynamic behaviour of the structures, and on the other hand the energy dissipation. However, the way to incorporate this dissipation remained very elementary and did not allow correct differentiation between the behaviour of the various materials and types of lateral resisting systems.

The current third generation of codes makes it possible on the one hand to specify the way to take the energy dissipation into account, according to the type of lateral resistance and the type of structural material used, and on the other hand to widen the scope of the codes, for instance by dealing with geotechnical aspects. Moreover, these new rules take into account the semi-probabilistic approach for verification of safety, as defined in EN 1990.

The appearance of displacement-based analysis methods makes it possible to foresee an evolution towards a fourth generation of seismic design codes, where the various components of the seismic behaviour will be better controlled, in particular those that relate to energy dissipation. From this point of view, in its present configuration, EC8 is at the junction between the third generation codes, of which it still forms part, and of fourth generation codes.

## 1.3 Implementation of EC8 in Member States

The clauses of Eurocodes are divided into two types, namely *Principles*, which are mandatory, and *Application Rules*, which are acceptable procedures to demonstrate compliance with the Principles. However, unless explicitly specified in the Eurocode, the use of alternative Application Rules to those given does not allow the design to be made in conformity with the code. Also, in a given *Member State*, the basic Eurocode text is accompanied for each of its parts by a *National Annex* specifying the values of certain parameters (*Nationally Determined Parameters (NDPs)*) to be used in this country, as well as the choice of methods when the Eurocode part allows such a choice. NDPs are ones that relate to the levels of safety to be achieved, and include for example partial factors for material properties.

In the absence of a National Annex, the recommended values given in the relevant Eurocode can be adopted for a specific project, unless the project documentation specifies otherwise.

For the structures and in the zones concerned, the application of EC8 involves that of other Eurocodes. EC8 only brings additional rules to those given in other Eurocodes, to which it refers. Guides or handbooks can also supplement EC8 as application documents for certain types of structural elements.

To allow the application of EC8 in a given territory, it is necessary to have a seismic zoning map and associated data defining peak ground accelerations and spectral shapes. This set of data, which constitutes an essential basis for analysis, can be directly introduced into the National Annex. However, in certain countries, seismic design codes are regulated by statute and, where this applies, zoning maps and associated data are defined separately by the national authorities.

## 1.4 Contents of EC8

EC8 comprises six parts relating to different types of structures (Table 1.1). Parts 1 and 5 form the basis for the seismic design of new buildings and their foundations; their rules are aimed both at protecting human life and also limiting economic loss. It is interesting to note that EC8 Part 1 also provides design rules for base isolated structures.

Particularly because of its overlap with other Eurocodes and the cross-referencing that this implies, EC8 presents some difficulties at first reading. Although these can be easily overcome by a good comprehension of the underlying principles, they point to the need for application manuals to assist the engineer in design of the most common types of structure.

*Table 1.1* Parts of Eurocode 8

| *Title* | *Reference* |
|---|---|
| Part 1: General Rules, Seismic Actions and Rules for Buildings | EN 1998-1:2004 |
| Part 2: Bridges | EN 1998-2:2005 |
| Part 3: Assessment and Retrofitting of Buildings | EN 1998-3:2005 |
| Part 4: Silos, Tanks and Pipelines | EN 1998-4:2006 |
| Part 5: Foundations, Retaining Structures and Geotechnical Aspects | EN 1998-5:2004 |
| Part 6: Towers, Masts and Chimneys | EN 1998-6:2005 |

## 1.5 Overview of this book

Seismic design of structures aims at ensuring, in the event of occurrence of a reference earthquake, the protection of human lives, the limitation of damage to the structures, and operational continuity of constructions important for civil safety. These goals are linked to seismic actions. Chapter 2 of this book provides a detailed review of methods used in determining seismic hazards and earthquake actions. It covers seismicity and ground-motion models, with specific reference to the stipulations of EC8.

To design economically a structure subjected to severe seismic actions, post elastic behaviour is allowed. The default method of analysis uses linear procedures, and post elastic behaviour is accounted for by simplified methods. More detailed analysis methods are normally only utilised in important or irregular structures. These aspects are addressed in Chapter 3, which presents a review of basic dynamics including the response of single- and multi-degree-of-freedom systems and the use of earthquake response spectra, leading to the seismic analysis methods used in EC8. This chapter also introduces an example building that is used throughout the book to illustrate the use of EC8 in practical building design. The structure was specifically selected to enable the presentation and examination of various provisions in EC8.

The design of buildings benefits from respecting certain general principles conducive to good seismic performance, and in particular to principles regarding structural regularity. The provisions relating to general consideration for the design of buildings are dealt with in Chapter 4. These relate to the shape and regularity of structures, the proper arrangement of the lateral resisting elements and a suitable foundation system. Chapter 4 also introduces the commonly adopted approach of design and dimensioning referred to as 'capacity design', which is used to control the yielding mechanisms of the structure and to organise the hierarchy of failure modes. The selected building introduced in Chapter 3 is then used to provide examples for the use of EC8 for siting as well as for assessing structural regularity.

Chapter 5 of this book focuses on the design of reinforced concrete structures to EC8. It starts by describing the design concepts related to structural types, behaviour factors, ductility provisions and other conceptual considerations. The procedures associated with the design for various ductility classes are discussed, with particular emphasis on the design of frames and walls for the intermediate (medium) ductility class. In order to illustrate the design of both frames and walls to EC8, the design of a dual frame/wall lateral resisting system is presented and discussed.

The design of steel structures is discussed in Chapter 6. The chapter starts by outlining the provisions related to structural types, behaviour factors, ductility classes and cross sections. This is followed by a discussion of the design procedures for moment and braced frames. Requirements related to

material properties, as well as the control of design and construction, are also summarised. The example building is then utilised in order to demonstrate the application of EC8 procedures for the design of moment and braced lateral resisting steel systems.

Due to the similarity of various design approaches and procedures used for steel and composite steel/concrete structures in EC8, Chapter 7 focuses primarily on discussing additional requirements that are imposed when composite dissipative elements are adopted. Important design aspects are also highlighted by considering the design of the example building used in previous chapters.

It is clearly necessary to ensure the stability of soils and adequate performance of foundations under earthquake loading. This is addressed in Chapters 8 and 9 for shallow and deep foundations, respectively. Chapter 8 provides background information on the behaviour of soils and on seismic loading conditions, and covers issues related to liquefaction and settlement. Focus is given to the behaviour and design of shallow foundations. The design of a raft foundation for the example building according to the provisions of EC8 is also illustrated. On the other hand, Chapter 9 focuses on the design of deep foundations. It covers the assessment of capacity of piled foundations and pile buckling in liquefied soils as well as comparison of static and dynamic performance requirements. These aspects of design are illustrated through numerical applications for the example building.

In the illustrative design examples presented in Chapters 3 through to 9 of this book, reference is made to the relevant rules and clauses in EC8, such that the discussions and calculations can be considered in conjunction with the code procedures. To this end, it is important to note that this publication is not intended as a complete description of the code requirements or as a replacement for any of its provisions. The purpose of this book is mainly to provide background information on seismic design in general, and to offer discussions and comments on the use of EC8 in the design of buildings and their foundations.

# 2 Seismic hazard and earthquake actions

*J.J. Bommer and P.J. Stafford*

## 2.1 Introduction

Earthquake-resistant design can be considered as the art of balancing the seismic capacity of structures with the expected seismic demand to which they may be subjected. In this sense, earthquake-resistant design is the mitigation of seismic risk, which may be defined as the possibility of losses (human, social or economic) due to the effects of future earthquakes. Seismic risk is often considered as the convolution of seismic hazard, exposure and vulnerability. Exposure refers to the people, buildings, infrastructure, commercial and industrial facilities located in an area where earthquake effects may be felt; exposure is usually determined by planners and investors, although in some cases avoidance of major geo-hazards may lead to relocation of new infrastructure. Vulnerability is the susceptibility of structures to earthquake effects and is generally defined by the expected degree of damage that would result under different levels of seismic demand; this is the component of the risk equation that can be controlled by engineering design. Seismic hazards are the potentially damaging effects of earthquakes at a particular location, which may include surface rupture, tsunami run-up, liquefaction and landslides, although the most important cause of damage on a global scale is earthquake-induced ground shaking (Bird and Bommer, 2004). The focus in this chapter is exclusively on this particular hazard and the definition of seismic actions in terms of strong ground motions. In the context of probabilistic seismic hazard analysis (PSHA), seismic hazard actually refers to the probability of exceeding a specific level of ground shaking within a given time.

If resources were unlimited, seismic protection would be achieved by simply providing as much earthquake resistance as possible to structures. In practice, it is not feasible to reduce seismic vulnerability to an absolute minimum because the costs would be prohibitive and certainly not justified since they would be for protection against a loading case that may not even occur during the useful life of the structure. Seismic design therefore seeks to balance the investment in provision of seismic resistance against the level of damage, loss or disruption that earthquake loading could impose. For this

reason, quantitative assessment and characterisation of the expected levels of ground shaking constitute an indispensable first step of seismic design, and it is this process of seismic hazard analysis that is introduced in this chapter.

The assessment of ground-shaking hazard due to future earthquakes invariably involves three steps: the development of a seismicity model for the location and size (and, if appropriate, the frequency) of future earthquakes in the region; the development of a ground-motion model for the prediction of expected levels of shaking at a given site as a result of any of these earthquake scenarios; and the integration of these two models into a model for the expected levels of shaking at the site of interest (Figure 2.1).

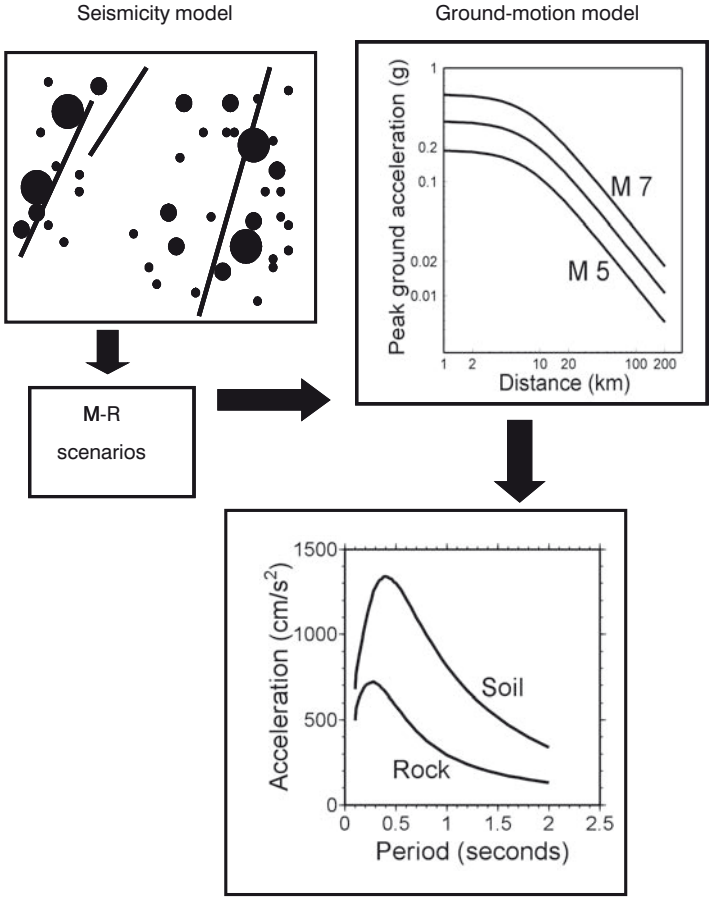



*Figure 2.1* Schematic overview of seismic hazard analysis. The seismicity model defines scenarios of earthquakes of magnitude, M, at a distance, R, from the site of interest, and the ground-motion model predicts the shaking parameter of interest for this M-R combination. The results in this case are expressed in terms of acceleration response spectra (see Chapter 3 for definition and detailed explanation of response spectra)

The first three sections of this chapter deal with the three steps illustrated in Figure 2.1, that is seismicity models (Section 2.2), ground-motion models (Section 2.3) and seismic hazard analysis (Section 2.4). The remaining two sections then explore in more detail specific representations of the ground motion for engineering analysis and design, namely response spectra (Section 2.5) and acceleration time-histories (Section 2.6), both with specific reference to the stipulations of EC8. The chapter closes with brief conclusions and recommendations regarding both the use of EC8 as the basis for defining seismic design loads and possible improvements to the code that could be made in future revisions.

## 2.2 Earthquake parameters and seismicity

An entire book, let alone a chapter, could be dedicated to the issue of seismicity models. Herein, however, a very brief overview, with key references, is presented, with the aim of introducing definitions for the key parameters and the main concepts behind seismicity models.

With the exception of some classes of volcanic seismicity and very deep events, earthquakes are generally produced by sudden rupture of geological faults, releasing elastic strain energy stored in the surrounding crust, which then radiates from the fault rupture in the form of seismic waves. The location of the earthquake is specified by the location of the focus or hypocentre, which is the point on the fault where the rupture initiates and from where the first seismic waves are generated. This point is specified by the geographical coordinates of the epicentre, which is the projection of the hypocentre on the Earth's surface, and the focal depth, which is the distance of the hypocentre below the Earth's surface, measured in kilometres. Although for the purposes of observatory seismology, using recordings obtained on sensitive instruments at distances of hundreds or thousands of kilometres from the earthquake, the source can be approximated as a point, it is important to emphasise that in reality the earthquake source can be very large. The source is ultimately the part of the crust that experiences relaxation as a result of the fault slip; the dimensions of the earthquake source are controlled by the length of the fault rupture and, to a lesser extent, the amount of slip on the fault during the earthquake. The rupture and slip lengths both grow exponentially with the magnitude of the earthquake, as shown in Figure 2.2. Two good texts on the geological origin of earthquakes and the nature of faulting are Yeats et al. (1997) and Scholz (2002).

The magnitude of an earthquake is in effect a measure of the total amount of energy released in the form of seismic waves. There are several different magnitude scales, each of which is measured from the amplitude of different waves at different periods. The first magnitude scale proposed was the Richter scale, generally denoted by $M_L$, where the subscript stands for local. Global earthquake catalogues generally report event size in terms of body-wave magnitude, $m_b$, or surface-wave magnitude, $M_s$, which will often give

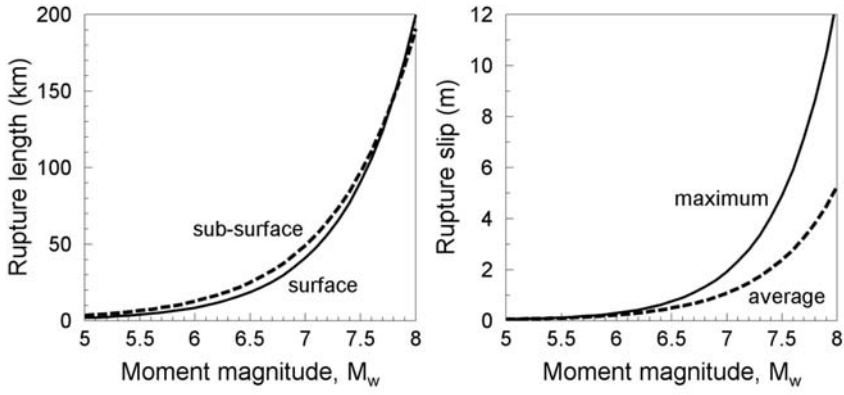



*Figure 2.2* Median predicted values of rupture length and slip from the empirical equations of Wells and Coppersmith (1994)

different values for the same earthquake. All of the scales mentioned so far share a common deficiency in that they saturate at a certain size and are therefore unable to distinguish the sizes of the very largest earthquakes. This shortcoming does not apply to moment magnitude, designated as $M_w$ or $M$, which is determined from the very long-period part of the seismic radiation. This scale is based on the parameter seismic moment, which is the product of the area of the fault rupture, the average slip on the fault plane and the rigidity of the crust.

A seismicity model needs to specify the expected location and frequency of future earthquakes of different magnitudes. A wide range of data can be used to build up seismicity models, generally starting with regional earthquake catalogues. Instrumental recordings of earthquakes are only available since the end of the nineteenth century and even then the sparse nature of early networks and low sensitivity of the instruments means that catalogues are generally incomplete for smaller magnitudes prior to the 1960s. The catalogue for a region can be extended through the study of historical accounts of earthquakes and the inference, through empirical relationships derived from twentieth-century earthquakes, of magnitudes. For some parts of the world, historical seismicity can extend the catalogue from 100 years to several centuries. The record can be extended even further through paleoseismological studies (McCalpin, 1996), which essentially means the field study of geological faults to assess the date and amplitude of previous co-seismic ruptures. Additional constraint on the seismicity model can be obtained from the tectonic framework and more specifically from the field study of potentially active structures and their signature on the landscape. Measurements of current crustal deformation, using traditional geodesy or satellite-based techniques, also provide useful input to estimating the total seismic moment budget (e.g. Jackson, 2001).

The seismicity model needs to first specify the spatial distribution of future earthquake events, which is achieved by the definition of seismic sources. Where active geological faults are identified and their degree of activity can be characterised, the seismic sources will be lines or planes that reflect the location of these structures. Since in many cases active faults will not have been identified and also because it is generally not possible to unambiguously assign all events in a catalogue to known faults, source zones will often be defined. These are general areas in which it is assumed that seismicity is uniform in terms of mechanism and type of earthquake, and that events are equally likely to occur at any location within the source. Even where fault sources are specified, these will generally lie within areal sources that capture the seismicity that is not associated with the fault.

Once the boundaries of the source zones are defined, which fixes the spatial distribution of the seismicity model, the next step is to produce a model for the temporal distribution of seismicity. These models are generally referred to as recurrence models as they define the average rates of occurrence of earthquakes of magnitude greater than or equal to a particular value. The most widely used model is that known as the Gutenberg–Richter (G–R) relationship, which defines a simple power law relationship between the number of earthquakes per unit time and magnitude. The relationship is defined by two parameters, the activity (i.e. the annual rate of occurrence of earthquakes of magnitude greater than or equal to zero or some other threshold level) and the *b*-value, which is the slope of the recurrence relation and defines the relative proportions of small and large earthquakes; *b*-values for large areas in much of the world are very often close to unity. The relationship must be truncated at an upper limit, $M_{max}$, which is the largest earthquake that the seismic source zone is considered capable of producing; this may be inferred from the dimensions of capable geological structures and empirical relations such as that shown in Figure 2.2 or simply by adding a small increment to the largest historical event in the earthquake catalogue. The typical form of the G–R relationship is illustrated in Figure 2.3.

For major faults, it is believed that the G–R recurrence relationship may not hold and that large magnitude earthquakes occur quasi-periodically with relatively little activity at moderate magnitudes. This leads to alternative models, also illustrated in Figure 2.3: if only large earthquakes occur, then the maximum magnitude model is adopted, whereas if there is also some activity in the smaller magnitude ranges then a model is adopted which combines a G–R relationship for lower magnitudes with the occurrence of larger characteristic earthquakes at higher rates than would be predicted by the extrapolation of the G–R relationship. The recurrence rate of characteristic events will generally be inferred from paleoseismological studies rather than from the earthquake catalogue, since such earthquakes are generally too infrequent to have multiple occurrences in catalogues. Highly recommended references on recurrence relationships include Reiter (1990), Utsu (1999) and McGuire (2004).

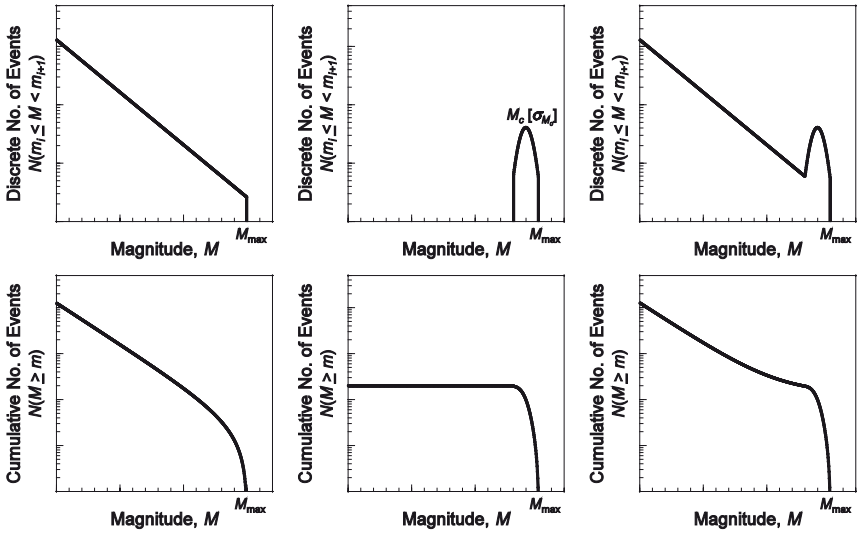



*Figure 2.3* Typical forms of earthquake recurrence relationships, shown in non-cumulative (*upper row*) and cumulative (*lower row*) forms. *From left to right*: Gutenberg–Richter model, maximum magnitude model, and characteristic earthquake model

## 2.3 Ground-motion characterisation and prediction

The crux of specifying earthquake actions for seismic design lies in estimating the ground motions caused by earthquakes. The inertial loads that are ultimately induced in structures are directly related to the motion of the ground upon which the structure is built. The present section is concerned with introducing the tools developed, and used, by engineering seismologists for the purpose of relating what occurs at the source of an earthquake to the ground motions that can be expected at any given site.

### *2.3.1 Accelerograms: recording and processing*

Most of the developments in the field of engineering seismology have spawned from the acquisition of high-quality recordings of strong ground motions using accelerographs. The first of these was not obtained until March 1933 during the Long Beach, California, earthquake but since that time thousands of strong-motion records have been acquired through various seismic networks across the globe. Prior to the acquisition of the first accelerograms, recordings of earthquake ground motions had been made using seismographs but the relatively high sensitivity of these instruments precluded truly strong ground motions from being recorded. It was not until the fine balance between creating a robust yet sensitive instrument

was achieved, through the invention of the accelerograph, that the field of engineering seismology was born.

Accelerographs currently come in two main forms: analogue and digital. The first instruments were analogue and, while modern instruments are now almost exclusively digital, many analogue instruments remain in operation and continue to provide important recordings of strong ground motions. The records obtained from both types of instrument must be processed before being used for most applications. Accelerographs simultaneously record accelerations with respect to time in three orthogonal directions (usually two in the horizontal plane and one vertical) yet, despite this configuration, it is never possible to fully capture the true three-dimensional motion of the ground as the instruments do not 'see' all of the ground motion. The acceleration time series that are recorded may be viewed in the frequency domain following a Fourier transform. Upon performing this operation and comparing the recorded Fourier amplitude spectrum with the spectrum associated with the background noise relevant for the instrument, one finds that all accelerographs have a finite bandwidth over which the signal-to-noise ratio is sufficiently high that one can be confident that the recorded motions are genuinely associated with earthquake-induced ground shaking. Beyond the lower and upper limits of this frequency range, and even at the peripheries if proper filtering is not performed, the record may become contaminated by noise. Boore and Bommer (2005) provide extensive guidance on how one should process accelerograms in order to ensure that the records are not contaminated. Boore and Bommer (2005) highlight the fundamental importance of applying an appropriate low-cut filter, particularly when using an accelerogram to obtain displacement spectral ordinates. However, the key issue is to identify the maximum period up to which the filtered data can be used reliably.

Akkar and Bommer (2006) explored the usable period ranges for processed analogue and digital accelerograms and concluded that for rock, stiff and soft soil sites, analogue recordings can be used for determining the elastic response at periods up to 0.65, 0.65 and 0.7 of the long-period filter cut-off respectively, whereas for digital recordings these limits increase to 0.8, 0.9 and 0.97. This issue is of great relevance as displacement-based design methods (Priestley et al., 2007), which rely upon the specification of long-period displacement spectral ordinates, become more widely adopted. An example of the influence of proper record processing is shown in Figure 2.4 in which both an analogue and a digital record are shown before and after processing – this example clearly shows how sensitive the displacement is to the presence of noise.

### *2.3.2 Ground-motion parameters*

Once an accelerogram has been recorded and properly processed, many quantitative parameters of the ground motion may be calculated (for a

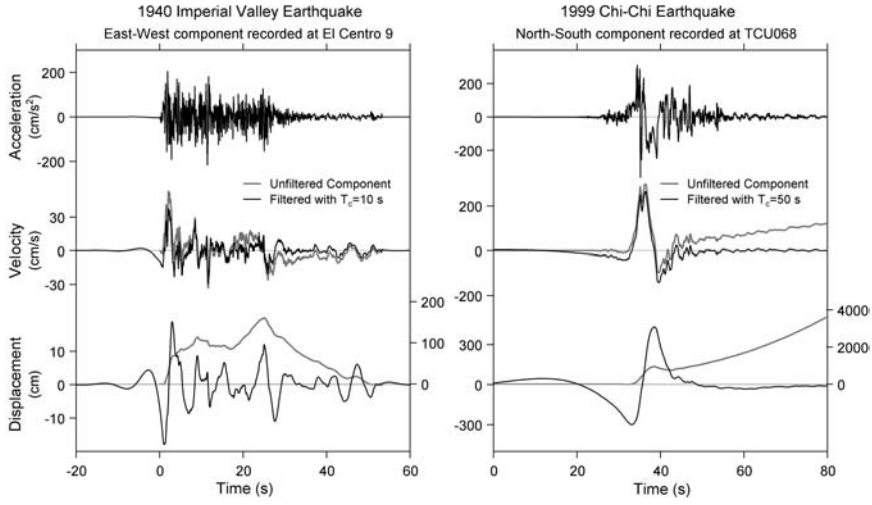



*Figure 2.4* Acceleration, velocity and displacement traces from analogue (left) and digital (right) recordings. Grey traces were obtained from the original records by removing the overall mean and the pre-event mean for the analogue and digital records respectively. The displacement axis labels for the unfiltered motions are given on the right-hand-side of the graphs. Modified from Boore and Bommer (2005)

description of many of these, see Kramer, 1996). Each of these parameters provides information about a different characteristic of the recorded ground motion. As far as engineering design is concerned, very few of these parameters are actually considered or used during the specification of design loads. Of those that may be calculated, peak ground acceleration, PGA, and ordinates of 5 percent damped elastic acceleration response spectra, $S_a(T,\xi=5\%)$, have been used by far the most frequently.

Figure 2.5 shows many of the possible ground-motion parameters that may be calculated for an individual earthquake record. Each one of these descriptive parameters provides some degree of information that may be used to help understand the demands imposed upon a structure. Although methodological frameworks are in place to simultaneously specify more than one ground-motion parameter (Bazzurro and Cornell, 2002) and to carry these parameters through to a structural analysis (Shome and Cornell, 2006), the additional complexity that is required for their implementation is excessively prohibitive without justifiable benefit in most cases. However, it is inevitable that earthquake engineers will seek to account for more characteristics of ground motions in the future.

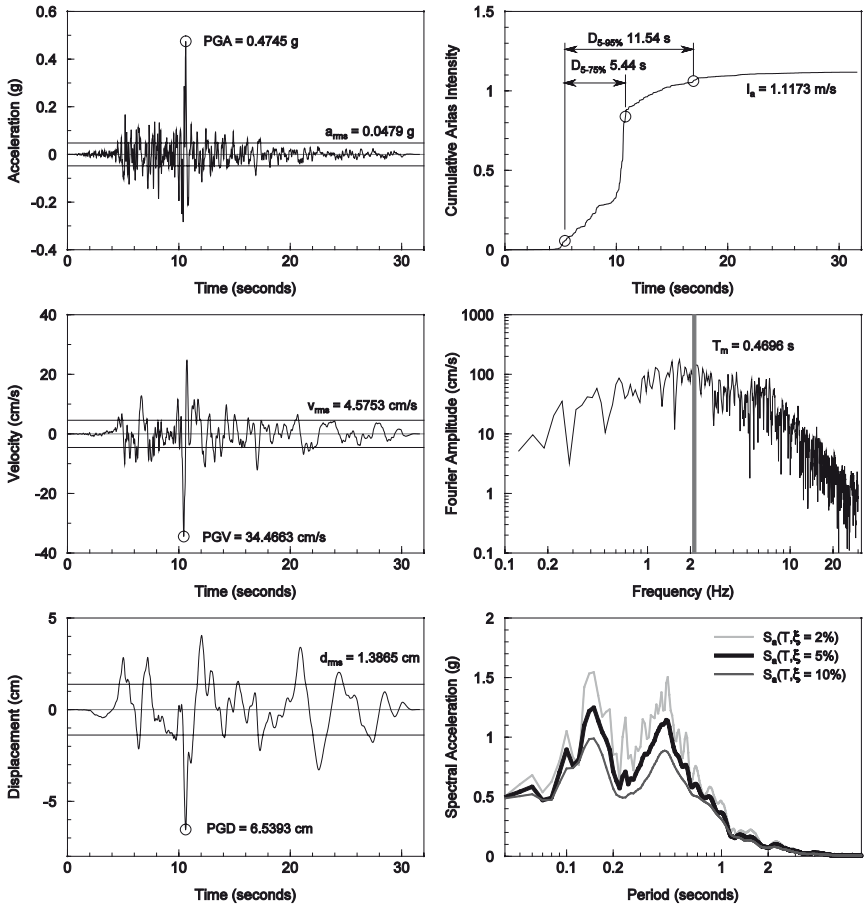



*Figure 2.5* Demonstration of the types of ground-motion parameters that may be calculated from a single record. The record in this case is the 020° component recorded during the 1994 Northridge earthquake at the Saturn St. station in Los Angeles. The three panels on the left show the acceleration, velocity, and displacement time-series as well as the peak and root-mean-square (*rms*) values. The panels on the right show, from top to bottom, a Husid plot of the build-up of Arias intensity as well as significant durations between 5–75% and 5–95% of the total Arias intensity, the Fourier amplitude spectrum along with the mean period and finally the acceleration response spectrum for damping levels of 2, 5, and 10% of critical

### *2.3.3 Empirical ground-motion prediction equations*

We have seen the numerous options that are available for describing characteristics of ground motions in the previous section. Now, given a large number of records, one can calculate values for any of these parameters and obtain a robust estimate of the correlation of these values with any other parameter relevant to this suite of records, such as the magnitude of the

earthquake from which they came. This type of reasoning is the basis for the development of empirical predictive equations for strong ground-motions. Usually, a relationship is sought between a suite of observed ground-motion parameters and an associated set of independent variables including a measure of the size of the earthquake, a measure of the distance from the source to the site, some classification of the style of faulting involved and some description of the geological and geotechnical conditions at the recording site. An empirical ground-motion prediction equation is simply a function of these independent variables that provides an estimate of the expected value of the ground-motion parameter in consideration as well as some measure of the distribution of values about this expected value.

Thus far the development of empirical ground-motion prediction equations has been almost exclusively focused upon the prediction of peak ground motions, particularly PGA and, to a far lesser extent, peak ground velocity (PGV), and ordinates of 5 percent damped elastic acceleration response spectra (Douglas, 2003; Bommer and Alarcón, 2006). Predictive equations have also been developed for most of the other parameters of the previous section, as well as others not mentioned, but as seismic design actions have historically been derived from PGA or $S_a(T)$ the demand for such equations is relatively weak. The performance of PGA (Wald et al., 1999) and, to a lesser extent, $S_a(T)$ (Priestley, 2003; Akkar and Özen, 2005), for the purposes of predicting structural damage has begun to be questioned. Improvements in the collaboration between engineering seismologists and structural earthquake engineers has prompted the emergence of research into what really are the key descriptors (such as inelastic spectral ordinates and elastic spectral ordinates for damping ratios other than 5 percent) of the ground motion that are of importance to structural response and to the assessment of damage in structures (Bozorgnia et al., 2006; Tothong and Cornell, 2006).

Regardless of the ground-motion measure in consideration, a ground-motion prediction equation can be represented as a generic function of predictor variables, $\mu(M,R,\boldsymbol{\theta})$, and a variance term, $\varepsilon\sigma_T$, as in Equation (2.1) where $y$ is the ground motion measure:

$$\log y = \mu(M,R,\boldsymbol{\theta}) + \varepsilon\sigma_T \tag{2.1}$$

Many developers of ground-motion prediction equations attempt to assign physical significance to the terms in the empirically derived function $\mu(M,R,\boldsymbol{\theta})$. In some cases it is possible to derive theoretical equations that may be used as the basis for selecting appropriate functional forms (e.g. Douglas, 2002). Although these theoretical considerations enable us to select appropriate functional forms, once the regression analysis has been conducted the actual values of regression coefficients should not be interpreted as having physical meaning as correlations of varying degrees always exist between the coefficients for different terms of the model.

For most ground-motion measures the values will increase with increasing magnitude and decrease with increasing distance. These two scaling effects form the backbone of prediction equations and many functional forms have been proposed to capture the variation of motions with respect to these two predictors (Douglas, 2003). For modern relationships distinctions are also made between ground motions that come from earthquakes having different styles of faulting, with reverse faulting earthquakes tending to generate larger ground motions than either strike-slip or normal faulting events (Bommer et al., 2003). Historically, account was also taken for site conditions by adding modifying terms similar to those used for the style-of-faulting effects – stiff soil sites have larger motions than rock, and soft soil sites have larger motions still. In Europe this use of dummy variables for generic site classes remains the adopted approach in the latest generation of prediction equations (Ambraseys et al., 2005; Akkar and Bommer, 2007a, 2007b), primarily due to the absence of more detailed site information. However, in the US, site response is now modelled using the average shear-wave velocity over the upper 30m, as introduced by Boore et al. (1997). Furthermore, the influence of non-linear site response, whereby weaker motions tend to be amplified more so that stronger motions due to the increased damping and reduced strength associated with the latter, is also taken into account (Abrahamson and Silva, 1997; Choi and Stewart, 2005). Figure 2.6 demonstrates the form of the non-linear site amplification functions adopted in two recent prediction equations developed as part of the Next Generation of Attenuation relations (NGA) project in the US. The

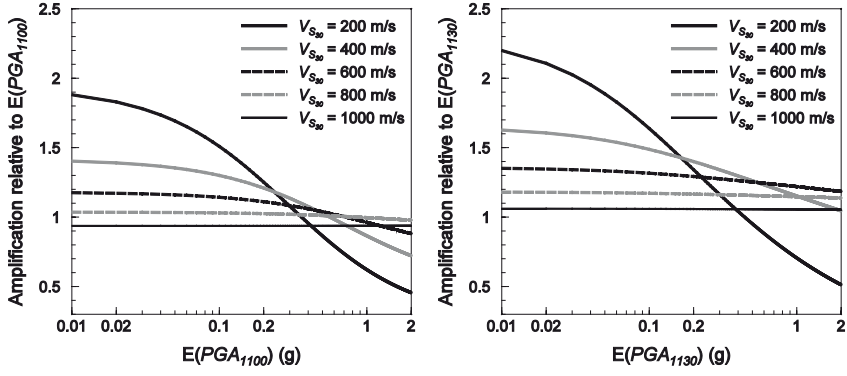



*Figure 2.6* Comparison of two nonlinear site response models for peak ground acceleration. Both models are from the NGA project with Abrahamson and Silva (2007) and Chiou and Youngs (2006) on the left and right respectively. The Abrahamson and Silva (2007) model shows amplification with respect to the expected value of PGA at a site with $V_{S30}$ = 1100 m/s while the Chiou and Youngs (2006) model shows the amplification with respect to expected motions on a site with $V_{S30}$ = 1130 m/s

difference in site amplification relative to rock for sites with differing shear-wave velocities and varying input rock ground motion is striking, with both models predicting de-amplification at strong levels of input rock motion.

In addition to the basic scaling of ground motions with magnitude, distance, site conditions, etc., there are additional situations that may result in modified ground motions that are commonly either omitted from developed equations or are later applied as correction factors to the base models. The most common examples include accounting for differences between sites located on the hangingwall or footwall of dip-slip fault sources (Abrahamson and Somerville, 1996; Chang et al., 2004), accounting for rupture directivity effects (Somerville et al., 1997; Abrahamson, 2000), including models for the velocity pulse associated with directivity effects (Bray and Rodriguez-Marek, 2004), basin effects (Choi et al., 2005) and topographic modifiers (Toshinawa et al., 2004). The most recent predictor variable to be included in prediction equations for peak ground motions and spectral ordinates is the depth to the top of the rupture (Kagawa et al., 2004; Somerville and Pitarka, 2006). Currently, none of these effects are incorporated into any predictive equations for ground motions in Europe, nor is any account made for non-linearity of site response. Again, this is primarily a result of the lack of well-recorded strong earthquakes in the region.

### *2.3.4 Ground-motion variability*

For any particular ground-motion record the total variance term given in Equation (2.1) may be partitioned into two components as in Equation (2.2):

$$\log y_{ij} = \mu(m_i, r_{ij}, \boldsymbol{\theta}_{ij}) + \delta_{e,i} + \delta_{a,ij} \tag{2.2}$$

The terms $\delta_{e,i}$ and $\delta_{a,ij}$ represent the inter-event and intra-event residuals respectively and quantify how far away from the mean estimate of $\log y_{ij}$ the motions from the *i*th event and the *j*th recording from the *i*th event are respectively (Abrahamson and Youngs, 1992). Alternatively, these terms may be expressed in terms of standard normal variates ($z_{e,i}$ and $z_{a,ij}$) and the standard deviations of the inter-event ($\tau$) and intra-event ($\sigma$) components, i.e. $\delta_{e,i} = z_{e,i}\tau$ and $\delta_{a,ij} = z_{a,ij}\sigma$. The total standard deviation for a predictive equation is obtained from the square root of the sum of the inter-event and intra-event variances, i.e. from $\sigma_T^2 = \tau^2 + \sigma^2$. Later, in Section 2.4 regarding PSHA, mention will be made of epsilon, $\varepsilon$, representing the number of total standard deviations from the median predicted ground motion. Often ground-motion modellers represent the terms $\delta_{e,i}$ and $\delta_{a,ij}$ by $\eta_i$ and $\varepsilon_{ij}$ respectively. Under this convention care must be taken to not confuse the epsilon, $\varepsilon$, with the intra-event residual, $\varepsilon_{ij}$, term – the two are related via the expression $\varepsilon = \left(\eta_i + \varepsilon_{ij}\right)/\sigma_T$, i.e. $\varepsilon = \left(\delta_{e,i} + \delta_{a,ij}\right)/\sigma_T$ using our notation.

Each of these components of variability may be modelled as functions of other parameters such as the magnitude of the earthquake (Youngs et al.,

1995), the average shear-wave velocity of the site (Abrahamson and Silva, 2007), or the amplitude of the ground motion (Campbell, 1997). Exactly how these components are calculated depends upon the regression methodology that is used to derive the equations. However, the most common approach is to adopt random effects procedures where the correlation between ground motions observed within any particular event is assumed to be the same across events and is equal to $\rho = \tau^2 / (\tau^2 + \sigma^2)$. This concept is shown schematically in Figure 2.7.

Many people think of ground-motion variability as a measure of the lack of fit of a particular predictive equation. However, in most cases it is better to think of a predictive equation as providing an estimate of the distribution of ground motions for a given set of predictor variables such as magnitude and distance. From this perspective, the real misfit of the model is related to how well the model's distribution represents the true distribution of ground motions rather than how large are the variance components. People tend not to like large variability, reasoning that this implies that we cannot predict this measure of ground motion with much certainty. However, this perspective

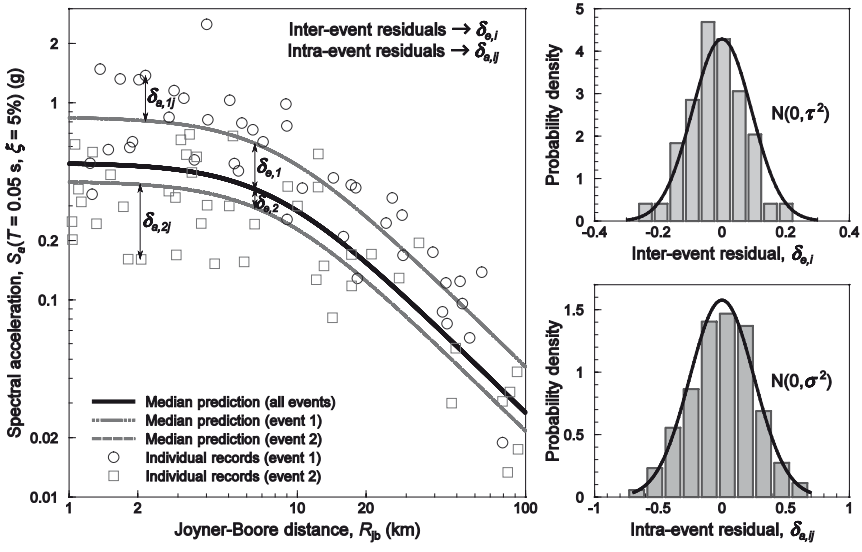



*Figure 2.7* Explanation of the variance components specified in ground-motion prediction equations. The left panel shows how the median prediction for an individual event may be higher or lower than the median prediction for all events – the inter-event residuals, $\delta_{e,i}$. About this median prediction for each event are random variations in ground motion – the intra-event residuals, $\delta_{a,ij}$. The histograms on the right show how both the inter- and intra-event residuals are normally distributed with zero means and variances of $\tau^2$ and $\sigma^2$ respectively. The median predictions are generated for an $M_w$ 6.5 earthquake with an $R_{JB}$ distance of 10km for strike-slip faulting and rock conditions using the equations of Akkar and Bommer (2007b); Figure based on a concept from Youngs et al. (1995)

is closely related to the paradigm that ground motions are ultimately predictable and that it is only through a result of inadequate modelling and incomplete knowledge that the apparent variability arises. If, on the other hand, one views ground motions as being inherently unpredictable (beyond a certain resolution) then one must view the variability not as a measure of the misfit, but rather as an additional part of the model that describes the range of observable ground motions given an event. Under this latter paradigm there is no reason to like or dislike a particular ground-motion measure simply because predictive equations for this measure have a broad distribution. The only rational basis for judging the importance of a ground-motion measure is to assess the ability of this measure to accurately predict structural response. That said, in most cases, less variability in the ground-motion estimate will translate into less variability in the response.

## 2.4 Seismic hazard analysis

The primary objective of engineering seismology is to enable seismic hazard analyses to be conducted. The two previous sections have provided most of the essential background required to understand seismic hazard analysis at its most basic level. As will soon be demonstrated, the mechanics of hazard analysis are relatively straightforward. However, a thorough understanding of the concepts laid out in the sections thus far, as well as many others, is a prerequisite for conducting a high-quality hazard analysis. Unfortunately, in current practice this prerequisite is all too often not met.

### *2.4.1 Probabilistic vs. deterministic approaches*

Bommer (2002) presents a comprehensive discussion of the differences and similarities between probabilistic and deterministic approaches to seismic hazard analysis. While the proponents of deterministic methods would like to perpetuate the conception that there is ongoing academic debate regarding which is the superior method, the truth of the matter is that deterministic seismic hazard analysis (DSHA) is simply a special case of probabilistic seismic hazard analysis (PSHA) in which only a small number of earthquake scenarios (combinations of magnitude, distance and epsilon) are considered. In contrast, in PSHA all possible scenarios that are deemed to be of engineering interest are considered (Abrahamson, 2006; Bommer and Abrahamson, 2006). Much of the discussion regarding PSHA and DSHA has focused on apparent issues that really stem from misunderstandings of the terminology that is often loosely used in PSHA. Bommer (2003) highlights some of the most common misunderstandings, particularly in relation to the treatment of uncertainty, and urges the proponents of DSHA to try to develop a consistent set of terminology for their approaches.

### *2.4.2 Basics of PSHA, hazard curves and return periods*

It is perhaps unfortunate that the mathematical formulation of PSHA is somewhat intimidating for some as the mechanics behind the framework are actually very simple. For example, imagine one wanted to know how often a particular level of some ground-motion measure is exceeded at a site. Now, suppose that there is a seismic source near this site that regularly generates earthquakes of a particular magnitude and further suppose that the rate at which these earthquakes occur may be quantified. Once this rate is obtained it may be combined with an estimate of how often the ground-motion level at the site is exceeded when this earthquake scenario occurs. For example, an event of magnitude *M* may occur once every six months and each time it does there is a 50 percent chance of exceeding a target ground motion – this target level is then exceeded by this scenario, on average, once every year. If one then considered another earthquake scenario, and repeated the above procedure, one would determine how often the ground-motion level in consideration was exceeded for this alternative scenario. If the first scenario resulted in an exceedance of the ground-motion level $\lambda_1$ times per year and the second $\lambda_2$ times per year, then for these two scenarios the ground-motion level is exceeded $\lambda_1 + \lambda_2$ times per year. This is how a PSHA is conducted: all one has to do to complete the process is to repeat the above steps for all of the possible earthquake scenarios that may affect the site, calculate the rates at which these scenarios result in ground motions above the target level, and then add them all up. Of course, it is not always straightforward to ascertain how often different earthquake scenarios occur, nor is it always obvious how to most appropriately determine the rate at which the ground motions are exceeded given these scenarios. However, none of these issues change the simplicity of the underlying framework that constitutes PSHA (Cornell, 1968, 1971). With this simple explanation firmly in mind, it is now timely to relate this to what is more commonly seen in the literature on this subject.

Formally, basic PSHA may be represented as in Equation (2.3) (Bazzurro and Cornell, 1999):

$$\lambda_{GM}(gm^*) = \sum_i \left\{ \int\int\int I\left[GM > gm^* | m,r,\varepsilon\right] \nu_i f_{M,R,E}(m,r,\varepsilon)_i \, dmdrd\varepsilon \right\} \quad (2.3)$$

where the capital letters represent random variables (*GM* = a chosen ground-motion parameter, *M* = magnitude, *R* = distance and *E* = epsilon) while their lower-case counterparts represent realisations of these random variables. The total rate at which earthquakes occur having a magnitude greater than the minimum considered for source *i* is denoted by $\nu_i$ (as this term is a constant for each source it may be taken outside of the triple integral, as is commonly done in many representations of this equation). The joint probability density function of magnitude, distance and epsilon is given by $f_{M,R,E}(m,r,\varepsilon)_i$ and $I\left[GM > gm^* | m,r,\varepsilon\right]$ is an indicator function equal

to one if $GM > gm^*$ and zero otherwise. Finally, and most importantly, $\lambda_{GM}(gm^*)$ is the total annual rate at which the target ground-motion value, $gm^*$, is exceeded. This is often the way that PSHA is presented in the literature; however, the nature of the joint probability density function in magnitude, distance and epsilon may be intractable for the non-cognoscenti and it is consequently worth spending some time to describe this key term of Equation (2.3). Using some basic concepts of probability theory we may decompose the joint probability density function (pdf) into more tractable parts as in Equation (2.4).

$$\underbrace{\nu_i f_{M,R,\mathrm{E}}(m,r,\varepsilon)_i}_{\substack{\text{how many times per year do}\\ \text{all possible levels of ground}\\ \text{motion occur from source } i?}} = \underbrace{\nu_i f_M\left(m|\mathbf{x}_{hyp}\right) f_{\mathbf{X}_{hyp}}\left(\mathbf{x}_{hyp}\right)}_{\substack{\text{how many times per year does an}\\ \text{earthquake of } M{=}m \text{ occur in source}\\ i \text{ with a hypocentre at } \mathbf{x}_{hyp}?}} \underbrace{f_R\left(r|m,\mathbf{x}_{hyp},\boldsymbol{\theta}_i\right)}_{\substack{\text{when this event occurs,}\\ \text{what sort of rupture does}\\ \text{it produce?}}} \underbrace{f_{\mathrm{E}}(\varepsilon)}_{\substack{\text{how likely are}\\ \text{the possible GM}\\ \text{values for this}\\ \text{scenario?}}} \tag{2.4}$$

Each of these components of the joint pdf, while already annotated, deserves some additional comment and explanation:

- $f_{\mathbf{X}_{hyp}}\left(\mathbf{x}_{hyp}\right)$ – the pdf for an event having a hypocentre equal to $\mathbf{x}_{hyp}$, where $\mathbf{x}_{hyp} = (longitude, latitude, depth)$ is any position within source $i$. A common assumption that is made, and that was made in Cornell's original presentation of PSHA, is that hypocentres are equally likely to occur anywhere within a seismic source. This assumption requires the least amount of information regarding the nature of activity for the seismic source.
- $f_M\left(m|\mathbf{x}_{hyp}\right)$ – the conditional pdf of magnitude given the hypocentral position. In many hazard analyses this term is not implicitly considered instead analysts simply take the previous assumption that earthquakes may occur with equal probability anywhere within a seismic source and also assume that these events may have the full range of magnitudes deemed possible for the source. In this case this term is not conditioned upon the hypocentre position and one simply recovers $f_M(m)$, the pdf of magnitude. However, some analysts may wish to address this problem more thoroughly and make alternative assumptions using analyses such as those of Somerville et al. (1999) and Mai et al. (2005). For example, it may be assumed that large earthquakes tend to have relatively deep hypocentres and the pdf may be modified accordingly. The pdf of magnitude is often assumed to follow a doubly-bounded exponential distribution for areal sources (Cornell and Vanmarcke, 1969); a modified form of the famous G–R equation (Gutenberg and Richter, 1944), and a characteristic distribution for fault sources (Schwartz and Coppersmith, 1984) as mentioned in Section 2.2. However, any distribution that relates the relative rates of occurrence of earthquakes of different sizes is permissible.
- $f_R\left(r|m,\mathbf{x}_{hyp},\boldsymbol{\theta}_i\right)$ – the conditional pdf of the distance measure used in the ground-motion prediction equation given the rupture surface of

the earthquake. The rupture surface depends upon the hypocentre, the size of the event and various other parameters encapsulated in $\boldsymbol{\theta}_i$ including the strike and dip of the fault plane (for fault sources), the depth boundaries of the seismogenic zone, the segment of the fault on which the rupture starts, etc. This term is important as it translates the assumptions regarding the potential locations of earthquakes into measures of distance that are appropriate for use in empirical prediction equations. Note that this term is necessarily different for each distance measure that is considered.

- $f_{\mathrm{E}}(\varepsilon)$ – the pdf of epsilon. It is important to note that this term is always simply the pdf of the standard normal distribution. For this reason it is not necessary to make this a conditional pdf with respect to anything else. Although standard deviations from ground-motion predictive equations may be dependent upon predictor variables such as magnitude, the pdf of epsilon remains statistically independent of these other variables (Bazzurro and Cornell, 1999).

Given this more complete representation of Equation (2.4) one must now also modify the integral to be expressed in terms of the relevant variables in Equation (2.3). In reality, this is not at all cumbersome as the integrals are not evaluated analytically anyway and all that is required is to discretise the range of possible parameter values and to determine the contribution to the hazard from each permissible set of these values. The general process alluded to in the introductory example and elaborated upon in the above is further represented schematically in Figure 2.8. In this figure, the method via which the probability that the ground motion exceeds the target level is represented two ways: 1) in a continuous manner through the use of the cumulative distribution function of the standard normal distribution, and 2) in a discrete manner whereby the range of epsilon values is discretised and the contribution to the total hazard is determined for each increment. Both of these approaches will give very similar answers but the latter approach offers advantages in terms of later representing the total hazard and also for the selection of acceleration time-histories to be used in seismic design (McGuire, 1995; Bazzurro and Cornell, 1999; Baker and Cornell, 2006).

Thus far we have only been concerned with calculating the rate at which a single target ground motion is exceeded. If we now select a series of target ground-motion levels and calculate the total rate at which each level is exceeded we may obtain a hazard curve, which is the standard output of a PSHA, i.e. a plot of $\lambda_{GM}(gm^*)$ against $gm^*$. Examples of the form of typical hazard curves are given in Figure 2.9 where the ground-motion measure in this case is PGA.

The curves shown in Figure 2.9 demonstrate the strong influence that the aleatory variability in the ground-motion prediction equation has on the results of a seismic hazard analysis. Bommer and Abrahamson (2006) have recently discussed this issue in detail, reviewing the historical development

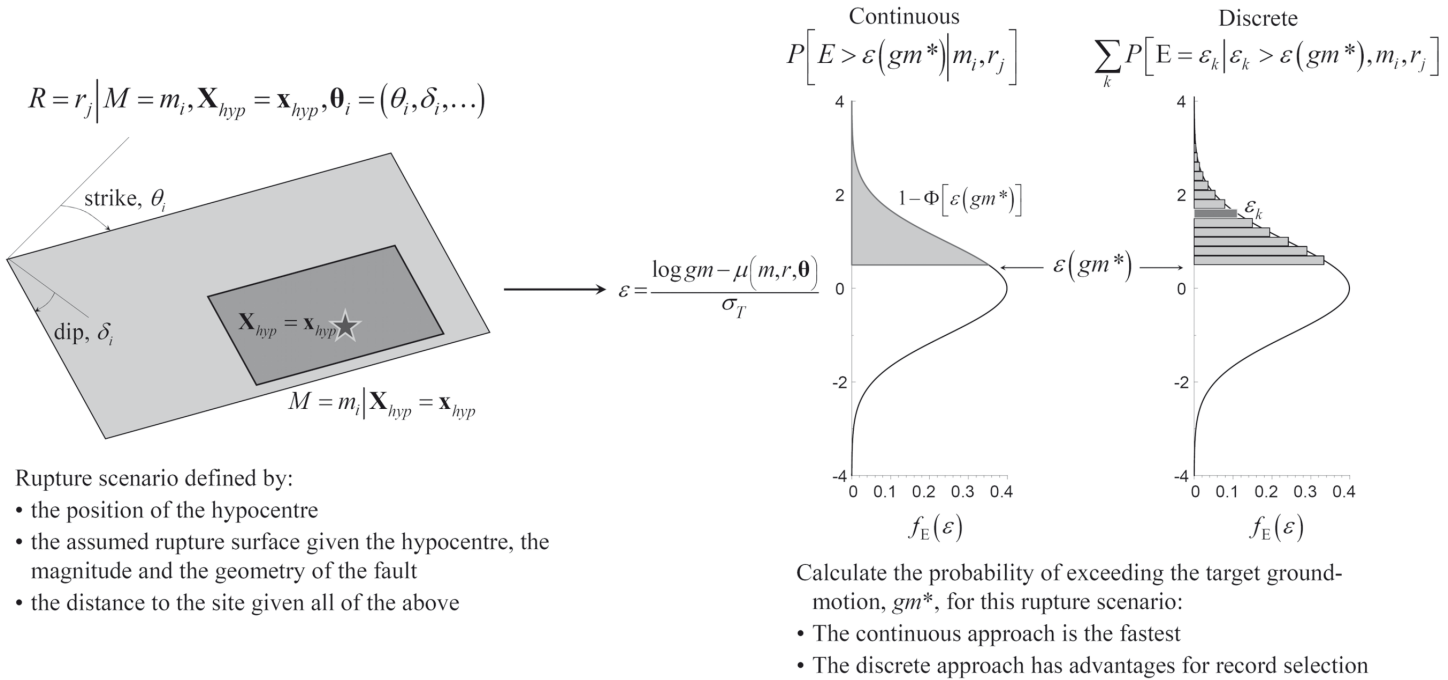



*Figure 2.8* Schematic representation of the PSHA process. On the left a portion (dark grey) of a fault source (light grey) ruptures about the hypocentral position given by the star. The geometry of this rupture surface depends upon various characteristics of the source as well as the magnitude of the earthquake. On the right, the probability of the target ground motion (gm*) being exceeded given this scenario is shown using two equivalent approaches

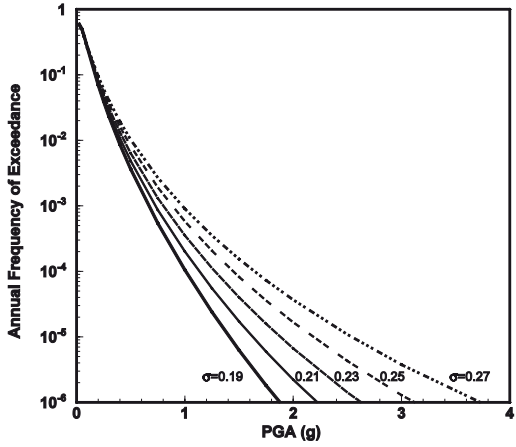



*Figure 2.9* Example hazard curves for a fictitious site. Each hazard curve is calculated using a different value for the total standard deviation for the ground-motion prediction equation; the values presented on the figure correspond to typical values for prediction equations using base 10 logarithms. From Bommer and Abrahamson (2006)

of PSHA as well as bringing to light the reason why modern hazard analyses often lead to higher hazard estimates. The answer to this question often lies in the inappropriate treatment, in early studies, of the aleatory variability in ground-motion prediction equations, with the worst practice being to simply ignore this component of PSHA in a manner akin to most deterministic hazard analyses.

Once a hazard curve has been developed the process of obtaining a design ground motion is straightforward. The hazard curve represents values of the average annual rate of exceedance for any given ground-motion value. Under the assumption that ground motions may be described by a Poisson distribution over time, the average rate corresponding to the probability of at least one exceedance within a given time period may be determined using Equation (2.5):

$$\lambda = \frac{-\ln(1-P)}{T} \tag{2.5}$$

For example, the ubiquitous, yet arbitrary (Bommer, 2006a) 475-year return period used in most seismic design codes throughout the world comes from specifying ground motions having a 10 percent chance of being exceeded at least once in any 50-year period. Inserting $P = 0.1$ and $T = 50$ years into Equation (2.5) yields the average annual rate corresponding to this condition, the reciprocal of which is the return period, that in this case is equal to 475 years. Note that because $\lambda$ is a function of both $P$ and $T$ there are infinitely many combinations of $P$ and $T$ that result in a 475-year return period. Once this design criterion is specified, one simply finds the level of

ground motion that corresponds to this rate on the hazard curve in order to obtain the design ground motion.

### *2.4.3 Uncertainty and logic trees*

The PSHA methodology laid out thus far is capable of accounting for all of the aleatory variability that exists within the process. However, there is another important component of uncertainty that must also be accounted for – the uncertainty associated with not knowing the applicability of available models. This type of uncertainty is known as epistemic uncertainty within the context of PSHA. Aleatory variability and epistemic uncertainty can further be partitioned into modelling and parametric components as is described in Table 2.1 (here the focus is on ground-motion modelling, but the concepts hold for any other component of the PSHA process). These distinctions are not just semantics, each aspect of the overall uncertainty must be treated prudently and each must be approached in a different manner. As implied in Table 2.1, the logic tree is the mechanism via which the epistemic uncertainty is accounted for in PSHA. As with any conceptual framework, practical application often reveals nuances that require further investigation and many such issues have recently been brought to light as a result of the PEGASOS project (Abrahamson et al., 2002). Aspects such as

*Table 2.1* Proper partitioning of the total uncertainty associated with ground-motion modelling into distinct modelling and parametric components of both aleatory variability and epistemic uncertainty. From Bommer and Abrahamson (2007)

| | *Aleatory Variability* | *Epistemic Uncertainty* |
|---|---|---|
| Modelling | Variability based on the misfit between model predictions and observed ground motions (unexplained randomness) $\sigma_m$ | Uncertainty that the model is correct. Relative weights given to alternative credible models. (Alternative estimates of median ground motions and $\sigma_m$) |
| Parametric | Variability based on propagating the aleatory variability of additional source parameters through a model (understood randomness) $\sigma_p$ | Uncertainty that the distribution of the additional source parameters is correct. Relative weights given to alternative models of the parameter distributions. (Alternative estimates of $\sigma_p$ for each model) |
| Total | $\sqrt{\sigma_m^2 + \sigma_p^2}$ (the modelling and parametric variabilities are uncorrelated) | Logic trees for both components (the modelling and parametric logic trees will be correlated) |

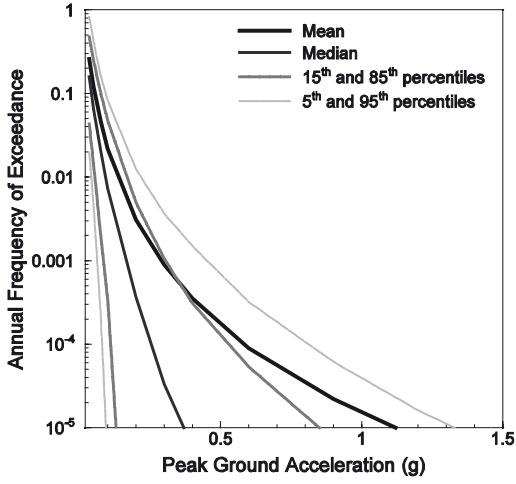



*Figure 2.10* Example of a suite of PGA hazard curves obtained from a logic tree for a fictitious site

model selection, model compatibility and the overall sensitivity of PSHA to logic-tree branches for ground-motion models have all been addressed (Scherbaum et al., 2004a, 2004b; Sabetta et al., 2005; Bommer et al., 2005; Scherbaum et al., 2005, Beyer and Bommer, 2006; Cotton et al., 2006) as have issues associated with how the outputs (suites of hazard curves) of the logic tree are harvested (Abrahamson and Bommer, 2005; McGuire et al., 2005; Musson, 2005).

Figure 2.10 shows a suite of hazard curves, including the mean, the median and four other fractiles, obtained from a hypothetical PSHA conducted using a logic tree. This figure highlights two important aspects associated with the outputs of logic trees: 1) the range of ground-motion values corresponding to a given hazard level may vary considerably across fractiles, and 2) as one moves to longer return periods the difference between the mean and median hazard curves may become very large. The first aspect reinforces the importance of taking into account different interpretations of the regional seismotectonics as well as different models or approaches to estimating ground motions (see Table 2.1), while the second aspect demonstrates that one must be clear about how the design ground motion is to be specified as the results corresponding to the mean hazard and various fractiles may differ considerably.

### *2.4.4 Hazard maps and zonations*

For the purpose of representing seismic hazard over a broad spatial region, separate hazard analyses are conducted at a sufficiently large number of points throughout the region such that contours of ground-motion parameters

may be plotted. Such maps could be used directly for the specification of seismic design loads, but what is more common is to take these maps and to identify zones over which the level of hazard is roughly consistent. If the hazard map is produced with a high enough spatial resolution, then changes in hazard over small distances are always relatively subtle. However, for zonation maps there will often be locations where small differences in position will mean the difference between being in one zone or another with the associated possibility of non-trivial changes in ground motions. Under such a circumstance regulatory authorities must take care in defining the boundaries of the relevant sources; common practice is to adjust the zone limits to coincide with political boundaries in order to prevent ambiguity.

Recently, with the introduction of EC8 looming, a comparative analysis of the state of national hazard maps within sixteen European countries was undertaken (García-Mayordomo et al., 2004). The study highlights the numerous methodological differences that exist between hazard maps developed for various countries across Europe. Many of the differences that exist do so as a result of the differing degrees of seismicity that exist throughout the region, but some of these differences are exacerbated as a result of parochialism despite geological processes not being concerned with man-made or political boundaries. There are, however, other examples of efforts that have been made to develop consistent seismic hazard maps over extended regions. The two primary examples of such efforts are the GSHAP (Giardini et al., 1999) and SESAME (Jiménez et al., 2001) projects that integrate national hazard information in order to develop continental or global-scale hazard maps. These examples of regional hazard maps may be viewed at the following URLs: the GSHAP map at www.seismo.ethz.ch/GSHAP/ and the SESAME map at http://wija.ija.csic.es/gt/earthquakes/.

For truly robust hazard maps to be developed the best of both approaches must be drawn upon. For example, ground-motion prediction equations developed from large regional datasets, such as those of Ambraseys et al. (2005) or Akkar and Bommer (2007a, 2007b), are likely to be more robust when applied within individual countries than those developed from a more limited national dataset (Bommer, 2006b). Furthermore, ground-motion modellers working in low-seismicity regions, such as in most parts of Europe, often make inferences regarding the scaling of ground motions with magnitude on the basis of the small magnitude data that is available to them. In doing so, researchers find apparent regional differences that exist when making comparisons between their data and the predictions of regional ground-motion models derived predominantly from recordings of larger magnitude earthquakes (i.e. Marin et al., 2004). Recent work has shown that such inferences may be unfounded and that particular care must be taken when extrapolating empirical ground-motion models beyond the range of magnitudes from which they were derived (Bommer et al., 2007). On the other hand, the detailed assessments of seismogenic sources that are often included for national hazard map and zonation purposes are often

not fully incorporated into regional studies where the spatial resolution is relatively poor.

## 2.5 Elastic design response spectra

Most seismic design is based on representing the earthquake actions in the form of an equivalent static force applied to the structure. These forces are determined from the maximum acceleration response of the structure under the expected earthquake-induced ground shaking, which is represented by the acceleration response spectrum. The starting point is an elastic response spectrum, which is subsequently reduced by factors that account for the capacity of the structure to dissipate the seismic energy through inelastic deformations. The definition of the elastic response spectrum and its conversion to an inelastic spectrum are presented in Chapter 3; this section focuses on how the elastic design response spectra are presented in seismic design codes, with particular reference to EC8.

The purpose of representing earthquake actions in a seismic design code such as EC8 is to circumvent the necessity of carrying out a site-specific seismic hazard analysis for every engineering project in seismically active regions. For non-critical structures it is generally considered sufficient to provide a zonation map indicating the levels of expected ground motions throughout the region of applicability of the code and then to use the parameters represented in these zonations, together with a classification of the near-surface geology, in order to construct the elastic design response spectrum at any given site.

### *2.5.1 Uniform hazard spectra and code spectra*

The primary output from a PSHA is a suite of hazard curves for response spectral ordinates for different response periods. A design return period is then selected – often rather arbitrarily as noted previously (e.g. Bommer, 2006a) – and then the response parameter at this return period is determined at each response period and used to construct the elastic response spectrum. A spectrum produced in this way, for which it is known that the return period associated with several response periods is the same, is known as a uniform hazard spectrum (UHS) and it is considered an appropriate probabilistic representation of the basic earthquake actions at a particular location. The UHS will often be an envelope of the spectra associated with different sources of seismicity, with short-period ordinates controlled by nearby moderate-magnitude earthquakes and the longer-period part of the spectrum dominated by larger and more distant events. As a consequence, the motion represented by the UHS may not be particularly realistic, if interpreted as being associated with some design scenario, and this becomes an issue when the motions need to be represented in the form of acceleration time-histories, as discussed in Section 2.6. If the only parameter of interest to

the engineer is the maximum acceleration that the structure will experience in its fundamental mode of vibration, regardless of the origin of this motion or any other of its features (such as duration), then the UHS is a perfectly acceptable format for the representation of the earthquake actions. In the following discussion it is assumed that the UHS is a desirable objective.

Until the late 1980s, seismic design codes invariably presented a single zonation map, usually for a return period of 475 years, showing values of a parameter that in essence was the PGA. This value was used to anchor a spectral shape specified for the type of site, usually defined by the nature of the surface geology, and thus obtain the elastic design spectrum. In many codes, the ordinates could also be multiplied by an importance factor, which would increase the spectral ordinates (and thereby the effective return period) for the design of structures required to perform to a higher level under the expected earthquake actions, either because of the consequences of damage (e.g. large occupancy or toxic materials) or because the facility would need to remain operational in a post-earthquake situation (e.g. fire station or hospital).

A code spectrum constructed in this way would almost never be a UHS. Even at zero period, where the spectral acceleration is equal to PGA, the associated return period would often not be the target value of 475 years since the hazard contours were simplified into zones with a single representative PGA value over the entire area. More importantly, this spectral construction technique did not allow the specification of seismic loads to account for the fact that the shape of response spectrum varies with earthquake magnitude as well as with site classification (Figure 2.11), with the result that even if the PGA anchor value was associated with the exact design return period, it is very unlikely indeed that the spectral ordinates at different periods would have the same return period (McGuire, 1977). Consequently, the

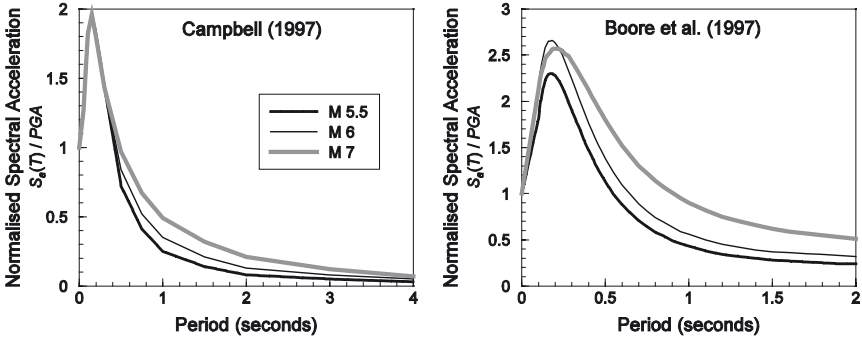



*Figure 2.11* Median predicted response spectra, normalised to PGA, for a rock site at 10 km from earthquakes of different magnitudes from the Californian equations of Campbell (1997) and Boore et al. (1997)

objective of a UHS is not met by anchoring spectral shapes to the zero-period acceleration.

Various different approaches have been introduced in order to achieve a better approximation to the UHS in design codes, generally by using more than one parameter to construct the spectrum. The 1984 Colombian and 1985 Canadian codes both introduced a second zonation map for PGV and in effect used PGA to anchor the short-period part of the spectrum and PGV for the intermediate spectral ordinates. Since the zonation maps for the two parameters were different, the shape of the resulting elastic design spectrum varied from place to place, reflecting the influence of earthquakes of different magnitude in controlling the hazard. The 1997 edition of the Uniform Building Code (UBC) used two parameters, $C_a$ and $C_v$, for the short- and intermediate-period portions of the spectra (with the subscripts indicating relations with acceleration and velocity) but curiously the ratio of the two parameters was the same in each zone with the result that the shape of the spectrum did not vary except with site classification.

In the Luso-Iberian peninsula, seismic hazard is the result of moderate-magnitude local earthquakes and large-magnitude earthquakes offshore in the Atlantic. The Spanish seismic code handles their relative influence by anchoring the response spectrum to PGA but then introducing a second set of contours, of a factor called the 'contribution coefficient', K, that controls the relative amplitude of the longer-period spectral ordinates; high values of K occur to the west, reflecting the stronger influence of the large offshore events. The Portuguese seismic code goes one step further and simply presents separate response spectra, with different shapes, for local and distant events. The Portuguese code is an interesting case because it effectively abandons the UHS concept, although it is noteworthy that the return period of the individual spectra is 975 years, in effect twice the value of 475 years associated with the response spectra in most European seismic design codes.

Within the drafting committee for EC8 there were extensive discussions about how the elastic design spectra should be constructed, with the final decision being an inelegant and almost anachronistic compromise to remain with spectral shapes anchored only to PGA. In order to reduce the divergence from the target UHS, however, the code introduced two different sets of spectral shapes (for different site classes), one for the higher seismicity areas of southern Europe (Type 1) and the other for adoption in the less active areas of northern Europe (Type 2). The Type 1 spectrum is in effect anchored to earthquakes of magnitude close to $M_s$ ~7 whereas the Type 2 spectrum is appropriate to events of $M_s$ 5.5 (e.g. Rey et al., 2002). (See Figure 2.12.) At any location where the dominant earthquake event underlying the hazard is different from one or other of these magnitudes, the spectrum will tend to diverge from the target 475-year UHS, especially at longer periods.

The importance of the vertical component of shaking in terms of the demand on structures is a subject of some debate (e.g. Papazoglou and

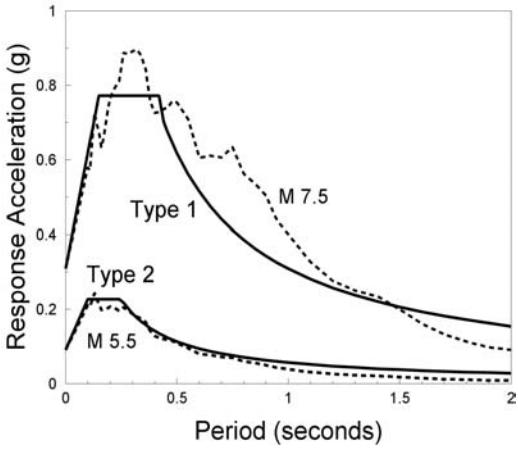



*Figure 2.12* Median predicted spectral ordinates from the European ground-motion prediction equations of Ambraseys et al. (1996) for rock sites at 10 km from small and large magnitude events, compared with the EC8 Type 1 and 2 rock spectra anchored to the median predicted PGA

Elnashai, 1996) but there are certain types of structures and structural elements, such as cantilever beams, for which the vertical loading could be important. Many seismic codes do not provide a vertical spectrum at all and those that do generally specify it as simply the horizontal spectrum with the ordinates reduced by one-third. Near-source recordings have shown that the short-period motions in the vertical direction can actually exceed the horizontal motion, and it has also been clearly established that the shape of the vertical response spectrum is very different from the horizontal components of motion (e.g. Bozorgnia and Campbell, 2004). In this respect, EC8 has some merit in specifying the vertical response spectrum separately rather than through scaling of the horizontal spectrum; this approach was based on the work of Elnashai and Papazoglou (1997). As a result, at least for a site close to the source of an earthquake, the EC8 vertical spectrum provides a more realistic estimation of the vertical motion than is achieved in many seismic design codes (Figure 2.13).

### *2.5.2 The influence of near-surface geology on response spectra*

The fact that locations underlain by soil deposits generally experience stronger shaking than rock sites during earthquakes has been recognised for many years, both from field studies of earthquake effects and from recordings of ground motions. The influence of surface geology on ground motions is now routinely included in predictive equations. The nature of the near-surface deposits is characterised either by broad site classes, usually defined by ranges of shear-wave velocities ($V_s$), or else by the explicit value of the $V_s$ over the uppermost 30 m at the site. Figure 2.14 shows the influence

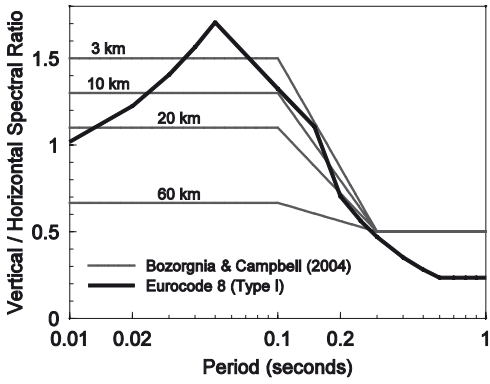



*Figure 2.13* The implied vertical-to-horizontal (V/H) ratio of the Type 1 spectra for soil sites in Eurocode 8 compared with the median ratios predicted by Bozorgnia and Campbell (2004) for soil sites at different distances from the earthquake source

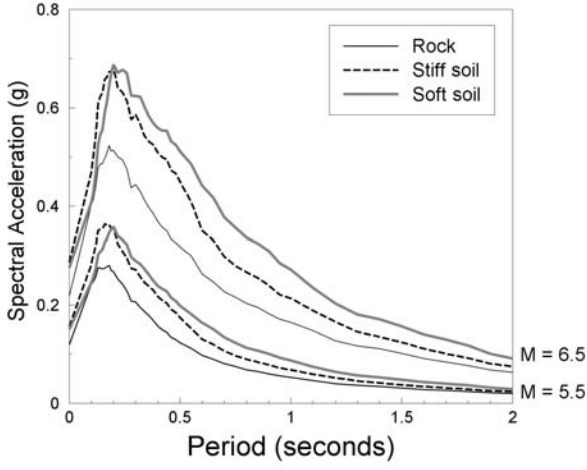



*Figure 2.14* Median predicted spectral ordinates from the equations of Bommer et al. (2003) for different site classes at 10 km from strike-slip earthquakes of Ms 5.5 and Ms 6.5 as indicated

of different soil classes on the predicted spectral ordinates from European attenuation equations and for two different magnitudes.

Code specifications of spectral shapes for different site classes generally reflect the amplifying effect of softer soil layers, resulting in increased spectral ordinates for such sites, and the effect on the frequency content, which leads to a wider constant acceleration plateau and higher ordinates at intermediate and long response periods. The EC8 Type 1 spectra for different site classes are illustrated in Figure 2.15.

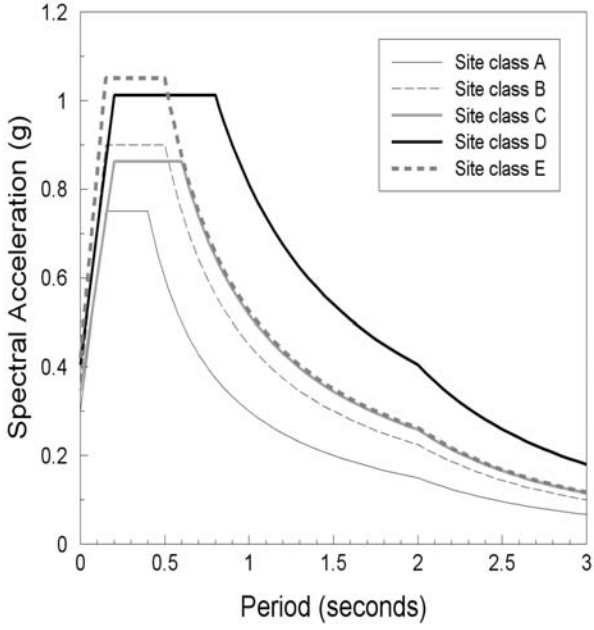



*Figure 2.15* Type 1 spectra from Eurocode 8 for different site classes, anchored for a PGA in rock of 0.3g

As previously mentioned in Section 2.3.3, the response of a soil layer to motions propagating upwards from an underlying rock layer depends on the strength of the incoming rock motions as a result of the non-linear response of soil (see Figure 2.6). The greater the shear strain in the soil, the higher the damping and the lower the shear modulus of the soil, whence weak input motion tends to be amplified far more than stronger shaking. As a rule-of-thumb, non-linear soil response can be expected to be invoked by rock accelerations beyond 0.1–0.2 g (Beresnev and Wen, 1996). In recent years, ground-motion prediction equations developed for California have included the influence of soil non-linearity with greater ratios of soft soil to rock motions for magnitude–distance combinations resulting in weaker rock motions than those for which strong shaking would be expected (Figure 2.6). Attempts to find evidence of non-linearity in the derivation of empirical equations using European data have not been conclusive, probably due to the lack of good-quality data on site characteristics and the relatively small number of recordings of genuinely strong motion in the European area (Akkar and Bommer, 2007a: Bommer et al., 2008). Some design codes, most notably the 1997 edition of UBC, have included the effects of soil non-linearity in the specification of amplification factors for spectral loads. The implied amplification factors for rock motions from a few attenuation equations and design regulations for intermediate-period spectral response ordinates are compared in Figure 2.16.

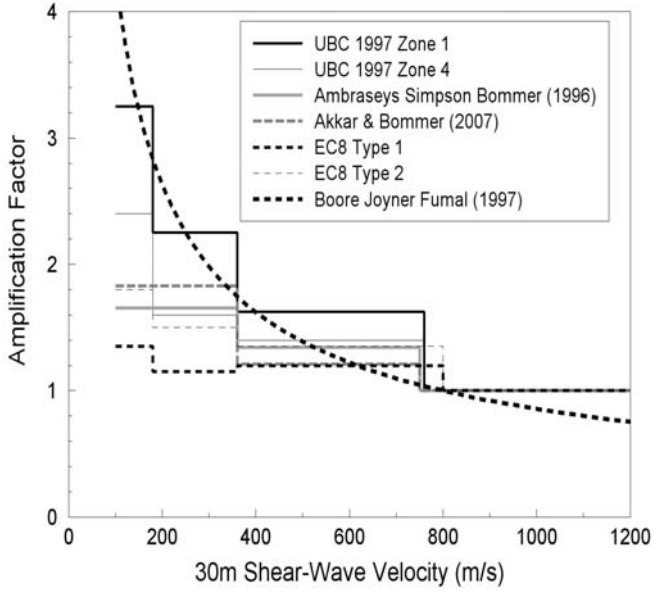



*Figure 2.16* Amplification factors for 1.0-second spectral acceleration for different site shear-wave velocity values relative to rock motions; for Boore et al. (1997), rock has been assigned a shear-wave velocity of 800 m/s

A number of interesting observations can be made regarding the curves in Figure 2.16, the first being the wide range of proposed amplification factors for different sites, especially those overlain by soft soil layers. The second observation that can be made is that amplification factors assigned to broad site classes will often be rather crude approximations to those obtained for specific sites where the $V_s$ profile is known. The UBC spectra for Zone 1 (low hazard) and Zone 4 (high hazard) have quite different amplification factors, with non-linear soil response leading to much lower soil amplification in the high hazard zone. A similar feature seems to be captured by the Type 1 and Type 2 spectra from EC8.

### *2.5.3 Displacement response spectra*

In recent years, exclusively force-based approaches to seismic design have been questioned, both because of the poor correlation between transient accelerations and structural damage, and also because for post-yield response the forces effectively remain constant and damage control requires limitation of the ensuing displacements. Most of the recently introduced performance-based design methodologies can be classified as being based either on displacement modification techniques or else equivalent linearisation. FEMA-440 (ATC, 2005) presents both approaches, allowing the designer to select the one felt to be more appropriate, acknowledging, in effect, that

opinions are currently divided as to which is the preferred approach. EC8 also envisages the potential application of these two general approaches to the computation of displacement demand, and provides guidelines on the appropriate seismic actions in informative annexes A and B.

The equivalent linearisation approach to displacement-based seismic design requires the characterisation of the design motions in the form of elastic displacement response spectra. The inelastic deformation of the structure is reflected in the longer effective period of vibration, which requires the spectral ordinates to be specified for a wider range of periods than has normally been the case in design codes. The dissipation of energy through hysteresis is modelled through an increased equivalent damping. Based on a proposal by Bommer et al. (2000), the EC8 acceleration spectrum can be transformed to a displacement spectrum by multiplying the ordinates by $T^2/4\pi^2$, where T is the natural period of vibration. The critical question is at which period should the constant displacement plateau begin, which, as can be discerned in Figure 2.17, was set at 2 seconds for the Type 1 spectrum in EC8. This value has since been recognised to be excessively small; the corner period of the spectrum increases with earthquake magnitude, and for the larger events expected in Europe (M ~7) the period could be expected to be in the order of 10 seconds (e.g. Bommer and Pinho, 2006). The inadequacy of the corner period, $T_D$, being set at 2 seconds has recently been demonstrated by new European equations for the prediction of response spectral ordinates up to 4 seconds (Akkar and Bommer, 2007b). Figure 2.17 compares the displacement spectra from EC8 with those from Akkar and Bommer (2007b)

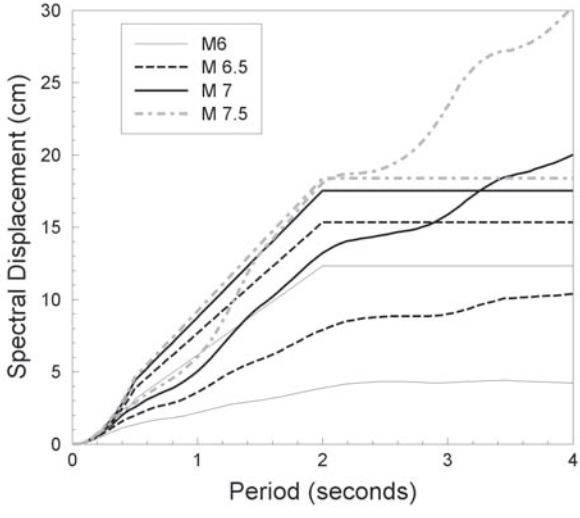



*Figure 2.17* Comparison of 5%-damped displacement response spectra for a stiff soil site at 10 km from earthquakes of different magnitudes from Akkar and Bommer (2007b) with the EC8 Type 1 spectra for the same conditions, anchored to the PGA value predicted by the equation presented in the same study

for stiff soil sites at 10 km from earthquakes of different magnitudes. In each case, the EC8 spectra have been anchored to the predicted median PGA from the equation of Akkar and Bommer (2007b). A number of interesting observations can be made, the first being that the fixed spectral shape of EC8 is unable to capture the influence of varying magnitude, with the result that the short-period spectral ordinates are severely over-predicted for the smaller magnitudes. The second observation is that the fixed corner period of 2 seconds is clearly inadequate and the dependence of this parameter on magnitude is very clear; for earthquakes of greater than magnitude 6, the corner period is longer than 2 seconds, and for the larger events greater than 4 seconds.

The spectral ordinates with damping ratios higher than the nominal 5 percent of critical are obtained by multiplying the spectral ordinates at intermediate periods by a factor, derived by Bommer et al. (2000), that is a function only of the target damping level. These factors replaced those in an early draft of EC8, and many other factors have since been proposed in the literature and in other seismic design codes. Bommer and Mendis (2005) explored the differences amongst the various factors and found that the amount of reduction of the 5 percent-damped ordinates required to match the ordinates at higher damping levels increases with the duration of the ground motion. Since the Type 2 spectrum in EC8 corresponds to relatively small magnitude earthquakes, which will generate motions of short duration, it was proposed that the existing scaling equation in EC8 to obtain spectral displacements, *SD*, and different damping values, $\xi$ :

$$SD(\xi) = SD(5\%)\sqrt{\frac{10}{5+\xi}} \tag{2.6}$$

should be retained for the Type 1 spectrum, whereas for the Type 2 spectrum this should be replaced by the following expression derived by Mendis and Bommer (2006):

$$SD(\xi) = SD(5\%)\sqrt{\frac{35}{30+\xi}} \tag{2.7}$$

## 2.6 Acceleration time-histories

Although seismic design invariably begins with methods of analysis in which the earthquake actions are represented in the form of response spectra, some situations require fully dynamic analyses to be performed and in these cases the earthquake actions must be represented in the form of acceleration time-histories. Such situations include the design of safety-critical structures, highly irregular buildings, base-isolated structures, and structures designed for a high degree of ductility. For such projects, the simulation of structural response using a scaled elastic response spectrum is not considered appropriate and suites of accelerograms are required for the dynamic analyses. The guidance given in the majority of seismic design codes on the selection and

scaling of suites of acceleration time-histories for such purposes is either very inadequate or else so prescriptive as to make it practically impossible to identify realistic accelerograms that meet the specified criteria (Bommer and Ruggeri, 2002). A point that cannot be emphasised too strongly is that time-histories should never be matched to a uniform hazard spectrum, but rather to a spectrum corresponding to a particular earthquake scenario. In the case of codes, this may be difficult since the code generally provides an approximation, albeit a crude one, to the UHS and offers no possibility to generate a disaggregated event-specific spectrum.

There are a number of options for obtaining suites of acceleration time-histories for dynamic analysis of structures, including the generation of spectrum-compatible accelerograms from white noise, a method that is now widely regarded as inappropriate because the resulting signals are so unlike earthquake ground motions. The most popular option is to use real accelerograms, which can be selected either on the basis of having response spectra similar, at least in shape, to the elastic design spectrum, or else matching an earthquake scenario in terms of magnitude, source-to-site distance and possibly also site geology (Bommer and Acevedo, 2004). The latter approach, however, is generally not feasible in the context of seismic design code applications, because information regarding the underlying earthquake scenarios is usually not available to the user. Selecting records from earthquakes of appropriate magnitude is only an issue if the duration of the shaking is considered an important parameter in determining the degree of seismic demand that the records impose, which is an issue of ongoing debate in the technical literature (Hancock and Bommer, 2006).

Once a suite of records has been selected, whether on the basis of the spectral shape or an earthquake scenario, the next question for the design engineer to address is how many records are needed. Most of the seismic design codes that address this issue, including EC8, specify that a minimum of 3 records should be used, and that if less than 7 records are used then the maximum structural response must be used as the basis for design, whereas if 7 or more time-histories are employed then the average structural response can be used. The use of the maximum inelastic response obtained from dynamic analyses may never be appropriate since the input accelerograms will in some sense have been adjusted to approximate to the elastic design spectrum, which, if determined from a probabilistic hazard assessment, will already include the influence of the ground-motion variability. The largest dynamic response will probably correspond to a record that is somewhat above the target spectrum, and in a sense the ground-motion variability is therefore being taken into account twice.

The key question then becomes how many records are required to obtain a stable estimate of the mean inelastic response, which will depend on how the records are adjusted so that their spectral ordinates approximate to those of the elastic design spectrum: the more closely the adjusted records match the target elastic design spectrum, the fewer analyses will be needed.

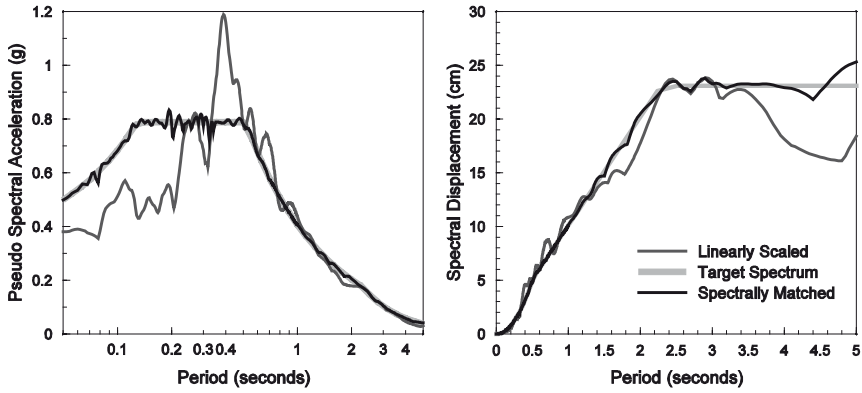



*Figure 2.18* Comparison of the difference between scaled and matched spectra. Modified from Hancock et al. (2006)

Options include scaling the records to match the design spectrum at the natural period of the structure or scaling to match or exceed the average ordinates over a period range around this value, the extended range accounting for both the contributions to the response from higher modes and also for the elongation of response period due to inelastic deformations. Scaling the records in amplitude is legitimate given that whilst the amplitude of the motion is highly dependent on distance – especially within a few tens of kilometres from the source – the shape of the response spectrum is actually rather insensitive to distance over the range of distances of normal engineering interest (Bommer and Acevedo, 2004). Although scaling limits of a factor of 2 were proposed at one time, and became embedded in the 'folklore' of engineering practice, much larger scaling factors can be applied (Watson-Lamprey and Abrahamson, 2006). Adjusting records by scaling the time axis, however, is to be avoided.

An alternative to linear scaling of the records is to make adjustments, using Fast Fourier Transform or wavelet transformations, to achieve a spectral shape that approximates to that of the target design spectrum (Bommer and Acevedo, 2004). The most elegant way to achieve this is using the wavelet transformation, which minimises the alteration of the original accelerogram but at the same time can achieve a very good spectral match (Hancock et al., 2006). An example of the difference between linearly scaling a record and matching spectra through wavelet transformations is given in Figure 2.18.

## 2.7 Conclusions and recommendations

For most engineering projects in seismic zones, the earthquake loading can be represented by an acceleration response spectrum, modified to account for inelastic deformation of the structure. The elastic design spectrum will

most frequently be obtained through probabilistic seismic hazard analysis, which provides the most rational framework for handling the large uncertainties associated with the models for seismicity and ground-motion prediction. Most seismic design codes present zonation maps and response spectra derived probabilistically, even though these design loads are often associated with a return period whose origin is a fairly arbitrary selection, and the resulting response spectrum is generally a poor approximation to the concept of a uniform hazard spectrum.

The main advantage that seismic codes offer in terms of earthquake loading is allowing the engineer to bypass the very considerable effort, expense and time required for a full site-specific hazard assessment. This should not, however, be interpreted to mean that the engineer should not be aware of the assumptions underlying the derivation and presentation of the earthquake actions, as well as their limitations.

EC8 is unique amongst seismic design codes in that it is actually a template for a code rather than a complete set of definitions of earthquake actions for engineering design. Each member state of the European Union will have to produce its own National Application Document, including a seismic hazard map showing PGA values for the 475-year return period, select either the Type 1 or Type 2 spectrum and, if considered appropriate, adapt details of the specification of site classes and spectral parameters. Interestingly, although the stated purpose of EC8 is harmonisation of seismic design across Europe, there could well be jumps in the level of seismic design loads across national borders as currently there is no official project for a community-wide hazard zonation map.

Although there are a number of innovative features in EC8 with regards to the specification of design earthquake actions, such as the separate definition of the vertical response spectrum and the provision of input for displacement-based design approaches, the basic mechanism for defining the horizontal elastic design spectrum is outdated and significantly behind innovations in recent codes from other parts of the world, most notably the US. It is to be hoped that the first major revision of EC8, which should be carried out 5 years after its initial introduction, will modify the spectral construction technique, incorporating at least one more anchoring parameter in addition to PGA. Several other modifications are also desirable, including to the long-period portion of the displacement spectrum and the adjustment for damping levels higher than 5 percent of critical.

Although seismic codes provide useful guidance for the earthquake-resistant design of many structures, there are cases where the code specifications will not be sufficient. Examples may include the following:

- projects located in proximity to active faults for which near-source directivity effects associated with the fault rupture need to be considered in the design (such effects are considered in the 1997 edition of UBC but not in EC8);

- projects in areas where active faults are known or suspected to be present, and for which surface displacements would be a critical consideration for the performance of the structure;
- projects on sites with deep and/or very soft soil deposits, for which the effects of the near-surface geology on the ground motions are unlikely to be well captured by the simplified site classes and corresponding spectral shapes in the code;
- projects for which return periods significantly longer than the nominal 475 years are considered appropriate;
- any project for which fully dynamic analysis is required (since the EC8 guidelines on preparing time-history input for such analyses is lacking in many respects).

If it is judged that a site-specific seismic hazard assessment is required, then this needs to be planned carefully and in good time – it should be considered as an integral part of the site investigation, and scheduled and budgeted accordingly. If investigations of active geological faults are to be part of the assessment, then the time and budget requirements are likely to increase very significantly.

Seismic hazard analysis is a highly specialised discipline that is constantly evolving and advancing, and in which a great deal of expert judgement is required. Nowadays it is fairly straightforward to obtain geological maps, satellite imagery, earthquake catalogues, published ground-motion prediction equations and software for performing hazard calculations, in many cases from the Internet and free of charge. The art of seismic hazard analysis, however, lies not primarily in accessing and analysing these resources but rather in judging their completeness and quality, and assessing the uncertainties associated with the data and the applicability of models to the specific region and site under consideration.

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# 3 Structural analysis

*M.S. Williams*

## 3.1 Introduction

This chapter presents a brief account of the basics of dynamic behaviour of structures; the representation of earthquake ground motion by response spectra; and the principal methods of seismic structural analysis.

Dynamic analysis is normally a two-stage process: we first estimate the dynamic properties of the structure (natural frequencies and mode shapes) by analysing it in the absence of external loads, and then use these properties in the determination of earthquake response.

Earthquakes often induce non-linear response in structures. However, most practical seismic design continues to be based on linear analysis. The effect of non-linearity is generally to reduce the seismic demands on the structure, and this is normally accounted for by a simple modification to the linear analysis procedure.

A fuller account of this basic theory can be found in Clough and Penzien (1993) or Craig (1981).

## 3.2 Basic dynamics

This section outlines the key properties of structures that govern their dynamic response, and introduces the main concepts of dynamic behaviour with reference to single-degree-of-freedom (SDOF) systems.

### *3.2.1 Dynamic properties of structures*

For linear dynamic analysis, a structure can be defined by three key properties: its stiffness, mass and damping. For non-linear analysis, estimates of the yield load and the post-yield behaviour are also required. This section will concentrate on the linear properties, with non-linearity introduced later on.

First, consider how mass and stiffness combine to give oscillatory behaviour. The mass, $m$, of a structure, measured in kg, should not be confused with its weight, $mg$, which is a force measured in N. Stiffness, $k$, is the constant of proportionality between force and displacement, measured in

N/m. If a structure is displaced from its equilibrium position then a restoring force is generated equal to stiffness × displacement. This force accelerates the structure back towards its equilibrium position. As it accelerates, the structure acquires momentum (equal to mass × velocity), which causes it to overshoot. The restoring force then reverses sign and the process is repeated in the opposite direction, so that the structure oscillates about its equilibrium position. The behaviour can also be considered in terms of energy – vibrations involve repeated transfer of strain energy into kinetic energy as the structure oscillates around its unstrained position.

In addition to the above, all structures gradually dissipate energy as they move, through a variety of internal mechanisms that are normally grouped together and known as damping. Without damping, a structure, once set in motion, would continue to vibrate indefinitely. There are many different mechanisms of damping in structures. However, analysis methods are based on the assumption of linear viscous damping, in which a viscous dashpot generates a retarding force proportional to the velocity difference across it. The damping coefficient, $c$, is the constant of proportionality between force and velocity, measured in Ns/m. Whereas it should be possible to calculate values of $m$ and $k$ with some confidence, $c$ is a rather nebulous quantity that is difficult to estimate. It is far more convenient to convert it to a dimensionless parameter $\xi$, called the damping ratio:

$$\xi = \frac{c}{2\sqrt{km}} \tag{3.1}$$

$\xi$ can be estimated based on experience of similar structures. In civil engineering it generally takes a value in the range 0.01 to 0.1, and an assumed value of 0.05 is widely used in earthquake engineering.

In reality, all structures have distributed mass, stiffness and damping. However, in most cases it is possible to obtain reasonably accurate estimates of the dynamic behaviour using *lumped parameter* models, in which the structure is modelled as a number of discrete masses connected by light spring elements representing the structural stiffness and dashpots representing damping.

Each possible displacement of the structure is known as a *degree of freedom*. Obviously a real structure with distributed mass and stiffness has an infinite number of degrees of freedom, but in lumped-parameter idealisations we are concerned only with the possible displacements of the lumped masses. For a complex structure the finite element method may be used to create a model with many degrees of freedom, giving a very accurate representation of the mass and stiffness distributions. However, the damping is still represented by the approximate global parameter, $\xi$.

### *3.2.2 Equation of motion of a linear SDOF system*

An SDOF system is one whose deformation can be completely defined by a single displacement. Obviously most real structures have many degrees of freedom, but a surprisingly large number can be modelled approximately as SDOF systems.

Figure 3.1 shows an SDOF system subjected to a time-varying external force, $F(t)$, which causes a displacement, $x$. The movement of the mass generates restoring forces in the spring and damper as shown in the free body diagram on the right.

By Newton's second law, resultant force = mass × acceleration:

$$F(t) - kx - c\dot{x} = m\ddot{x} \qquad \text{or} \qquad m\ddot{x} + c\dot{x} + kx = F(t) \tag{3.2}$$

where each dot represents one differentiation with respect to time, so that $\dot{x}$ is the velocity and $\ddot{x}$ is the acceleration. This is known as the equation of motion of the system. An alternative way of coming to the same result is to treat the term $m\ddot{x}$ as an additional internal force, the *inertia force*, acting on the mass in the opposite direction to the acceleration. The equation of motion is then an expression of *dynamic equilibrium* between the internal and external forces:

inertia force + damping force + stiffness force = external force

In an earthquake, there is no force applied directly to the structure. Instead, the ground beneath it is subjected to a (predominantly horizontal) time-varying motion as shown in Figure 3.2.

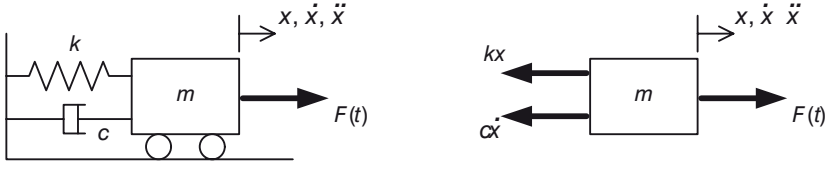



*Figure 3.1* Dynamic forces on a mass-spring-damper system

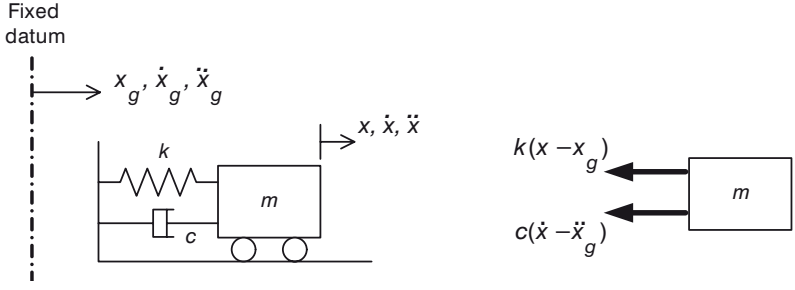



*Figure 3.2* Mass-spring-damper system subjected to base motion

In the absence of any external forces Newton's second law now gives:

$$-k(x-x_g)-c(\dot{x}-\dot{x}_g)=m\ddot{x} \qquad \text{or} \qquad m\ddot{x}+c(\dot{x}-\dot{x}_g)+k(x-x_g)=0 \quad (3.3)$$

Note that the stiffness and damping forces are proportional to the relative motion between the mass and the ground, while the inertia force is proportional to the absolute acceleration experienced by the mass. Let the relative displacement between the mass and the ground be $y = x - x_g$, with similar expressions for velocity and acceleration. The equation of motion can then be written as:

$$m\ddot{y}+c\dot{y}+ky=-m\ddot{x}_g \quad (3.4)$$

So a seismic ground motion results in a similar equation of motion to an applied force, but in terms of motion relative to the ground and with the forcing function proportional to the ground acceleration.

Before looking at solutions to this equation we will look at the free vibration case (i.e. no external excitation). This will provide us with the essential building blocks for solution of the case when the right-hand side of Equations (3.2) or (3.4) is non-zero.

### *3.2.3 Free vibrations of SDOF systems*

Consider first the theoretical case of a simple mass-spring system with no damping and no external force. The equation of motion is simply:

$$m\ddot{x}+kx=0 \quad (3.5)$$

If the mass is set in motion by giving it a small initial displacement $x_0$ from its equilibrium position, then it undergoes free vibrations at a rate known as the natural frequency. The solution to Equation (3.5) is:

$$x(t)=x_0\cos\omega_n t \quad \text{where} \quad \omega_n=\sqrt{\frac{k}{m}} \quad (3.6)$$

$\omega_n$ is called the circular natural frequency (measured in rad/s). It can be thought of as the angular speed of an equivalent circular motion, such that one complete revolution of the equivalent motion takes the same time as one complete vibration cycle. More easily visualised parameters are the natural frequency, $f_n$ (measured in cycles per second, or Hz) and the natural period, $T_n$ (the time taken for one complete cycle, measured in s). These are related to $w_n$ by:

$$f_n=\frac{\omega_n}{2\pi}=\frac{1}{2\pi}\sqrt{\frac{k}{m}} \quad (3.7)$$

$$T_n = \frac{1}{f_n} = 2\pi\sqrt{\frac{m}{k}} \tag{3.8}$$

Next consider the vibration of an SDOF system with damping included but still with no external force, again set in motion by applying an initial displacement $x_0$. The equation of motion is:

$$m\ddot{x} + c\dot{x} + kx = 0 \tag{3.9}$$

The behaviour of this system depends on the relative magnitudes of $c$, $k$ and $m$. If $c = 2\sqrt{km}$ the system is said to be critically damped and will return to its equilibrium position without oscillating. In general $c$ is much smaller than this, giving an underdamped system. Critical damping is useful mainly as a reference case against which others can be scaled to give the damping ratio defined earlier in Equation (3.1):

$$\xi = \frac{c}{2\sqrt{km}}$$

For an underdamped system the displacement is given by:

$$x = x_0 e^{-\xi\omega_n t} \cos\sqrt{1-\xi^2}\,\omega_n t \tag{3.10}$$

An example is given in Figure 3.3, which shows the response of SDOF systems with natural period 1 s and different damping ratios, when released from an initial unit displacement. This damped response differ, from the underdamped case in two ways: first the oscillations are multiplied by an exponential decay term, so that they die away quite quickly; second, the natural frequency has been altered by the factor $\sqrt{1-\xi^2}$. However for practical values

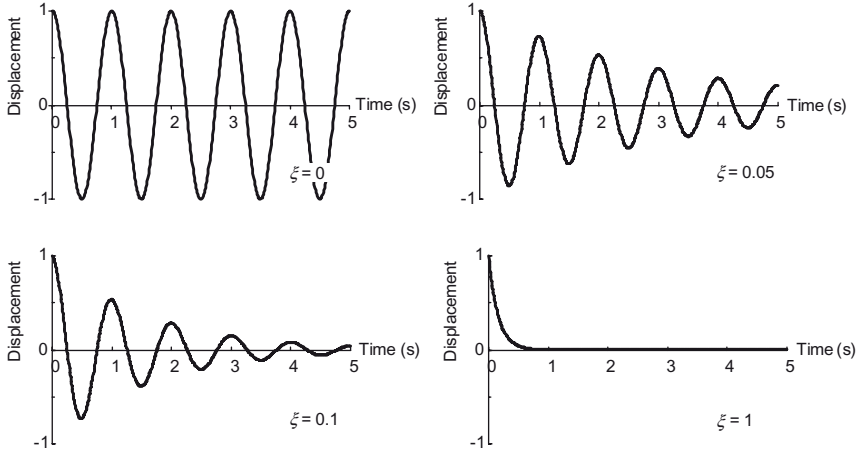



*Figure 3.3* Effect of damping on free vibrations

of damping this factor is very close to unity. It is therefore acceptable to neglect damping when calculating natural frequencies.

Using the relationships between $\omega_n$, $\xi$, $m$, $c$ and $k$, the Equation (3.4) can conveniently be written as:

$$\ddot{y} + 2\xi\omega_n\dot{y} + \omega_n^2 y = -\ddot{x}_g \tag{3.11}$$

### *3.2.4 Response to a sinusoidal base motion*

Suppose first of all that the ground motion varies sinusoidally with time at a circular frequency, $\omega$, with corresponding period $T = 2\pi / \omega$:

$$x_g = X_g \sin \omega t \tag{3.12}$$

Of course, a real earthquake ground motion is more complex, but this simplification serves to illustrate the main characteristics of the response.

Equation (3.11) can be solved by standard techniques and the response computed. Figure 3.4 shows the variation of structural acceleration, $\ddot{x}$, with time for a structure with a natural period of 0.5 s and 5 per cent damping, for a variety of frequencies of ground shaking. Three regimes of structural response can be seen:

a. The ground shaking is at a much slower rate than the structure's natural oscillations, so that the behaviour is *quasi-static*; the structure simply moves with the ground, with minimal internal deformation and its absolute displacement amplitude is approximately equal to the ground displacement amplitude.
b. When the ground motion period and natural period are similar, resonance occurs and there is a large dynamic amplification of the motion. In this region the stiffness and inertia forces at any time are approximately equal and opposite, so that the main resistance to motion is provided by the damping of the system.
c. If the ground motion is much faster than the natural oscillations of the structure then the mass undergoes less motion than the ground, with the spring and damper acting as vibration absorbers.

The effect of the loading rate on the response of an SDOF structure is summarised in Figure 3.5, for different damping levels. Here the peak absolute displacement of the structure $X$ (normalised by the peak ground displacement, $X_g$) is plotted against the ratio of the natural period $T_n$ to the period of the sinusoidal loading $T$.

The same three response regimes are evident in this figure, with the structural motion equal to the ground motion at the left-hand end of the graph, then large resonant amplifications at around $T_n/T = 1$, and finally very low displacements when $T_n/T$ is large. At pure resonance ($T_n/T = 1$)

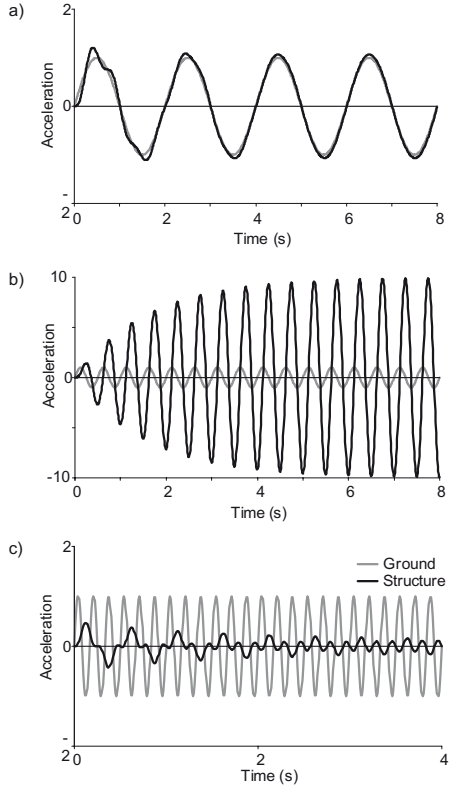



*Figure 3.4* Acceleration (in arbitrary units) of a 0.5 s natural period SDOF structure subject to ground shaking at a period of: a) 2 s, b) 0.5 s, c) 0.167 s

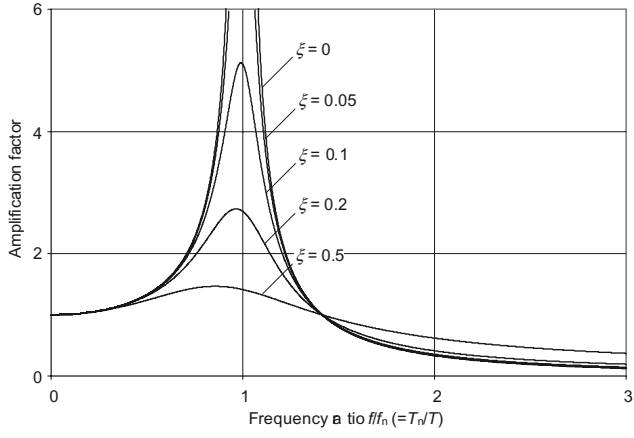



*Figure 3.5* Displacement amplification curves for an SDOF structure subject to sinusoidal ground shaking

the ratio $X/X_g$ roughly equals $1/(2\xi)$. The peak displacement at resonance is thus very sensitive to damping, and is infinite for the theoretical case of zero damping. For a more realistic damping ratio of 0.05, the displacement of the structure is around ten times the ground displacement.

This illustrates the key principles of dynamic response, but it is worth noting here that the dynamic amplifications observed under real earthquake loading are rather lower than those discussed above, both because an earthquake time-history is not a simple sinusoid, and because it has a finite (usually quite short) duration.

## 3.3 Response spectra and their application to linear structural systems

We now go on to consider the linear response of structures to realistic earthquake time-histories. An earthquake can be measured and represented as the variation of ground acceleration with time in three orthogonal directions (N-S, E-W and vertical). An example, recorded during the 1940 El Centro earthquake in California, is shown in Figure 3.6. Obviously, the exact nature of an earthquake time-history is unknown in advance, will be different for every earthquake, and indeed will vary over the affected region due to factors such as local ground conditions, epicentral distance etc.

### *3.3.1 Earthquake response*

The time-domain response to an earthquake ground motion can be determined by a variety of techniques, all of which are quite mathematically complex. For example, in the Duhamel's integral approach, the earthquake

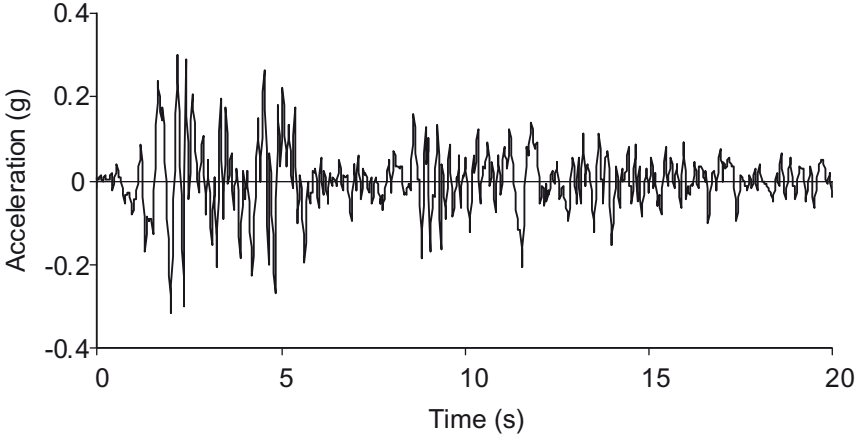



*Figure 3.6* Accelerogram for 1940 El Centro earthquake (N-S component).

record is treated as a sequence of short impulses, and the time-varying responses to each impulse are summed to give the total response.

Although the method of evaluation is rather complex, the behaviour under a general dynamic load can be quite easily understood by comparison with the single-frequency, sinusoidal load case discussed in Section 3.2.4. In that case, we saw that large dynamic amplifications occur if the loading period is close to the natural period of the structure. Irregular dynamic loading can be thought of as having many different components at different periods. Often the structure's natural period will lie within the band of periods contained in the loading. The structure will tend to pick up and amplify those components close to its own natural period just as it would with a simple sinusoid. The response will therefore be dominated by vibration at or close to the natural period of the structure. However, because the loading does not have constant amplitude and is likely to have only finite duration, the amplifications achieved are likely to be much smaller than for the sinusoidal case. An example is shown in Figure 3.7, where a 0.5 s period structure is subjected to the El Centro earthquake record plotted in Figure 3.6. The earthquake contains a wide band of frequency components, but it can be seen that the 0.5 s component undergoes a large amplification and dominates the response.

### *3.3.2 Response spectrum*

The response of a wide range of structures to a particular earthquake can be summarised using a response spectrum. The time domain response of

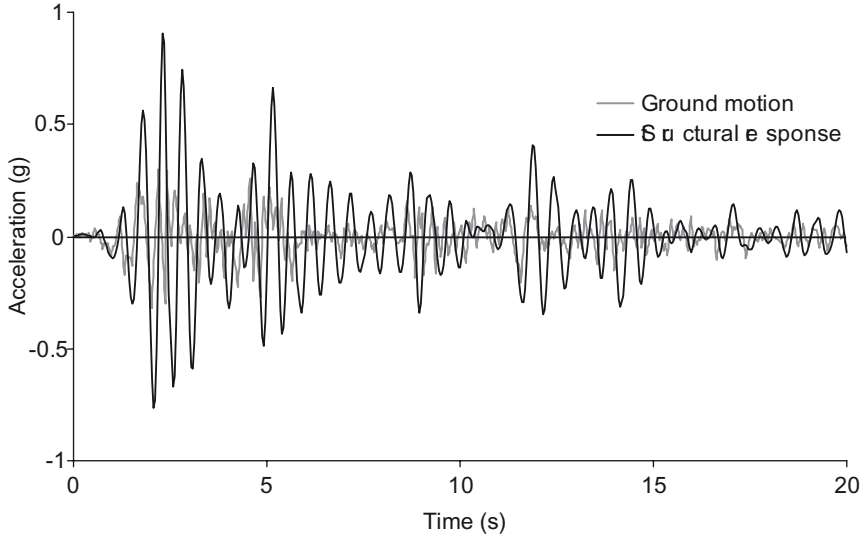



*Figure 3.7* Acceleration of 0.5 s period SDOF structure subject to the El Centro (N-S) earthquake record

numerous SDOF systems having different natural periods is computed, and the maximum absolute displacement (or acceleration, or velocity) achieved is plotted as a function of the SDOF system period. If desired, a range of curves can be plotted for SDOF systems having different damping ratios.

So the response spectrum shows *the peak response of an SDOF structure to a particular earthquake, as a function of the natural period and damping ratio of the structure*. For example, Figure 3.8 shows the response spectrum for the El Centro (N-S) accelerogram in Figure 3.6, for SDOF structures with 5 per cent damping.

A key advantage of the response spectrum approach is that earthquakes that look quite different when represented in the time domain may actually contain similar frequency contents, and so result in broadly similar response spectra. This makes the response spectrum a useful design tool for dealing with a future earthquake whose precise nature is unknown. To create a design spectrum, it is normal to compute spectra for several different earthquakes, then envelope and smooth them, resulting in a single curve that encapsulates the dynamic characteristics of a large number of possible earthquake accelerograms.

Figure 3.9 shows the elastic response spectra defined by EC8 (2004). EC8 specifies two categories of spectra: type 1 for areas of high seismicity (defined as $M_s > 5.5$), and type 2 for areas of moderate seismicity ($M_s \leq 5.5$). Within each category, spectra are given for five different soil types: A – rock; B – very dense sand or gravel, or very stiff clay; C – dense sand or gravel, or stiff clay; D – loose-to-medium cohesionless soil, or soft-to-firm cohesive soil; E – soil profiles with a surface layer of alluvium of thickness 5–20 m. The vertical axis is the peak, or *spectral acceleration* of the elastic structure,

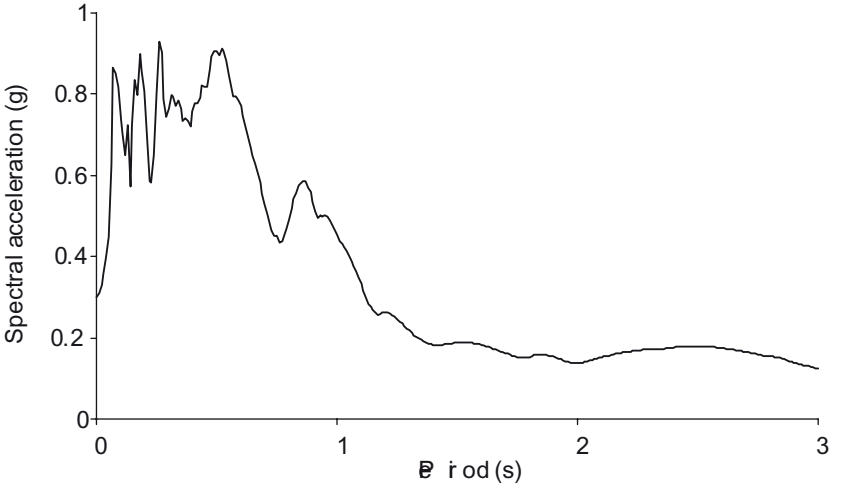



*Figure 3.8* 5% damped response spectrum for 1940 El Centro earthquake (N-S component)

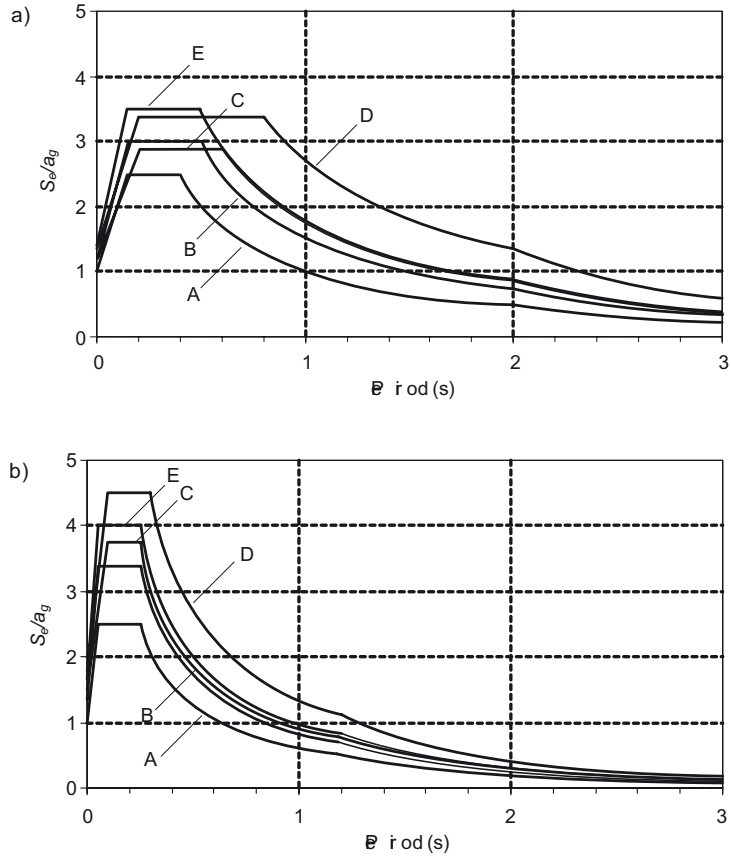



*Figure 3.9* EC8 5% damped, elastic spectra, a) type 1, b) type 2

denoted by $S_e$, normalised by $a_g$, the design peak ground acceleration on type A ground. The spectra are plotted for an assumed structural damping ratio of 5 per cent. See EC8 Cl. 3.2.2.2 for mathematical definitions of these curves and EC8 Table 3.1 for fuller descriptions of ground types A–E.

As with the harmonic load case, there are three regimes of response. Very stiff, short period structures simply move with the ground. At intermediate periods there is dynamic amplification of the ground motion, though only by a factor of 2.5–3, and at long periods the structure moves less than the ground beneath it. In the region of the spectra between $T_B$ and $T_C$ the spectral acceleration is constant with period. The region between $T_C$ and $T_D$ represents constant velocity and beyond $T_D$ is the constant displacement region.

It can be seen that in the high seismicity events (Type 1 spectra) the spectral amplifications tend to occur at longer periods, and over a wider period range, than in the moderate seismicity events. It is also noticeable

that the different soil types give rise to varying levels of amplification of the bedrock motions, and affect the period range over which amplification occurs. The EC8 values for $T_D$ have caused some controversy – it has been argued that the constant velocity region of the spectra should continue to higher periods, which would result in a more onerous spectral acceleration for long-period (e.g. very tall) structures.

### *3.3.3 Application of response spectra to elastic SDOF systems*

In a response spectrum analysis of an SDOF system, we generally wish to determine the force to which the structure is subjected, and its maximum displacement. We start by estimating the natural period, $T_n$, and damping ratio, $\xi$, The peak (spectral) acceleration, $S_e$, experienced by the mass can then be read directly from the response spectrum. Now the maximum acceleration in a vibrating system occurs when it is at its point of extreme displacement, at which instant the velocity (and therefore the damping force) is zero. The peak force is then just equal to the inertia force experienced by the mass:

$$F = mS_e \tag{3.13}$$

This must be in dynamic equilibrium with the stiffness force developed within the structure. If we define the spectral displacement, $S_D$, as the peak absolute displacement corresponding to the spectral acceleration, $S_e$, then we must have $kS_D = mS_e$, which, using the relationships between mass, stiffness and natural period given in Equation (3.8), leads to:

$$S_D = \frac{F}{k} = mS_e \cdot \frac{T_n^2}{4\pi^2 m} = \frac{S_e T_n^2}{4\pi^2} \tag{3.14}$$

Note that, while the force experienced depends on the mass, the spectral acceleration and displacement do not – they are functions only of the natural period and damping ratio.

It should be remembered that the spectral acceleration is *absolute* (i.e. it is the acceleration of the mass relative to the ground plus the ground acceleration, hence proportional to the inertia force experienced by the mass), but the spectral displacement is the displacement of the mass *relative* to the ground (and hence proportional to the spring force).

While elastic spectra are useful tools for design and assessment, they do not account for the inelasticity that will occur during severe earthquakes. In practice, energy absorption and plastic redistribution can be used to reduce the design forces significantly. This is dealt with in EC8 by the modification of the elastic spectra to give *design spectra* $S_d$, as described in Section 3.4.2.

### *3.3.4 Analysis of linear MDOF systems*

Not all structures can be realistically modelled as SDOF systems. Structures with distributed mass and stiffness may undergo significant deformations in several modes of vibration and therefore need to be analysed as multi-degree-of-freedom (MDOF) systems. These are not generally amenable to hand solution and so computer methods are widely used – see e.g. Hitchings (1992) or Petyt (1998) for details.

For a system with $N$ degrees of freedom it is possible to write a set of equations of motion in matrix form, exactly analogous to Equation (3.4):

$$\mathbf{m}\ddot{\mathbf{y}} + \mathbf{c}\dot{\mathbf{y}} + \mathbf{k}\mathbf{y} = \mathbf{m}\iota\ddot{x}_g \tag{3.15}$$

where **m**, **c** and **k** are the mass, damping and stiffness matrices (dimensions $N \times N$), **y** is the relative displacement vector and $\iota$ is an $N \times 1$ influence vector containing ones corresponding to the DOFs in the direction of the earthquake load, and zeroes elsewhere. **k** is derived in the same way as for a static analysis and is a banded matrix.

**m** is most simply derived by dividing the mass of each element between its nodes. This results in a *lumped* mass matrix, which contains only diagonal terms. To get a sufficiently detailed description of how the mass is distributed, it may be necessary to divide the structure into smaller elements than would be required for a static analysis. Alternatively, many finite element programs give the option of using a *consistent* mass matrix, which allows a more accurate representation of the mass distribution without the need for substantial mesh refinement. A consistent mass matrix includes off-diagonal terms.

In practice **c** is very difficult to define accurately and is not usually formulated explicitly. Instead, damping is incorporated in a simplified form. We shall see how this is done later.

### *3.3.5 Free vibration analysis*

As with SDOF systems, before attempting to solve Equation (3.15) it is helpful to consider the free vibration problem. Because it has little effect on free vibrations, we also omit the damping term, leaving:

$$\mathbf{m}\ddot{\mathbf{y}} + \mathbf{k}\mathbf{y} = 0 \tag{3.16}$$

The solution to this equation has the form

$$\mathbf{y} = \boldsymbol{\varphi}\sin\omega t \tag{3.17}$$

where $\boldsymbol{\varphi}$ is the mode shape, which is a function solely of position within the structure. Differentiating and substituting into Equation (3.16) gives:

$$(\mathbf{k} - \omega^2 \mathbf{m})\boldsymbol{\varphi} = 0 \tag{3.18}$$

This can be solved to give $N$ circular natural frequencies $\omega_1$, $\omega_2 \ldots \omega_i \ldots \omega_N$, each associated with a mode shape $\boldsymbol{\varphi}_i$. Thus an $N$-DOF system is able to vibrate in $N$ different modes, each having a distinct deformed shape and each occurring at a particular natural frequency (or period). The modes of vibration are system properties, independent of the external loading. Figure 3.10 shows the sway modes of vibration of a four-storey shear-type building (i.e. one with relatively stiff floors, so that lateral deformations are dominated by shearing deformation between floors), with the modes numbered in order of ascending natural frequency (or descending period).

Often approximate formulae are used for estimating the fundamental natural period of multi-storey buildings. EC8 recommends the following formulae. For multi-storey frame buildings:

$$T_1 = C_t H^{0.75} \tag{3.19}$$

where $T_1$ is measured in seconds, the building height, $H$, is measured in metres and the constant, $C_t$, equals 0.085 for steel moment-resisting frames, 0.075 for concrete moment-resisting frames or steel eccentrically braced frames, and 0.05 for other types of frame. For shear-wall type buildings:

$$C_t = \frac{0.075}{\sqrt{A_c}} \tag{3.20}$$

where $A_c$ is the total effective area of shear walls in the bottom storey, in m$^2$.

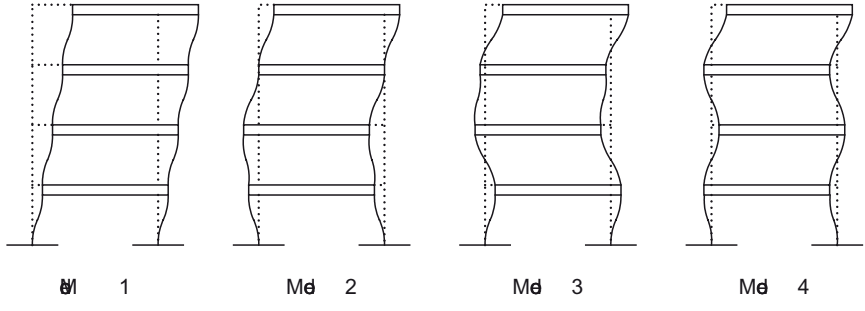



*Figure 3.10* Mode shapes of a four-storey building

### *3.3.6 Multi-modal response spectrum analysis*

Having determined the natural frequencies and mode shapes of our system, we can go on to analyse the response to an applied load. Equation (3.15) is a set of $N$ coupled equations in terms of the $N$ degrees of freedom. This can be most easily solved using the principle of modal superposition, which states that any set of displacements can be expressed as a linear combination of the mode shapes:

$$\mathbf{y} = Y_1\boldsymbol{\varphi}_1 + Y_2\boldsymbol{\varphi}_2 + Y_3\boldsymbol{\varphi}_3 + \ldots\ldots + Y_N\boldsymbol{\varphi}_N = \sum_i Y_i\boldsymbol{\varphi}_i \tag{3.21}$$

The coefficients $Y_i$ are known as the *generalised* or *modal displacements*. The modal displacements are functions only of time, while the mode shapes are functions only of position. Equation (3.21) allows us to transform the equations of motion into a set of equations in terms of the modal displacements rather than the original degrees of freedom:

$$\mathbf{M}\ddot{\mathbf{Y}} + \mathbf{C}\dot{\mathbf{Y}} + \mathbf{K}\mathbf{Y} = \boldsymbol{\varphi}^T\mathbf{m}\iota\,\ddot{x}_g \tag{3.22}$$

where $\mathbf{Y}$ is the vector of modal displacements, and $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are the modal mass, stiffness and damping matrices. Because of the orthogonality properties of the modes, it turns out that $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are all diagonal matrices, so that the $N$ equations in Equation (3.22) are *uncoupled*, i.e. each mode acts as an SDOF system and is independent of the responses in all other modes. Each line of Equation (3.22) has the form:

$$M_i\ddot{Y}_i + C_i\dot{Y}_i + K_iY_i = L_i\ddot{x}_g \tag{3.23}$$

or, by analogy with Equation (3.11) for an SDOF system:

$$\ddot{Y}_i + 2\xi\omega_i\dot{Y}_i + \omega_i^2Y = \frac{L_i}{M_i}\ddot{x}_g \tag{3.24}$$

where

$$L_i = \sum_j m_j\varphi_{ij} \tag{3.25}$$

$$M_i = \sum_j m_j\varphi_{ij}^2 \tag{3.26}$$

Here the subscript $i$ refers to the mode shape and $j$ to the degrees of freedom in the structure. So $\varphi_{ij}$ is the value of mode shape $i$ at DOF $j$. $L_i$ is an earthquake excitation factor, representing the extent to which the earthquake tends to excite response in mode $i$. $M_i$ is called the modal mass. The dimensionless factor $L_i/M_i$ is the ratio of the response of a MDOF

structure in a particular mode to that of an SDOF system with the same mass and period.

Note that Equation (3.24) allows us to define the damping in each mode simply by specifying a damping ratio, $\xi$, without having to define the original damping matrix **c**.

While Equation (3.24) could be solved explicitly to give $Y_i$ as a function of time for each mode, it is more normal to use the response spectrum approach. For each mode we can read off the spectral acceleration, $S_{ei}$, corresponding to that mode's natural period and damping – this is the peak response of an SDOF system with period, $T_i$, to the ground acceleration, $\ddot{x}_g$. For our MDOF system, the way we have broken it down into separate modes has resulted in the ground acceleration being scaled by the factor $L_i/M_i$. Since the system is linear, the structural response will be scaled by the same amount. So the acceleration amplitude in mode $i$ is $(L_i/M_i).S_{ei}$ and the maximum acceleration of DOF $j$ in mode $i$ is:

$$\ddot{x}_{ij}(\max) = \frac{L_i}{M_i} S_{ei} \varphi_{ij} \tag{3.27}$$

Similarly for displacements, by analogy with Equation (3.14):

$$y_{ij}(\max) = \frac{L_i}{M_i} S_{ei} \varphi_{ij} \cdot \frac{T_i^2}{4\pi^2} \tag{3.28}$$

To find the horizontal force on mass $j$ in mode $i$ we simply multiply the acceleration by the mass:

$$F_{ij}(\max) = \frac{L_i}{M_i} S_{ei} \varphi_{ij} m_j \tag{3.29}$$

and the total horizontal force on the structure (usually called the base shear) in mode $i$ is found by summing all the storey forces to give:

$$F_{bi}(\max) = \frac{L_i^2}{M_i} S_{ei} \tag{3.30}$$

The ratio $L_i^2/M_i$ is known as the *effective modal mass*. It can be thought of as the amount of mass participating in the structural response in a particular mode. If we sum this quantity for all modes of vibration, the result is equal to the total mass of the structure.

To obtain the overall response of the structure, in theory we need to apply Equations (3.27) to (3.30) to each mode of vibration and then combine the results. Since there are as many modes as there are degrees of freedom, this could be an extremely long-winded process. In practice, however, the scaling factors $L_i/M_i$ and $L_i^2/M_i$ are small for the higher modes of vibration. It is therefore normally sufficient to consider only a subset of the modes. EC8 offers a variety of ways of assessing how many modes need to be included in the response analysis. The normal approach is either to include sufficient modes that the sum of their effective modal masses is at least 90 per cent

of the total structural mass, or to include all modes with an effective modal mass greater than 5 per cent of the total mass. If these conditions are difficult to satisfy, a permissible alternative is that the number of modes should be at least $3\sqrt{n}$ where $n$ is the number of storeys, and should include all modes with periods below 0.2 s.

Another potential problem is the combination of modal responses. Equations (3.27) to (3.30) give only the peak values in each mode, and it is unlikely that these peaks will all occur at the same point in time. Simple combination rules are used to give an *estimate* of the total response. Two methods are permitted by EC8. If the difference in natural period between any two modes is at least 10 per cent of the longer period, then the modes can be regarded as independent. In this case, the simple SRSS method can be used, in which the peak overall response is taken as the Square Root of the Sum of the Squares of the peak modal responses. If the independence condition is not met, then the SRSS approach may be non-conservative and a more sophisticated combination rule should be used. The most widely accepted alternative is the Complete Quadratic Combination, or CQC method (Wilson et al, 1981), which is based on calculating a correlation coefficient between two modes. Although it is more mathematically complex, the additional effort associated with using this more general and reliable method is likely to be minimal, since it is built into many dynamic analysis computer programs.

In conclusion, the main steps of the mode superposition procedure can be summarised as follows:

a. Perform free vibration analysis to find natural periods, $T_i$, and corresponding mode shapes, $\boldsymbol{\varphi}_i$ . Estimate damping ratio $\xi$.
b. Decide how many modes need to be included in the analysis.
c. For each mode:
   - compute the modal properties $L_i$ and $M_i$ from Equation (3.25) and (3.26);
   - read the spectral acceleration from the design spectrum;
   - compute the desired response parameters using Equations (3.27) to (3.30).
d. Combine modal contributions to give estimates of total response.

### *3.3.7 Equivalent static analysis of MDOF systems*

A logical extension of the process of including only a subset of the vibrational modes in the response calculation is that, in some cases, it may be possible to approximate the dynamic behaviour by considering only a single mode. It can be seen from Equation (3.29) that, for a single mode of vibration, the force at level $j$ is proportional to the product of mass and mode shape at level $j$, the other terms being modal parameters that do not vary with position.

If the structure can reasonably be assumed to be dominated by a single (normally the fundamental) mode then a simple static analysis procedure can be used that involves only minimal consideration of the dynamic behaviour. For many years this approach has been a mainstay of earthquake design codes. In EC8 the procedure is as follows.

Estimate the period of the fundamental mode, $T_1$ – usually by some simplified approximate method rather than a detailed dynamic analysis (e.g. Equation (3.19)). It is then possible to check whether equivalent static analysis is permitted – this requires that $T_1 < 4T_C$ where $T_C$ is the period at the end of the constant-acceleration part of the design response spectrum. The building must also satisfy the EC8 regularity criteria. If these two conditions are not met, the multi-modal response spectrum method outlined above must be used.

For the calculated structural period, the spectral acceleration $S_e$ can be obtained from the design response spectrum. The base shear is then calculated as:

$$F_b = \lambda m S_e \tag{3.31}$$

where $m$ is the total mass. This is analogous to Equation (3.30), with the ratio $L_i^2/M_i$ replaced by $\lambda m$. $\lambda$ takes the value 0.85 for buildings of more than two storeys with $T_1 < 2T_C$, and is 1.0 otherwise. The total horizontal load is then distributed over the height of the building in proportion to (mass × mode shape). Normally this is done by making some simple assumption about the mode shape. For instance, for simple, regular buildings EC8 permits the assumption that the first mode shape is a straight line (i.e. displacement is directly proportional to height). This leads to a storey force at level $k$ given by:

$$F_k = F_b \frac{z_k m_k}{\sum_j z_j m_j} \tag{3.32}$$

where $z$ represents storey height. Finally, the member forces and deformations can be calculated by static analysis.

## 3.4 Practical seismic analysis to EC8

### *3.4.1 Ductility and behaviour factor*

Designing structures to remain elastic in large earthquakes is likely to be uneconomic in most cases, as the force demands will be very large. A more economical design can be achieved by accepting some level of damage short of complete collapse, and making use of the ductility of the structure to reduce the force demands to acceptable levels.

Ductility is defined as the ability of a structure or member to withstand large deformations beyond its yield point (often over many cycles) without fracture. In earthquake engineering, ductility is expressed in terms of demand and supply. The *ductility demand* is the maximum ductility that the structure experiences during an earthquake, which is a function of both the structure and the earthquake. The *ductility supply* is the maximum ductility the structure can sustain without fracture. This is purely a structural property.

Of course, if one calculates design forces on the basis of a ductile response, it is then essential to ensure that the structure does indeed fail by a ductile mode well before brittle failure modes develop, i.e. that ductility supply exceeds the maximum likely demand – a principle known as *capacity design*. Examples of designing for ductility include:

- ensuring plastic hinges form in beams before columns;
- providing adequate confinement to concrete using closely spaced steel hoops;
- ensuring that steel members fail away from connections;
- avoiding large irregularities in structural form;
- ensuring flexural strengths are significantly lower than shear strengths.

Probably the easiest way of defining ductility is in terms of displacement. Suppose we have an SDOF system with a clear yield point – the displacement ductility is defined as the maximum displacement divided by the displacement at first yield.

$$= \frac{x_{\max}}{x_y} \tag{3.33}$$

Yielding of a structure also has the effect of limiting the peak force that it must sustain. In EC8 this force reduction is quantified by the behaviour factor, $q$:

$$q = \frac{F_{el}}{F_y} \tag{3.34}$$

where $F_{el}$ is the peak force that would be developed in an SDOF system if it responded to the earthquake elastically, and $F_y$ is the yield load of the system.

A well-known empirical observation is that, at long periods ($>T_C$), yielding and elastic structures undergo roughly the same peak displacement. It follows that, for these structures, the force reduction is simply equal to the ductility (see Figure 3.11). At shorter periods, the amount of force reduction achieved for a given ductility reduces. EC8 therefore uses the following expressions:

$$\begin{aligned} &= q && \text{for } T \geq T_C \\ &= 1 + (q-1)\frac{T_C}{T} && \text{for } T < T_C \end{aligned} \tag{3.35}$$

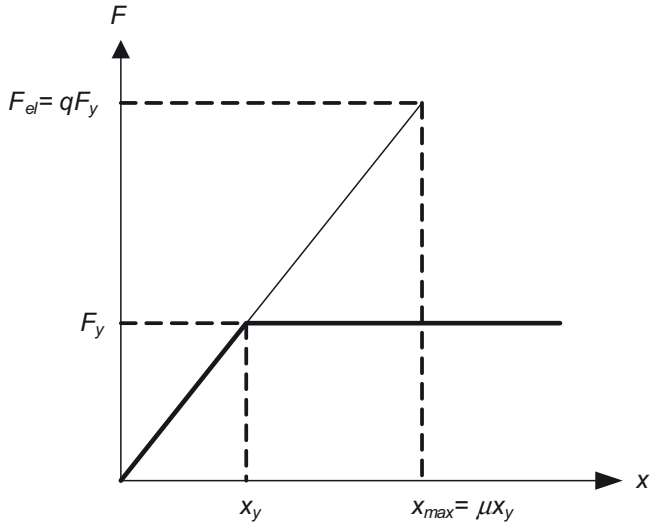



*Figure 3.11* Equivalence of ductility and behaviour factor with equal elastic and inelastic displacements

When designing structures taking account of non-linear seismic response, a variety of analysis options are available. The simplest and most widely used approach is to use the linear analysis methods set out above, but with the design forces reduced on the basis of a single, global behaviour factor, *q*. EC8 gives recommended values of *q* for common structural forms. This approach is most suitable for regular structures, where inelasticity can be expected to be reasonably uniformly distributed.

In more complex cases, the *q*-factor approach can be become inaccurate and a more realistic description of the distribution of inelasticity through the structure may be required. In these cases, a fully non-linear analysis should be performed, using either the non-linear static (*pushover*) approach, or non-linear time-history analysis. Rather than using a single factor, these methods require representation of the non-linear load-deformation characteristics of each member within the structure.

### *3.4.2 Ductility-modified response spectra*

To make use of ductility requires the structure to respond non-linearly, meaning that the linear methods introduced above are not appropriate. However, for an SDOF system, an approximate analysis can be performed in a very similar way to above by using a ductility-modified response spectrum. In EC8 this is known as the *design spectrum*, $S_d$. Figure 3.12 shows EC8 design spectra based on the Type 1 spectrum and soil type C, for a range of behaviour factors. Over most of the period range (for $T \geq T_B$) the spectral

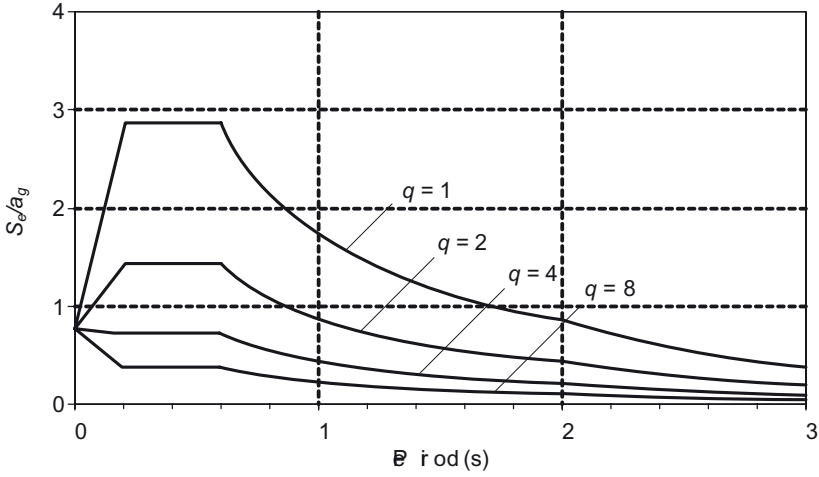



*Figure 3.12* EC8 design response spectra (Type 1 spectrum, soil type C)

accelerations $S_d$ (and hence the design forces) are a factor of $q$ times lower than the values $S_e$ for the equivalent elastic system. For a theoretical, infinitely stiff system (zero period), ductility does not imply any reduction in spectral acceleration, since an infinitely stiff structure will not undergo any deformation and will simply move with the ground beneath it. Therefore, the curves all converge to the same spectral acceleration at zero period. A linear interpolation is used between periods of zero and $T_B$.

When calculating displacements using the design spectrum, it must be noted that the relationship between peak displacement and acceleration in a ductile system is different from that derived in Equation (3.14) for an elastic system. The ductile value is given by:

$$S_D(\text{ductile}) = \mu \frac{F_y}{k} = \mu m S_d . \frac{T_n^2}{4\pi^2 m} = \mu \frac{S_d T_n^2}{4\pi^2} \tag{3.36}$$

Comparing with Equation (3.14) we see that the ratio between spectral displacement and acceleration is  times larger for a ductile system than for an elastic one. Thus, the seismic analysis of a ductile system can be performed in exactly the same way as for an elastic system, but with spectral accelerations taken from the design spectrum rather than the elastic spectrum, and with the calculated displacements scaled up by the ductility factor,  .

For long period structures ($T > T_C$) the result of this approach will be that design forces are reduced by the factor $q$ compared to an elastic design, and the displacement of the ductile system is the same as for an equivalent elastic system (since $q =$  in this period range). For $T_B < T < T_C$ the same force reduction will be achieved but displacements will be slightly greater than the elastic case. For very stiff structures ($T < T_B$) the benefits of ductility are

reduced, with smaller force reductions and large displacements compared to the elastic case.

Lastly, it should be noted that the use of ductility-modified spectra is reasonable for SDOF systems, but should be applied with caution to MDOF structures. For elastic systems we have seen that an accurate dynamic analysis can be performed by considering the response of the structure in each of its vibration modes, then combining the modal responses. A similar approach is widely used for inelastic structures, i.e. each mode is treated as an SDOF system and its ductility-modified response determined as above. The modal responses are then combined by a method such as SRSS. While this approach forms the basis of much practical design, it is important to realise that it has no theoretical justification. For linear systems, the method is based on the fact that any deformation can be treated as a linear combination of the mode shapes. Once the structure yields, its properties change and these mode shapes no longer apply.

When yielding is evenly spread throughout the structure, the deformed shape of the plastic structure is likely to be similar to the elastic one, and the ductility-modified response spectrum analysis may give reasonable (though by no means precise) results. If, however, yielding is concentrated in certain parts of the structure, such as a soft storey, then this procedure is likely to be substantially in error and one of the non-linear analysis methods described below should be used.

### *3.4.3 Non-linear static analysis*

In recent years there has been a substantial growth of interest in the use of non-linear static, or *pushover*, analysis (Lawson et al., 1994; Krawinkler and Seneviratna, 1998; Fajfar, 2002) as an alternative to the ductility-modified spectrum approach. In this approach, appropriate lateral load patterns are applied to a numerical model of the structure and their amplitude is increased in a stepwise fashion. A non-linear static analysis is performed at each step, until the building forms a collapse mechanism. A pushover curve (base shear against top displacement) can then be plotted. This is often referred to as the *capacity curve* since it describes the deformation capacity of the structure. To determine the demands imposed on the structure by the earthquake, it is necessary to equate this to the demand curve (i.e. the earthquake response spectrum) to obtain the peak displacement under the design earthquake – termed the *target displacement*. The non-linear static analysis is then revisited to determine member forces and deformations at this point.

This method is considered a step forward from the use of linear analysis and ductility-modified response spectra, because it is based on a more accurate estimate of the distributed yielding within a structure, rather than an assumed, uniform ductility. The generation of the pushover curve also

provides the engineer with a good feel for the non-linear behaviour of the structure under lateral load. However, it is important to remember that pushover methods have no rigorous theoretical basis, and may be inaccurate if the assumed load distribution is incorrect. For example, the use of a load pattern based on the fundamental mode shape may be inaccurate if higher modes are significant, and the use of a fixed load pattern may be unrealistic if yielding is not uniformly distributed, so that the stiffness profile changes as the structure yields.

The main differences between the various pushover analysis procedures that have been proposed are (i) the choices of load patterns to be applied and (ii) the method of simplifying the pushover curve for design use. The EC8 method is summarised below.

First, two pushover analyses are performed, using two different lateral load distributions. The most unfavourable results from these two force patterns should be adopted for design purposes. In the first, the acceleration distribution is assumed proportional to the fundamental mode shape. The inertia force $F_k$ on mass $k$ is then:

$$F_k = \frac{m_k \varphi_k}{\sum_j m_j \varphi_j} F_b \tag{3.37}$$

where $F_b$ is the base shear (which is increased steadily from zero until failure), $m_k$ the $k$th storey mass and $\mathrm{f}_k$ the mode shape coefficient for the $k$th floor. If the fundamental mode shape is assumed linear then $\varphi_k$ is proportional to storey height $z_k$ and Equation (3.37) then becomes identical to Equation (3.32), presented earlier for equivalent static analysis. In the second case, the acceleration is assumed constant with height. The inertia forces are then given by:

$$F_k = \frac{m_k}{\sum_j m_j} F_b \tag{3.38}$$

The output from each analysis can be summarised by the variation of base shear, $F_b$, with top displacement, $d$, with maximum displacement, $d_m$. This can be transformed to an equivalent SDOF characteristic ($F^*$ vs $d^*$) using:

$$F^* = \frac{F_b}{\Gamma}, \quad d^* = \frac{d}{\Gamma} \tag{3.39}$$

where

$$\Gamma = \frac{\sum_j m_j \varphi_j}{\sum_j m_j \varphi_j^2} \tag{3.40}$$

The SDOF pushover curve is likely to be piecewise linear due to the formation of successive plastic hinges as the lateral load intensity is increased, until a collapse mechanism forms. For determination of the seismic demand from a response spectrum, it is necessary to simplify this to an equivalent elastic-perfectly plastic curve as shown in Figure 3.13. The yield load, $F_y^*$, is taken as the load required to cause formation of a collapse mechanism, and the yield displacement, $d_y^*$, is chosen so as to give equal areas under the actual and idealised curves. The initial elastic period of this idealised system is then estimated as:

$$T^* = 2\pi\sqrt{\frac{m^* d_y^*}{F_y^*}} \tag{3.41}$$

The target displacement of the SDOF system under the design earthquake is then calculated from:

$$\begin{aligned} d_t^* &= S_e\left(\frac{T^*}{2\pi}\right)^2 && T^* \geq T_C \\ d_t^* &= S_e\left(\frac{T^*}{2\pi}\right)^2 \frac{1}{q_u}\left[1+(q_u-1)\frac{T_C}{T^*}\right] && T^* < T_C \end{aligned} \tag{3.42}$$

where $q_u = \dfrac{S_e}{(F_y^*/m^*)}$ and $m^* = \sum_{j=1}^{n} m_j \varphi_j$.

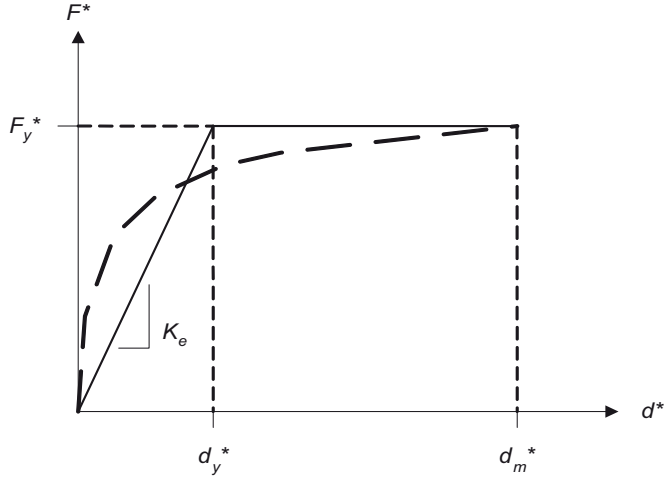



*Figure 3.13* Idealisation of pushover curve in EC8

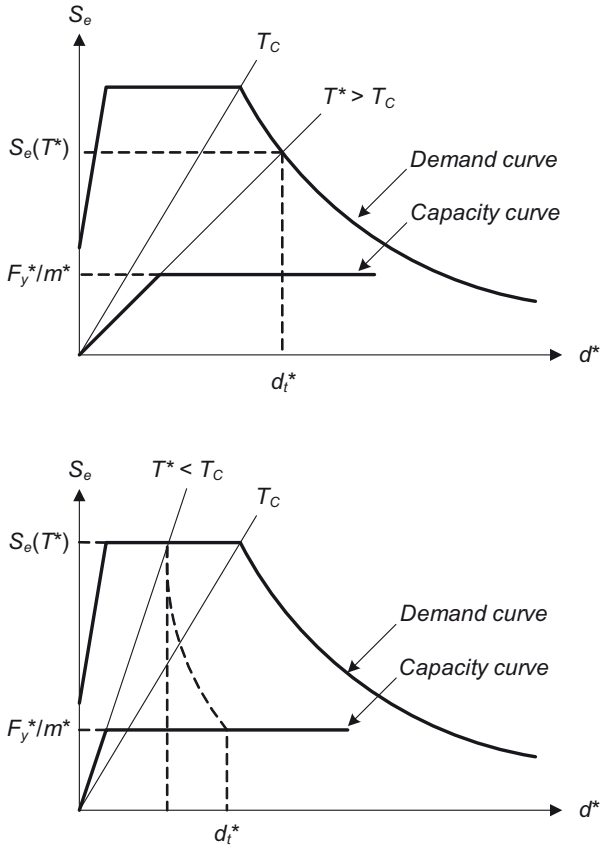



*Figure 3.14* Determination of target displacement in pushover analysis for a) long-period structure, b) short-period structure

Equation (3.42) is illustrated schematically in Figure 3.14, in which the design response spectrum has been plotted in acceleration vs displacement format rather than the more normal acceleration vs period. This enables both the spectrum (i.e. the demand curve) and the capacity curve to be plotted on the same axes, with a constant period represented by a radial line from the origin. For $T^* \geq T_C$ the target displacement is based on the equal displacement rule for elastic and inelastic systems. For shorter period structures, a correction is applied to account for the more complex interaction between behaviour factor and ductility – see Equation (3.35).

Having found the target displacement for the idealised SDOF system, this can be transformed back to that of the original MDOF system using Equation (3.39), and the forces and deformations in the structure can be checked by considering the point in the pushover analysis corresponding to this displacement value.

The EC8 procedure is simple and unambiguous, but can be rather conservative. Some other guidelines (mainly ones aimed at assessing existing structures rather than new construction) recommend rather more complex procedures that may give more accurate results. For example, ASCE 41-04 (2006) allows the use of adaptive load patterns, which take account of load redistribution due to yielding, and simplifies the pushover curve to bilinear with a positive post-yield stiffness.

### *3.4.4 Non-linear time-history analysis*

A final alternative, which remains comparatively rare, is the use of full non-linear dynamic analysis. In this approach a non-linear model of the structure is analysed under a ground acceleration time-history whose frequency content matches the design spectrum. The time-history is specified as a series of data points at time intervals of the order of 0.01 s, and the analysis is performed using a stepwise procedure usually referred to as direct integration. This is a highly specialised topic that will not be covered in detail here – see Clough and Penzien (1993) or Petyt (1998) for a presentation of several popular time integration methods and a discussion of their relative merits.

Since the design spectrum has been defined by enveloping and smoothing spectra corresponding to different earthquake time-histories, it follows that there are many (in fact, an infinite number of) time-histories that are compatible with the spectrum. These may be either recorded or artificially generated – specialised programs exist, such as SIMQKE, for generating suites of spectrum-compatible accelerograms. Different spectrum-compatible time-histories may give rise to quite different structural responses, and so it is necessary to perform several analyses to be sure of achieving representative results. EC8 specifies that a minimum of three analyses under different accelerograms must be performed. If at least seven different analyses are performed then mean results may be used, otherwise the most onerous result should be used.

Beyond being compatible with the design spectrum, it is important that earthquake time-histories should be chosen whose time-domain characteristics (e.g. duration, number of cycles of strong motion) are appropriate to the regional seismicity and local ground conditions. Some guidance is given in Chapter 2, but this is a complex topic for which specialist seismological input is often needed.

## 3.5 Concluding summary

A seismic analysis must take adequate account of dynamic amplification of earthquake ground motions due to resonance. The normal way of doing this is by using a response spectrum.

The analysis of the effects of an earthquake (or any other dynamic loadcase) has two stages:

a. Estimation of the dynamic properties of the structure – natural period(s), mode shape(s), damping ratio – these are structural properties, independent of the loading. The periods and mode shapes may be estimated analytically or using empirical formulae.
b. A response calculation for the particular loadcase under consideration. This calculation makes use of the dynamic properties calculated in a), which influence the load the structure sustains under earthquake excitation.

Methods based on linear analysis (either multi-modal response analysis or equivalent static analysis based on a single mode of vibration) are widely used. In these cases non-linearity is normally dealt with by using a ductility-modified response spectrum.

Alternative methods of dealing with non-linear behaviour (particularly static pushover methods) are growing in popularity and are permitted in EC8.

## 3.6 Design example

### *3.6.1 Introduction*

An example building structure has been chosen to illustrate the use of EC8 in practical building design. It is used to show the derivation of design seismic forces in the remaining part of this chapter, and the same building is used in subsequent chapters to illustrate checks for regularity, foundation design and alternative designs in steel and concrete. It is important to note that the illustrative examples presented herein and in subsequent chapters do not attempt to present complete design exercises. The main purpose is to illustrate the main calculations and design checks associated with seismic design to EC8 and to discussions of related approaches and procedures.

The example building represents a hotel, with a single-storey podium housing the public spaces of the hotel, surmounted by a seven-storey tower block, comprising a central corridor with bedrooms to either side. Figure 3.15 provides a schematic plan and section of the building, while Figure 3.16 gives an isometric view.

The building is later shown to be regular in plan and elevation (see Section 4.9). EC8 then allows the use of a planar structural model and the equivalent static analysis approach. There is no need to reduce $q$ factors to account for irregularity. The calculation of seismic loads for equivalent static analysis can be broken down into the following tasks:

a. estimate self-weight and seismic mass of building;
b. calculate seismic base shear in x-direction;
c. calculate distribution of lateral loads and seismic moment;
d. consider how frame type and spacing influence member forces.

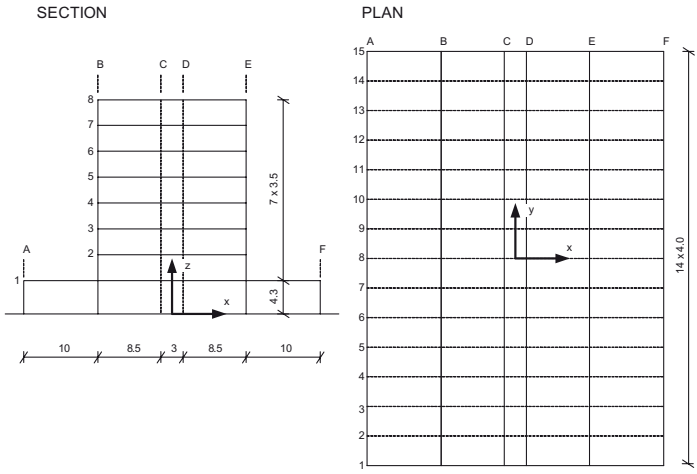



*Figure 3.15* Schematic plan and section of example building

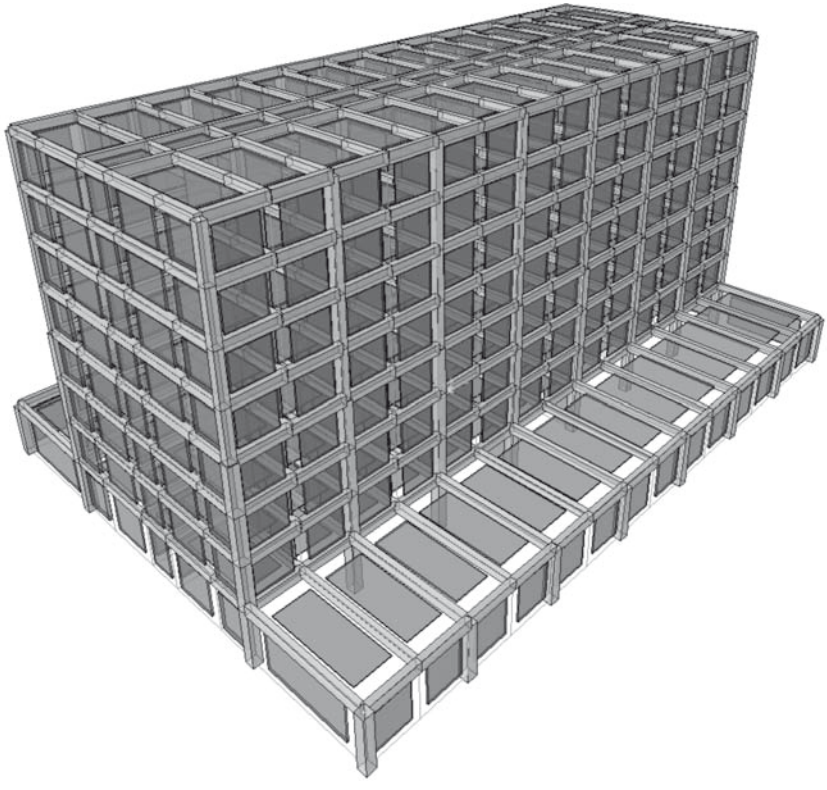


*Figure 3.16* Isometric view of example building

### *3.6.2 Weight and mass calculation*

#### *3.6.2.1 Dead load*

For this preliminary load estimate, neglect weight of frame elements (resulting in same weight/mass for steel and concrete frame structures). Assume:

a. 150 mm concrete floor slabs throughout: $0.15 \times 24 = 3.6$ kN/m$^2$.
b. Outer walls – brick/block cavity wall, each 100 mm thick, 12 mm plaster on inside face:
   - brick: $0.1 \times 18 = 1.8$;
   - block: $0.1 \times 12 = 1.2$;
   - plaster: $0.012 \times 21 = 0.25$;
   - total = 3.25 kN/m$^2$.
c. Internal walls – single leaf 100 mm blockwork, plastered both sides:
   - block: $0.1 \times 12 = 1.2$;
   - plaster: $0.024 \times 21 = 0.5$;
   - total = 1.7 kN/m$^2$.
d. Ground floor perimeter glazing: 0.4 kN/m$^2$.
e. Floor finishes etc: 1.0 kN/m$^2$.

*Table 3.1* Dead load calculation

| *Level* | | *Calculation* | *Load (kN)* | *Total (kN)* |
|---|---|---|---|---|
| 8 | Slab | $(56 \times 20) \times 3.6$ | 4032 | |
| | Finishes | $(56 \times 20) \times 1.0$ | 1120 | 5152 |
| 2–7 | Slab | $(56 \times 20) \times 3.6$ | 4032 | |
| | Finishes | $(56 \times 20) \times 1.0$ | 1120 | |
| | Outer walls | $(2 \times (56 + 20) \times 3.5) \times 3.25$ | 1729 | |
| | Internal walls (gl 2–14) | $(26 \times 8.5 \times 3.5) \times 1.7$ | 1315 | |
| | Internal walls (gl C, D) | $(2 \times 56 \times 3.5) \times 1.7$ | 666 | 8862 |
| 1 | Tower section (gl B–E) | As levels 2–7 | 8862 | |
| | Slab (gl A–B, E–F) | $(56 \times 20) \times 3.6$ | 4032 | |
| | Finishes (gl A–B, E–F) | $(56 \times 20) \times 1.0$ | 1120 | |
| | External glazing | $(2 \times (56 + 40) \times 4.3) \times 0.4$ | 330 | 14344 |
| Total dead load, *G* | | | | 72668 |

*3.6.2.2 Imposed load*

*Table 3.2* Imposed load calculation

| *Level* | | *Calculation* | *Load (kN)* | *Total (kN)* |
|---|---|---|---|---|
| 8 | Roof | (56 × 20) × 2.0 | 2240 | 2240 |
| 2–7 | Corridors etc. | ((56 × 3) + (8.5 × 4) + (8.5 × 8)) × 4.0 | 1080 | |
| | Bedrooms | ((56 × 20) – 270) × 2.0 | 1700 | 2780 |
| 1 | Tower area | As levels 2–7 | 2780 | |
| | Roof terrace | (56 × 20) × 4.0 | 4480 | 7260 |
| Total imposed load, $Q$ | | | | 26180 |

*3.6.2.3 Seismic mass*

Cl. 3.2.4 states that the masses to be used in a seismic analysis should be those associated with the load combination:

$G + \psi_{E,i}Q$

Take $\psi_{E,i}$ to be 0.3.

The corresponding building weight is 8208 × 9.81 = 80,522 kN.

### *3.6.3 Seismic base shear*

First, define design response spectrum. Use Type 1 spectrum (for areas of high seismicity) soil type C. Spectral parameters are (from EC8 Table 3.2):

$S = 1.15$, $T_B = 0.2$ s, $T_C = 0.6$ s, $T_D = 2.0$ s

The reference peak ground acceleration is $a_{gR} = 3.0$ m/s$^2$. The importance factor for the building is $\gamma_I = 1.0$, so the design ground acceleration $a_g = \gamma_I a_{gR} = 3.0$ m/s$^2$. The resulting design spectrum is shown in Figure 3.17 for $q = 1$ and $q = 4$, and design spectral accelerations can also be obtained from the equations in Cl. 3.2.2.5 of EC8.

The framing type has not yet been considered, so we will calculate base shear for three possible options:

*Table 3.3* Seismic mass calculation

| *Level* | *G (kN)* | *Q (kN)* | $G + \psi_{E,i}Q$ *(kN)* | *Mass (tonne)* |
|---|---|---|---|---|
| 8 | 5152 | 2240 | 5824 | 593.7 |
| 2–7 | 8862 | 2780 | 9696 | 988.4 |
| 1 | 14344 | 7260 | 16522 | 1684.2 |
| Total seismic mass | | | | 8208.3 |

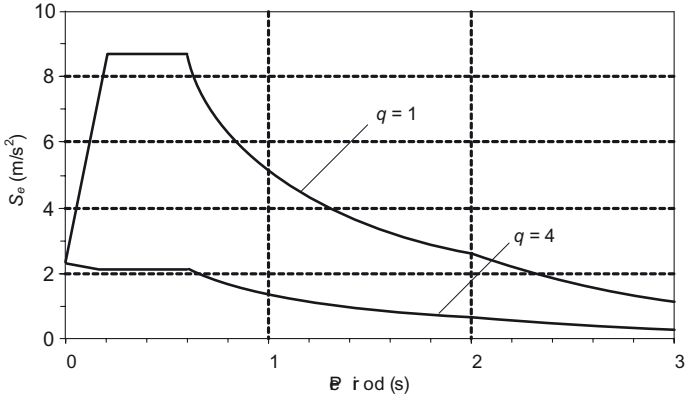



*Figure 3.17* Design spectrum

- steel moment-resisting frame (MRF);
- concrete moment-resisting frame;
- dual system (concrete core with either concrete or steel frame).

The procedure follows EC8 Cl. 4.3.3.2.2.

*3.6.3.1 Steel MRF*

Estimate natural period, EC8 Equation (4.6): $T_1 = C_t H^{0.75}$

For steel MRF $C_t = 0.085$, hence: $T_1 = 0.085 \times 28.8^{0.75} = 1.06$ s

$T_C \leq T_1 \leq T_D$ so EC8 Equation (3.15) applies: $S_d = a_g S \dfrac{2.5}{q} \dfrac{T_C}{T_1}$

EC8 Table 6.2: assuming ductility class medium (DCM), $q = 4$

Therefore: $S_d = 3.0 \times 1.15 \times \dfrac{2.5}{4} \dfrac{0.6}{1.06} = 1.22\,\text{m/s}^2$

EC8 Equation (4.5): $F_b = \lambda m S_d$

In this case $T_1 < 2\,T_C$ so $\lambda = 0.85$

Therefore: $F_b = 0.85 \times 8208 \times 1.22 = 8{,}515$ kN

Net horizontal force is 100 × 8,515/80,522 = 10.6% of total building weight.

*3.6.3.2 Concrete MRF*

Estimate natural period, EC8 Equation (4.6): $T_1 = C_t H^{0.75}$

For concrete MRF $C_t = 0.075$, hence: $T_1 = 0.075 \times 28.8^{0.75} = 0.93$ s

$T_C \leq T_1 \leq T_D$ so EC8 Equation (3.15) applies: $$S_d = a_g S \frac{2.5}{q} \frac{T_C}{T_1}$$

EC8 Table 5.1: assuming DCM, $$q = 3.0 \frac{\alpha_u}{\alpha_1}$$

where $\alpha_u$ is the load factor to cause overall instability due to plastic hinge formation, and $\alpha_1$ is the load factor at first yield in the structure.

Where these values have not been determined explicitly, for regular buildings, EC Cl. 5.2.2.2 allows default values of the ratio $\alpha_u/\alpha_1$ to be assumed. For our multi-storey, multi-bay frame, $\alpha_u/\alpha_1 = 1.3$, hence $q = 3 \times 1.3 = 3.9$.

Therefore: $$S_d = 3.0 \times 1.15 \times \frac{2.5}{3.9} \frac{0.6}{0.93} = 1.43 \text{m/s}^2$$

EC8 Equation (4.5): $F_b = \lambda m S_d$
In this case $T_1 < 2\,T_C$ so $\lambda = 0.85$
Therefore: $F_b = 0.85 \times 8028 \times 1.43 = 9{,}954$ kN
Net horizontal force is $100 \times 9{,}954/80{,}522 = 12.4\%$ of total building weight.

*3.6.3.3 Dual system (concrete core with either concrete or steel frame)*

Estimate natural period, EC8 Equation (4.6): $T_1 = C_t H^{0.75}$

For structures other than MRFs, EC8 gives $C_t = 0.05$, hence:

$T_1 = 0.05 \times 28.8^{0.75} = 0.62$ s

(For buildings with shear walls, EC8 Equation (4.7) gives a permissible alternative method of evaluating $C_t$ based on the area of shear walls in the lowest storey. This is likely to give a slightly shorter period than that calculated above. However, as the calculated value is very close to the constant-acceleration part of the response spectrum ($T_C = 0.6$ s), the lower period would result in very little increase in the spectral acceleration or the design base shear. This method has therefore not been pursued here.)

$T_C \leq T_1 \leq T_D$ so EC8 Equation (3.15) applies: $$S_d = a_g S \frac{2.5}{q} \frac{T_C}{T_1}$$

For dual systems, DCM, EC8 Table 5.1 gives: $$q = 3.0 \frac{\alpha_u}{\alpha_1}$$

and EC Cl. 5.2.2.2 gives a default value of the ratio $\alpha_u/\alpha_1 = 1.2$ for a wall-equivalent dual system. Hence $q = 3 \times 1.2 = 3.6$.

Therefore: $S_d = 3.0 \times 1.15 \times \frac{2.5}{3.6}\frac{0.6}{0.62} = 2.32\,\text{m/s}^2$

EC8 Equation (4.5): $F_b = \lambda m S_d$

In this case $T_1 < 2\,T_C$ so $\lambda = 0.85$

Therefore: $F_b = 0.85 \times 8208 \times 2.32 = 16{,}176$ kN

Net horizontal force is 100 × 16,176/80,522 = 20.1% of total building weight.

### *3.6.4 Load distribution and moment calculation*

The way the base shear is distributed over the height of the building is a function of the fundamental mode shape. For a regular building, EC8 Cl. 4.3.3.2.3 permits the assumption that the deflected shape is linear. With this assumption, the inertia force generated at a given storey is proportional to the product of the storey mass and its height from the base.

Since the assumed load distribution is independent of the form of framing chosen, and of the value of the base shear, we will calculate a single load distribution based on a base shear of 1000 kN. This can then simply be scaled by the appropriate base shear value from above.

EC8 Equation (4.11) gives the force on storey *k* to be:

$$F_k = F_b \frac{z_k m_k}{\sum_j z_j m_j}$$

*Table 3.4* Lateral load distribution using linear mode shape approximation

| *Level* k | *Height* $z_k$ *(m)* | *Mass* $m_k$ *(t)* | $z_k\,m_k$ *(m.t)* | *Force* $F_k$ *(kN)* | *Moment* = $F_k z_k$ *(kNm)* |
|---|---|---|---|---|---|
| 8 | 28.8 | 593.7 | 17098 | 139.6 | 4020 |
| 7 | 25.3 | 988.4 | 25006 | 204.1 | 5165 |
| 6 | 21.8 | 988.4 | 21547 | 175.9 | 3835 |
| 5 | 18.3 | 988.4 | 18087 | 147.7 | 2702 |
| 4 | 14.8 | 988.4 | 14628 | 119.4 | 1767 |
| 3 | 11.3 | 988.4 | 11169 | 91.2 | 1030 |
| 2 | 7.8 | 988.4 | 7709 | 62.9 | 491 |
| 1 | 4.3 | 1684.2 | 7242 | 59.2 | 254 |
| Totals | – | 8208.5 | 122,486 | 1000.0 | 19265 |

The ratio of the total base moment to the base shear gives the effective height of the resultant lateral force:

$$h_{\text{eff}} = \frac{19{,}265}{1{,}000} = 19.3\text{m above the base, and } h_{\text{eff}}/h = 19.3/28.8 = 0.67.$$

### *3.6.5 Framing options*

Although not strictly part of the loading and analysis task, it is helpful at this stage to consider the different possible ways of framing the structure.

#### *3.6.5.1 Regularity and symmetry*

The general structural form has already been shown to meet the EC8 regularity requirements in plan and elevation. A regular framing solution needs to be adopted to ensure that there is no large torsional eccentricity. Large reductions in section size with height should be avoided. If these requirements are satisfied, the total seismic loads calculated above can be assumed to be evenly divided between the transverse frames.

#### *3.6.5.2 Steel or concrete*

Either material is suitable for a structure such as this, and the choice is likely to be made based on considerations other than seismic performance. The loads calculated above are based on a seismic mass that has neglected the mass of the main frame elements. These will tend to be more significant for a concrete structure, which may therefore sustain somewhat higher loads than the initial estimates calculated here.

#### *3.6.5.3 Frame type – moment-resisting, dual frame/shear wall system or braced frame*

In the preceding calculations both frame and dual frame/shear wall systems have been considered. In practice, it is likely to be advantageous to make use of the shear wall action of the service cores to provide additional lateral resistance. It can be seen that this reduces the natural period of the structure, shifting it closer to the peak of the response spectrum and thus increasing the seismic loads. However, the benefit in terms of the additional resistance would outweigh this disadvantage.

In general, MRFs provide the most economic solution for low-rise buildings, but for taller structures they tend to sustain unacceptably large deflections and some form of bracing or shear wall action is then required. The height of this structure is intermediate in this respect, so that a variety of solutions are worth considering.

The load distributions for each of the frame types considered can be obtained by scaling the results from 3.6.4 by the base shears from 3.6.3.

*Table 3.5* Total lateral forces for different frame types

| *Level* | *Total lateral forces (kN)* | | |
|---|---|---|---|
| | *Steel MRF* | *Concrete MRF* | *Dual system* |
| 8 | 1189 | 1390 | 2258 |
| 7 | 1738 | 2032 | 3302 |
| 6 | 1498 | 1751 | 2845 |
| 5 | 1258 | 1470 | 2389 |
| 4 | 1017 | 1189 | 1931 |
| 3 | 777 | 908 | 1475 |
| 2 | 536 | 626 | 1017 |
| 1 | 504 | 589 | 958 |
| Base shear (kN) | 8515 | 9954 | 16176 |
| Base moment (MNm) | 164.0 | 191.8 | 311.6 |

Clearly the dual structure gives rise to significantly larger forces (because its lower period puts it closer to the peak of the response spectrum). However, it also provides a more efficient lateral load-resisting system, so it will not necessarily be uneconomic.

Steel braced frames have not been considered explicitly here. They would give rise to similar design forces to the dual system, since EC8 recommends the use of the same $C_t$ value in the period calculation, and allows use of a slightly higher $q$ factor (4 instead of 3.6).

### *3.6.5.4 Frame spacing*

In the short plan ($x$) dimension, it is likely that columns would be provided at each of gridlines B, C, D and E, ensuring regularity and symmetry, and limiting beam spans to reasonable levels. For vertical continuity, the framing of the tower should be continued down to ground level. It may then be desirable to pin the first floor roof terrace beams to the tower structure, so as to prevent them from picking up too much load.

In the long plan ($y$) dimension the choice is between providing a frame at every gridline (i.e. at 4 m spacing) or at alternate gridlines (8 m spacing). With 8 m spacing, the seismic loads to be carried by a typical internal frame are simply those given above scaled by 8/56. With a 4 m spacing, these values would be halved.

### *3.6.5.5 Ductility class and its influence on q factor*

All calculations so far have assumed DCM. If instead the structure is designed with high ductility (DCH) then higher $q$-factors may be used, further reducing the seismic loads. Since in all cases we are on the long-period part

of the response spectrum, the spectral acceleration, and hence all seismic loads, are simply divided by $q$.

The design and detailing requirements to meet the specified ductility classes will be discussed in depth in the concrete design and steel design chapters. At this stage, it is worth noting that the EC8 DCH requirements for concrete are rather onerous and are unlikely to be achieved with the construction skills available. For steel, designing for DCH is likely to be more feasible.

Consider the effect of designing to DCH for the three frame types (refer to Tables 5.1 and 6.2 of EC8, and associated text).

For the steel MRF, EC8 Table 6.2 specifies that for DCH $q = 5\alpha_u/\alpha_1$ . A default value of $\alpha_u/\alpha_1$ of 1.3 may be assumed, or a value of up to 1.6 may be used if justified by a static pushover analysis. Thus $q$ may be taken as up to 6.5 by default, or up to 8.0 based on analysis. If we use a value of 6.5 (compared to 4.0 for DCM) then all seismic loads calculated above can be scaled by 4.0/6.5, i.e. reduced by 38 per cent.

For the concrete MRF, EC8 Table 5.1 specifies that for DCH $q = 4.5\alpha_u/\alpha_1$ . A default value of $\alpha_u/\alpha_1$ of 1.3 may be assumed, or a value of up to 1.5 may be used if justified by a static pushover analysis. Thus $q$ may be taken as up to 5.85 by default, or up to 6.75 based on analysis. If we use a value of 5.85 (compared to 3.9 for DCM) then the seismic loads calculated above can be scaled by 3.9/5.85, i.e. reduced by 33 per cent. A similar proportional reduction in loads can be achieved for the dual system.

If a steel concentrically braced frame were used, EC8 Table 6.2 specifies a maximum $q$ value of 4.0 for both DCM and DCH, so changing to DCH would offer no benefit in terms of design loads.

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# 4 Basic seismic design principles for buildings

*E. Booth and Z. Lubkowski*

## 4.1 Introduction

Fundamental decisions taken at the initial stages of planning a building structure usually play a crucial role in determining how successfully the finished building achieves its performance objectives in an earthquake. This chapter describes how EC8 sets out to guide these decisions, with respect to siting considerations, foundation design and choice of superstructure.

## 4.2 Fundamental principles

### *4.2.1 Introduction*

In EC8, the fundamental requirements for seismic performance are set out in Section 2. There are two main requirements. The first is to meet a 'no collapse' performance level, which requires that the structure retains its full vertical load bearing capacity after an earthquake with a recommended return period of 475 years; longer return periods are given for special structures, for example casualty hospitals or high risk petrochemical installations. After the earthquake, there should also be sufficient residual lateral strength and stiffness to protect life even during strong aftershocks. The second main requirement is to meet a 'damage limitation' performance level, which requires that the cost of damage and associated limitations of use should not be disproportionately high, in comparison with the total cost of the structure, after an earthquake with a recommended return period (for normal structures) of 95 years. Note that Section 2 of EC8 (and hence these basic requirements) applies to all types of structure, not just buildings.

EC8's rules for meeting the 'no collapse' performance level in buildings are given in Section 4 of Part 1 with respect to analysis procedures and in Sections 5 to 9 of Part 1 with respect to material specific procedures to ensure sufficient strength and ductility in the structure. The rules for meeting the 'damage limitation' performance level in buildings are given in Section 4 of Part 1; they consist of simple restrictions on deflections to limit structural

and non-structural damage, and some additional rules for protecting non-structural elements.

EC8 Part 1 Section 4.2.1 sets out some aspects of seismic design specifically for buildings, which should be considered at conceptual design stage, and which will assist in meeting the 'no collapse' and 'damage limitation' requirements. It is not mandatory that they should be satisfied, and indeed since they are qualitative in nature, it would be hard to enforce them, but they are sound principles that deserve study. Related, but quantified, rules generally appear elsewhere in EC8; for example, the structural regularity rules in Section 4.2.3 supplement the uniformity and symmetry principles given in Section 4.2.1. Six guiding principles are given EC8 Part 1 Section 4.2.1 as follows, and these are now discussed in turn.

- Structural simplicity.
- Uniformity, symmetry and redundancy.
- Bi-directional resistance and stiffness.
- Torsional resistance and stiffness.
- Adequacy of diaphragms at each storey level.
- Adequate foundations.

### *4.2.2 Structural simplicity*

This entails the provision of a clear and direct load path for transmission of seismic forces from the top of a building to its foundations. The load path must be clearly identified by the building's structural designer, who must ensure that all parts of the load path have adequate strength, stiffness and ductility.

Direct load paths will help to reduce uncertainty in assessing both strength and ductility, and also dynamic response. Complex load paths, for example involving transfer structures, tend to give rise to stress concentrations and make the assessment of strength, ductility and dynamic response more difficult. Satisfactory structures may still be possible with complex load paths but they are harder to achieve.

### *4.2.3 Uniformity, symmetry and redundancy*

Numerous studies of earthquake damage have found that buildings with a uniform and symmetrical distribution of mass, strength and stiffness in plan and elevation generally perform much better than buildings lacking these characteristics.

Uniformity in plan improves dynamic performance by suppressing torsional response, as discussed further in Section 4.2.5 below. Irregular or asymmetrical plan shapes such as L or T configurations may be improved by dividing the building with joints to achieve compact, rectangular shapes (Figure 4.1), but this introduces a number of design issues that must be

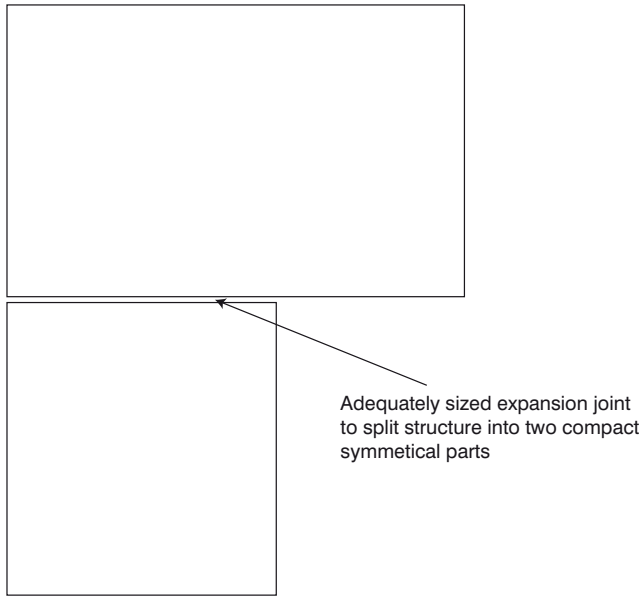



*Figure 4.1* Introduction of joints to achieve uniformity and symmetry in plan

solved; these are avoiding 'buffeting' (impact) across the joint, and detailing the finishes, cladding and services that cross the joint to accommodate the associated seismic movements.

Uniformity of strength and stiffness in elevation helps avoid the formation of weak or soft storeys. Non-uniformity in elevation does not always lead to poor performance, however; for example, seismically isolated buildings are highly non-uniform in elevation but are found to perform very well in earthquakes.

Redundancy implies that more than one loadpath is available to transmit seismic loads, so that if a particular loadpath becomes degraded in strength or stiffness during an earthquake, another is available to provide a backup. Redundancy should therefore increase reliability, since the joint probability of two parallel systems both having lower than expected capacity (or greater than expected demand) should be less than is the case for one system separately. Redundant systems, however, are inherently less 'simple' than determinate ones, which usually makes their assessment more complex.

### *4.2.4 Bi-directional resistance and stiffness*

Unlike the situation that often applies to wind loads on buildings, seismic loads are generally similar along both principal horizontal axes of a building. Therefore, similar resistance in both directions is advisable. Systems such as cross-wall construction found in some hotel buildings, where there are many partition walls along the short direction but fewer in the long direction,

work well for wind loading, which is greatest in the short direction, but tend to be unsatisfactory for seismic loads.

### *4.2.5 Torsional resistance and stiffness*

Pure torsional excitation in an earthquake may arise in a site across which there is significantly varying soils, but significant torsional excitations on buildings are unusual. However, coupled lateral-torsional excitation, arising from an eccentricity between centres of mass and stiffness, is common and is found to increase damage in earthquakes. Such response may be inadequately represented by a linear dynamic analysis, because yielding caused by lateral-torsional response can reduce the stiffness on one side of a building structure and further increase the eccentricity between mass and stiffness centres.

Minimising the eccentricity of mass and stiffness is one important goal during scheme design, and achieving symmetry and uniformity should help to satisfy it. However, some eccentricity is likely to remain, and may be significant due to a number of effects that may be difficult for the structural designer to control; they may arise from uneven mass distributions, uneven stiffness contributions from non-structural elements or non-uniform stiffness degradation of structural members during a severe earthquake. Therefore, achieving good torsional strength and stiffness is an important goal. Stiff and resistant elements on the outside the building, for example in the form of a perimeter frame, will help to achieve this, while internal elements, such as a central core, contribute much less. Quantified rules are provided later in Section 4 of EC8 Part 1, as discussed in Section 4.5 of this chapter.

### *4.2.6 Adequacy of diaphragms at each storey level*

Floor diaphragms perform several vital functions. They distribute seismic inertia loads at each floor level back to the main vertical seismic resisting elements, such as walls or frames. They act as a horizontal tie, preventing excessive relative deformations between the vertical elements, and so helping to distribute seismic loads between them. In masonry buildings, they act to restrain the walls laterally. At transfer levels, for example between a podium and a tower structure, they may also serve to transfer global seismic forces from one set of elements to another.

Floor diaphragms that have very elongated plan shapes, or large openings, are likely to be inefficient in distributing seismic loads to the vertical elements. Precast concrete floors need to have adequate bearing to prevent the loss of bearing and subsequent floor collapse observed in a number of earthquakes. In masonry buildings, it is especially important to ensure a good connection between floors and the masonry walls they bear onto in order to provide lateral stability for the walls.

### *4.2.7 Adequate foundations*

EC8 Part 1 Section 4.2.1.6 states that 'the design and construction of the foundations and of the connection to the superstructure shall ensure that the whole building is subjected to a uniform seismic excitation'. To achieve this, it recommends that a rigid cellular foundation should usually be provided where the superstructure consists of discrete walls of differing stiffnesses. Where individual piled or pad foundations are employed, they should be connected by a slab or by ground beams, unless they are founded on rock.

The interaction of foundations with the ground, in addition to interaction with the superstructure, is of course vital to seismic performance. Part 5 of EC8 gives related advice on conceptual seismic design of foundations, and this is further discussed in Chapters 8 and 9.

## 4.3 Siting considerations

The regional seismic hazard is not the only determinant of how strongly a building may be shaken. (Regional seismic hazard is defined here as the ground shaking expected on a rock site as a function of return period). Within an area of uniform regional hazard, the level of expected ground shaking is likely to vary strongly, and so is the threat from other hazards related to seismic hazard, such as landsliding or fault rupture, for reasons described in the next paragraph. Choice of the exact location of a building structure may not always be within a designer's control, but sometimes even quite small changes in siting can make a dramatic difference to the seismic hazard.

The most obvious cause of local variation in hazard arises from the soils overlying bedrock, which affect the intensity and period of ground motions. It is not only the soils immediately below the site that affect the hazard; the horizontal profiles of soil and rock can also be important, due to 'basin effects'. Soil amplification effects are discussed in Chapters 8 and 9. Topographic amplification of motions may be significant near the crest of steep slopes. Fault rupture, slope instability, liquefaction and shakedown settlement are other hazards associated with seismic activity that may also need to be considered. Figure 4.2 shows just a few examples where a failure to assess these phenomena has impinged on the performance of structures during a major earthquake.

Section 3 of EC8 Part 1 addresses soil amplification, Annex A of EC8 Part 5 addresses topographical amplification and Section 4 of EC8 Part 5 addresses the other siting considerations. By ensuring these potential hazards at a site are identified, the designer can take appropriate actions to minimise those hazards. In some cases, choice of a different site may be the best (or indeed only satisfactory) choice, for example to avoid building on an unstable slope or crossing a fault assessed as potentially active. If the hazard cannot be avoided, appropriate design measures must be taken to accommodate or




*(a)* Fault rupture – Luzon, Philippines 1990



*(b)* Liquefaction – Adapazari, Turkey 1999



*(c)* Slope instability – Niigata, Japan 2004

*Figure 4.2* Examples of poorly sited structures

mitigate it. For example, ground improvement measures may be one option for a site assessed as susceptible to liquefaction, and suitable articulation to accommodate fault movements may be possible for extended structures such as pipelines and bridges.

## 4.4 Choice of structural form

The most appropriate structural material and form to use in a building is influenced by a host of different factors, including relative costs, locally available skills, environmental, durability, architectural considerations, and so on. Some very brief notes on the seismic aspects are given below; further discussion is given in text books such as Booth and Key (2006); Chen and Scawthorn (2003); and Taranath (1998).

Steel has high strength to mass ratio, a clear advantage over concrete because seismic forces are generated through inertia. It is also easy to make steel members ductile in both flexure and shear. Steel moment frames can be highly ductile, although achieving adequate seismic resistance of connections can be difficult, and deflections may govern the design rather than strength. Braced steel frames are less ductile, because buckling modes of failure lack ductility, but braced frames possess good lateral strength and stiffness, which serves to protect non-structural as well as structural elements. Eccentrically braced frames (EBFs), where some of the bracing members are arranged so that their ends do not meet concentrically on a main member, but are separated to meet eccentrically at a ductile shear link, possess some of the advantages of both systems. More recently, buckling-restrained braces (also known as unbonded braces) have found more favour than EBFs in California; these consist of concentrically braced systems where the braces are restrained laterally but not longitudinally by concrete filled tubes, which results in a response in compression that is as ductile as that in tension (Hamburger and Nazir, 2003). Buckling-restrained braces combine ductility and stiffness in a similar way to EBFs.

Concrete has an unfavourably low strength to mass ratio, and it is easy to produce beams and columns that are brittle in shear, and columns that are brittle in compression. However, with proper design and detailing, ductility in flexure can be excellent, ductility in compression can be greatly improved by provision of adequate confinement steel, and failure in shear can be avoided by 'capacity design' measures. Moreover, brittle buckling modes of failure are much less likely than in steel. Although poorly built concrete frames have an appalling record of collapse in earthquakes, well built frames perform well. Moreover, concrete shear wall buildings have an excellent record of good performance in earthquakes, even where design and construction standards are less than perfect, and are relatively straightforward to build.

Seismic isolation involves the introduction of low lateral stiffness bearings to detune the building from the predominant frequencies of an earthquake;

it has proved highly effective in the earthquakes of the past decade. Seismic isolation is a 'passive' method of response control; more radically, active and semi-active systems seek to change structural characteristics in real time during an earthquake to optimise dynamic response. At present, they have been little used in practice.

## 4.5 Evaluating regularity in plan and elevation

### *4.5.1 General*

EC8 Part 1 Section 4.2.3 sets out quantified criteria for assessing structural regularity, complementing the qualitative advice on symmetry and uniformity given in Section 4.2.1. Note that irregular configurations are allowed by EC8, but lead to more onerous design requirements.

A classification of 'non-regularity' in plan requires the use of modal analysis, as opposed to equivalent lateral force analysis, and (generally) a 3D as opposed to a 2D structural model. For a linear analysis, a 3D model would usually be chosen for convenience, even for regular structures. However, a non-linear static (pushover) analysis becomes much less straightforward with 3D analysis models, and should be used with caution if there is plan irregularity, because of the difficulty in capturing coupled lateral-torsional modes of response. Other consequences of non-regularity in plan are the need to combine the effects of earthquakes in the two principal directions of a structure and for certain structures (primarily moment frame buildings) the q factor must be reduced by up to 13 per cent. Moreover, in 'torsionally flexible' concrete buildings, the q value is reduced to 2 for medium ductility and 3 for high ductility, with a further reduction of 20 per cent if there is irregularity in elevation. 'Torsionally flexible' buildings are defined in the next section.

A classification of 'non-regular' in elevation also requires the use of modal analysis, and leads to a reduced q factor, equal to the reference value for regular structures reduced by 20 per cent.

The EC8 Manual (ISE/AFPS 2009) proposes some simplified methods of evaluating regularity that are suitable for preliminary design purposes.

Section 4.9 provides a worked example of assessing the regularity in plan and elevation of the demonstration building structure adopted for this book.

### *4.5.2 Regularity in plan*

Classification as regular in plan requires the following:

1 'Approximately' symmetrical distribution of mass and stiffness in plan.
2 A 'compact' shape, i.e. one in which the perimeter line is always convex, or at least encloses not more than 5 per cent re-entrant area (Figure 4.3).

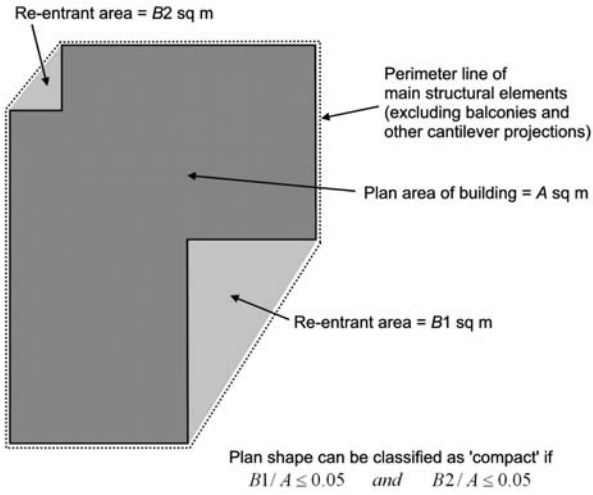



*Figure 4.3* Definition of compact shapes

3 The floor diaphragms shall be sufficiently stiff in-plane not to affect the distribution of lateral loads between vertical elements. EC8 warns that this should be carefully examined in the branches of branched systems, such as L, C, H, I and X plan shapes.
4 The ratio of longer side to shorter sides in plan does not exceed 4.
5 The torsional radius $r_x$ in the $x$ direction must exceed 3.33 times $e_{ox}$, the eccentricity between centres of stiffness and mass in the $x$ direction. Similarly, $r_y$ must exceed 3.33 times $e_{oy}$. The terms $r_x$, $r_y$ , $e_{ox}$ and $e_{oy}$ are defined below.
6 $r_x$ and $r_y$ must exceed the radius of gyration $l_s$, otherwise the building is classified as 'torsionally flexible', and the $q$ values in concrete buildings are greatly reduced. The term $l_s$ is defined below.

The torsional radius, $r_x$, is the square root of the ratio of torsional stiffness (rotation per unit moment) to lateral stiffness in the $x$ direction (deflection per unit force). A similar definition applies to $r_y$. $e_{ox}$ and $e_{oy}$ are the distances between the centre of stiffness and centre of mass in the $x$ and $y$ directions respectively.

These are not exact definitions for a multi-storey building, since only approximate definitions of centre of stiffness and torsional radius are possible; they depend on the vertical distribution of lateral force and moment assumed. Approximate values may be obtained, based on the moments of inertia (and hence lateral stiffness) of the individual vertical elements comprising the lateral force resisting system; see Figure 4.4 and Equations (4.1) and (4.2). These equations are not reliable where the lateral load resisting system consists of elements that assume different deflected shapes under lateral loading, for example unbraced frames combined with

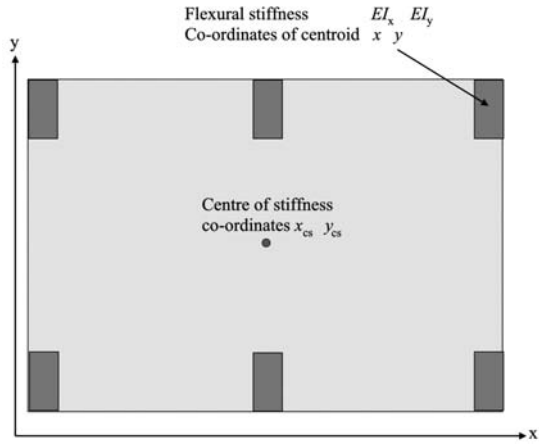



*Figure 4.4* Approximate calculation of torsional radii (see also Equations 4.1 and 4.2)

shear walls. Alternatively, using a computer analysis, values can be obtained from the deflections and rotations at each floor level found from the application of unit forces and torsional moments applied to a 3D model of the structure; various vertical distributions of forces and moments may need to be considered. A worked example is provided in Section 4.9 below. Further advice is provided in the EC8 Manual (ISE/AFPS 2009).

$$x_{cs} \approx \sum \frac{(xEI_y)}{EI_y} \quad y_{cs} \approx \sum \frac{(yEI_x)}{EI_x} \tag{4.1}$$

$$r_x \approx \sqrt{\sum \frac{(x^2EI_y + y^2EI_x)}{\sum (EI_y)}} \quad r_y \approx \sqrt{\sum \frac{(x^2EI_y + y^2EI_x)}{\sum (EI_x)}} \tag{4.2}$$

The radius of gyration, $l_s$, is the square root of the ratio of the polar moment of inertia to the mass, the polar moment of inertia being calculated about the centre of mass. For a rectangular building of side lengths $l$ and $b$, and a uniform mass distribution, Equation (4.3) applies.

$$l_s = \sqrt{(l^2 + b^2)/12} \tag{4.3}$$

The requirement for torsional radius $r_x$ to exceed 3.33 times the mass-stiffness eccentricity $e_{ox}$ (item 5 on the list at the beginning of this section) relates the torsional resistance to the driving lateral-torsional excitation, correctly favouring configurations with stiff perimeter elements and penalising those relying on central elements for lateral resistance. It is very similar to a requirement that has appeared for many years in the Japanese code.

The requirement for $r_x$ to exceed radius of gyration, $l_s$ (item 6 on the list at the beginning of this section), ensures that the first torsional mode of vibration does not occur at a higher period than the first translational mode in either direction, and demonstrating that this applies is an alternative way of showing that 'torsional flexibility' is avoided (EC8 Manual, (ISE/AFPS 2009)).

### *4.5.3 Regularity in elevation*

A building must satisfy all the following requirements to be classified as regular in elevation.

1 All the vertical load resisting elements must continue uninterrupted from foundation level to the top of the building or (where setbacks are present – see 4 below) to the top of the setback.
2 Mass and stiffness must either remain constant with height or reduce only gradually, without abrupt changes. Quantification is not provided in EC8; the EC8 Manual (ISE/AFPS 2009) recommends that buildings where the mass or stiffness of any storey is less than 70 per cent of that of the storey above or less than 80 per cent of the average of the three storeys above should be classified as irregular in elevation.
3 In buildings with moment-resisting frames, the lateral resistance of each storey (i.e. the seismic shear initiating failure within that storey, for the code-specified distribution of seismic loads) should not vary 'disproportionately' between storeys. Generally, no quantified limits are stated by EC8, although special rules are given where the variation in lateral resistance is due to masonry infill within the frames. The EC8 Manual (ISE/AFPS 2009) recommends that buildings where the strength of any storey is less than 80 per cent of that of the storey above should be classified as irregular in elevation.
4 Buildings with setbacks (i.e. where the plan area suddenly reduces between successive storeys) are generally irregular, but may be classified as regular if less than limits defined in the code. The limits broadly speaking are a total reduction in width from top to bottom on any face not exceeding 30 per cent, with not more than 10 per cent at any level compared to the level below. However, an overall reduction in width of up to half is permissible within the lowest 15 per cent of the height of the building.

## 4.6 Capacity design

EC8 Part 1 Section 2.2.4 contains some specific design measures for ensuring that structures meet the performance requirements of the code. These apply to all structures, not just buildings, and a crucial requirement concerns capacity design, which determines much of the content of the material-

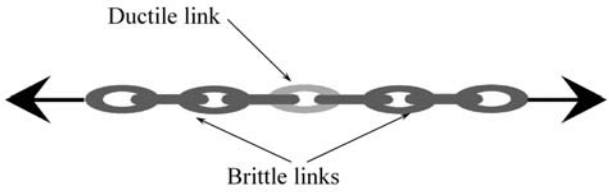



*Figure 4.5* Capacity design – ensuring that ductile links are weaker than brittle ones

specific rules for concrete, steel and composite buildings in Sections 5, 6 and 7 of EC8 Part 1.

Clause 2(P) of Section 2.2.4.1 states:

> In order to ensure an overall dissipative and ductile behaviour, brittle failure or the premature formation of unstable mechanisms shall be avoided. To this end, where required in the relevant Parts of EN 1998, resort shall be made to the capacity design procedure, which is used to obtain the hierarchy of resistance of the various structural components and failure modes necessary for ensuring a suitable plastic mechanism and for avoiding brittle failure modes.

Professor Paulay's 'ductile chain' illustrates the principle of capacity design – see Figure 4.5. The idea is that the ductile link yields at a load that is well below the failure load of the brittle links. Although most building structures are somewhat less straightforward than the chain used in Tom Paulay's example, one of the great strengths of the capacity design principle is that it relies on simple static analysis to ensure good performance, and is not dependent on the vagaries of a complex dynamic calculation.

Ensuring that columns are stronger than beams in moment frames, concrete beams are stronger in shear than in flexure, and steel braces buckle before columns, are three important examples of capacity design. A general rule for all types of frame building given in EC8 Part 1 Section 4.4.2.3 is that the moment strength of columns connected to a particular node should be 30 per cent greater than the moment strength of the beams:

$$\sum M_{Rc} \geq 1.3 \sum M_{Rb} \tag{4.4}$$

The rule must be satisfied for concrete buildings, but the alternative capacity design rules given in EC8 Section 6.6.3 may apply to steel columns (see Chapter 6 of this book).

One feature of capacity design is that it ensures that designers identify clearly which parts of the structure will yield in a severe earthquake (the 'critical' regions) and which will remain elastic. An important related clause is given by Clause 3(P) of Section 2.2.4.1 of EC8.

Since the seismic performance of a structure is largely dependent on the behaviour of its critical regions or elements, the detailing of the structure in general and of these regions or elements in particular, shall be such as to maintain the capacity to transmit the necessary forces and to dissipate energy under cyclic conditions. To this end, the detailing of connections between structural elements and of regions where non-linear behaviour is foreseeable should receive special care in design.

## 4.7 Other basic issues for building design

### *4.7.1 Load combinations*

Basic load combinations are given in EN 1990: *Basis for design*, and for seismic load combinations are as follows:

$$\underset{\text{Design action effect}}{E_d} = \underset{\text{Permanent}}{\sum G_{kj}} + \underset{\text{Earthquake}}{A_{Ed}} + \underset{\text{Reduced variable load}}{\sum \psi_{2i} Q_{ki}} \tag{4.5}$$

$\psi_{2i}$ is the factor defined in EN 1990, which reduces the variable (or live) load from its characteristic (upper bound) value to its 'quasi-permanent' value, expected to be present for most of the time. It is typically in the range 0.0 to 0.8, depending on the variability of the loading type.

### *4.7.2 'Seismic' mass*

The mass taken when calculating the earthquake loads should comprise the full permanent (or dead) load plus the variable (or live) load multiplied by a factor $\psi_{Ei}$. EC8 Part 1 Section 4.2.4 quantifies this as the factor $\psi_{2i}$ defined in Section 4.7.1 above multiplied by a further reduction factor $\varphi$ that allows for the incomplete coupling between the structure and its live load:

$$\psi_{Ei} = \psi_{2i}\varphi \tag{4.6}$$

Typical values of $\varphi$ are in the range 0.5 to 1, depending on the loading type.

### *4.7.3 Importance classes and factors*

Four importance classes are recognised, as shown in Table 4.1, which also shows the recommended $\gamma_I$ factor; this is, however, a 'Nationally Determined Parameter' (NDP), which may be varied in the National Annex.

Note that whereas in US practice the importance factors are applied to the seismic loads, in EC8 they are applied to the input motions. This makes an important difference when non-linear analysis is employed, since increasing the ground motions by X per cent may cause an increase of less than X per

*Table 4.1* Importance classes

| *Importance class* | *Buildings* | $\gamma_I$ |
|---|---|---|
| I | Buildings of minor importance for public safety, e.g. agricultural buildings, etc. | 0.8 |
| II | Ordinary buildings, not belonging in the other categories. | 1.0 (NB: not an NDP) |
| III | Buildings whose seismic resistance is of importance in view of the consequences associated with a collapse, e.g. schools, assembly halls, cultural institutions, etc. | 1.2 |
| IV | Buildings whose integrity during earthquakes is of vital importance for civil protection, e.g. hospitals, fire stations, power plants, etc. | 1.4 |

cent in forces, due to yielding of elements, but (possibly) more than X per cent in deflections, due to plastic strains and P-delta effects.

### *4.7.4 Primary and secondary members*

EC8 Part 1 Section 4.2.2 distinguishes between primary and secondary elements. Primary elements are those that contribute to the seismic resistance of the structure. Some structural elements can, however, be designated as 'secondary' elements, which are taken as resisting gravity loads only. Their contribution to seismic resistance must be neglected. These elements must be shown to be capable of maintaining their ability to support the gravity loads under the maximum deflections occurring during the design earthquake. This may be done by showing that the actions (moments, shears, axial forces) that develop in them under the calculated seismic deformations do not exceed their design strength, as calculated in EC2. Otherwise no further seismic design or detailing requirements are required.

An example of the use of secondary elements occurring in a frame building is the following arrangement (Figure 4.6). The perimeter frame is considered as the primary seismic resisting element and is designed for high ductility, while the internal members are considered secondary. This gives considerable architectural freedom for the layout of the internal spaces; the column spacing can be much greater than would be efficient in a moment resisting frame, while close spaced columns on the perimeter represents much less obstruction. This arrangement is (or was) favoured in US but not Japanese practice.

### *4.7.5 Other design measures in EC8 Part 1 Section 2.2.4*

The need for an adequate structural model for analysis is identified and, where necessary, soil deformability, the influence of non-structural elements

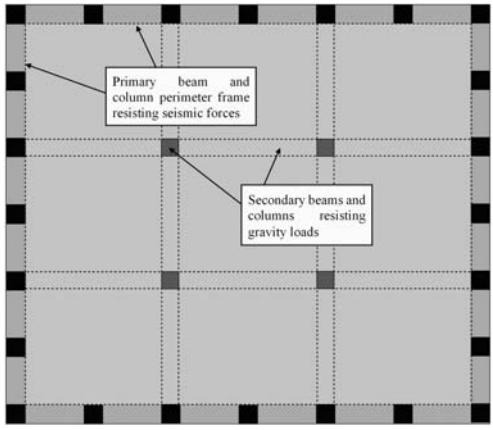



*Figure 4.6* Building with external primary perimeter frame and internal secondary members

and adjacent structures should be included in the analysis (Clause 2.2.4 1(4) P). More detailed advice on analysis is given in the EC8 Manual (ISE/AFPS 2009).

The need for quality control is discussed and, in particular, a formal quality system plan is specified for areas of high seismicity and structures of special importance (Section 2.2.4.3). Where a formal quality plan is applied to concrete buildings, a reduction in q values (and hence lateral strength requirements) is permitted – see Section 5.2.2.2(10).

## 4.8 Worked example for siting of structures

### *4.8.1 Introduction*

For this example, four sites (A, B, C and D) are postulated to be available for construction of the demonstration hotel structure. Preliminary site investigation was carried out at all the sites. Borehole data and SPT (Standard Penetration Test) and field vane shear tests were carried out at each site. This information is shown in Figure 4.7.

### *4.8.2 Notes on key aspects of each site*

Site A: Loose sands below water table imply a high liquefaction risk. Piled foundations likely to be necessary; piling through liquefiable material poses serious design problems associated with ensuring pile integrity and/or pile settlements.

Site B: Strong stiffness contrast between top 5 m of soft clay and stiff clay below implies a high amplification of ground motions, especially around the

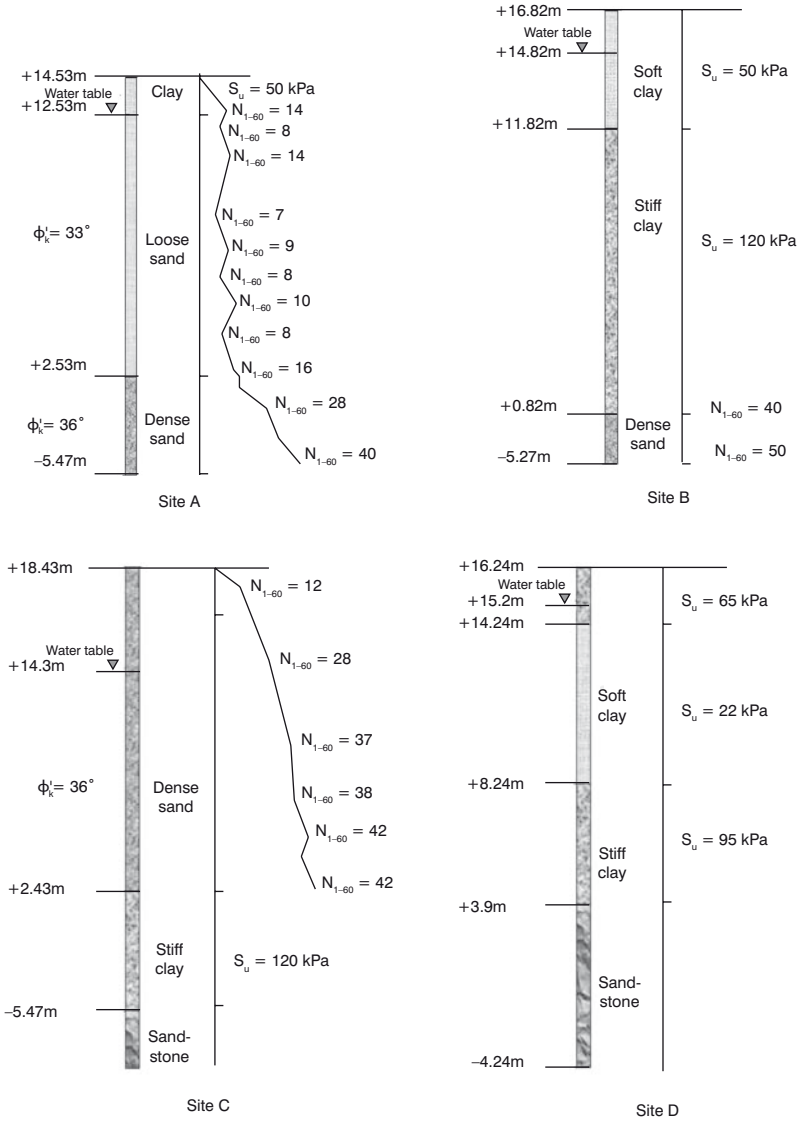



*Figure 4.7* Example borehole logs for possible sites for the building

0.5 second ($=4H/V_s$) period. Founding would need to be on to stiff clay, via piles or a deep basement.

Site C: Stiff materials throughout form good foundation material with lowered potential for ground motion amplification. Shallow foundation is feasible.

Site D: 6 m strata of soft clay may give rise to significant amplification of ground motions. Piling would be likely to be necessary into sandstone layer; relatively high shear strain differential between soft clay strata and stiffer strata above and below would probably result in plastic hinge formation in the piles.

### *4.8.3 Site selected for the hotel*

Choose 'Site C' for shallow foundation design.
Reasons:

a. good, dense sand layer with 16 m thickness with high SPT numbers, overlying stiff clay;
b. angle of internal friction is 36°;
c. above the water table.

## 4.9 Worked example for assessing structural regularity

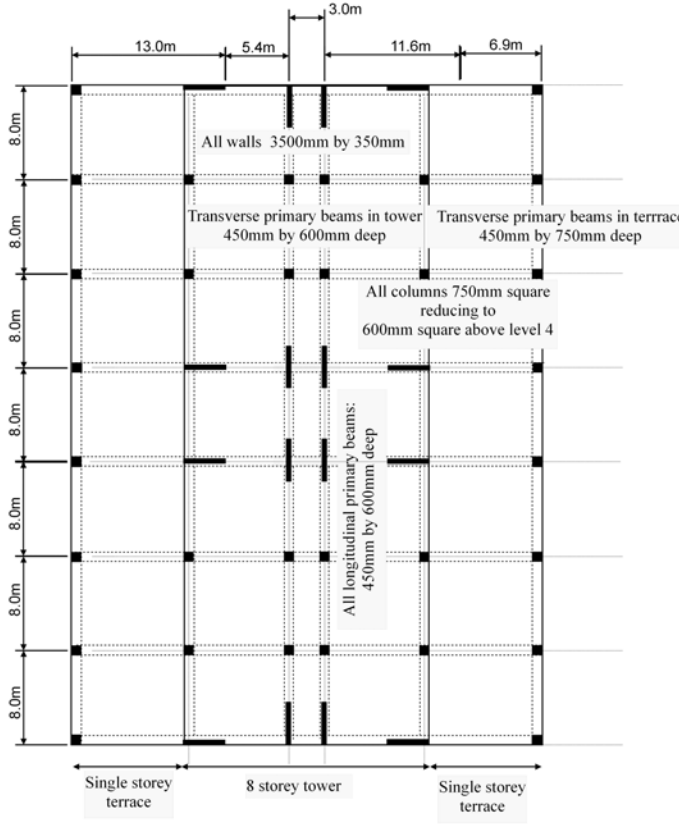
### *4.9.1 Introduction*

The structural layout shown in Figure 4.8 is now checked for regularity in plan and elevation. A concrete frame and shear wall scheme has been adopted.

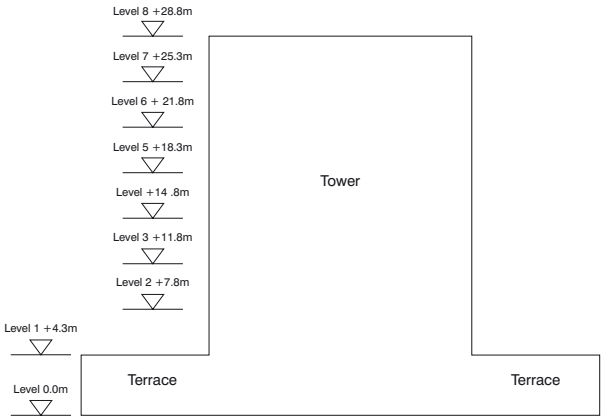
### *4.9.2 Regularity in plan*

All the following conditions must be met.

1 'Approximately' symmetrical distribution of mass and stiffness in plan. By inspection, it can be seen that a symmetrical distribution of stiffness has been achieved in plan, and there is no indication from the brief that significantly asymmetrical distributions of mass are to be expected.
2 A 'compact' shape, i.e. one in which the perimeter line is always convex, or at least encloses not more than 5 per cent re-entrant area (Figure 4.3). There are no re-entrant corners.
3 The floor diaphragms shall be sufficiently stiff in-plane not to affect the distribution of lateral loads between vertical elements. EC8 warns that this should be carefully examined in the branches of branched systems, such as L, C, H, I and X plan shapes.




*Plan*




*Elevation*

*Figure 4.8* Structural layout taken for regularity checks

The floor slabs in the tower are rectangular, without branches, and have an aspect ratio in the tower (see 4 below) of 56 m/20 m = 2.8, which is relatively compact. Given the uniform distribution of mass and lateral load resisting elements (i.e. the frames and shear walls) in the long direction, a continuous concrete solid slab or topping slab over precast elements of at least 70 mm would not be expected to give rise to uneven load distributions, unless there were substantial openings in the slabs.

4 The ratio of longer side to shorter sides in plan does not exceed 4. The ratio in the tower is 2.8 (see above).

5 The torsional radius, $r_x$, in the X (short) direction must exceed 3.33 times $e_{ox}$, the eccentricity between centres of stiffness and mass in the X direction. Similarly, $r_y$ must exceed 3.33 times $e_{oy}$.

The EC8 Manual (ISE/AFPS 2009) gives conservative but simplified rules for satisfying this condition for some standard cases, but does not cover that of a uniform space frame with isolated shear walls, as here. The well distributed layout of shear walls and frames suggests that the structure should possess adequate torsional stiffness. A 3D computer analysis was carried out to perform a detailed check, as follows.

| | |
|---|---|
| Top deflection at top of building in X (short) direction under 1000 kN load applied at stiffness centre in X direction: | 7.35 mm |
| Top deflection at top of building in Y (long) direction under 1000 kN load applied at stiffness centre in Y direction: | 7.14 mm |
| Top rotation at top of building about Z (vertical) axis under 1000 kNm moment about Z axis: | 8.18 E-6 radians |

Note: The building is taken as perfectly symmetrical, and so the geometric centre, the centre of stiffness and the centre of mass all coincide. For cases where the stiffness and mass centre do not coincide with the geometric centre, see the example calculation in Appendix A of the EC8 Manual (ISE/AFPS 2009).

| | | |
|---|---|---|
| X stiffness = | $1000/(7.35E^{-3})$ = | 136E3 kN/m |
| Y stiffness = | $1000/(7.14E^{-3})$ = | 140E3 kN/m |
| Torsional stiffness = | $1000/(8.18E^{-6})$ = | 122E6 kNm/radian |
| $r_x$ = | $(122E6/140E3)^{1/2}$ = | 29.5 m |
| $0.3r_x$ = | 0.3*29.5 = | 8.9 m |
| $r_y$ = | $(61.7E6/137E3)^{1/2}$ = | 30.0 m |
| $0.3r_y$ = | 0.3*30 = | 9.0 m |

Therefore, the separation between centres of mass and stiffness needs to be less than about 9 m.

$r_x$ and $r_y$ must exceed the radius of gyration, $l_s$, otherwise the building is classified as 'torsionally flexible', and the $q$ values in concrete buildings are greatly reduced.

The radius of gyration, assuming a uniform mass distribution, is calculated as follows. It can be seen that the requirement for regularity is satisfied.

$$l_s = [(56^2 + 20^2)/12]^{1/2} = 17.2 \text{ m} < r_x\ (=29.5 \text{ m}) \text{ and } < r_y\ (=30 \text{ m}) - \text{OK}.$$

The EC8 Manual (ISE/AFPS 2009) notes that an alternative demonstration that this condition is satisfied is to show that the first predominantly torsional mode has a lower period than either of the first predominantly translational modes in the two principal directions. A 3D computer analysis, which assumed that the mass and stiffness centres coincided, gave the following values, confirming that this applies to the present structure. The period of the first torsional mode is well below that of the first two translational modes, reflecting the large excess of $r_x$ and $r_y$ over $l_s$ calculated previously.

| | |
|---|---|
| Period of first Y translational mode | 0.90 s |
| Period of first X translational mode | 0.88 s |
| Period of first torsional mode | 0.62 s |

Hence all the conditions for regularity in plan are satisfied.

### *4.9.3 Regularity in elevation*

The following conditions must be met:

1 All the vertical load resisting elements must continue uninterrupted from foundation level to the top of the building or (where setbacks are present – see 4 below) to the top of the setback.

Satisfied by inspection.

2 Mass and stiffness must either remain constant with height or reduce only gradually, without abrupt changes. Quantification is not provided in EC8; the EC8 Manual (ISE/AFPS 2009) recommends that buildings where the mass or stiffness of any storey is less than 70 per cent of that of the storey above or less than 80 per cent of the average of the three storeys above should be classified as irregular in elevation.

The ground floor has a storey height of 4.3 m, compared with 3.5 m for the upper storeys, which tends to reduce stiffness by a factor of approximately $(3.5/4.3)^2 = 66$ per cent, which is a bit less than the 70 per cent or 80 per cent proposed above. However, there are more columns in the ground floor

– an additional 50 per cent – which offsets this, as does the base fixity of the ground floor columns and shear walls. Overall, this suggests that the stiffness ratio is within limits.

A 3D computer analysis shows that under earthquake loading, the ground floor storey drift is significantly less than that of the first floor, confirming that the stiffness check is satisfied. There is a stiffness change where the columns reduce in section at the fifth floor, but this is a reduction in stiffness so the regularity condition is met.

The assumption that there is similar use of the floors in the tower at all levels above ground level leads to the conclusion that the mass at one level is always less that of the level below.

Hence the 'soft storey' check is satisfied.

3 In buildings with moment-resisting frames, the lateral resistance of each storey (i.e. the seismic shear initiating failure within that storey, for the code-specified distribution of seismic loads) should not vary 'disproportionately' between storeys. Generally, no quantified limits are stated by EC8, although special rules are given where the variation in lateral resistance is due to masonry infill within the frames. The ISE Manual on EC8 (ISE/AFPS 2009) recommends that buildings where the strength of any storey is less than 80 per cent of that of the storey above should be classified as irregular in elevation.

It is unlikely that any viable design would violate this condition. It cannot, of course, be checked without knowledge of the reinforcement in the beams, columns and walls.

4 Buildings with setbacks (i.e. where the plan area suddenly reduces between successive storeys) are generally irregular, but may be classified as regular if less than limits defined in the code. The limits broadly speaking are a total reduction in width from top to bottom on any face not exceeding 30 per cent, with not more than 10 per cent at any level compared to the level below. However, an overall reduction in width of up to half is permissible within the lowest 15 per cent of the height of the building.

The reduction in building width between the ground and first floors, as the tower rises above the podium, constitutes a setback. Since the ground floor height, at 4.3 m, is less than 15 per cent of the total height of 28.8 m (28.8 times 0.15 = 4.32 m), and the reduction in width is from 40 m to 20 m (= 50 per cent reduction), the setback remains (just) within 'regular' limits.

Hence all the conditions for regularity in elevation are satisfied.

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# 5 Design of concrete structures

*A. Campbell and M. Lopes*

## 5.1 Introduction

As noted in earlier chapters, EC8 aims to ensure life safety in a large earthquake together with damage limitation following a more frequent event. Whilst the code allows these events to be resisted by either dissipative (ductile) or non-dissipative (essentially elastic) behaviour, there is a clear preference for resisting larger events through dissipative behaviour. Hence, much of the code is framed with the aim of ensuring stable, reliable dissipative performance in predefined 'critical regions', which limit the inertial loads experienced by other parts of the structure. The design and detailing rules are formulated to reflect the extent of the intended plasticity in these critical regions, with the benefits of reduced inertial loads being obtained through the penalty of more stringent layout, design and detailing requirements.

This is particularly the case for reinforced concrete structures where such performance can only be achieved if strength degradation during hysteretic cycling is suppressed by appropriate detailing of these critical zones to ensure that stable plastic behaviour is not undermined by the occurrence of brittle failure modes such as shear or compression in the concrete or buckling of reinforcing steel.

With this in mind, three dissipation classes are introduced:

- Low (ductility class low (DCL)) in which virtually no hysteretic ductility is intended and the resistance to earthquake loading is achieved through the strength of the structure rather than its ductility.
- Medium (DCM) in which quite high levels of plasticity are permitted and corresponding design and detailing requirements are imposed.
- High (DCH) where very large inelastic excursions are permitted accompanied by even more onerous and complex design and detailing requirements.

In this chapter, the primary focus is on DCM structures, which are likely to form the most commonly used group in practice. However, the limited provisions for DCL structures and the additional requirements for DCH

structures are briefly introduced. Only the design of in-situ reinforced concrete buildings to EC8 Part 1 is addressed here. Rules for the design of precast concrete structures are included in Section 5.11 of the code and guidance on their use in standard building structures is given in the Institution of Structural Engineers' manual on the application of EC8 (Institution of Structural Engineers/SECED/AFPS, 2009). Prestressed concrete structures, although not explicitly excluded from the scope of EC8 Part 1, are implicitly excluded as dissipative structures since the rules for detailing of critical regions are limited to reinforced concrete elements. Prestressed components could still be used within dissipative structures but should then be designed as protected elements, as discussed later.

## 5.2 Design concepts

### *5.2.1 Energy dissipation and ductility class*

EC8 is not a stand-alone code but relies heavily on the material Eurocodes to calculate resistance to seismic actions. EC2 (BS EN 1992-1-1:2004 in the UK) fulfils this function for concrete structures. For DCL structures, EC8 imposes very limited material requirements in addition to the EC2 provisions, whereas for DCM and DCH structures, increasingly more onerous material requirements are imposed, together with geometrical constraints, capacity design provisions and detailing rules tied to local ductility demand.

These rules are aimed at the suppression of brittle failure modes, provision of capacity to withstand non-linear load cycles without significant strength degradation, and improving the ability of defined critical regions to undergo very high local rotational ductility demands in order to achieve the lower global demands. Typically, this includes:

- Ensuring flexural yielding prior to shear failure.
- Providing stronger columns than beams to promote a more efficient beam sidesway mode of response and avoid soft storey failure.
- Retention of an intact concrete core within confining links.
- Prevention of buckling of longitudinal reinforcement.
- Limiting flexural tension reinforcement to suppress concrete crushing in the compression zone.

These detailed requirements build upon the guidelines in Section 4 of EC8 Part 1 on:

- Regularity of structural arrangement, aiming to promote an even distribution of ductility demand throughout the structure.
- Providing adequate stiffness, both to limit damage in events smaller than the design earthquake and to reduce the potential for significant secondary P-$\delta$ effects.

### *5.2.2 Structural types*

EC8 Part 1 classifies concrete buildings into the following structural types:

- frame system
- dual system, which may be either frame or wall equivalent
- ductile wall system
- system of large, lightly reinforced walls
- inverted pendulum system
- torsionally flexible system.

Apart from torsionally flexible systems, buildings may be classified as different systems in the two orthogonal directions.

Frame systems are defined as those systems where moment frames carry both vertical and lateral loads and provide resistance to 65 per cent or more of the total base shear.

Conversely, buildings are designated as wall systems if walls resist 65 per cent or more of the base shear. Walls may be classed as either ductile walls, which are designed to respond as vertical cantilevers yielding just above a rigid foundation, or as large lightly reinforced walls. Ductile walls are further subdivided into coupled or uncoupled walls. Coupled walls comprise individual walls linked by coupling beams, shown in Figure 5.1, resisting lateral loads through moment and shear reactions in the individual walls together with an axial tensile reaction in one wall balanced by an axial compressive reaction in the other to create a global moment reaction. The magnitude of these axial

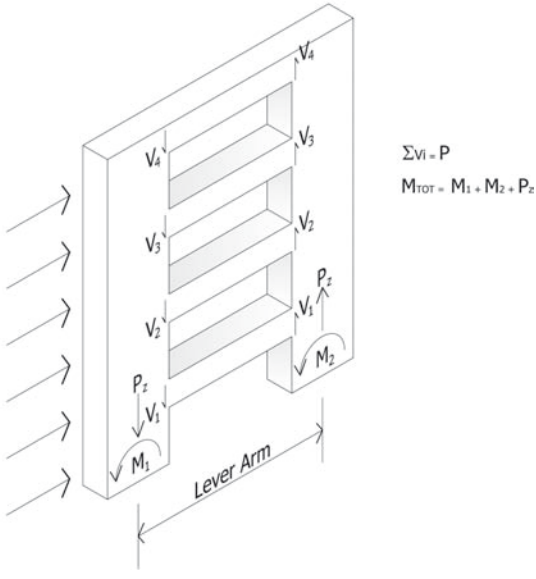



*Figure 5.1* Coupled Wall System

loads is limited by the shear forces that can be transferred across the coupling beams. In order to qualify as a coupled wall system, the inclusion of coupling beams must cause at least a 25 per cent reduction in the base moments of the individual walls from that which would have occurred in the uncoupled case. As coupled walls dissipate energy, not only in yielding at the base but also in yielding of the coupling beams, buildings with coupled walls may be designed for lower inertial loads than buildings with uncoupled walls to reflect their greater ductility and redundancy.

Large lightly reinforced walls are a category of structure introduced in EC8 and not found in other national or international seismic codes. These walls are assumed to dissipate energy, not through hysteresis in plastic hinges, but by rocking and uplift of the foundation, converting kinetic energy into potential energy of the structural mass and dissipating this through radiation damping. The dimensions of these walls or their fixity conditions or the presence of stiff orthogonal walls effectively prevent plastic hinging at the base. These provisions are likely to find wide application in heavy concrete industrial structures. However, since this book is concerned primarily with conventional building structures, this type of structure is not considered further here.

Dual systems are structural systems in which vertical loads are carried primarily by structural frames but lateral loads are resisted by both frame and wall systems. From the earlier definitions, it is clear that, to act as a dual system, the frame and wall components must each carry more than 35 per cent but less than 65 per cent of the total base shear. When more than 50 per cent of the base shear is carried by the frames, it is designated a frame-equivalent dual system. Conversely, it is termed a wall-equivalent dual system when walls carry more than 50 per cent of the base shear.

Torsionally flexible systems are defined as those systems where the radius of gyration of the floor mass exceeds the torsional radius in one or both directions. An example of this type of system is a dual system of structural frames and walls with the stiffer walls all concentrated near the centre of the building on plan.

Inverted pendulum systems are defined as systems where 50 per cent of the total mass is concentrated in the upper third of the height of the structure or where energy dissipation is concentrated at the base of a single element. A common example would normally be one-storey frame structures. However, single storey frames are specifically excluded from this category provided the normalised axial load, $\upsilon_d$, does not exceed 0.3.

$$\upsilon_d = N_{Ed}/(A_c * f_{cd}) \tag{5.1}$$

where $N_{Ed}$ is the applied axial load in the seismic design situation, $A_c$ is the area of the column and $f_{cd}$ is the design compressive strength of the concrete (i.e. the characteristic strength divided by the partial material factor, which can usually be taken as 1.5).

The treatment of both torsionally flexible and inverted pendulum systems within EC8 is discussed further in Section 5.4.

### *5.2.3 q Factors for concrete buildings*

Table 5.1 shows the basic values of *q* factors for reinforced concrete buildings. These are the factors by which the inertial loads derived from an elastic response analysis may be reduced to account for the anticipated non-linear response of the structure, together with associated aspects such as frequency shift, increased damping, overstrength and redundancy. The factor, $\alpha_u/\alpha_1$, represents the ratio between the lateral load at which structural instability occurs and that at which first yield occurs in any member. Default values of between 1.0 and 1.3 are given in the code with an upper limit of 1.5. Higher values than the default figures may be utilised but need to be justified by pushover analysis.

For walls or wall-equivalent dual systems, the basic value of the behaviour factor then needs to be modified by a factor, $k_w$, which accounts for the prevailing failure mode of the wall, the *q* factors being reduced on squat walls where more brittle shear failure modes tend to govern the design.

$$k_w = (1 + \alpha_0)/3 \tag{5.2}$$

where $\alpha_0$ is the prevailing aspect ratio, $h_w/l_w$, of the walls.

A lower limit of 0.5 is placed on $k_w$ for walls with an aspect ratio of 0.5 or less, with the basic q factor being applied unmodified to walls with an aspect ratio of 2 or more.

The basic $q_0$ factors tabulated are for structures that satisfy the EC8 regularity criteria, the basic factors needing to be reduced by 20 per cent for structures that are irregular in elevation according to the criteria given earlier in Chapter 4.

*Table 5.1* Basic value of the behaviour factor, $q_0$, for systems regular in elevation

| *Structural type* | *DCM* | *DCH* |
|---|---|---|
| Frame system, dual system, coupled wall system | $3.0\alpha_u/\alpha_1$ | $4.5\alpha_u/\alpha_1$ |
| Uncoupled wall system | 3.0 | $4.0\alpha_u/\alpha_1$ |
| Torsionally flexible system | 2.0 | 3.0 |
| Inverted pendulum system | 1.5 | 2.0 |

### *5.2.4 Partial factors*

In checking the resistance of concrete elements, the partial factors for material properties, $\gamma_c$ and $\gamma_s$, for concrete and reinforcement respectively, are generally taken as those for the persistent and transient design situation

rather than for the accidental design situation that may initially appear to be more in keeping with an infrequent event, such as the design earthquake. Hence, a value of 1.5 is adopted for $\gamma_c$ and 1.15 for $\gamma_s$ in the UK, the values being defined in the National Annex to EC2 for each country. This practice is based upon an implicit assumption that the difference between the partial factors for the persistent and transient situation and those for the accidental situation are adequate to cater for possible strength degradation due to cyclic deformations. The use of these material factors has the added benefit to the design process that standard EC2 design charts can be used.

## 5.3 Design criteria

### *5.3.1 Capacity design*

Capacity design is the basic concept underpinning the EC8 design philosophy for ductile structures (DCM and DCH). Therefore it is important to fully understand this basic principle in order to place in context the design rules aimed at implementing it.

This concept can be exemplified considering the chain, introduced by Paulay (1993) and represented in Figure 5.2, in which link 1 is ductile and all other links are brittle.

According to standard design procedures for quasi-static loading (termed 'direct design'), as the applied force is the same for all links the design force is also equal on all links. Therefore, assuming there are no reserve strengths, the yield capacity of all links is the same. In this situation the system cannot resist any force above $F_y$, at which rupture of the brittle links takes place. Therefore with direct design the overall increase in length of the chain at rupture is:

$$\delta u = 4\delta y \tag{5.3}$$

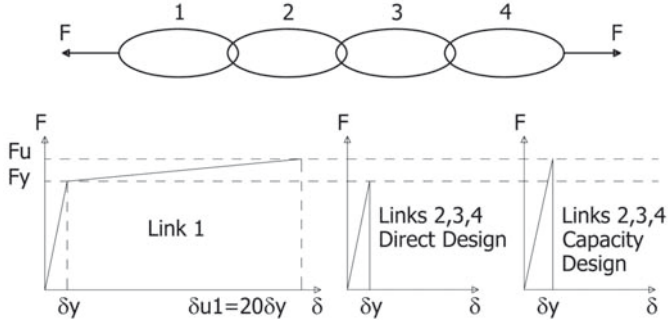



*Figure 5.2* Ductility of chain with brittle and ductile links

According to capacity design principles, to maximise the ductility of the chain, some links have to be chosen to have ductile behaviour and be designed with that purpose. The rest of the structure must be designed with excess strength in order to remain elastic during the plastic deformations of the ductile links. For this purpose the design force of the brittle links must be equal to the maximum resistance of the ductile links after yielding, that is, a force equal or above $F_u$. The ductile link behaves like a fuse, which does not allow the applied force acting on the brittle links to increase above their maximum resistance. Therefore the force applied on the chain can increase above $F_y$ up to the value $F_u$, but cannot exceed this value. At this stage the chain collapses at a displacement much higher than the chain designed with the direct design methodology, as follows:

$$\delta u = 3\delta y + \delta u_1 = 3\delta y + 20\ \delta y = 23\ \delta y \tag{5.4}$$

Hence, the brittle links must be designed for a force different from the ductile link, which is a function not of the notional applied load but of the capacity of the ductile link, in order to prevent the premature failure of the brittle links before the deformation capacity of the ductile links is exhausted. The fact that the design action effects in predefined 'protected' elements are a function of the resistance of other key elements is a basic characteristic of capacity design, and is an important difference to standard design procedures for quasi-static loading.

This highlights the fact that the indiscriminate provision of excess strength, which is usually considered positive according to standard design procedures, may adversely affect the non-linear behaviour of a structural system, as it may prevent an intended ductile link from acting like a fuse. Hence, if after designing a ductile frame, the flexural reinforcement of beams or of the base section of walls is increased, this is not necessarily a 'safe' change since it may increase the forces transmitted to other parts of the structure.

Whilst capacity design is an important concept for seismic design in all materials, it is included here because it is particularly relevant to reinforced concrete structures, which can potentially exhibit brittle failure modes unless attention is paid to suppressing these modes in the design and detailing.

In the case of reinforced concrete elements the best way to dissipate energy is by flexural yielding, as shear and axial forces tend to induce brittle behaviour. Therefore, the ductility of a structure can generally be optimised by enforcing flexural yielding at specific locations (ductile links), called plastic hinge zones, avoiding any type of shear or axial compressive failure (brittle links) and designing the rest of the structure to remain elastic throughout the development of the plastic hinges.

The approach adopted by EC8 to promote capacity design of reinforced concrete structures is to choose critical regions of the structure (the plastic hinge zones referred to above) that are designed to yield in flexure when

subject to the design earthquake loading, modified by the q factor appropriate to the structural system. These critical regions are then detailed to undergo large, inelastic cyclic deformations and fulfil the role of structural 'fuses', limiting the inertial loads that can be transferred to the remaining 'protected' parts of the structure, which can then be designed to normal EC2 provisions.

The capacity design rules in EC8 are discussed in more detail later but primarily cover:

- Derivation of shear forces in members from the flexural capacity of their critical regions.
- Promotion of the strong column/weak beam hierarchy in frame structures, evaluating column moments as a function of the capacity of the beams framing into them.

In both cases, in the design of notionally elastic parts of the structure, an allowance for overstrength of the critical regions is made, a greater allowance being made for DCH than for DCM structures.

### *5.3.2 Local ductility provisions*

The EC8 design rules take account of the fact that, to achieve the global response reductions consistent with the q factor chosen, much greater local ductility has to be available within the critical regions of the structure. Design and detailing rules for these critical regions are therefore formulated with the objective of ensuring that:

- Sufficient curvature ductility is provided in critical regions of primary elements.
- Local buckling of compressed steel within plastic hinge regions is prevented.

This is fulfilled by special rules for confinement of critical regions, particularly at the base of columns, within beam/column joints and in boundary elements of ductile walls, which depend, in part, on the local curvature ductility factor $\mu_\Phi$. This is related to the global $q$ factor as follows:

$$\mu_\Phi = 2q_0 - 1 \qquad \text{if } T_1 \geq T_C \tag{5.5}$$

$$\mu_\Phi = 1 + 2(q_0 - 1)T_C/T_1 \qquad \text{if } T_1 < T_C \tag{5.6}$$

where $q_0$ is the basic behaviour factor given in Table 5.1 before any reductions are made for lack of structural regularity or low aspect ratio of walls. $T_1$ is the fundamental period of the building and $T_C$ is the period at the upper end of the constant acceleration zone of the input spectrum as described in Section 3.2 of EC8 Part 1.

Additionally, if Class B reinforcement is chosen rather than Class C in DCM structures, the value of $\mu_\Phi$ should be at least 1.5 times the value given by Equations 5.5 and 5.6, whichever is applicable.

### *5.3.3 Primary and secondary members*

Primary elements are specified as being those elements that contribute to the seismic resistance of the structure and are designed and detailed to the relevant provisions of EC8 for the designated ductility class. Elements that are not part of the main system for resisting seismic loading can be classed as secondary elements. They are assumed to make no contribution to seismic resistance, and secondary concrete elements are designed to EC2 to resist gravity loads together with imposed seismic displacements derived from the response of the primary system. In this case, no special detailing requirements are imposed upon these elements.

A common problem in seismic design is that of unintentional stiffening of the designated seismic load resisting system by secondary or non-structural elements (e.g. masonry partition walls) leading to a higher frequency of response and generally increased inertial loads. To guard against this, EC8 specifies that the contribution of secondary elements to the lateral stiffness should be no more than 15 per cent of that of the primary elements. If secondary elements do not meet this criterion, one option is to provide flexible joints to prevent stiffening of the primary system by these elements.

Whilst this stiffness limit protects against the global effects of unintentional stiffening, the designer also needs to be aware of potentially adverse local effects such as:

- Local changes to the intended load paths, potentially leading to increased loads on members not designed to cater for them or introducing a lack of regularity into discrete areas of the structure, modifying their dynamic response.
- Stiffening of parts of individual members (e.g. columns restrained by masonry panels over part of their height) preventing the intended ductile flexural response from occurring and resulting in a brittle shear failure.

Guidance on local stiffening issues associated with the most common case of masonry infill panels is given in Sections 4.3.6 and 5.9 of the code.

### *5.3.4 Stiffness considerations*

Apart from its major influence in determining the magnitude of inertial loads, dealt with in earlier chapters, structural stiffness is important in meeting the damage limitation provisions of EC8 Part 1 (Clause 4.4.3) and in assessing the significance of *P*-δ effects as per Clause 4.4.2.2 (2) to (4).

Both effectively place limits on storey drift, the former explicitly albeit for a lower return period earthquake, and the latter implicitly through the inter-storey drift sensitivity coefficient, θ. In both cases, the relative displacements between storeys, $d_{e,r}$, if obtained from a linear analysis, should be multiplied by a displacement behaviour factor, $q_d$, to obtain the plastic realtive displacements, $d_r$. When the period of response of the structure is greater than $T_C$ (i.e. on the constant displacement or constant velocity portion of the response spectrum), $q_d$, is equal to the behaviour factor $q$, so that the plastic displacement is equal to the elastic displacement obtained from the unreduced input spectrum. However, $q_d$ exceeds $q$ at lower periods as defined in Appendix B of the code.

In calculating displacements, EC8 requires that the flexural and shear stiffness of concrete structures reflect the effective stiffness consistent with the level of cracking expected at the initiation of yield of the reinforcement. If the designer does not take the option of calculating the stiffness reduction directly through pushover analysis, for example, the code allows the effective stiffness to be based upon half of the gross section stiffness [Clause 4.3.1 (7)] to account for softening of the structure at the strain levels consistent with reinforcement yield. It is acknowledged that the true stiffness reduction would probably be greater than this but the value chosen is a compromise; lower stiffness being more onerous for *P*-δ effects but less onerous for calculation of inertial loading on the structure. The EC8 approach, whilst similar to performance-based methodologies elsewhere, differs in applying a uniform stiffness reduction independent of the type of element considered. Paulay and Priestley (1992) and Priestley (2003) propose greater stiffness reductions in beams than in columns, reflecting the weak beam/strong column philosophy and the beneficial effects of compressive axial loads.

Checks on damage limitation aim to maintain the maximum storey drifts below limiting values set between 0.5 per cent and 1 per cent of the storey height, dependent upon the ductility and fixity conditions of the non-structural elements. The amplified displacements for the design earthquake are modified by a reduction factor, ν, of either 0.4 or 0.5, varying with the importance class of the building, to derive the displacements applicable for the more frequent return period earthquake considered for the damage limitation state.

The inter-storey drift sensitivity coefficient, θ, used to take account of *P*-δ effects, is defined in Equation (5.7) below:

$$\theta = P_{tot} * d_r / (V_{tot} * h) \tag{5.7}$$

$P_{tot}$ is the total gravity load at and above the storey, $V_{tot}$ the cumulative seismic shear force acting at each storey and h the storey height. If the maximum value of θ at any level is less than 0.1, then *P*-δ effects may be ignored. If θ exceeds 0.3, then the frame is insufficiently stiff and an alternative solution is required.

For values of θ between 0.1 and 0.2, an approximate allowance for *P*-δ effects may be made by increasing the analysis forces by a factor of 1/ (1– θ) whilst, for values of θ of between 0.2 and 0.3, a second order analysis is required.

### *5.3.5 Torsional effects*

A simplified approach towards catering for the increase in seismic forces due to accidental eccentricity in regular structures is given in Clause 4.3.3.2.4 of EC8 Part 1. Loads on each frame are multiplied by a factor, δ, equal to $[1 + 0.6(x/L_e)]$ where $x$ is the distance of the frame from the centre of mass and $L_e$ is the distance between the two outermost load resisting elements.

Hence, for a building where the mass is uniformly distributed, the forces and moments on the outermost frames are increased by 30 per cent. Fardis et al (2005) note that this simplified method is conservative by a factor of 2 on average for structures with the stiffness uniformly distributed in plan. Where this is judged to be excessive, the general approach of Clause 4.3.2 may be applied within a 3D analysis.

However, as the expression for δ was derived for structures with the stiffness uniformly distributed in plan, it may produce unsafe results for structures with a large proportion of the lateral stiffness concentrated at a single location. Therefore, it should not be applied to torsionally flexible systems.

## 5.4 Conceptual Design

As already referred to in Chapter 4, EC8 provides guidance on the basic principles of good conception of building structures for earthquake resistance. These principles apply to all types of buildings and are qualitative and not mandatory. However, in Section 5 'Specific rules for concrete buildings', besides providing guidance and rules for the design of several types of reinforced concrete building structures, EC8 clearly encourages designers to choose the most adequate structural types. Next, the most important quantitative aspects and clauses of Section 5 of EC8 that condition the choice of structural types are highlighted. These are:

- reduction of the *q* factors assigned to the less adequate structural types and to irregular structures;
- the control of inter-storey drifts, which tends to penalize more flexible structures.

The reduction of the *q* factors is apparent in Table 5.1. The torsionally flexible system and the inverted pendulum system are clearly penalized with *q* factors that can be less than half of the ones prescribed for the frame, dual or coupled wall systems. Buildings with walls may fall under the classification

of torsionally flexible if the walls are concentrated at a single location in plan. Buildings with several walls closer to the periphery of the floor plans tend to meet the criterion that avoids this type of classification.

Buildings with irregularities in plan or along the height are penalized as the irregularities tend to induce concentration of ductility demands at some locations of the structure as opposed to the more uniform spread of ductility demands in regular buildings. In particular the interruption of vertical elements that are important for the resistance to horizontal inertia forces (including both columns and walls, the latter being particularly important) before reaching the foundations, is a type of irregularity that the observation of past earthquakes shows is more likely to lead to catastrophic failure. EC8 only includes a moderate reduction of 20 per cent in the behaviour factor for structures with this type of irregularity, but designers are cautioned to avoid it if at all possible.

The above means there is a price to pay for the use of these systems and buildings, both in terms of increasing amounts of reinforcement and dimensions of structural elements, which in regions of medium and high seismicity may create problems of compatibility with architectural requirements.

Frame or frame-equivalent dual structures may not be stiff enough to meet the requirements for the control of inter-storey drifts especially in the cases of tall buildings in regions of medium and high seismicity, prompting designers to conceive coupled wall or wall-equivalent dual structures. These structural types generally present a better combination of stiffness and ductility characteristics, important for the seismic behaviour of reinforced concrete buildings.

Other requirements of design, related to the application of capacity design, may influence the overall structural conception of the buildings in order to make it possible to satisfy those requirements. The most important is the weak-beam/strong-column design of frames, referred to in Section 5.6.2, aimed at preventing the formation of soft-storey mechanisms. For this purpose it is necessary that the sum of the flexural capacity of the columns converging at a joint is greater than the flexural capacity of the beams converging at the same joint. In practical terms this implies that the dimension of the columns in the bending plane must not be much smaller than the dimension of the beams. This is not too difficult to enforce in a single plan direction, but its implementation in two orthogonal plan directions simultaneously may imply that both dimensions of the columns have to be large. This is likely to create difficulties in compatibility with architectural requirements as, in many cases, architects wish columns to protrude as little as possible from inner partition walls and exterior walls. However, the weak-beam/strong column requirement is not mandatory in all ductile structures with frames: the inclusion of walls with reasonable stiffness and strength in the horizontal resisting system of reinforced concrete buildings is enough for the prevention of the formation of soft-storey mechanisms, associated

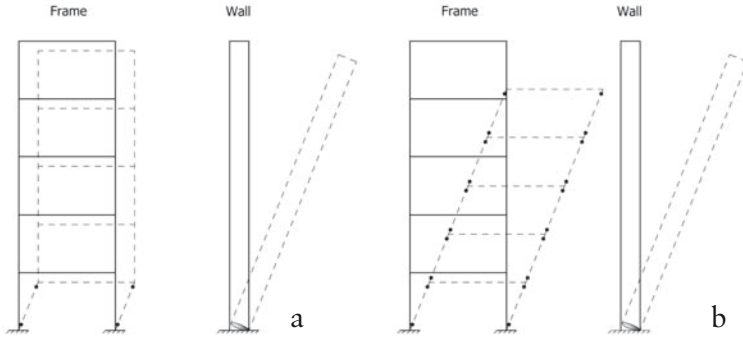



*Figure 5.3* Kinematic incompatibility between wall deformation and soft-storey

with hinging of both extremities of all columns at one floor. This is because there is a kinematic incompatibility between the wall deformation and the deformation of the frames at the formation of the soft-storey mechanism, as illustrated in Figure 5.3.a. Figure 5.3.b shows that in dual systems if the hinges develop at the columns a mechanism can only develop if plasticity spreads throughout the height of the building

In order to quantify how 'reasonable' the wall stiffness and strength is, EC8 establishes that for the above purpose the walls must absorb at least 50 per cent of the total base shear in the seismic design situation. Therefore in wall or wall-equivalent dual structures the walls are considered stiff enough to prevent the formation of the soft-storey mechanisms, regardless of frame design.

This allows designers to solve the above-mentioned problem of compatibility between the weak-beam/strong column design with architectural requirements by providing at least in one direction walls stiff enough to take at least half of the global seismic shear in that direction. And the fact there is no need to enforce the weak-beam/strong column requirement simplifies the design process. This adds more advantages to the choice of wall or wall-equivalent dual structures.

The control of inter-storey drifts due to the presence of walls in dual or wall structures, in particular the ones designed for the lower levels of lateral stiffness and strength (DCH), also helps to limit the possible consequences of effects associated with the presence of secondary structural elements or non-structural elements, as already referred to in Section 5.3.3. For this reason the additional measures prescribed by EC8 to account for the presence of masonry infills apply only to frame or frame-equivalent dual structures (Clause 4.3.6.1) of DCH structures.

Another requirement, with possible implications for all vertical structural elements in all types of structures, is the limitation of the axial force, aimed at restricting the negative effects of large compressive forces on the available ductility. In order to meet this requirement, in some cases designers may be

forced to provide columns with cross-section areas larger than desirable for compatibility with architectural requirements.

## 5.5 Design for DCL

As noted earlier, EC8 permits the design of structures for non-dissipative behaviour. If this option is taken, then standard concrete design to EC2 should be carried out, the only additional requirement being that reasonably ductile reinforcing steel, Class B or C as defined in EC2, must be used. A q factor of up to 1.5 is permitted, this being regarded as effectively an overstrength factor. However, other than for design of secondary elements, the DCL option is only recommended for areas of low seismicity as defined by Clause 3.2.1(4) of EC8 Part 1.

## 5.6 Frames – design for DCM

### *5.6.1 Material and geometrical restrictions*

There are limited material restrictions for DCM structures. In addition to the requirement to use Class B or C reinforcement, as for DCL, only ribbed bars are permitted as longitudinal reinforcement of critical regions and concrete of Class C16/20 or higher must be used.

Geometrical constraints are also imposed on primary elements.

#### *5.6.1.1 Beams*

In order to promote an efficient transfer of moments between columns and beams, and reduce secondary effects, the offset of the beam centre line from the column centre line is limited to less than a quarter of the column width.

Also, to take advantage of the favourable effect of column compression on the bond of reinforcement passing through the beam/column joint:

| | |
|---|---|
| Width of beam | ≤ (column width + depth of beam) |
| | ≤ twice column width if less |

This requirement makes the use of flat slabs in ductile frames inefficient as the slab width that contributes to the stiffness and strength of the frame is reduced. Their use as primary elements is further discouraged by Clause 5.1.1(2).

#### *5.6.1.2 Columns*

The cross-sectional dimension should be at least 1/10th distance between the point of contraflexure and the end of the column, if the inter-storey drift sensitivity coefficient θ is larger than 0.1.

### *5.6.2 Calculation of action effects*

Action effects are calculated initially from analytical output, for elements and effects associated with non-linear ductile behaviour, and then from capacity design principles for effects that are to be resisted in the linear range.

In frame structures, the starting point is the calculation of beam flexural reinforcement to resist the loads output from the analysis for the relevant gravity load and seismic combination with the seismic loads reduced by the applicable q factor and factored as appropriate to account for P-δ effects and accidental eccentricity.

The shear actions on the beam should then be established from the flexural capacity for the actual reinforcement arrangement provided.

The shear force is calculated from the shear that develops when plastic hinges develop in the critical regions at each end of the beam. This equates to the sum of the negative yield moment capacity at one end and the positive yield moment capacity at the other, divided by the clear span, to which the shear due to gravity loads should be added. The yield moment is calculated from the design flexural strength, multiplied by an overstrength factor, $\gamma_{Rd}$, but this is taken as 1.0 in DCM beams. In calculating the hogging capacity of the beam, the slab reinforcement within an effective flange width, defined in Clause 5.4.3.1.1(3), needs to be included. If the reinforcement differs at opposite ends of the beam, the calculation must be repeated to cater for sway in both directions.

The shear may be reduced in cases where the sum of the column moment strengths at the joint being considered is less than the sum of the beam moment strengths. This will not generally apply because of the provisions encouraging a strong column/weak beam mechanism, and only usually occurs in the top storeys of multi-storey frames, or in single storey frames.

The principle is illustrated in Figure 5.4 below, following the rules of EC8.

The moments that should then be applied to the columns are also calculated from capacity design principles to meet the strong column/weak beam requirement.

$$\sum M_{Rc} >= 1.3 \sum M_{Rb} \tag{5.8}$$

where $\sum M_{Rc}$ is the sum of the column strengths provided at the face of the joint and $\sum M_{Rb}$ is the sum of the beam strengths provided at the face of the joint.

This rule need not be observed in the top storeys of multi-storey frames, or in single storey frames. It is also not necessary to apply this rule in frames belonging to wall or wall-equivalent dual structures.

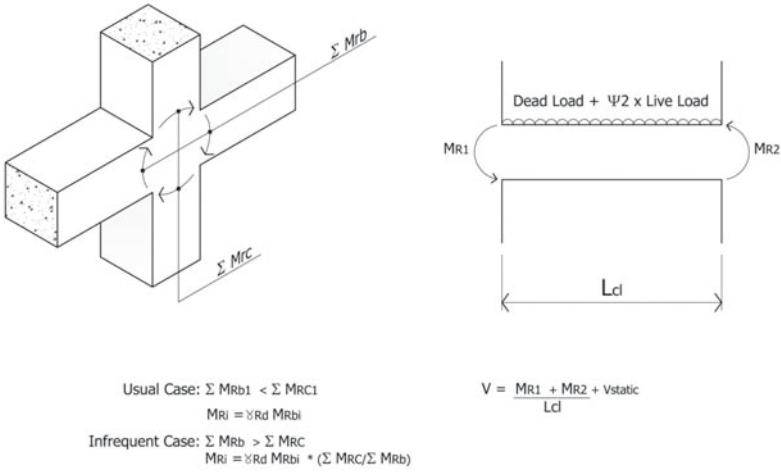



*Figure 5.4* Capacity design values of shear forces in beams, from EC8

Therefore, if a structure is classified as a frame system or frame-equivalent dual system in only one vertical plane of bending, there is no need for this rule to be satisfied in the orthogonal plane.

The proportion of the summed beam moments to be resisted by the column sections above and below the beam/column joint should be allocated in accordance with the relative stiffnesses. Fardis et al. (2005) suggest that for columns of equal proportions and spans, 45 per cent of the total moment should be assigned to the column above the joint and 55 per cent to that below. This aims at constant column reinforcement, allowing for the flexural capacity of the column generally increasing with axial compression.

Having obtained the flexural demand on the column, its capacity to carry combined flexure and axial load can be checked against standard EC2 interaction charts.

The shear load to be applied to the column is then derived from the flexural capacity in a similar way to that described above for beams, as shown in Figure 5.5 below. Generally, there will be significant axial loads in the column, which affects the moment strength. Also, there will not usually be significant lateral loading within the length of the column, so there is no additional term analogous to the gravity loading applied to the beams. However, in all cases the moment strength at each end of the column is factored by $\gamma_{Rd}$ (equal to 1.1 for DCM columns) and may also be factored by $\dfrac{\sum M_{Rb}}{\sum M_{Rc}}$ provided this ratio is less than 1. In most cases, following the 'strong column/weak beam' rule, the ratio of column to beam strength will be at least 1.3 so the capacity design shear can be reduced accordingly. As is

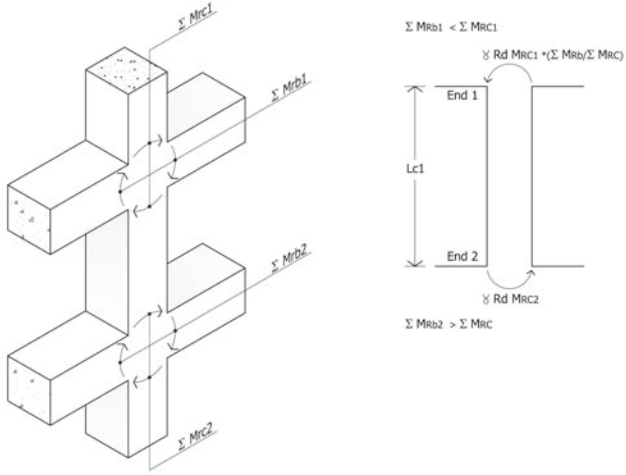



*Figure 5.5* Capacity design values of shear forces in columns

the case for beams, the calculation must be done for sway in both directions; this will mainly affect the influence of axial load on the bending strength, since the column might be in tension for the positive direction of seismic load, and in compression for the negative direction, and this has a large influence on bending strength.

### *5.6.3 Strength verification*

Having derived the design shear and bending actions in the structural members, the resistances are then calculated according to EC2. If the partial material factors are chosen as discussed in Section 5.2.4 to cater for potential strength degradation, then the design process is simplified. Standard design aids for strength such as Narayanan and Beeby (2005) or guidance available on the Internet (e.g. www.concretecentre.com) can then be used for seismic design. However, EC8 allows National Authorities to choose more complex options.

An additional restriction in columns is that the normalised axial compression force $\upsilon_d$ must be less than 0.65:

$$\upsilon_{\mathrm{d}} = N_{Ed}/A_c f_{cd} \leq 0.65 \tag{5.9}$$

This is intended to limit the adverse effects of cover spalling and avoid the situation, characteristic of members subject to high levels of axial stress, where only limited ductility is available.

For DCM frames, biaxial bending is allowed to be taken into account in a simplified way, by carrying out the checks separately in each direction but with the uniaxial moment of resistance reduced by 30 per cent.

### *5.6.4 Design and detailing for ductility*

Special detailing is required in the 'critical' regions, where plastic hinges are expected to form. These requirements are a mixture of standard prescriptive measures outlining a set of rules to be followed for all structures in a given ductility class and numerically based measures, where the detailing rules are dependent upon the calculated local ductility demand. The latter are typically required at the key locations for assurance of ductile performance such as hinge regions at the base of columns, beam-column joints and boundary elements of ductile walls, the detailing provisions becoming progressively more onerous as the ductility demand is increased.

In frame structures, specific requirements are outlined for beams, columns and beam/column joints, addressed in turn below.

#### *5.6.4.1 Beams*

CRITICAL REGIONS

The critical regions are defined as extending a length $h_w$ away from the face of the support, and a distance of $h_w$ to either side of an anticipated hinge position (e.g. where a beam supports a discontinued column), where $h_w$ is the depth of the beam.

MAIN (LONGITUDINAL) STEEL

Although flexural response of reinforced concrete beams to seismic excitation is generally deformation-controlled, abrupt brittle failure can occur if the area of reinforcement provided is so low that the yield moment is lower than the concrete cracking moment. In this situation, when the concrete cracks and tensile forces are transferred suddenly to the reinforcement, the beam may be unable to withstand the applied bending moment. To guard against this, EC8 requires a minimum amount of tension steel equal to:

$$\rho_{\min} = 0,5\left(\frac{f_{\text{ctm}}}{f_{\text{yk}}}\right) \tag{5.10}$$

along the entire length of the beam (and not just in the critical regions). In this expression, $f_{ctm}$ is the mean value of concrete tensile strength as defined in Table 3.1 of EC2 and $f_{yk}$ is the characteristic yield strength of the reinforcement.

To ensure that yielding of the flexural reinforcement occurs prior to crushing of the compression block, the maximum amount of tension steel provided, $\rho_{\max}$, is limited to:

$$\rho_{\max} = \rho + \frac{0.0018}{\mu_\phi \varepsilon_{sy,d}} \cdot \frac{f_{cd}}{f_{yd}} \tag{5.11}$$

Here, $\varepsilon_{sy,d}$ is the design value of reinforcement strain at yield, $\rho'$ is the compression steel ratio in the beam and $f_{cd}$ and $f_{yd}$ are the design compressive strength of the concrete and design yield strength of the reinforcement respectively.

The development of the required local curvature ductility, $\mu_\phi$, is also promoted by specifying that the area of steel in the compression zone should be no less than half of the steel provided in the tension zone in addition to any design compression steel.

Since bond between concrete and reinforcement becomes less reliable under conditions of repeated inelastic load cycles, no splicing of bars should take place in critical regions according to Clause 8.7.2(2) of EC2. All splices must be confined by specially designed transverse steel as defined in Equations 5.51 and 5.52 of EC8.

Another area where particular attention needs to be paid to bond stresses is in beam/column joints of primary seismic frames, due to the high rate of change of reinforcement stress, generally varying from negative to positive yield on either side of the joint. To cater for this, the diameter of bars passing through the beam/column joint region is limited according to Equations 5.50a and 5.50b of EC8 Part 1. For DCM structures, these become:

$$\frac{d_{bL}}{h_c} \le 7.5.f_{ctm} / f_{yd})(1 + 0.8\nu_d) / (1 + 0.5\rho / \rho_{max}) \text{ for interior columns} \quad (5.12)$$

$$\frac{d_{bL}}{h_c} \le 7.5.f_{ctm} / f_{yd})(1 + 0.8\nu_d) \text{ for exterior columns.} \quad (5.13)$$

where $d_{bL}$ is the longitudinal bar diameter and $h_c$ is the depth of the column in the direction of interest.

### HOOP (TRANSVERSE) STEEL

Many of the detailing provisions in EC8 revolve around the inclusion of transverse reinforcement to provide a degree of triaxial confinement to the concrete core of compression zones and restraint against buckling of longitudinal reinforcement. As confinement increases the available compressive capacity, in terms of both strength and more pertinently strain, it has enormous benefits in assuring the availability of local curvature ductility in plastic hinge regions. EC2 gives relationships for increased compressive strength and available strain associated with triaxial confinement, illustrated in Figure 5.6. These indicate that for the minimum areas of confinement reinforcement required at column bases and in beam column joints, the ultimate strain available would be between about two and four times that of the unconfined situation, dependent on the effectiveness of the confinement arrangement, as defined later.

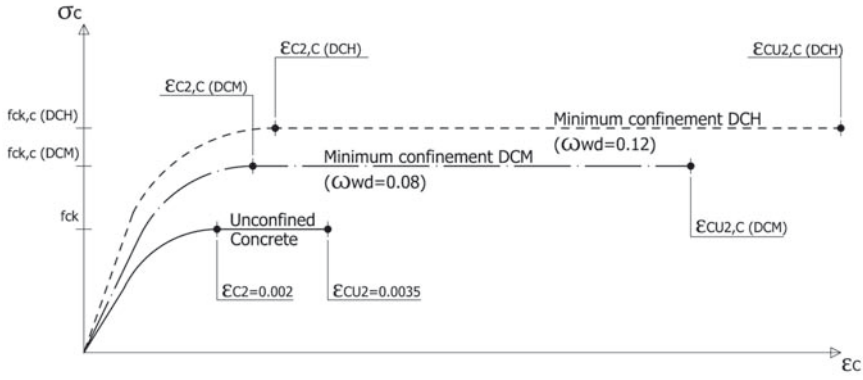



*Figure 5.6* Stress-strain relationships for confined concrete

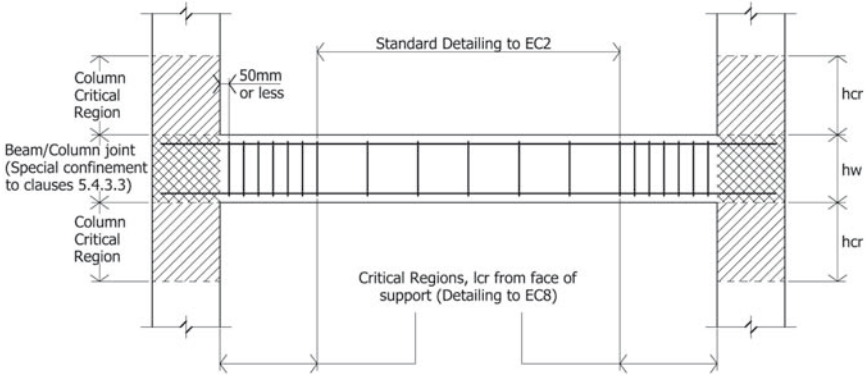



*Figure 5.7* Transverse reinforcement in beams, from Eurocode 8 Part 1

The requirements set out in EC8 to achieve this through detailing of critical regions are briefly summarised below.

- Hoops of at least 6 mm diameter $d_{bw}$ must be provided.
- The spacing, $s$, of hoops should be less than the minimum of: $h_w/4$; $24d_{bw}$; 225 mm or $8d_{bL}$.
- The first hoop should be placed not more than 50 mm from the beam end section as shown in Figure 5.7.

Hoops must have 10 bar diameters anchorage length into the core of the beam.

### *5.6.4.2 Columns*

CRITICAL REGIONS

These are the regions adjacent to both end sections of all primary seismic columns. The length of the critical region (where special detailing is required) is the largest of the following:

- $h_c$
- $l_{cl}/6$
- 0.45 m.

where $h_c$ is the largest cross-section dimension of the column and $l_{cl}$ is the clear length of the column.

The whole length of the column between floors is considered a critical region:

- if $(h_c/l_{cl})$ is less than 3
- for structures with masonry infills:
  - if it is a ground floor column
  - if the height of adjacent infills is less than the clear height of the column
  - if there is a masonry panel on only one side of the column in a given plane.

MAIN (LONGITUDINAL) STEEL

The longitudinal reinforcement ratio must be between 0.01 and 0.04.

- Symmetric sections must be symmetrically reinforced.
- At least one intermediate bar is required along each side of the column.
- Full tension anchorage lengths must always be provided, and 50 per cent additional length supplied if the column is in tension under any seismic load combination.
- As for beams, no splicing of bars is allowed in the critical regions and where splices are made, they must be confined by specially designed transverse steel.

TRANSVERSE STEEL (HOOPS AND TIES)

The amount of transverse steel supplied in the critical regions at the base of columns must satisfy Equation 5.14 below:

$$\alpha\omega_{wd} \geq 30\mu_{\Phi}\nu_{d}.\varepsilon_{sy,d}.\, b_c/b_0 - 0.035 \qquad (5.14)$$

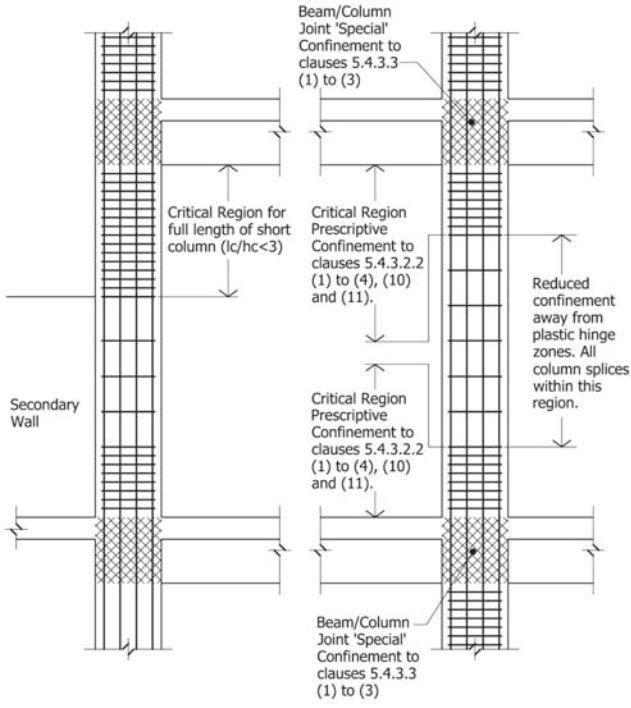



*Figure 5.8* Typical column details – elevation

where $\omega_{wd}$ is (volume of confining hoops$*f_{yd}$)/(volume of concrete core$*f_{cd}$), $b_0$ is the minimum dimension of concrete core and $\alpha$ is a confinement effectiveness factor, depending on concrete core dimensions, confinement spacing and the arrangement of hoops and ties. It is defined in Equations 5.16a to 5.16c and 5.17a to 5.17c of the code (EC8) for various cross sections.

In the critical region at the base of columns, a minimum value of $\omega_{wd}$ of 0.08 is specified. However, for structures utilising low levels of ductility ($q$ of 2 or less) and subject to relatively low compressive stresses ($\nu_d$<0.2), this requirement is waived and the normal EC2 provisions apply. In all other critical regions of columns the following applies:

- Minimum hoop and tie diameter is 6 mm.
- Maximum spacing of hoops and ties in the critical region is the least of:
  - $\boldsymbol{b}_0/2$
  - 175 mm
  - $8d_{bL}$.
- The maximum distance between restrained longitudinal bars should be 200 mm.

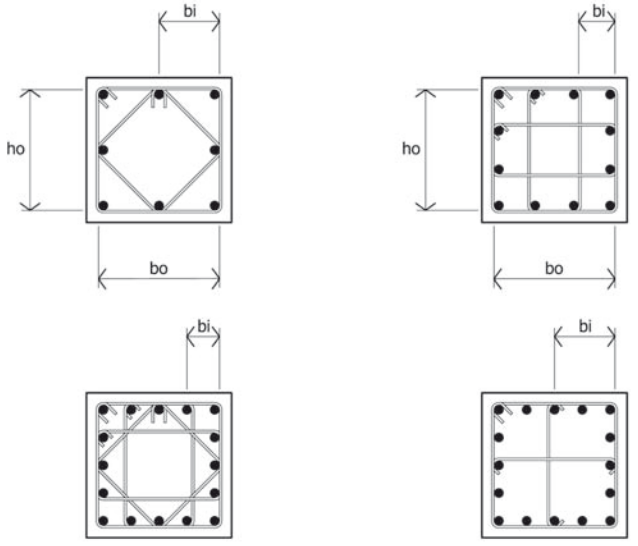



*Figure 5.9* Typical column details – cross section

- Hoops must have 10 bar diameter anchorage length into the core of the column. Typical detailing requirements in critical regions of columns are illustrated in Figures 5.8 and 5.9.

*5.6.4.3 Beam-column joints*

The beam-column joints of frames represent a highly stressed region with quite complex reinforcement detailing. The design requirements in EC8 are much more straightforward for DCM than for DCH.

In DCM joints, no explicit calculation of shear resistance is required, provided the following rules are satisfied:

a. To ensure that there is adequate bond between reinforcement and concrete, the diameter of the main beam bars passing through the joint must be limited as given earlier in Equations 5.12 and 5.13.
b. At least one intermediate column bar is provided between each of the corners of the columns.
c. Hoops must continue unreduced through the joint from the critical region of the column, or must meet the confinement requirements of Equation 5.14 if greater, unless the joint is confined on all four sides by beams. In this case, the hoop spacing may be doubled (but must not exceed 150 mm).

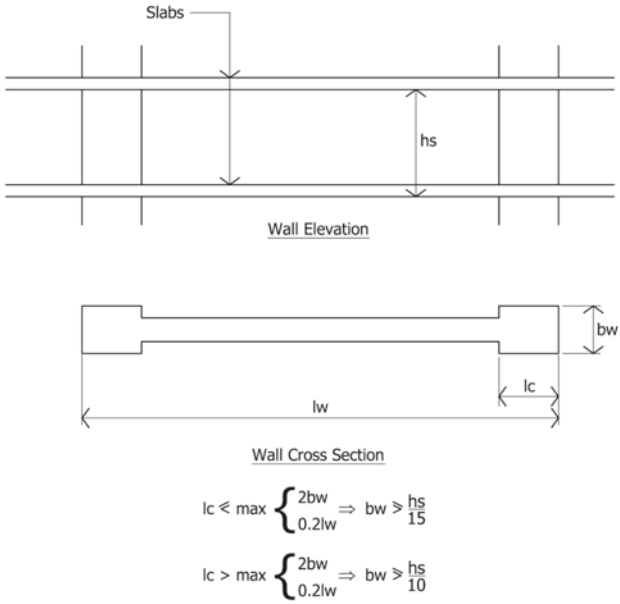



*Figure 5.10* Minimum thickness of wall boundary elements, following the rules of EC8

## 5.7 Ductile walls – design for DCM

Reinforced concrete walls are defined as vertical elements in which one of the dimensions of its cross section is at least four times the other dimension. In these elements the flexural resistance is provided by two boundary elements at section extremities, where flexural reinforcement is concentrated and concrete is confined, with the web in between providing most of the shear resistance. The association of intersecting rectangular wall segments that develop in different directions may give rise to a three-dimensional element, which must be analysed as an integral unit.

### *5.7.1 Geometrical restrictions*

The thickness of the web of reinforced concrete walls, $b_{wo}$, must be larger than $h_s/20$ (in which $h_s$ is the clear storey height), with a minimum of 0.15 m. The width of the boundary elements, $b_w$, must not be less than 0.20 m. If the length of the boundary element, $l_c$, is restricted to no more than twice its thickness, $b_w$, and one fifth of the wall cross-section length, $l_w$, then $b_w$ must be greater than or equal to $h_s/15$. If the boundary element length exceeds the above value, then the thickness of the boundary element must be higher than $h_s/10$. Figure 5.10 illustrates the above restrictions.

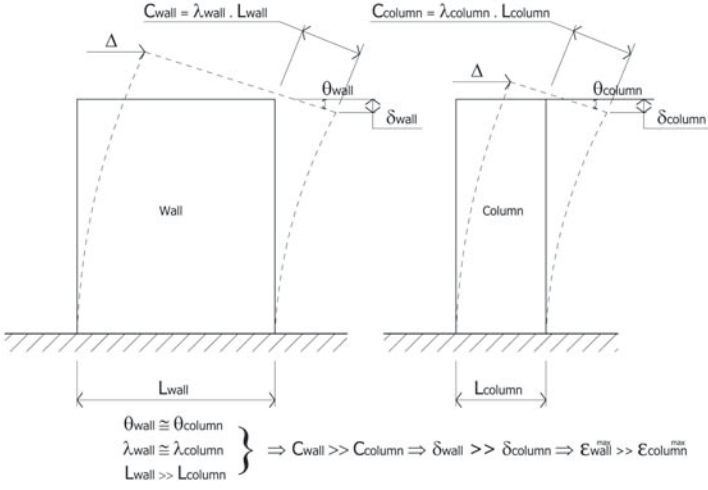



*Figure 5.11* Unavoidability of wall hinging

### *5.7.2 Calculation of action effects*

Figure 5.3 highlights that for the best seismic performance the walls must act as vertical cantilevers and only be allowed to develop a single plastic hinge at the base. The formation of this hinge is practically unavoidable, which can be explained as follows: both walls and frames have to withstand similar displacements at floor levels, and therefore both types of elements have to withstand similar curvatures. As shown in Figure 5.11, this induces higher axial strains in wall sections due to their larger cross-section dimensions. The fact that this derives only from kinematic compatibility, and is therefore independent of flexural design, implies that wall hinging is practically unavoidable for structures designed to resist earthquakes in the inelastic range and tend to occur before hinges develop in the frames.

However, besides avoiding the soft-storey mechanism there are other advantages in maintaining elastic behaviour in the rest of the wall, by preventing the formation of plastic hinges in the wall at the upper stories: the elastic part of the wall tends to behave almost as a rigid body above the flexible zone of the hinge at the base, maintaining relatively uniform inter-storey drifts throughout the height of the building. This tends to minimise the local ductility demand in the frames and hence the extent of non-structural damage, for the same global ductility of the structure.

In order that walls act as vertical cantilevers, the length of their cross sections must be significantly greater than the height of the beams to which they are connected in the plane that contains the larger wall dimension. For this purpose Fardis et al. (2005) recommend a minimum value of wall length $l_w$ = 1.5 m in low-rise buildings and $l_w$ = 2.0 m in medium and high-rise buildings.

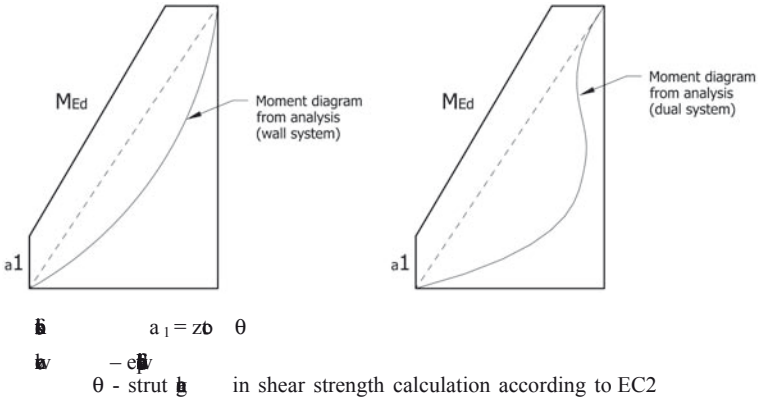



*Figure 5.12* Design bending moment for RC walls (EC8)

The non-linear behaviour of an uncoupled reinforced concrete wall is governed by a single plastic hinge at its base. This section must be designed in flexure for the bending moment that results from the analysis for the seismic design situation. However, unlike the frames, the bending moment diagram does not change sign between successive floor levels, which creates uncertainties in moment distribution along the wall. In order to avoid yielding above the base hinge, EC8 prescribes that the design bending moment diagram is based on an envelope derived from the bending moment diagram obtained from analysis, as shown in Figure 5.12.

Turning to shear design, since the bending moment diagrams in the wall at the various stages of hinge development are not known, it is not possible to derive shear forces based on equilibrium equations. In order to avoid shear failure considering (i) possible bending moments at the base section being greater than the design value used to derive the flexural reinforcement, (ii) possible variations of the distribution of inertia forces in the non-linear range and (iii) effects of higher vibration modes, EC8 prescribes that the evaluation of design shear forces should comprise the magnification of the shear forces from analysis by a factor of 1.5. Besides the overstrength of the plastic hinge, the flexural capacity can easily increase above the design value used to derive the flexural reinforcement if seismic action effects in the wall include a reasonably significant axial force. In this situation the amount of flexural reinforcement at the base is probably conditioned by the situation in which the seismic axial force is tensile, a higher flexural capacity corresponding to the case in which the seismic axial force is compressive. The EC8 approach for DCM structures is clearly a simplified and generous one, as the magnifying factor is constant. Therefore designers should be aware that it may not cover all the factors that may increase shear forces above the value obtained from analysis.

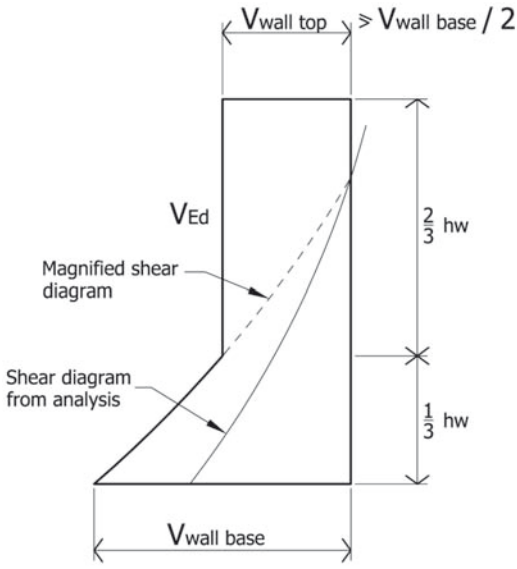



*Figure 5.13* Design envelope of shear forces of dual systems

In the case of dual structures and due to the larger influence of the frames in the overall behaviour of the structure at the upper floors, EC8 prescribes an envelope of shear forces, as shown in Figure 5.13, which accounts for higher shear forces than predicted at upper floor levels.

### *5.7.3 Strength verification*

Just as for the frames, both flexural and shear verifications for walls of DCM structures follow standard EC2 procedures. Flexural reinforcement should be concentrated at the extremities of the section, in the zones furthest away from the neutral axis. This is also the most efficient distribution of the flexural reinforcement in terms of curvature ductility. The minimum longitudinal reinforcement ratio in the boundary elements is 0.005.

In three-dimensional elements, resulting from the intersection of rectangular wall segments, for the purpose of evaluating the flexural capacity, the effective flange width on each side of a web should be the minimum of:

- the actual flange width
- half the distance to an adjacent web
- 25 per cent of the height of the wall above that level.

Just as for columns, EC8 imposes an upper limit to the normalised axial compression force on the walls, as follows:

$$\nu_d = N_{Ed} / A_c f_{cd} \leq 0.4 \tag{5.15}$$

where $A_c$ is the cross-section area of the wall.

### *5.7.4 Design and detailing for ductility*

It is not necessary to design the boundary elements for ductility if the normalised axial force is below 0.15 $(\upsilon_d \leq 0.15)$. In this situation the transverse reinforcement can be evaluated as prescribed in EC2. If $0.15 \leq \upsilon_d \leq 0.20$ the design of the transverse reinforcement can also follow EC2 if the $q$ factor is reduced by 15 per cent. If $\upsilon_d > 0.20$ the ductility of the rectangular wall plastic hinges is achieved by confinement of the wall boundary elements, according to EC8 prescriptions, as follows:

a. Height of confined boundary elements ($h_{cr}$)

$$h_{cr} = \max[l_w, h_w/6] \tag{5.16}$$

where $l_w$ is the length of wall section (largest dimension) and $h_w$ is the total height of the wall above the foundation or top basement floor, but $h_{cr}$ need not be greater than:

$$h_{cr} \leq \begin{cases} 2l_w \\ h_s \quad \text{for } n \leq 6 \text{ storeys} \\ 2h_s \quad \text{for } n \geq 7 \text{ storeys} \end{cases} \tag{5.17}$$

where $h_s$ is the clear storey height.

b. Length of confined boundary element

The confined boundary element must extend throughout the zone of the section where the axial strain exceeds the code limit for unconfined concrete $\varepsilon_{cu2} = 0.0035$. Therefore, for rectangular sections, it must extend at least to a distance from the hoop centreline on the compressive side of

$$x_u(1-\varepsilon_{cu2}/\varepsilon_{cu2,c}) \tag{5.18}$$

where $x_u$ is the depth of compressive zone and $\varepsilon_{cu2,c}$ is the maximum strain of confined concrete.

The values of $x_u$ and $\varepsilon_{cu2,c}$ can be evaluated as follows:

$$x_u = (\upsilon_d + \omega_v)\frac{l_w b_c}{b_0} \tag{5.19}$$

$$\varepsilon_{cu2,c} = 0.0035 + 0.1\,\alpha\,\omega_{wd} \tag{5.20}$$

$$\omega_v = (A_{sv}/h_c\, b_c) f_{yd}/f_{cd} \tag{5.21}$$

where $N_{Ed}$ is the design axial force, $b_c$ -is the width of web, $b_0$ is the width of confined boundary element (measured to centrelines of hoops), $h_c$ is the largest dimension of the web and $A_{sv}$ is the amount of vertical web reinforcement.

The value of $\alpha\,\omega_{wd}$ can be evaluated as follows:

$$\alpha\,\omega_{wd} \geq 30\mu_{\varphi}\left(\upsilon_d + \omega_v\right)\varepsilon_{sy,d}\frac{b_c}{b_0} - 0.035 \tag{5.22}$$

in which $\mu_{\varphi}$ is the local curvature ductility factor, evaluated by Equations (5.5) or (5.6), with the basic value of the $q$ factor, $q_o$, replaced by the product of $q_o$ times the maximum value of the ratio $M_{Ed}/M_{Rd}$ at the base of the wall.

Regardless of the above, EC8 specifies that the length of boundary elements should not be smaller than $0.15l_w$ or $1.5b_w$, with $b_w$ being the width of the wall.

c. Amount of confinement reinforcement

This is calculated from the mechanical volumetric ratio of confinement reinforcement, $\omega_{wd}$, evaluated according to Equation (5.22). The minimum value of $\omega_{wd} = 0.08$.

Sections with barbells, flanges or sections consisting of several intersecting rectangular segments, can be treated as rectangular sections with the width of the barbell or flange provided that all the compressive zone is within the barbell or flange. If the depth of the compressive zone exceeds the depth of the barbell or flange the designer may:

1 increase the depth of the barbell or flange in order that all the zone under compression is within the barbell or flange;
2 if the width of the barbell or flange is not much higher than the width of the web, design the section as rectangular with the width of the web, and confine the entire barbell or flange similarly to the web, or;
3 verify if the available curvature ductility exceeds the curvature ductility demand by non-linear analysis of the section, including the effect of confinement, after full detailing of the section.

In cases of three-dimensional elements consisting of several intersecting rectangular wall segments, parts of the section that act as the web for bending moments about one axis may act as flanges for the bending moment acting on an orthogonal axis. Therefore it is possible that in some of these cases the entire cross section may need to be designed for ductility as boundary elements.

General detailing rules regarding the diameter, spacing and anchorage of hoops and ties for wall boundary elements designed for ductility according

to EC8 are the same as for columns. The maximum distance between longitudinal bars is also 200 mm.

## 5.8 Design for DCH

The rules for DCH structures build upon those for DCM and, in certain instances, introduce additional or more onerous design checks. These are briefly introduced below. Additionally, the option of large lightly reinforced walls is removed, this type of system not being considered suitable for DCH performance.

### *5.8.1 Material and geometrical restrictions*

The major differences from DCM are:

- Concrete must be Class C20/25 or above.
- Only Class C reinforcement must be used.
- The potential overstrength of reinforcement is limited by requiring the upper characteristic (95 per cent fractile) value of the yield strength to be no more than 25 per cent higher than the nominal value.
- Additional limitations on the arrangement of ductile walls and minimum dimensions of beams and columns.

### *5.8.2 Derivation of actions*

The capacity design approach used in DCM structures is reinforced as follows:

- Overstrength factors are increased to 1.2 on beams, 1.3 on columns and 1.2 on beam/column joints.
- An additional requirement for calculating the shear demand on beam/column joints is introduced in Clause 5.5.2.3.
- The shear demand on ductile walls is generally greater, the enhancement of the shear forces output from the analysis increasing from a constant factor of 1.5 to a factor of between 1.5 and q, determined from Equation 5.25 of EC8 Part 1. An overstrength factor of 1.2 is introduced for this purpose.
- Additional requirements are introduced for calculating the shear demand on squat walls.

### *5.8.3 Resistances and detailing*

The main changes and additions are as follows:

- The assumed strut inclination in checking the shear capacity of beams to EC2 is limited to 45° and additional shear checks are introduced when almost full reversal of shear loading can occur.
- The maximum permissible normalised axial force is reduced from 0.65 to 0.55 in columns and from 0.4 to 0.35 in walls.
- Wall boundary elements need to be designed for ductility according to EC8, regardless of the level of the normalised axial force.
- The length, $l_{cr}$, of the critical regions of beams, columns and walls is increased and the spacing of confinement reinforcement reduced.
- Confinement requirements are extended to a length of $1.5l_{cr}$ for columns in the bottom two storeys of buildings.
- The minimum value of $\omega_{wd}$ in the critical region at the base of columns and in the boundary elements of ductile walls is increased from 0.08 to 0.12.
- The maximum distance between column and wall longitudinal bars restrained by transverse hoops or ties is reduced from 200 mm to 150 mm.
- More comprehensive and complex checks for the shear resistance and confinement requirements at beam/column joints are introduced.
- Much more stringent checks on the resistance to shear by diagonal tension and diagonal compression are introduced, namely the limitation of the strut inclination to 45° and the reduction of the resistance to diagonal compression of the web in the critical region to 40 per cent of the resistance outside the critical region. A different verification is also introduced of the resistance against shear failure by diagonal tension in walls with shear ratio $\alpha_s = M_{Ed}/(V_{Ed}\, l_w)$ below 2 as is verification against sliding shear.
- Special provisions for short coupling beams ($l/h$ <3) are included, effectively comprising confined diagonal reinforcement cages as proposed by Park and Paulay (1974).

## 5.9 Concrete design example – wall-equivalent dual structure

### *5.9.1 Introduction*

The concrete design example is based on a dual frame solution for the eight-storey hotel introduced in earlier chapters. For clarity, the example only considers the critical transverse direction with primary frames at 8 m spacing. The frames on GLs 1, 7, 9 and 15 incorporate structural walls whereas those on GLs 3, 5, 11 and 13 are moment frames.

In a typical frame, transverse beams support masonry cross-walls between GLs B and C, and D and E, primary beams fulfilling this function on odd gridlines and secondary beams on even gridlines. The masonry walls are effectively isolated from the frames so as not to stiffen the primary structural system.

At all levels in the moment frames, primary columns are situated on gridlines B, C, D and E, whilst at Level 1 additional primary columns are located on GLs A and F.

Longitudinal beams are also continuous along these gridlines: the beams on GLs B and E supporting the external cavity wall, those on GLs C and D supporting internal corridor walls and those on GLs A and F supporting the external glazing.

The structure has been analysed for both gravity loads and seismic-equivalent lateral force loading derived as in Chapter 3 but with an additional allowance for the mass of the concrete frame.

The primary members are as follows:

| | |
|---|---|
| Slabs | 150 mm thick |
| *Frames including structural walls* | |
| Walls | 2 No – 3,500 mm x 350 mm, with the outer edge on GL B and E respectively |
| Primary beams | 450 mm x 600 mm |
| Columns | 600 mm square (upper four storeys)<br>750 mm square (lower four storeys) |
| *Moment frames* | |
| Primary beams | 450 mm x 750 mm |
| Columns | 600 mm square (upper four storeys)<br>750 mm square (lower four storeys) |
| Secondary transverse beams on intermediate frames | 350 mm x 500 mm |
| Longitudinal beams | 350 mm x 600 mm |

All reinforcement is Class C with characteristic yield strength of 500 N/mm$^2$ and concrete is Class C30/37.

Cover to main bars is 45 mm in beams, columns and walls and 30 mm in slabs.

The slabs span longitudinally between the main transverse frames and secondary beams on the intermediate gridlines.

They have been designed for the dominant gravity load condition, and the longitudinal reinforcement comprises Φ12–300 T&B (754 mm$^2$/m).

For the purposes of the example, the structure can be assumed to be braced in the longitudinal direction and is to be designed and detailed for ductility class DCM.

Sample checks are carried out on a structural wall and on typical frame members and are intended to illustrate the requirements for design and detailing of critical regions.

A dual system has been chosen in order to illustrate key aspects of the code for both frame and wall elements. Hence, compromises have been made to the member sizes to remain within the limitations for the proportion of total

base shear carried by the frame and wall members. In particular, to reduce structural displacements for the damage limitation requirement, a wall-equivalent dual system is provided but this has the potential disadvantage of increasing the level of acceleration applied to the stiffer structure. Thus, in the initial design, the spectral acceleration is derived based on a $C_t$ coefficient of 0.05 [from EC8 Clause 4.3.3.2.2(3) for structures other than moment resistant space frames or eccentrically braced steel frames] as in Chapter 3. This results in a conservative estimate of the inertial loads for the relatively flexible dual structure and a comparison with an approach based on modal response spectrum analysis, which results in significantly lower inertial loads, is given later to justify the performance in the damage limitation case.

### *5.9.2 Layout*

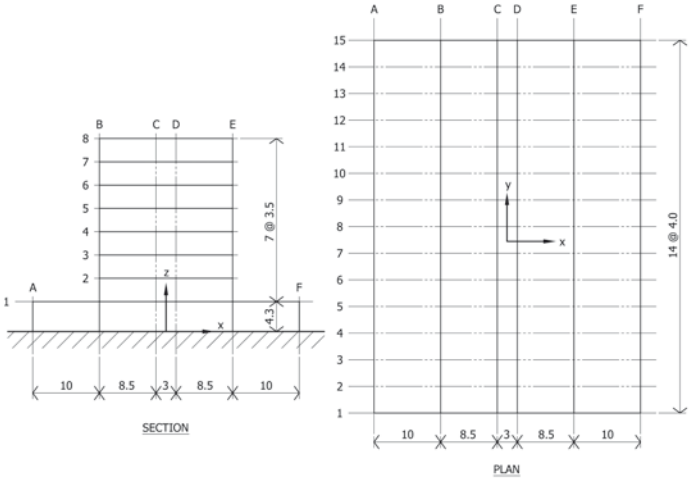



*Figure 5.14* Structural layout

### *5.9.3 Evaluation of the q factor*

$$q = q_o \, k_w \tag{5.23}$$

According to Table 5.1 of EC8:

DCM + dual system $\Rightarrow q_o = 3.0 \, \alpha_u/\alpha_1$

The ratio $\alpha_u/\alpha_1$ depends on the classification of the structure. For multi-storey, multi-bay wall-equivalent dual structures, and unless a more accurate value is obtained by pushover analysis, EC8 allows the assumption that $\alpha_u/\alpha_1 = 1.2$.

$$q_o = 3 \text{ x } 1.2 = 3.6 \tag{5.24}$$

$$k_w = (1 + h_w/l_w)/3 \leq 1$$

$$k_w = (1 + 28.8/3.5)/3 = 3.06$$

Therefore: $k_w = 1$

$$q = 3.6 \text{ x } 1.0 = 3.6$$

The lateral loads imposed on the structure are based on those derived in Chapter 3 for structures other than moment frames. Thus, the spectral acceleration derived from the empirical period calculation is 2.32 m/s$^2$ as per Chapter 3 but the masses to which it is applied are increased to reflect the frame weight. Hence, the applied lateral loads at each level are as listed below. It will be shown later that use of modal response spectrum analysis to calculate the structural frequencies results in significantly reduced inertial forces but the more conservative equivalent lateral force approach is retained initially for the preliminary design.

- Level 8 3448 kN
- Level 7 4592 kN
- Level 6 3952 kN
- Level 5 3328 kN
- Level 4 2720 kN
- Level 3 2112 kN
- Level 2 1456 kN
- Level 1 1384 kN

*5.9.3.1 Part of the total base shear taken by the walls*

$$V_{total} = 22992\text{kN}$$

$$V_{frames} = 8218\text{kN}$$

$$V_{walls} = 14774\text{kN}$$

$$\frac{V_{wall}}{V_{total}} = \frac{14774}{22992} = 0.64 \geq 0.5 \leq 0.65$$

Based on the above the structure is classified as a wall-equivalent dual structure.

*5.9.3.2 Verification of P-δ effects and inter-storey drifts*

The stability index, θ, needs to be checked to see if *P*-δ effects can be ignored or covered by an approximate method [Clause 4.4.2.2 (2)].

*Table 5.2* Horizontal displacements and inter-storey drift sensitivity coefficient

| *Level* | $d_e$ *(mm)* | $d_{er}$ *(mm)* | $d_r$ *(mm)* | $d_r$*ν | $P_{tot}$ | $V_{tot}$ | $h$ *(mm)* | θ |
|---|---|---|---|---|---|---|---|---|
| 8 | 149 | – | – | – | – | – | | |
| 7 | 130 | 19 | 68.4 | 34.2 | 12660 | 3448 | 3500 | 0.072 |
| 6 | 110 | 20 | 72.0 | 36.0 | 25482 | 8040 | 3500 | 0.065 |
| 5 | 88 | 22 | 79.2 | 39.6 | 38304 | 11992 | 3500 | 0.072 |
| 4 | 65 | 23 | 82.8 | 41.4 | 51126 | 15320 | 3500 | 0.079 |
| 3 | 44 | 21 | 75.6 | 37.8 | 64317 | 18040 | 3500 | 0.077 |
| 2 | 24 | 20 | 72.0 | 36.0 | 77508 | 20152 | 3500 | 0.079 |
| 1 | 9 | 15 | 54.0 | 27.0 | 90699 | 21608 | 3500 | 0.065 |
| | 0 | 9 | 32.4 | 16.2 | 113690 | 22992 | 4300 | 0.037 |

$$\theta = \frac{P_{tot} \times d_r}{V_{tot} \times h}$$ [Clause 4.4.2.2(2), Equation 4.28]

This should be based on the displacements of the structure that have been output from an analysis based on a stiffness of $0.5 * E * I_g$ [Clause 4.3.1 (7)].

$E$ for grade C30/37 concrete = 33 x $10^6$ kN/m$^2$ (EC2 Table 3.1).

Table 5.2 shows the values necessary to verify EC8 rules for P-δ effects and inter-storey drifts.

$d_e$ is the average horizontal displacement from analysis (based on the design response spectrum) in the transverse direction at each floor level;

$d_{er}$ is the relative displacement between storeys based on the design response spectrum;

$d_r = q_d * d_{er}$ is the relative displacement between storeys accounting for the ductility-modified spectrum;

ν is the reduction factor that accounts for the lower return period of the seismic action associated with the damage limitation requirement;

$P_{tot}$ is the total gravity load at and above the storey considered at the seismic design situation;

$V_{tot}$ is the total seismic storey shear;

As $\theta_{max} = 0.079 < 0.1 \Rightarrow$ no need to increase action effects to cater for *P*-δ effects.

The adequacy to meet the damage limitation case is addressed later.

### *5.9.4 Design of wall elements*

#### *5.9.4.1 Allowance for torsion*

In the preliminary design, it was decided to increase the action effects on the stiff wall elements using the simplified conservative allowance for torsion given in Section 4.3.3.2.4 of the EC8 code. Because the structure contains stiff perimeter elements that provide good resistance to torsional effects and would thus reduce the likelihood of significant torsional response, no increase is applied to the frame elements at this stage. It would be expected that the final actions taking full account of torsion introduced through accidental eccentricity would be derived in the final design using the more rigorous approach of Section 4.3.2 and that this would confirm the assumptions of the preliminary design.

$$\delta = 1 + 0.6\frac{x}{L}$$ (eq, 4.12 of EC8) (Equation 4.12)

For GLs 7 and 9

$x = 1 \times 4 = 4$

$L = 14 \times 4 = 56$

$\delta = 1 + 0.6 \times \frac{4}{56} = 1.04$

For GLs 1 and 15

$x = 7 \times 4 = 28$

$\delta = 1 + 0.6 \times \frac{28}{56} = 1.3$

Increase forces output from the analysis by a factor of 1.3 to account for torsional effects on GL 1 and 15, and by 1.04 on GL 7 and 9.

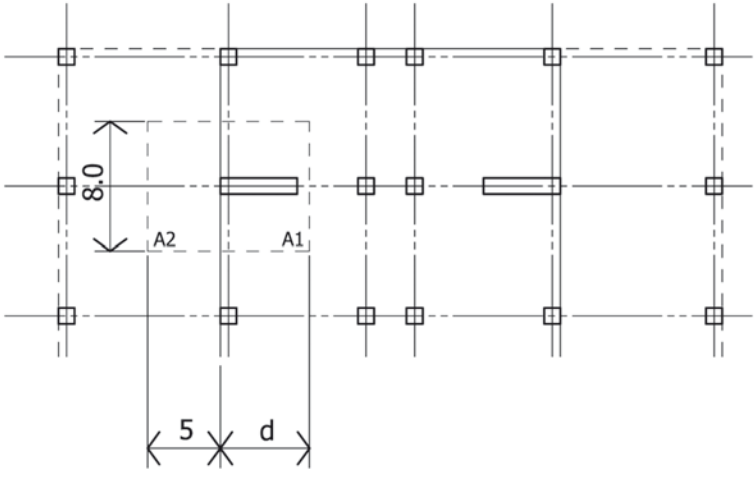



*Figure 5.15* Area of influence of the walls

*5.9.4.2 Design of the wall base section*

ACTION EFFECTS

Axial force at the base of the wall due to vertical actions ($G + 0.3Q$). These were evaluated using the area of influence of the wall.

$$d = 3.5 + (8.5 - 3.5)/2 = 6.0\text{m}$$

$$A_1 = 6.0 \times 8 = 48\text{m}^2$$

$$A_2 = 5 \times 8 = 40\text{m}^2$$

$$A_{tower} = A_{terrace} = 56 \times 20 = 1120\text{m}^2$$

General loading (from Chapter 3):

Tower section $G_k$ = 5152 + 7*8862 = 67186 kN

Terrace section $G_k$ = 5482 kN

Tower section $0.3Q_k$ = 672 + 7*834 = 6510 kN

Terrace section $0.3Q_k$ = 2178 – 834 = 1344 kN

*GL 1 and 15 (half-bay tributary area)*

$$N(G_k) = (67186) \times \frac{24}{1120} + (5482) \times \frac{20}{1120} = 1440 + 98 = 1538\text{kN}$$

$$N(G_k + 0.3Q_k) = (67186 + 6510) \times \frac{24}{1120} + (5482 + 1344) \times \frac{20}{1120} = 1579 + 122 = 1701kN$$

Allowance for weight of frame

Wall + transverse beams + longitudinal beams = 1068 kN

*GL 7 and 9 (full-bay tributary area)*

$$N(G_k) = (67186) \times \frac{48}{1120} + (5482) \times \frac{40}{1120} = 2879 + 196 = 3075\text{kN}$$

$$N(Gk + 0.3Qk) = (67186 + 6510) \times \frac{48}{1120} + (5482 + 1344) \times \frac{40}{1120} = 3158 + 244 = 3402\text{kN}$$

Allowance for weight of frame

Wall + transverse beams + longitudinal beams = 1288 kN

*GL 7 and 9*

$N(G_k)_{\text{total}} = 3075 + 1288 = 4363$ kN

$N(G_k + 0.3Q_k)_{\text{total}} = 3402 + 1288 = 4690$ kN

*GL 1 and 15*

$N(G_k)_{\text{total}} = 1538 + 1068 = 2606$ kN

$N(G_k + 0.3Q_k)_{\text{total}} = 1701 + 1068 = 2769$ kN

Action effects due to the seismic action (from analysis):

$M = \pm 19793 kN.m$

$V = \pm 1847 kN$

$N = \pm 1217 kN$

To obtain the design action effects due to the seismic action, the above values must be multiplied by $\xi = 1.3$ on GL 1 and 15 and 1.04 on GL 7 and 9.

*GL 7 and 9*

$M_{Ed1} = 19793 \times 1.04 = 20585$kN.m

$V_{Ed} = 1847 \times 1.04 = 1921$kN

$N_{Ed} = 1217 \times 1.04 = 1266$kN

*GL 1 and 15*

$M_{Ed1} = 19793 \times 1.3 = 25731$kN.m

$V_{Ed} = 1847 \times 1.3 = 2401$kN

$N_{Ed} = 1217 \times 1.3 = 1582$kN

Wall design – base section

$$M_{Ed} = M_{Ed1} + \frac{h}{30} \times N_{Ed}$$

$\frac{h}{30} = \frac{3.5}{30} = 0.117$m is the maximum eccentricity according to EC2 Clause 6.1 (4)P.

MAXIMUM AXIAL FORCE

*GL 7 and 9*

$$N_{Ed} = 4690 + 1266 = 5956\text{kN}$$
$$M_{Ed} = 20585 + 0.117 \times 5956 = 21282\text{kN.m}$$

*GL 1 and 15*

$$N_{Ed} = 2769 + 1582 = 4351kN$$
$$M_{Ed} = 25731 + 0.117 \times 4351 = 26240\text{kN.m}$$

MINIMUM AXIAL FORCE

*GL 7 and 9*

$$N_{Ed} = 4363 - 1266 = 3097\text{kN}$$
$$M_{Ed} = 20585 + 0.117 \times 3097 = 20947\text{kN.m}$$

*GL 1 and 15*

$$N_{Ed} = 2606 - 1582 = 1024kN$$
$$M_{Ed} = 25731 + 0.117 \times 1024 = 25851\text{kN.m}$$

The proportion of live load to be included in the gravity load component in the seismic combination is an area that is open to judgement by the designer based on the use of the building, the make-up of the live load and the potential consequences of failure. In this case, the minimum vertical load calculated above includes no live load in the gravity component although the lateral loads are based on 30 per cent of the characteristic live load being present. The rationale for this is that whilst 30 per cent of the live load may be present globally, this will not be distributed evenly around the floor slab and the tributary load local to individual elements may have more or less live load than this. The full characteristic live load is unlikely to be present during an earthquake whereas it is feasible that certain parts of the structure may be empty. Since it is the minimum axial load that tends to govern wall reinforcement design, this conservative approach has been adopted in deriving the minimum loads but the maximum loads have been based on only 30 per cent of the characteristic live load consistent with the global horizontal loads.

*5.9.4.3 Flexural design*

- use design charts or design programme;
- assume symmetric reinforcement $d_1/d = 0.1$ ($d_1$ being the distance from the centre of tensile reinforcement to the edge of wall section);
- steel constitutive relationship with an horizontal top branch [EC2 Section 3.2.7(2)];

- use partial factors for the persistent and transient design situations [Clause 5.2.4(2)].

$\gamma_s = 1.15 \qquad \gamma_c = 1.5$

Check normalised axial load for $N_{m\,x}$

$$\nu = \frac{N}{bhf_{cd}} = \frac{5956 \times 10^3}{350 \times 3500 \times 20} = 0.243 \qquad \text{[Clause 5.4.3.4.1(2)]}$$

$\nu_{m\,x} < 0.4 \Rightarrow$ the design axial force does not exceed the maximum limit for DCM structures.

$\upsilon_{m\,x} > 0.2 \Rightarrow$ it is necessary to design the boundary elements explicitly for ductility according to EC8 Clause 5.4.3.4.2(12).

Situation with $N_{max}$: design using Concrete Centre charts (from www.concretecentre.com). Note: these are based on characteristic concrete strength $f_{ck}$ rather than design strength $f_{cd}$.

*GL 7 and 9*

$$\frac{N}{bhf_{ck}} = \frac{5956 \times 10^3}{350 \times 3500 \times 30} = 0.162$$

$$\frac{M}{bh^2 f_{ck}} = \frac{21282 \times 10^6}{350 \times 3500^2 \times 30} = 0.165$$

*GL 1 and 15*

$$\frac{N}{bhf_{ck}} = \frac{4351 \times 10^3}{350 \times 3500 \times 30} = 0.118$$

$$\frac{M}{bh^2 f_{ck}} = \frac{26240 \times 10^6}{350 \times 3500^2 \times 30} = 0.204$$

Situation with $N_{\min}$:

*GL 7 and 9*

$$\frac{N}{bhf_{ck}} = \frac{3097 \times 10^3}{350 \times 3500 \times 30} = 0.084$$

$$\frac{M}{bh^2 f_{ck}} = \frac{20947 \times 10^6}{350 \times 3500^2 \times 30} = 0.163$$

*GL 1 and 15*

$$\frac{N}{bhf_{ck}} = \frac{1024 \times 10^3}{350 \times 3500 \times 30} = 0.028$$

$$\frac{M}{bh^2 f_{ck}} = \frac{25851 \times 10^6}{350 \times 3500^2 \times 30} = 0.201$$

$$\frac{A_{s,tot} f_{yk}}{bh f_{ck}} = 0.58$$

$A_s =$ total area of flexural reinforcement in the boundary elements of the wall section.

$$A_{S,tot} = bh \frac{f_{ck}}{f_{yk}} 0.58 = 350 \times 3500 \times \frac{30}{500} \times 0.58$$
$$= 42630 \text{mm}^2 \Rightarrow 2 \times 21315 \text{mm}^2$$

Before detailing the number and diameter of the flexural reinforcement bars, the length of the boundary elements will be evaluated. This is because the flexural reinforcement on the boundary elements can not be distributed arbitrarily. Even though it is convenient to concentrate it near the extremities, in practice it is necessary to spread part of it along the faces of the boundary element because the minimum diameter of longitudinal bars is related to the spacing of the confinement reinforcement (hoops and ties) according to Clauses 5.4.3.4.2(9) and 5.4.3.2.2(11), and because the spacing of flexural steel bars and of confinement reinforcement is relevant for the evaluation of the effectiveness of confinement, according to Equations 5.16a and 5.17a.

Minimum length of the boundary elements:

$0.15 l_w = 0.15 \times 3.5 = 0.525\text{m}$ [Clause 5.4.3.4.2(6)]

($l_w$ – length of wall cross section.)

$1.5 b_w = 1.5 \times 0.35 = 0.525\text{m}$ [Clause 5.4.3.4.2(6)]

($b_w$ – width of wall cross section.)

The length of the boundary elements ($h_0$) may be evaluated as follows:

$h_0 = x_u \times \left(1 - \varepsilon_{cu2} / \varepsilon_{cu2,c}\right)$ [Clause 5.4.3.4.2(6)]

$\varepsilon_{cu2} = 0.0035$ [Clause 5.4.3.4.2(6)]

$\varepsilon_{cu2,c} = 0.0035 + 0.1 \alpha\omega_{wd}$ [Clause 5.4.3.4.2(6)]

$x_u = \left(\nu_d + \omega_v\right) \frac{l_w b_c}{b_0}$ [Clause 5.4.3.4.2(5)a Equation 5.21]

$\alpha\omega_{wd} \geq 30 \mu_\varphi \left(\nu_d + \omega_v\right) \varepsilon_{sy,d} \frac{b_c}{b_0} - 0.035$ (Equation 5.20)

$b_0$ is the minimum dimension of concrete core, measured to centreline of the hoops
$x_u$ is the depth of the compressive zone
$\varepsilon_{cu2}$ is the maximum strain of unconfined concrete
$\varepsilon_{cu2,c}$ is the maximum strain of confined concrete
$\alpha$ is the confinement effectiveness factor
$\omega_{wd}$ is the mechanical ratio of confinement reinforcement $\omega_{wd} = \dfrac{V_{conf\,re\,inf}}{V_{concrete\,core}} \dfrac{f_{yd}}{f_{cd}}$

Assuming a concrete cover of 45mm to the main flexural reinforcement and $\varphi$ = 10 mm hoops:

$b_0 = 350 - 2 \times 45 + 10 = 270\text{mm}$

$b_c = 350\text{mm}$

$\rho_v - A_{s,v} / A_c$ ratio of vertical web reinforcement

Minimum amount of vertical web reinforcement [EC2 Clause 9.6.2(1)]

$A_{s,v\,\min} = 0.002 A_c = 0.002 \times 350 \times 1000 = 700 mm^2/m \Rightarrow$ 2 legs T10 at 200mm spacing = 785mm/m²

785 mm²/m (T>T$_C$) (Equation 5.4)

$\mu_\varphi = 2 \times 3.6 - 1 = 6.2$ (assuming that $M_{Rd} \approx M_{Ed}$)
[Clause 5.4.3.4.2(2)]

$$\varepsilon_{sy,d} = \frac{435}{200000} = 0.002175$$

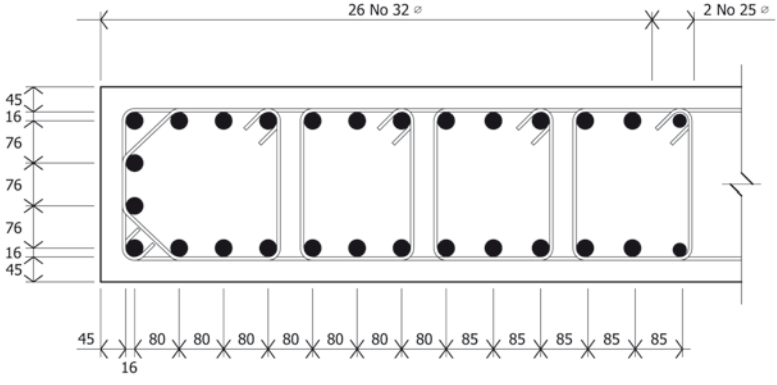



*Figure 5.16* Possible detailing of wall boundary elements

$\omega_v$ – mechanical ratio of vertical web reinforcement

$$\omega_v = \rho_v \frac{f_{yd,v}}{f_{cd}} = \frac{785}{350 \times 1000} \frac{435}{20} = 0.049$$

$$\alpha\omega_{wd} = 30 \times 6.2 \times (0.243 + 0.049) \times 0.002175 \frac{350}{270} - 0.035 = 0.118$$

$$\varepsilon_{cu2,c} = 0.0035 + 0.1 \times 0.118 = 0.0153$$

$$x_u = (0.243 + 0.049) \frac{3500 \times 350}{270} = 1325\text{mm}$$

Length of boundary elements

$$h_0 = 1325 \times (1 - 0.0035/0.0153) = 1022\text{mm}$$

Knowing the amount of flexural reinforcement and the length of the boundary elements it is possible to make a first detail of the boundary elements. Figure 5.16 shows a possible solution:

$$h_0 = 7 \times 80 + 5 \times 85 + \frac{10}{2} + \frac{32}{2} + \frac{10}{2} + \frac{25}{2} = 1024\text{mm}$$

In the evaluation of the dimension of the boundary elements it was assumed the diameter of the stirrups and hoops is φ = 10mm.

The proposed detail of the boundary elements meets EC8 and EC2 requirements. According to Clause 5.3.4.3.2(9) of EC8, it is only necessary that 'every other longitudinal bar is engaged by a hoop or cross-tie' and according to Clause 5.4.3.2.2(11) 'the distance between consecutive longitudinal bars engaged by hoops and cross ties does not exceed 200 mm'. EC2 states that 'No bar within a compressive zone should be further than 150 mm from a restrained bar' [Clause 9.5.3(6)]. However it should be noted that there are several options to meet EC2 and EC8 requirements for the design of the boundary elements, as will be discussed at a later stage.

Minimum concrete cover to main vertical reinforcement [EC2 Clause 8.1(2)]

$$c_{nom} = c_{\min} + \Delta c_{dev} \quad \text{(EC2 Equation 4.1)}$$

$$c_{\min} = \varphi = 32\text{mm}$$

$$\Delta c_{dev} = 10\text{mm}$$

$$c_{nom} = 42\text{mm} < 45\text{mm}$$

Minimum distance between flexural bars [EC2 Clause 8.2(2)]

$$\text{Max. of } \begin{cases} k_1\varphi = 1\times 32 = 32\text{mm} \\ d_g + k_2 = 25+5 = 30\text{mm} \\ 20\text{mm} \end{cases}$$

$d_g$ – maximum aggregate size.

*5.9.4.4 Shear design*

Shear failure associated with compressive failure of diagonal struts:

$$V_{Rd,m\,x} = \frac{\alpha_{cw} b_w z \nu_1 f_{cd}}{\cot g\theta + tg\theta} \qquad \text{(EC2 Equation 6.9)}$$

$\alpha_{cw} = 1$ for non-prestressed structures

$b_w = 0.35\text{m}$

$z = 0.9d = 0.9\times(0.9\times 3.5) = 2.835\text{m} \qquad d = 0.9\times 3.5$

$$\nu_1 = 0.6\times\left[1-\frac{f_{ck}}{250}\right] = 0.6\times\left[1-\frac{30}{250}\right] = 0.528 \qquad \text{(EC2 Equation 6.6N)}$$

According to Clause 6.2.3(2) of EC2 the limiting values for use in each country can be found in the respective National Annex of EC2. The limiting values recommended in EC2 are $1 \le cotg\,\theta \le 2.5$.

$\cot g\theta = 2.5 \qquad tg\theta = 0.4$

$$V_{Rd,m\,x} = \frac{1\times 350\times 2835\times 0.528\times 20}{2.5+0.4} = 3613158\text{N} \approx 3613\text{kN}$$

$\cot g\theta = 1 \qquad tg\theta = 1$

$$V_{Rd,m\,x} = \frac{1\times 350\times 2835\times 0.528\times 20}{1+1} = 5239080\text{N} \approx 5239\text{kN}$$

The design value of the shear force must be obtained by multiplying the shear force obtained from the global structural analysis by the magnification factor referred to in Section 5.7.2, as follows:

$$V_{Ed} = 1.5\times 2401 = 3602\text{kN} \qquad \text{[Clause 5.4.2.4(7)]}$$

If $V_{Ed} \rangle V_{Rd,\max}$ (associated with the $\cot g\theta = 2,5$) it would be necessary to adopt a lower value of $\cot g\theta$ until $V_{Ed} \le V_{Rd,\max}$. This would obviously lead to a larger amount of stirrups, according to Equation 6.8.

Shear resistance associated with failure in shear by diagonal tension:

$$V_{Rd,s} = \frac{A_s}{s} z.f_{ywd}.\cot g\theta \qquad \text{(EC2 Equation 6.8)}$$

(Equation 6.8 allows the evaluation of the amount of stirrups.)

According to Equation 6.8 of EC2 the higher the value of $\cot g\theta$ ($\theta$ – inclination of diagonal compressive struts) the lower is the necessary amount of stirrups.

Evaluate the amount of stirrups (assume $\cot g\theta = 2.5$ and apply Equation 6.8 of EC2):

$$\frac{A_s}{s} = \frac{V_{Ed}}{z.f_{ywd}.\cot g\theta} = \frac{3602\times10^3}{2835\times435\times2.5} = 1.17\text{mm}^2/\text{mm} = 1170\text{mm}^2/\text{m}$$

⇒ 2 legs Φ10 at 125 mm spacing (1256mm²/mm)

Verification of minimum wall horizontal reinforcement:

$$A_{sh,\min}\begin{cases}0.25\times A_{sv,\min} = 0.25\times785 = 197\text{mm}^2/\text{m}\\ 0.001A_c = 0.001\times350\times1000 = 350\text{mm}^2/\text{m}\end{cases}$$

The above design represents the most economic design that respects the limits for cotgθ recommended by EC8. However, the shear capacity of RC members depends on factors that are not explicitly accounted for in Equation 6.8, namely the level of axial force and the formation and development of plastic hinges. It may be considered that in some situations Equation 6.8 does not provide enough protection against shear failure. Considering the potentially catastrophic consequences of brittle shear failure of RC walls, if designers want to adopt a more conservative approach in shear wall design the following suggestion is offered: in the zones outside the plastic hinge adopt $\theta \geq 30^o$, and in the plastic hinge adopt $\theta \geq 38^o$ if the design axial force is compressive and $\theta = 45^o$ if the design axial force is tensile. This is less stringent than what is required for DCH structures but it reduces the gap between DCM and DCH requirements for shear design. This gap may be considered excessive, in particular for RC walls, as these elements are more prone to shear failure than beams and columns.

If this suggestion had been adopted, and since the design axial force is always compressive, the necessary amount of shear reinforcement would be:

$$\frac{A_s}{s} = \frac{3602\times10^3}{2835\times435\times\cot g38} = 2.282\text{mm}^2/\text{mm} = 2282\text{mm}^2/\text{m}$$

(e.g. 2 legs of Φ16@175 or Φ12@100)

*5.9.4.5 Detailing for local ductility*

Height of the plastic hinge above the base of the wall for the purpose of providing confinement reinforcement:

$$h_{cr} = \max\left[l_w, h_w/6\right] \qquad \text{(Equation 5.19a)}$$

$$h_{cr} = \max\left[3.5, 28.8/6\right] = 4.8\text{m}$$

$$\begin{cases} h_{cr} \le 2.l_w = 2 \times 3.5 = 7\text{m} \\ h_{cr} \le 2.h_s = 2 \times 4.3 = 8.6\text{m} \end{cases} \quad h_{cr} \le 7\text{m}$$

$$h_{cr} = 4.8m$$

EVALUATION OF CONFINEMENT REINFORCEMENT IN THE BOUNDARY ELEMENTS

According to Equation 5.20:

$$\alpha.w_{wd} \ge 0.118$$

$$\alpha = \alpha_n \alpha_s$$

$$\alpha_n = 1 - \sum_i b_i^2 / \left(6.b_0 h_0\right) \qquad \text{(Equation 5.16a)}$$

$$\alpha_s = \left(1 - \frac{s}{2b_0}\right) \times \left(1 - \frac{s}{2.h_0}\right) \qquad \text{(Equation 5.17a)}$$

All distances ($b_i, b_0, h_0, s$) are measured to centrelines of hoops or flexural reinforcement. The values $b_i$ are based on the detail of the edge members and represent the distance between consecutive engaged bars. The reason for this is that confining stresses are transferred from the steel cage to the concrete, essentially at the intersection of flexural engaged bars with the hoops and cross ties that engage them. These are the points at which the outwards movement of the steel cage is strongly restricted. The points where the flexural reinforcement is only connected to sides of rectangular hoops, as shown in Figure 5.17a, are restricted against outward movement in a much



a Deformation of hoops

b Confining stresses

*Figure 5.17* Efficiency of rectangular hoops

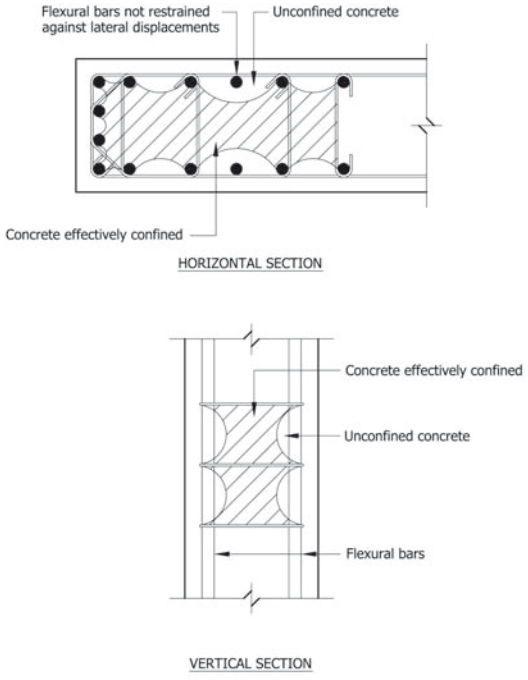



*Figure 5.18* Effect of confinement between layers by arch action

less efficient manner. This is because rectangular hoops work efficiently under tension and not under flexure and therefore restrict the outwards deformation of flexural bars and transfer confining stresses to concrete essentially at the corners and not along straight sides, as illustrated in Figure 5.17b.

Since with straight hoops confinement stresses are transferred to the concrete at discrete locations (with circular or spiral hoops, the distribution of confinement stresses takes place continuously along the length of the hoops), in between those locations the effect of confinement is felt essentially by arch action. This effect takes place both on the vertical and horizontal planes, leading to a reduction of the zone effectively confined between hoops layers, as shown in Figure 5.18.

The reduction of the zone effectively confined, away from the points in which most of the confining stresses are transferred to the concrete, is considered by means of the confinement effectiveness factor, $\alpha$, which corresponds to the ratio of the smallest area effectively confined by the area of the concrete core, of rectangular shape with dimensions $b_0 \times h_0$ in this case. Therefore the factor $\alpha$ is evaluated as $\alpha = \alpha_n.\alpha_s$, in which the term $\alpha_s$ accounts for the loss of confined area due to arch action in the vertical plane and $\alpha_n$ for the loss of confined area due to arch action in the horizontal plane. Therefore both the spacing between flexural engaged bars, as well

as the vertical spacing between hoop layers, are critical parameters for the effectiveness of the confinement. For this reason both these spacings must be kept below the smallest dimension of the confined concrete core. In the case of circular hoops or spirals, arch action only takes place in the vertical plane, therefore $\alpha_n = 1$ and the spacing between flexural bars is irrelevant.

The longitudinal bars pointed out with arrows at Figure 5.18 are not considered for the evaluation of the effectiveness of confinement, as they are not engaged by hoops or cross ties [Clause 5.4.3.2.2(8))]. This leads to values of $b_i$ = 160 mm being adopted instead of pairs of values of $b_i$ = 80mm. According to Figure 5.16 the value of $\alpha_n$ must be evaluated as follows:

$$\alpha_n = 1 - \left[2 \times \left(80^2 + 160^2 + 80^2 + 160^2 + 80^2 + 170^2 + 85^2 + 170^2\right) + 3 \times 76^2\right] / (6 \times 270 \times 1024) = 0.826$$

According to Equation 5.18 hoop spacing should not exceed any of the following values:

$$s = \min\left(\frac{b_0}{2}; 175; 8d_{bl}\right) \qquad \text{(Equation 5.18)}$$

$b_0$ = 270mm (width of confined boundary element)

$d_{bl}$ – diameter of flexural reinforcement

$$s = \min\left(\frac{270}{2}; 175; 8 \times 20\right) = 135\text{mm}$$

In order to match the stirrup spacing, $s = 0.125m$ can be adopted. Note that with the adopted hoop spacing of 125 mm, the minimum diameter of the longitudinal flexural bars within the boundary elements would be 16 mm. If a spacing of 150 mm had been adopted, the minimum diameter of the longitudinal bars would be 20 mm. The need to minimise the spacing of the longitudinal bars as far as practicable, coupled with the need to avoid longitudinal bars with small diameters in the boundary elements, forces the spread of a reasonable amount of flexural reinforcement along the faces of the boundary elements, as previously mentioned.

Assuming $s = 0.125m$ initially:

$$\alpha_s = \left(1 - \frac{125}{2 \times 270}\right) \times \left(1 - \frac{125}{2 \times 1024}\right) = 0.72$$

$$\alpha = 0.826 \times 0.72 = 0.59$$

$$\alpha.w_{wd} = 0.118 \Rightarrow 0.59 \times w_{wd} = 0.118$$

$w_{wd} = 0.20$

$w_{wd} \geq w_{wd,\min} = 0.08$ [Clause 5.4.3.2.2(9)]

EVALUATION OF $\omega_{wd}$ FOR THE ADOPTED DETAIL OF THE BOUNDARY ELEMENTS

Length of confining hoops:
Exterior hoops = $270 + 2 \times 1024 = 2318$mm
Interior hoops =

$$2 \times (2 \times 80 + 32 + 10) + (2 \times 85 + 32 + 10) +$$
$$(2 \times 85 + 28.5 + 10) + 7 \times 270 + 76 + 2 \times$$
$$\sqrt{(76 + 32 + 10/2)^2 + (80 + 32/2 + 10/2)^2} = 3094\text{mm}$$

Exterior hoops (=stirrups, which also contribute to confine the concrete)
$\varphi = 10$ mm

Assuming inner hoops $\varphi = 10$mm
Volume of hoops/m

$$V = \frac{1}{0.125} \times (2318 + 3094) \times 78.54 = 3398736\text{mm}^2/\text{m}$$

$$w_{wd} = \frac{3398736}{1024 \times 270 \times 1000} \frac{435}{20} = 0.267$$

If the diameter of the inner hoops is reduced to $\varphi = 8$mm

$$V = \frac{1}{0.125}(2318 \times 78.5 + 3094 \times 50) = 2693304\text{mm}^2/\text{m}$$

$$w_{wd} = 0.21$$

Adopt exterior hoops (stirrups from one edge of the wall section to the other) 2 legs Φ10 at 125 mm spacing.

Adopt inner hoops (according to the detail of the boundary elements) Φ8 at 12 5mm spacing.

### *5.9.4.6 Improvements to the detail of the boundary elements*

Designers will generally have several options for the design of walls' boundary elements. In this section some possible improvements of the detail of the boundary elements are analysed.

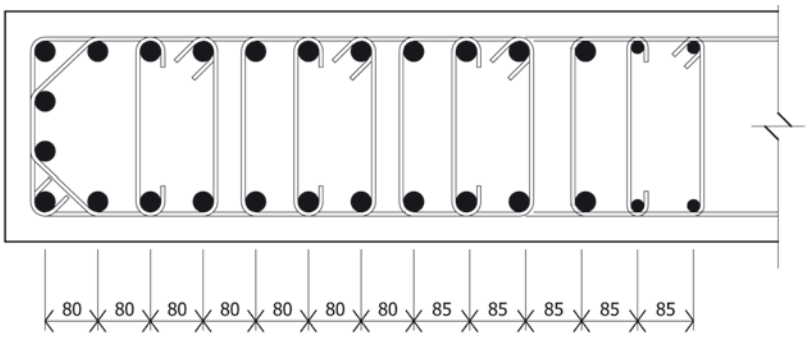



*Figure 5.19* Detail of boundary element with all flexural bars engaged by hoops or cross ties

HOOPS AND CROSS TIES

It was noted previously that flexural bars such as the ones pointed out in Figure 5.18 are not engaged and are inefficient from the point of view of confinement. Besides improving the effectiveness of confinement, engaging these bars with corner hoops or cross ties would provide additional restraint against buckling of those flexural bars.

Therefore the detailing of the boundary elements can be improved by additional hoops or cross ties that engage the eight flexural bars not engaged in the detail of Figure 5.16 to increase the efficiency of the confinement and give additional restraint against buckling of these bars. One simple way of doing this would be by adding cross ties, as shown in Figure 5.19.

With this detail the value of factor $\alpha_n$ (Equation 5.16a) would be as follows:

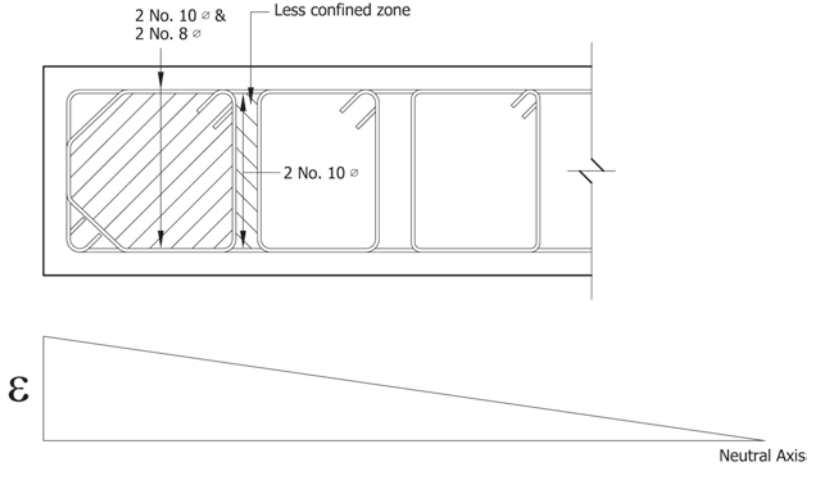



*Figure 5.20* Zones with different confinement within the wall edge member

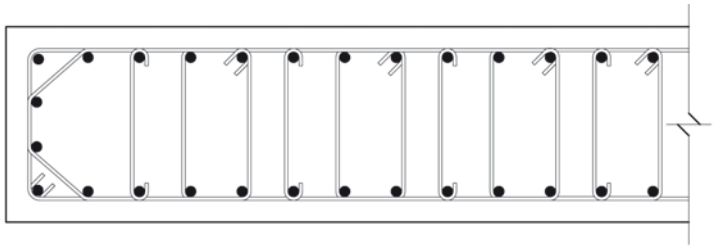


*Figure 5.21* Detail of boundary element with overlapping hoops

$$\alpha_n = 1 - \left(2 \times 7 \times 80^2 + 3 \times 76^2 + 2 \times 5 \times 85^2\right) / \left(6 \times 270 \times 1024\right) = 0.89$$

This represents an increase of around 8 per cent in the efficiency of confinement.

The layout of the inner stirrups is also less efficient than it could be. Figure 5.20 shows that the concrete between the inner hoops is less confined than the concrete within these hoops: the expansion of the concrete within the inner hoops is restricted in the direction of the largest dimension of the wall cross section by the stirrups (2T10) and the inner hoops (2T8), while the expansion of the concrete between the inner hoops in the same direction is restricted only by the stirrups. This is not consistent with the underlying EC8 design philosophy for the provision of confinement. Note that the boundary elements are analysed as integral units since the amount of confinement reinforcement is evaluated for the whole boundary element (by means of a single value of $\omega_{wd}$) and not parts of it.

Even though EC8 does not account for situations with different levels of confinement within the edge members of the walls, the relative importance of the above situation decreases if the zone with less confinement is closer to the neutral axis, as the strain demand on the concrete is less than near the section extremity. Anyway the inconvenience of having zones with different

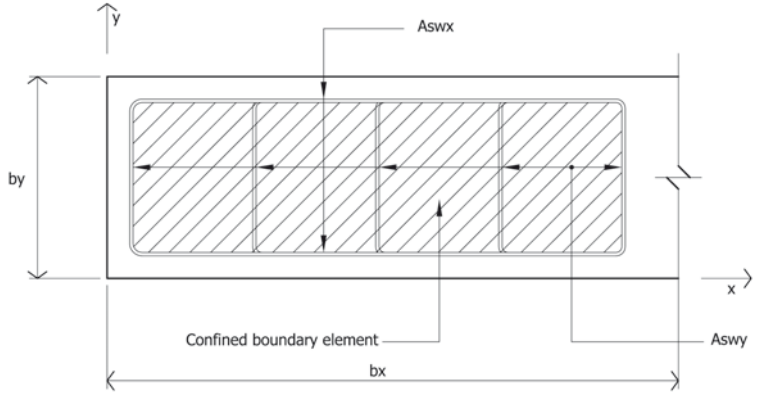



*Figure 5.22* Confined boundary elements

levels of confinement near the extremity of the wall can be avoided by overlapping the inner hoops, as shown in Figure 5.21.

This last detail is equivalent to having four hoops instead of two in the largest dimension of the wall cross section throughout the boundary element, increasing the confining stress in the weaker zones, shown in Figure 5.20, and thus providing a uniform distribution of the available strain ductility in the edge member. This is an improvement as compared with the detail of Figure 5.20, but its relative importance and efficiency will vary from case to case.

In rectangular or elongated sections the confining stress in two orthogonal directions may be different, but it is good design practice to make them similar, as the concrete is only properly confined if it is confined in all directions. This is illustrated in Annex E of EC8, Part 2, according to which in situations with different confining stresses ($\sigma_x, \sigma_y$) in orthogonal directions, an effective confining stress can be evaluated as $\sigma_e = \sqrt{\sigma_x . \sigma_y}$. In order that the orthogonal confining stresses are similar, the ratio of confinement reinforcement should be similar in both directions. In the case of the details of the boundary elements previously referred to, these values are as follows:

$$\rho_{swx} = \frac{A_{swx}}{s.b_y}$$

$$\rho_{swy} = \frac{A_{swy}}{s.b_x}$$

(*s* is the longitudinal spacing of hoops and cross ties).
For detail shown in Figure 5.16:

$$A_{swx}\,(2T10) = 157\text{mm}^2 \qquad \rho_{swx} = \frac{157}{125 \times 270} = 0.00465$$

$$A_{swy}\,(1T10+8T8) = 481\text{mm}^2 \quad \rho_{swy} = \frac{481}{125 \times 1024} = 0.00376$$

For detail shown in Figure 5.19:

$$A_{swx}\,(2T10) = 157\text{mm}^2 \quad \rho_{swx} = 0.00465$$

$$A_{swy}\,(1T10+12T8) = 681\text{mm}^2 \quad \rho_{swy} = \frac{681}{125 \times 1024} = 0.00532$$

For detail shown in Figure 5.21:

$$A_{swx}\,(2T10+2T8) = 258\text{mm}^2 \quad \rho_{swx} = \frac{258}{125 \times 270} = 0.00764$$

$$A_{swy}\,(1T10+12T8) = 681\text{mm}^2 \quad \rho_{swy} = 0.00532$$

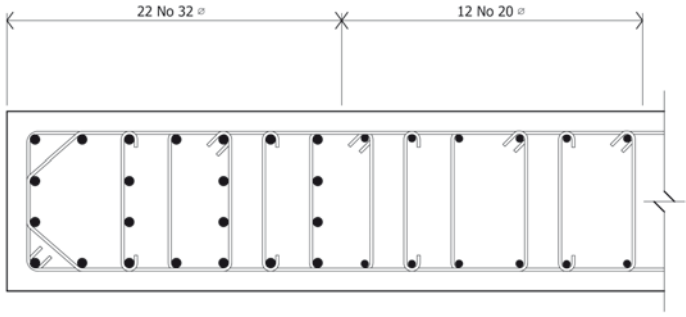



*Figure 5.23* Detail with flexural reinforcement closer to the extremity of the section

The overlapping hoops are clearly a better detail than non-overlapping hoops, and the efficiency is higher in the cases in which it increases the smallest confining stress, according to the equation for the effective stress. Therefore the recommended detail for the hoops and cross ties of the boundary elements of the example wall would be the detail of Figure 5.21.

FLEXURAL REINFORCEMENT

The distribution of the flexural reinforcement within the edge member shown in Figure 5.16 was done with steel bars with diameters 32 and 25 mm distributed along the periphery of the boundary element. However, this can be optimised by concentrating the reinforcement closer to the extremity of the wall section, leading to higher flexural capacity (for the same amount of reinforcement) and higher curvature ductility. This is due to the fact that the concentration of the flexural reinforcement at section extremities leads to the reduction of the dimension of the compressive zone, a feature of behaviour not accounted for in Equation 5.21.

The concentration of the flexural reinforcement closer to the wall extremities can be achieved for instance by placing some flexural reinforcement in the middle of the boundary elements, as shown in Figure 5.23.

The inner vertical bars can be maintained in their position during casting by tying them to the hoops and cross ties. In order to maintain the spacing between flexural bars, to retain the effectiveness of confinement, the position of the φ32 bars that were moved closer to the extremity of the section, is taken by smaller flexural bars. In order not to increase the total amount of steel, four of the φ32 bars and the two φ25 bars were replaced by φ20 bars. This meets the requirement that the maximum spacing of confinement reinforcement should not be higher than eight times the diameter of flexural bars (according to Equation 5.21). For the chosen spacing of hoops and

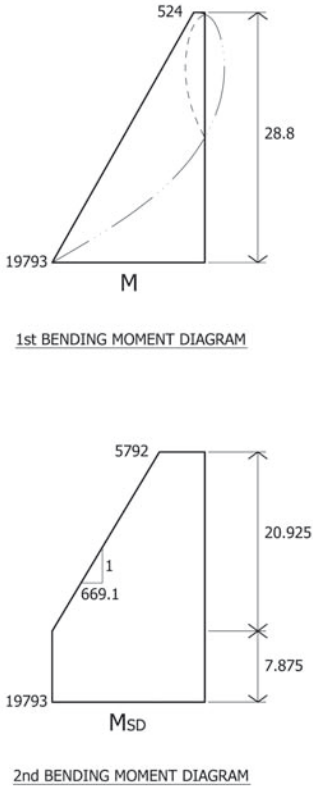



*Figure 5.24* Bending moment diagrams

cross ties of 125 mm, the minimum diameter of the flexural reinforcement is 16 mm.

### *5.9.5 Design of the wall above the plastic hinge*

The design of the wall above the plastic hinge at the base is different from the design of the plastic hinge in two main features:

1 It is based on the provisions of EC2, since all these zones are supposed to remain in the elastic range throughout the seismic action. There is no need to provide confinement reinforcement.
2 In order to ensure that the wall remains elastic above the base hinge considering the uncertainties in the structure dynamic behaviour, the design bending moments and shear forces obtained from analysis are magnified.

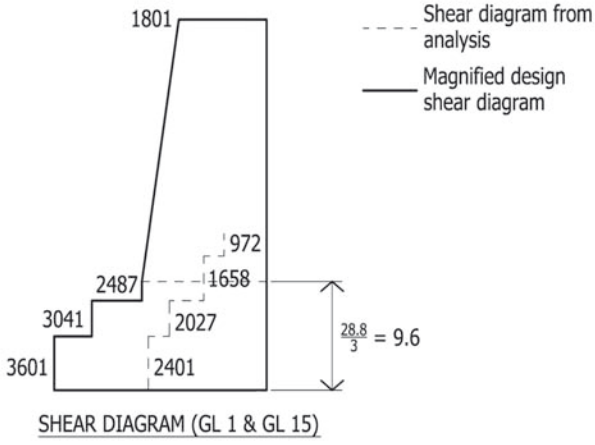



*Figure 5.25* Shear force diagrams

From the bending moment diagram in Figure 5.24 obtained from analysis the following linear envelope can be established.

This diagram must be shifted upwards a distance $a_1$, designated tension shift in EC8 [Clause 5.4.2.4(5)], consistent with the strut inclination adopted in the Ultimate Limit State verification for shear.

$$a_1 = d.\cot g\theta = 3150 \times 2.5 = 7875\text{mm} = 7.875\text{m}$$

The design bending moment diagram ($M_{Sd}$) in Figure 5.24(b) is obtained for the design of the wall above the plastic hinge.

$$M_{\text{top}} = 19793 - 669.1 \times (28.8 - 7.875) = 5792\text{kN.m}$$

The values above are the basic values prior to applying the factor accounting for torsional effects since these are dependent on location. Both the base and design moments need to be increased by the appropriate factor before being used in the design (e.g. for GL 7 and 9, $M_{\text{base}}$ = 19793*1.04 = 20585 kNm and $M_{top}$ = 5792*1.04 = 6024 kNm).

Shear force design diagrams are illustrated in Figure 5.25.

$$V_{\text{wall,top}} = \frac{V_{\text{wall,base}}}{2} = \frac{3602}{2} = 1801\text{kN}$$

The approach to design of the elastic sections at the higher levels is:

- Choose a level at which first curtailment of flexural reinforcement would be appropriate (say at the third-floor level in this instance).

- Carry out the design for moment and shear as described previously using the values from Figures 5.24 and 5.25 and the axial load appropriate for the level chosen.
- There is no requirement to detail boundary elements above the height of the critical region other than EC2 prescriptions.

### *5.9.6 Design of frame elements*

#### *5.9.6.1 Torsional effects*

The forces applied to the shear walls have been increased by a factor, δ, to account for accidental eccentricity (Clause 4.3.3.2.4).

$$\delta = 1 + 0.6\, x/L$$

However, as noted earlier, no increase has been applied to the frame elements in this preliminary analysis since torsional effects due to accidental eccentricity will tend to be controlled by the stiff perimeter walls and the simplified allowance for accidental eccentricity is considered to be quite conservative.

As previously, since the inter-storey drift sensitivity coefficient, θ, is less than 0.1 at all levels, no increase is required for P-δ effects.

#### *5.9.6.2 Design forces*

The beam flexural design is based on the maximum moments in the lower four storeys. The remainder of the design then follows from capacity design principles.

From the analysis output:

$M_{\text{hogging}}$ (max) = 1241 kNm (Level 3)

$M_{\text{sagging}}$ (max) = 1189 kNm (Level 3)

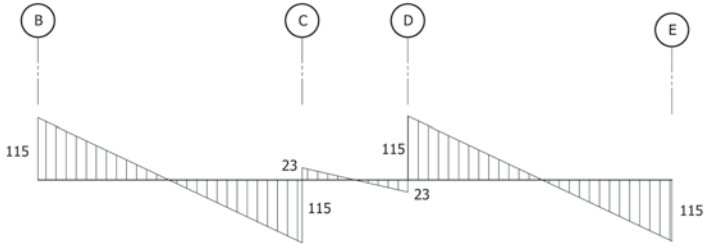



*Figure 5.26* Shear force diagram for gravity sub-frame analysis ($1.0G_k + 0.3Q_k$)

*Table 5.3* Gravity and seismic combinations – selected analysis output

a) Sample applied moments

| *Element* | *Level/ location** | *Applied moment (kNm)* | | *Combined static and seismic moment (kNm)* | |
|---|---|---|---|---|---|
| | | *Static* | *Seismic* | *Hogging* | *Sagging* |
| 114 | 1 Outer | 191 | ±482 | 673 | –291 |
| 115 | 1 Inner | 30 | ±751 | 781 | –721 |
| 118 | 2 Outer | 158 | ±799 | 957 | –641 |
| 119 | 2 Inner | 29 | ±1069 | 1098 | –1040 |
| 121 | 3 Outer | 160 | ±931 | 1091 | –771 |
| 122 | 3 Inner | 27 | ±1214 | 1241 | –1189 |
| 124 | 4 Outer | 158 | ±925 | 1083 | –767 |
| 125 | 4 Inner | 34 | ±1015 | 1049 | –981 |
| 127 | 5 Outer | 149 | ±825 | 974 | –676 |
| 128 | 5 Inner | 45 | ±659 | 704 | –614 |

* Location: 'Outer' refers to the beams between GL B and C or D and E; 'Inner' refers to the span between GL C and D

b) Sample axial loads

| *Element* | *Level/ location** | *Axial load (kN)* | | *Combined static and seismic axial load (kN)* | |
|---|---|---|---|---|---|
| | | *Static* | *Seismic* | *Maximum* | *Minimum* |
| 80 | 1 Outer | 3362 | ±1303 | 4665 | 2059 |
| 88 | 1 Inner | 3420 | ±2558 | 5978 | 862 |
| 81 | 2 Outer | 2677 | ±1265 | 3942 | 1412 |
| 89 | 2 Inner | 2977 | ±2164 | 5141 | 813 |
| 82 | 3 Outer | 2266 | ±1083 | 3349 | 1183 |
| 90 | 3 Inner | 2527 | ±1638 | 4165 | 889 |
| 83 | 4 Outer | 1855 | ±870 | 2725 | 985 |
| 91 | 4 Inner | 2078 | ±1041 | 3119 | 1037 |
| 84 | 5 Outer | 1443 | ±660 | 2103 | 783 |
| 92 | 5 Inner | 1629 | ±575 | 2204 | 1054 |

*Location: 'Outer' refers to the columns on GL B and E; 'Inner' refers to the columns on GL C and D

*5.9.6.3 Beam design*

Initially, treat as rectangular – add flange reinforcement later for capacity design.

Because of the shape of the bending moment diagram in the short span, it is assumed that no redistribution will take place.

$M_1$ (Hogging) = 1241 kNm

$M_2$ (Sagging) = 1189 kNm

Design for DCM – bending and shear resistances from EC2

[Clause 5.4.3.1.1(1)]

As the example is aimed at demonstrating the application of the seismic engineering principles of EC8, reference is made to design aids where standard design to EC2 is carried out as part of the verification. The EC2 design aids referenced here are the 'How to' sheets produced by the Concrete Centre and downloadable from www.concretecentre.com.

$K = M/bd^2 f_{ck}$

$z = 0.5 * d\{1 + (1 - 3.53\ K)^{½}\}$

HOGGING

Assume two layers of 32 mm diameter bars, 45 mm cover to main reinforcement:

$d = 750 - 45 - 32 - (32/2) = 657$ mm

$K = 1241 \times 10^6 / (450 * 657^2 * 30) = 0.212$

No redistribution – K´ = 0.205

K>K´ Therefore, compression reinforcement is required.

$z = 0.5*657\{1 + (1 - 3.53*0.205)^{1/2}\}$

z = 501 mm

Use partial factors for the persistent and transient design situations

[Clause 5.2.4(2)]

$\gamma_s = 1.15 \quad \gamma_c = 1.5$

Compression reinforcement

$x = 2(d–z) = 2\ (657–501) = 312$ mm

$$A_{s2} = \frac{(K - K')f_{ck}bd^2}{f_{sc}(d - d_2)}$$

$$f_{sc} = \frac{700(x - d_2)}{x} = \frac{700(312 - 93)}{312} = 491 \text{ N/mm}^2 > f_{yd}$$

$$f_{sc} = f_{yd} = 500/1.15 = 434.8 \text{ N/mm}^2$$

$$A_{s2} = \frac{(0.212 - 0.205)30 * 450 * 657^2}{434.8(657 - 93)} = 166 \text{ mm}^2$$

Nominal – will be enveloped by reinforcement provided for sagging moment in reverse cycle.

Tension reinforcement

$$As1 = \frac{K' f_{ck}bd^2}{f_{yd}z} + \frac{A_{s2}f_{sc}}{f_{yd}} = \frac{0.205 * 30 * 450 * 657^2}{434.8 * 501} + \frac{166 * 434.}{434.8} = 5650 \text{ mm}^2$$

Use 7–Φ32 (5628 mm² – 1.9 per cent) plus longitudinal slab reinforcement within effective width (see later).

Note: it is often recommended in the UK that $K'$ is limited to 0.168 to ensure a ductile failure. If the calculation above were to be repeated with this limit applied, the resulting areas of tension and compression reinforcement would be:

$A_{s1} = 5540 \text{ mm}^2$ and $A_{s2} = 1046 \text{ mm}^2$

(i.e. a similar area of tension reinforcement and significantly increased compression reinforcement is required. In this case, because of the much

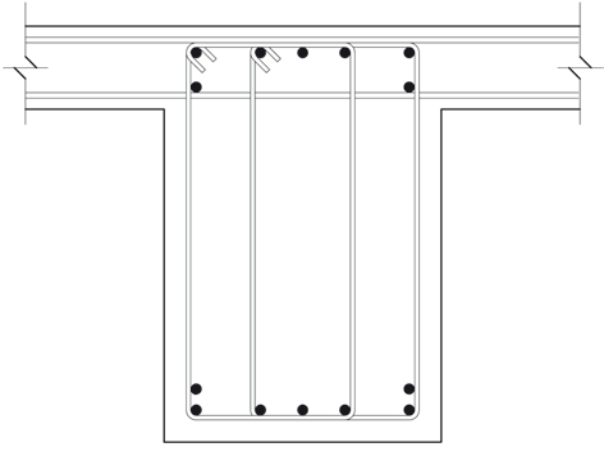


*Figure 5.27* Reinforcement arrangement in critical region of beam

larger area of reinforcement provided in the bottom face to cater for the reverse loading cycle, it has no practical effect on the solution).

SAGGING

$K = 1189 \times 10^6/(450 * 657^2 * 30) = 0.204 < 0.205$
Therefore, singly reinforced.

$z = 0.5*d\{1 + (1–3.53K)^{1/2}\}$
$= 0.5*657\{1+(1–3.53*0.204)^{1/2}\} = 502$ mm

$A_{s2} = 1189 \times 10^6/502*434.8 = 5447$ mm$^2$

7 – Φ32 (5628 mm$^2$ – 1.9 per cent)

Spacing = (450 – (2*45) – 32)/4 = 82 mm

Clear space between bars = 82 – 32 = 50 mm

Minimum clear space between bars = bar diameter OR aggregate size + 5 mm OR 20 mm OK.

$\rho min = 0.5 * (f_{ctm}/f_{yk}) = 0.5 * 2.9/500 = 0.0029$
(0.29 per cent cf 1.9 per cent provided)

$f_{ctm}$ from EC2 Table 3.1.

*5.9.6.4 Derive shear demand from flexural capacity*

Internal column connection framed by orthogonal beams.

Calculate hogging capacity:

$b_{eff} = b_w + 8 * h_f$ [Clause 5.4.3.1.1(3)]

Slab width to be considered = 8 * 0.15 = 1.2 m

Slab reinforcement = Φ12–300 T&B (754 mm$^2$/m in total)

$A_{s1} = 5628 + 754 * 1.2 = 6533$ mm$^2$

$A_{s1}$ (required) = 5650 mm$^2$ for an applied moment of 1241 kNm

Hogging capacity = 1241 * 6533/5650 = 1435 kNm

Calculate sagging capacity:

$A_{s2} = 5628$ mm$^2$

$A_{s2}$ (required) = 5447 mm$^2$ for an applied moment of 1189 kNm

Sagging capacity = 1189 * 5628/5447 = 1229 kNm

$V_{E,d} = \gamma_{Rd} * [M_{Rd}\,(\text{top}) + M_{Rd}\,(\text{bottom})]/l_{cl} + V_g$

(a) Long outer spans

For DCM structures, $\gamma_{Rd} = 1.0$ in beams

From gravity load analysis $V_g = 115$kN

$l_{cl} = 8.5 - 0.75 = 7.75$

$V_{E,d} = 115 + (1435 + 1229)/7.75 = 459$ kN

(b) Short central span

From gravity load analysis $V_g = 23$ kN

$l_{cl} = 3.0 - 0.75 = 2.25$

$V_{E,d} = 23 + (1435 + 1229)/2.25 = 1207$ kN

CHECK SHEAR RESISTANCE TO EC2 FOR DEMAND BASED ON FLEXURAL CAPACITY

As previously, where standard design to EC2 forms part of the verification, reference is made to the design aids downloadable from www.concretecentre.com.

(a) Outer spans

$v_{Ed} = 459*10^3/450*657 = 1.55$ N/mm$^2$

Assume Cot$\theta = 2.5$

$v_{Rd,\max} = 3.64$ N/mm$^2$ (from 'How to' Sheet 4: Beams – Table 7)

$A_{sw}/s = V_{E,d}/(z * f_{ywd} * \text{Cot}\theta) = 459 * 10^3/501 * 434.8 * 2.5 = 0.84$

Assume 8 mm links.

In a critical region, $s = \min\{h_w/4=188:\ 24d_{bw} = 192;\ 225;\ 8d_{bL}=256\}$ (Equation 5.13)

Use 175 mm spacing of links

$A_{sw} = 0.84 *175 = 147$ mm$^2$

Use 4 legs of $\Phi$8 (201 mm$^2$) as shown on Figure 5.27

$\rho_w$ (min) is OK from EC2 Equations 9.4 and 9.5.

(b) Short central span

$v_{Ed} = 1207*10^3/450*657 = 4.08$ N/mm$^2 > 3.64$

Cot$\theta$ is less than 2.5

$\theta = 0.5*\text{Sine}^{-1}[v_{Ed}/0.2f_{ck}*(1-f_{ck}/250)]$ (Concrete Centre 'How to' Guide 4 – Beams)

$\theta = 0.5*\text{Sine}^{-1}[4.08/0.2*30*(1-30/250)] = 25.3°$

$\text{Cot}\theta = 2.1$

$A_{sw}/s = V_{E,d}/(z * f_{ywd} * \text{Cot}\theta) = 1207 * 10^3/501* 434.8 * 2.1 = 2.64$

Assume links at 150 mm spacing.

$A_{sw} = 2.64*150 = 396$ mm$^2$ – Use 4 legs of T12 (452 mm$^2$)

*5.9.6.5 Check local ductility demand*

$\rho_{max} = \rho' + 0.0018 * f_{cd}/(\mu_\varphi * \varepsilon_{syd} * f_{yd})$ (EC8 Part 1 Equation 5.11)

$\mu_\varphi = 2* q - 1$

$\mu_\varphi = 2*3.6 - 1 = 6.2$

$f_{cd} = 30/1.5 = 20$ N/mm$^2$

$\varepsilon_{syd} = 434.8/200E3 = 0.0022$

Area of reinforcement in the compression zone = 5628 mm$^2$

$\rho' = 5628/450 * 657 = 0.019$

$\rho_{max} = \rho' + 0.0018*20/(6.2*0.0022_*434.8) = \rho' + 0.006$

By inspection, because $\rho$ only exceeds $\rho'$ by the nominal slab reinforcement, the expression is satisfied.

Check maximum diameter of flexural bars according to Equation 5.50a

$\rho_{max} = \rho + 0.006 = 0.025$

At Level 4, just above the critical node, $N_{static} = 2078$ kN and $N_{seismic} = \pm 1041$ kN

$N_{min} = 1037$ kN

$$\nu_d = \frac{1037 \times 10^3}{750^2 \times 20} = 0.092$$

Equation 5.50a $$\frac{d_{bL}}{h_c} \le \frac{7,5 f_{ctm}}{\gamma_{Rd} f_{yd}} \frac{1 + 0,8\nu_d}{1 + 0,75 k_D \rho' / \rho_{max}}$$

According to Clause 5.6.2.2.b, for DCM structures:
$\gamma_{Rd} = 1,0$

$$\frac{d_{bL}}{750} \le \frac{7,5 \times 2,9}{1,0 \times 434,8} \frac{1 + 0,8 \times 0,092}{1 + 0,75 \times \frac{2}{3} \times \frac{0,0190}{0,025}} \Rightarrow d_{bL} \le 29.2\text{mm}$$

Hence, the bond requirements across the column joint are not satisfied by the reinforcement arrangement proposed in the preliminary design, illustrating the difficulty in meeting the EC8 bond provisions. In the final design, this could be addressed through:

- modification of the reinforcement arrangement (providing 12–25 mm diameter bars in two layers would satisfy spacing requirements). This would be reduced further (to only eight bars) if the reduced inertial loads from the response spectrum analysis are considered rather than the equivalent lateral force approach (see the later calculations on the damage limitation requirement);
- increasing the concrete grade to C35/45 ($f_{ctm}$ would become 3.2 N/mm$^2$, which would result in a permitted bar diameter of 32.2 mm); or increasing the column size.

*5.9.6.6 Column design*

If the frame was to be designed as a moment frame in both directions, it may be designed for uniaxial bending about each direction in turn rather than considering biaxial bending, provided the uniaxial capacity is reduced by 30 per cent [Clause 5.4.3.2.1].

In this case, the frame is assumed braced in the longitudinal direction. Therefore, no reduction in capacity is taken.

$0.01 < \rho_l < 0.04$ [Clause 5.4.3.2.2 (1)]

Definition of critical regions:

$l_{cr} = \max \{h_c: l_{cl}/6; 0.45\}$ (EC8 Part 1 Equation 5.14)

$l_{cr} = \max \{0.75; 2.75/6 = 0.46; 0.45\}$

$l_{cr} = 0.75$ m

Consider the position of maximum moment at Level 3. In frame structures or frame-equivalent dual structures, it is necessary to design for a strong column/weak beam mechanism and satisfy EC8 Part 1 Equation 4.29:

$\Sigma M_{Rc} > 1.3 \Sigma M_{Rb}$ (EC8 Part 1 Equation 4.29)

However, since the walls carry greater than 50 per cent of the base shear and the structure is therefore classified as a wall-equivalent dual system, this requirement is waived. Thus, the designer may design the columns for the moments output from the analysis. Even though it is implicit within the code that soft-storey mechanisms are prevented by the presence of sufficient stiff walls in wall-equivalent dual systems, their inelastic behaviour

is more uncertain than pure wall or frame systems, as noted by Fardis et al. (2005). To cater for this, the designer may decide to reduce the probability of extensive plasticity in the columns by continuing to relate the column moments to the capacities of the beams framing into them. In this case, the beam capacities need not be increased by the 1.3 factor of Equation 4.29. The output from the analysis shows a maximum value of 1389 kNm and a value of 1465 kNm is derived from the beam capacities. These values are similar and the calculations proceed using the higher value derived from the beam capacities.

Assume 45 per cent/55 per cent split between the column sections above and below the joint.

DESIGN LOWER SECTION

$\Sigma M_{Rb} = (1435 + 1229) = 2664$ kNm

$M_{Rc1} = 0.55 * 2664 = 1465$ kNm

Axial load from analysis:

$N_{static} = 2527$ kN $\quad N_{seismic} = \pm 1638$ kN

Maximum compression: $N = 2527 + 1638 = 4165$ kN

Minimum compression: $N = 2527 - 1638 = 889$ kN

CHECK NORMALISED AXIAL COMPRESSION

$\nu_d < 0.65$ for DCM [Clause 5.4.3.2.1(3)]

$\nu_d = 4165 \times 10^3 / 750 * 750 * (30/1.5) = 0.37$ OK

CHECK COLUMN RESISTANCES

Design resistances to EC2 [Clause 5.4.3.2.1(1)]

From the Concrete Centre 'How To' Sheet 5 – Columns

Using Design Chart for C30/37 concrete and $d_2/h = 0.1$

Assume 32 mm diameter main steel; $d_2 = 45 + 32/2 = 61$ mm

$d_2/h = 61/750 = 0.08$ Chart for $d_2/h = 0.1$ is most appropriate.

Maximum compression: $N/(b*h*f_{ck}) = 4165 * 10^3/750*750*30 = 0.25$

Minimum compression: $N/(b*h*f_{ck}) = 889 * 10^3/750*750*30 = 0.05$

Flexure: $M/(b*h^2*f_{ck}) = 1465 * 10^6/(750^3 * 30) = 0.12$

Maximum compression: $A_s{*}f_{yk} / b{*}h{*}f_{ck} = 0.2$

Minimum compression: $A_s{*}f_{yk} / b{*}h{*}f_{ck} = 0.3$

$A_s / b{*}h = 0.3 * 30 / 500 = 0.018$ (1.8 per cent – within prescribed limits)

$A_s = 0.018 * 750 * 750 = 10125$ mm$^2$

Use 16 Φ32 – (5Φ32 in EF + 3 in each side) – [12864 mm$^2$]

CHECK CAPACITY FOR MAXIMUM COMPRESSION

$$\frac{A_s f_{yk}}{bhf_{ck}} = \frac{12864 * 434.8}{750 * 750 * 30} = 0.33$$

For $N_{max} = 4165$kN, $M/bh^2f_{ck} = 0.18$

$M_{cap} = 0.18{*}750^3{*}30{*}10^{-6} = 2278$ kNm

CHECK SHEAR – APPROACH AS FOR BEAMS BUT WITHOUT LATERAL LOAD BETWEEN SUPPORTS

For a conservative design, the column shear could be based upon the flexural capacity at maximum compression calculated above. However, EC8 Equation 5.9 allows the column flexural capacities to be multiplied by the ratio $\Sigma M_{R,b}/\Sigma M_{R,c}$ on the basis that yielding may develop initially in the beams and hence does not allow the development of the column overstrength moments.

$V_{E,d} = \gamma_{Rd} * (\Sigma M_{R,b}/\Sigma M_{R,c}){*}(M_{c,\text{top}} + M_{c,\text{bottom}}) / l_{cl}$

For DCM columns, $\gamma_{Rd} = 1.1$

$l_{cl} = 3.5 - 0.75 = 2.75$ m

$$V_{E,d} = 1.1 * \frac{2664}{2 * 2278} * 2 * 2278 * \frac{1}{2.75} = 1066kN$$

$d = 750 - 45 - 32/2 = 689$

$v_{E,d} = 1066 \text{ x } 10^3 / 689{*}750 = 2.06$ N/mm$^2$ < 3.64 N/mm$^2$

As previously, Cotθ = 2.5 and $f_{ywd} = f_{yk}/1.15$

$A_{sw}/s = V_{E,d} / (z * f_{ywd} * \text{Cot}\theta)$

$z$ can be taken as $0.9d$ for a steel couple

$A_{sw}/s = 1066 * 10^3 / (0.9 * 689 * 434.8 * 2.5) = 1.58$ mm$^2$/mm

Although the structural analysis shows that the flexural demand is lower at the lower levels, check normalised axial compression at the position of maximum axial load (on GL C and D at the base).

$N_{static}$ = 3420 kN

$N_{seismic}$ = ±2558 kN

$N_{max}$ = 5978 kN

$\nu_d$ = 5978*10$^3$/750*750*20 = 0.53<0.65

Therefore, the normalised axial compression is satisfactory.

DETAILING

For the critical regions of DCM columns:

$s_{max}$ = min{ $b_0$/2; 175; 8$d_{bL}$} (EC8 Part 1 Equation 5.18)

For columns, take 45 mm cover to the main reinforcement

$b_0 = h_0$ = 750 – 2 * 45 +10 = 670 mm (centre to centre of link)

$b_0$/2 = 335 mm

8*$d_{bL}$ = 8*32 = 256 mm

$s_{max}$ = 175 mm

$A_{sw}$ = 1.58*175 = 277 mm$^2$ (5 legs of Φ10 – 392 mm$^2$)

Provide five legs of Φ10 hoops/ties at 175 mm spacing within the critical region, 750 mm from the underside of the beam as shown in Figure 5.28.

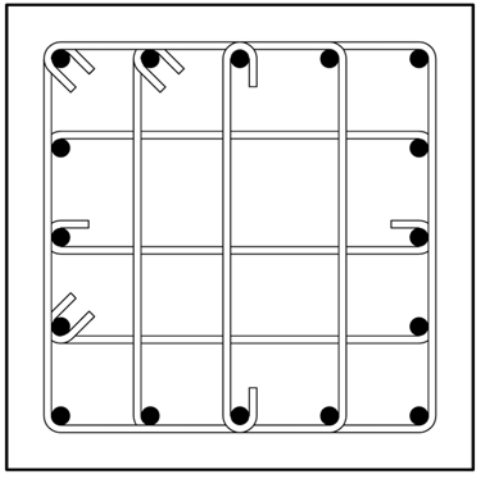


*Figure 5.28* Arrangement of column reinforcement

Distance between restrained main bars = (750 – 2*45 – 32)/4
= 157 mm < 200mm OK [EC8 Part 1 Clause 5.4.3.2.2(11)]

*5.9.6.7 Confinement reinforcement*

For DCM, confinement reinforcement within a beam/column joint and in the critical regions at the base of a column must meet the provisions of Clauses 5.4.3.2.2 (8) to (11).

Clauses 5.4.3.2.2 (10) and (11) are satisfied by the detailing requirements outlined above for all critical regions of the column.

Therefore, the additional requirements of Clauses 5.4.3.2.2 (8) and (9) need to be checked.

$\alpha * \omega_{wd} \geq 30\mu_{\Phi}\nu_d\varepsilon_{sy,d} * (b_c/b_0) - 0.035$ [Clause 5.4.3.2.2(8)]

AND $\omega_{wd} \geq 0.08$ [Clause 5.4.3.2.2(9)]

$\omega_{wd} = [(A_{svx}/b_0 * s) + (A_{svy}/h_0 * s)] * (f_{yd}/f_{cd})$ [Clause 5.4.3.2.2(8)]

$\alpha = \alpha_n * \alpha_s$

$\alpha_n = 1 - \sum(b_i)^2 / 6b_0h_0$ (EC8 Part 1 Equation 5.16a)

$\alpha_s = (1 - s/2b_0) * (1 - s/2h_0)$ (EC8 Part 1 Equation 5.17a)

$b_0 = h_0 = 670$ mm

Since the normalised axial compression is greatest at the base of the column, consider the detailing of this region to check the feasibility of the design.

All main column bars are equally spaced:

$b_i = (660 - 32)/4 = 157$ mm

$\alpha_n = 1 - (16*157^2)/6*670*670 = 0.85$

$\alpha_s = (1 - s/1340)*(1–s/1340)$

For $s = 100$ mm, $\alpha_s = 0.85$

For $s = 125$ mm, $\alpha_s = 0.82$

For $s = 150$ mm, $\alpha_s = 0.78$

As before:

$\mu_\varphi = 2* q - 1 = 2*3.6 - 1 = 6.2$

$\varepsilon_{syd} = 434.8/200E3 = 0.0022$

$\nu_d = 0.53$

$b_c/b_0 = 750/670 = 1.12$

$30\mu_\Phi \nu_d \varepsilon_{sy,d} * (b_c/b_0) - 0.035 = 30*6.2*0.53*0.0022*1.12 - 0.035 = 0.208$

Try hoops/ties at 100 mm spacing: $\alpha = 0.85*0.85 = 0.72$

$\omega_{wd} = 0.208/0.72 = 0.29 >$ minimum of 0.08

[EC8 Part 1 Clause 5.4.3.2.2(9)]

$\omega_{wd} = [(A_{svx}/b_0*s) + (A_{svy}/h_0*s)]*(f_{yd}/f_{cd})$

$\omega_{wd} = [2*(392/670*100)]*434.8/20 = 0.25 < 0.29$ Not sufficient

Consider 12 mm diameter hoops and ties $A_s = 565$ mm$^2$

$\omega_{wd} = [2*(565/670*100)]*434.8/20 = 0.37 > 0.29$ OK

For beam-column joints, the density of confinement reinforcement may be reduced up the height of the building as the normalised axial compression reduces.

Also, the internal 600 mm square columns in the upper four storeys have beams of three quarters of the column width that frame into them on all four sides. In these cases, the calculated confinement spacing may be doubled but may not exceed a limit of 150 mm [EC8 Clause 5.4.3.3(2)].

*5.9.6.8 Damage limitation case*

From Table 5.2, it can be seen that the maximum value of storey drift in the damage limitation event is $d_r \times \upsilon = 41.4mm$.

This is above the maximum inter-storey drift for buildings having non-structural elements fixed in a way so as not to interfere with structural deformations, which is 0.01h = 0.01 x 3500 = 35 mm (h is storey height).

However, as noted earlier, the lateral loads on the structure were initially calculated based on a standard formula that is applicable to a wide range of structures and, by necessity, this is quite conservative in its calculation of the period of response. Although it is wall-equivalent, the dual structure chosen is relatively flexible compared to typical shear wall structures and therefore might be expected to attract lower inertial loads. Therefore, a more realistic approach was adopted calculating the period using modal analysis with the stiffness of the structure based on $0.5*E_c*I_g$ as per the deflection calculation.

The modal analysis gives a fundamental period of 1.2 seconds with 67 per cent mass participating (compared to 0.62 seconds using the generic formula) together with significant secondary modes at periods of 0.32 seconds (18 per cent mass participating) and 0.14 seconds (9 per cent mass participating).

The spectral acceleration associated with the fundamental mode is only 1.35 ms$^{-2}$ rather than 2.32 ms$^{-2}$ previously obtained. Also, despite the higher spectral accelerations of the higher modes, their low mass participation

means that the effective acceleration consistent with the SRSS combination of the individual modal inertial loads is lower than taking the fundamental mode acceleration with 100 per cent mass participation in this case.

Hence, inertial loads would be less than 60 per cent of those used in the initial analysis.

This gives a maximum storey drift of 0.6*41.4 = 24.8 mm, well within the EC8 limit.

It can therefore be seen that the structure possesses adequate stiffness and the feasibility of the design is confirmed. The final design should proceed on the basis of these lower inertial loads, resulting in reduced quantities of reinforcement but the member sizes should remain unaltered to meet damage limitation requirements.

## References

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# 6 Design of steel structures

*A.Y. Elghazouli and J.M. Castro*

## 6.1 Introduction

In line with current seismic design practice, steel structures may be designed to EC8 according to either non-dissipative or dissipative behaviour. The former, through which the structure is dimensioned to respond largely in the elastic range, is normally limited to areas of low seismicity or to structures of special use and importance; it may also be feasible if vibration reduction devices are incorporated. Otherwise, codes aim to achieve economical design by employing dissipative behaviour in which considerable inelastic deformations can be accommodated under significant seismic events. In the case of irregular or complex structures, detailed non-linear dynamic analysis may be necessary. However, dissipative design of regular structures is usually performed by assigning a structural behaviour factor (i.e. force reduction or modification factor) that is used to reduce the code-specified forces resulting from idealised elastic response spectra. This is carried out in conjunction with the capacity design concept, which requires an appropriate determination of the capacity of the structure based on a predefined plastic mechanism, often referred to as failure mode, coupled with the provision of sufficient ductility in plastic zones and adequate overstrength factors for other regions.

This chapter focuses on the dissipative seismic design of steel frame structures according to the provisions of EN 1998-1 (2004), particularly Section 6 (Specific Rules for Steel Buildings). After giving an outline of common configurations and the associated behaviour factors, the seismic performance of the three main types of steel frame is discussed. Brief notes on material requirements and control of design and construction are also included. The chapter concludes with illustrative examples for the use of EC8 in the preliminary design of lateral resisting frames for the eight-storey building dealt with in previous chapters of this book.

## 6.2 Structural types and behaviour factors

There are essentially three main structural steel frame systems used to resist horizontal seismic actions, namely moment resisting, concentrically braced

and eccentrically braced frames. Other systems such as hybrid and dual configurations can be used and are referred to in EC8, but are not dealt with in detail herein. It should also be noted that other configurations such as those incorporating buckling restrained braces or special plate shear walls, which are covered in the most recent North American Provisions (AISC, 2005), are not directly addressed in the current version of EC8.

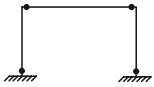
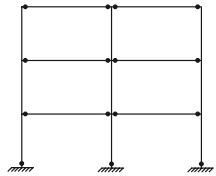
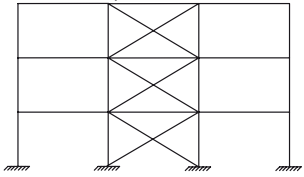
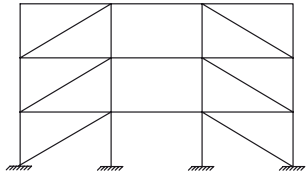
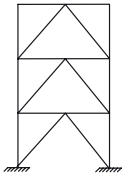
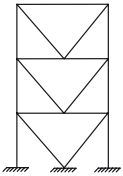
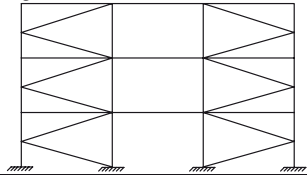
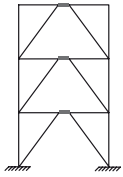
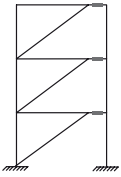
As noted before, unless the complexity or importance of a structure dictates the use of non-linear dynamic analysis, regular structures are designed using the procedures of capacity design and specified behaviour factors. These factors (also referred to as force reduction factors) are recommended by codes of practice based on background research involving extensive analytical and experimental investigations. Before discussing the behaviour of each type of frame, it is useful to start by indicating the structural classification and reference behaviour factors ($q$) stipulated in EC8 as this provides a general idea about the ductility and energy dissipation capability of various configurations. Table 6.1 shows the main structural types together with the associated dissipative zones according to the provisions and classification of EC8 (described in Section 6.3 of EN 1998-1). The upper values of $q$ allowed for each system, provided that regularity criteria are met, are also shown in Table 6.1. The ability of the structure to dissipate energy is quantified by the behaviour factor; the higher the behaviour factor, the higher is the expected energy dissipation as well as the ductility demand on critical zones.

The multiplier $\alpha_u/\alpha_1$ depends on the failure/first plasticity resistance ratio of the structure. A reasonable estimate of this value may be determined from conventional non-linear 'pushover' analysis, but should not exceed 1.6. In the absence of detailed calculations, the approximate values of this multiplier given in Table 6.1 may be used. If the building is irregular in elevation, the listed values should be reduced by 20 per cent.

The values of the structural behaviour factor given in the code should be considered as an upper bound even if in some cases non-linear dynamic analysis indicates higher $q$ factors. For regular structures in areas of low seismicity having standard structural systems with sections of standard sizes, a behaviour factor of 1.5–2.0 may be adopted (except for K-bracing) by satisfying only the resistance requirements of EN 1993-1 (2005, EC3).

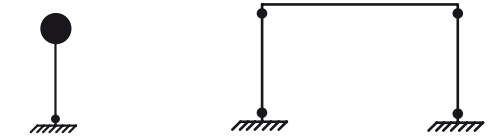
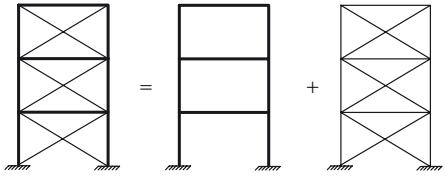
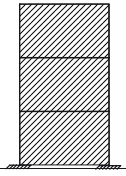
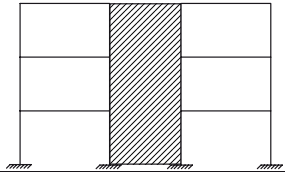
Although a direct code comparison between codes can only be reliable if it involves the full design procedure, the reference $q$ factors in EC8 appear to be generally lower than $R$ values in US provisions (ASCE/SEI, 2005) for similar frame configurations. It is also important to note that the same force-based behaviour factors ($q$) are proposed as displacement amplification factors ($q_d$). This is not the case in US provisions where specific seismic drift amplification factors ($C_d$) are suggested; these values are generally lower than the corresponding R factors for all frame types.

*Table 6.1* Structural types and behaviour factors

| Structural Type | q-factor | |
|---|---|---|
| | DCM | DCH |
| Moment-resisting frames<br>  <br>$\alpha_u/\alpha_1=1.1$<br>  <br>$\alpha_u/\alpha_1=1.2$ (1 bay)<br>$\alpha_u/\alpha_1=1.3$ (multi-bay)<br>dissipative zones in beams and column bases | 4 | $5\alpha_u/\alpha_1$ |
| Concentrically braced frames<br>  <br>  <br>dissipative zones in tension diagonals | 4 | 4 |
| V-braced frames<br>  <br>  <br>dissipative zones in tension and compression diagonals | 2 | 2.5 |
| Frames with K-bracings<br>  | Not allowed in dissipative design | |
| Eccentrically braced frames<br>  <br>  <br>$\alpha_u/\alpha_1=1.2$<br>dissipative zones in bending or shear links | 4 | $5\alpha_u/\alpha_1$ |

Continued…

Table 6.1 continued

| Structural Type | | q-factor DCM | q-factor DCH |
|---|---|---|---|
| Inverted pendulum structures<br>  <br>$\alpha_u/\alpha_1=1.0$ $\alpha_u/\alpha_1=1.1$<br>dissipative zones in column base, or column ends ($N_{Ed}/N_{pl,Rd}<0.3$) | | 2 | $2\alpha_u/\alpha_1$ |
| Moment-resisting frames with concentric bracing<br>  <br>$\alpha_u/\alpha_1=1.2$<br>dissipative zones in moment frame and tension diagonals | | 4 | $4\alpha_u/\alpha_1$ |
| Moment frames with infills<br>  | Unconnected concrete or masonry infills, in contact with the frame | 2 | 2 |
| | Connected reinforced concrete infills | See concrete rules | |
| | Infills isolated from moment frame | 4 | $5\alpha_u/\alpha_1$ |
| Structures with concrete cores or walls<br>  | | See concrete rules | |

## 6.3 Ductility classes and rules for cross sections

To achieve some consistency with other parts of the code, the most recent version of EC8 explicitly addresses three ductility classes namely DCL, DCM and DCH referring to low, medium and high dissipative structural behaviour, respectively. For DCL, global elastic analysis and the resistance of the members and connections may be evaluated according to EC3 without any additional requirements. The recommended reference *q* factor for DCL is 1.5–2.0. For buildings that are not seismically isolated or incorporating effective dissipation devices, design to DCL is only recommended for low

seismicity situations. In contrast, structures in DCM and DCH need to satisfy specific requirements primarily related to ensuring sufficient ductility in the main dissipative zones. Some of these requirements are general rules that apply to most structural types whilst others are more relevant to specific configurations.

The application of a behaviour factor larger than 1.5–2.0 must be coupled with sufficient local ductility within the critical dissipative zones. For elements in compression or bending (under any seismic loading scenario), this requirement is ensured in EC8 by restricting the width-to-thickness (b/t) ratios to avoid local buckling. An increase of b/t ratio results in lower element ductility due to the occurrence of local buckling (as illustrated in Figure 6.1) leading to a reduction in the energy dissipation capacity, which is expressed by a lower $q$ factor. The classification used in EC3 is adopted but with restrictions related to the value of $q$ factor as given in Table 6.2 (Section 6.5.3 and Table 6.3 of EN 1998-1). It is worth noting that the seismic cross-section requirements in US practice imply more strict limits for certain section types.

The cross-section requirements apply to all types of frame considered in EC8. These provisions implicitly account for the relationship between local buckling and rotational ductility of steel members that has been extensively

*Table 6.2* Cross-section requirements based on ductility class and reference $q$-factor

| *Ductility Class* | *Reference q-factor* | *Cross-Section Class* |
|---|---|---|
| DCM | $1.5 < q \leq 2$ | Class 1, 2 or 3 |
| | $2.0 < q \leq 4$ | Class 1 or 2 |
| DCH | $q > 4$ | Class 1 |

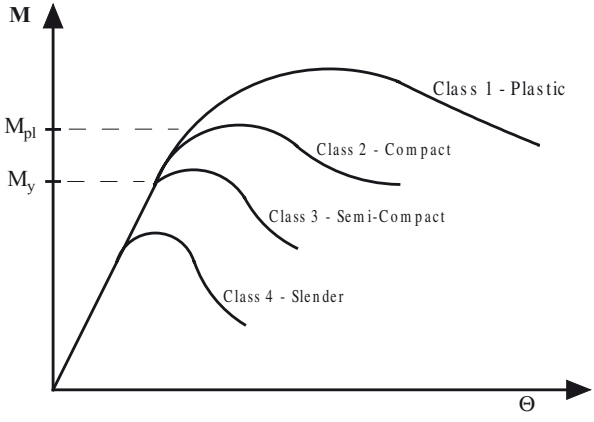



*Figure 6.1* Moment-rotation characteristics for different cross section classes

investigated by several researchers (e.g. Lay and Galambos, 1967; Kato and Akiyama, 1982; Kato, 1989).

In subsequent sections, the behaviour of the three main configurations of steel frame structure, namely moment resisting, concentrically and eccentrically braced frames, is discussed. Whereas moment-frames exhibit relatively ductile behaviour under earthquake loading, their low lateral stiffness may, in some situations, result in high storey drifts, thus leading to unacceptable damage to non-structural components and possible stability problems. On the other hand, concentrically braced frames may provide relatively higher stiffness, but can often suffer from reduced ductility once the compression braces buckle. Eccentrically braced frames have the potential of providing adequate ductility as well as stiffness, provided that the shear or bending links are carefully designed and detailed to withstand the substantial inelastic demands that are imposed on these dissipative zones.

## 6.4 Moment resisting frames

### *6.4.1 Frame characteristics*

Moment resisting frames (MRFs) are designed such that plastic hinges occur predominantly in beams rather than in columns (weak beam/strong column design) as shown in Figure 6.2. This provides favourable performance, compared to strong beam/weak column behaviour through which significant deformation and second order effects may arise in addition to the likelihood of premature storey collapse mechanisms. The only exception to this requirement is at the base of the ground floor columns, where plastic hinges may form.

Due to the spread of plasticity through flexural plastic hinges, MRFs usually possess high ductility as reflected in the high reference $q$ assigned in

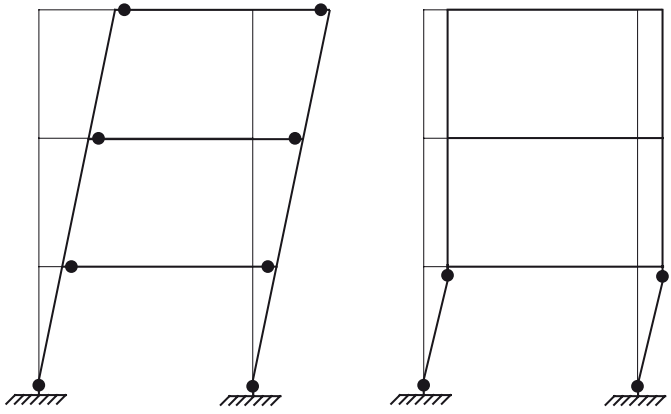


*Figure 6.2* Weak beam/strong column and weak column/strong beam behaviour in moment-resisting frames

EC8. Nevertheless, due to their inherent low stiffness, lateral deformation effects need careful consideration.

### *6.4.2 Capacity design requirements*

In EC8, the 'weak beam/strong column' concept is typically required, with plastic hinges allowed at the base of the frame, at the top floor of multi-storey frames and for single-storey frames. The most recent version of EC8 also allows dissipative zones to be located in the connections provided adequate behaviour can be demonstrated. Rules for moment-resisting frames are described mainly in Section 6.6 of EN 1998-1.

To obtain ductile plastic hinges in the beams, checks are made that the full plastic moment resistance and rotation is not reduced by coexisting compression and shear forces. To satisfy this for each critical section, the applied moment ($M_{Ed}$) should not exceed the design plastic moment capacity ($M_{pl,Rd}$) (i.e. $M_{Ed}/M_{pl,Rd} \leq 1.0$), the applied axial force ($N_{Ed}$) should not exceed 15 per cent of the plastic axial capacity ($N_{pl,Rd}$) (i.e. $N_{Ed}/N_{pl,Rd} \leq 0.15$). Also, the shear force ($V_{Ed}$) due to the application of the plastic moments with opposite signs at the extremities of the beam should not exceed 50 per cent of the design plastic shear resistance ($V_{pl,Rd}$) of the section (i.e. $V_{Ed}/V_{pl,Rd} \leq 0.5$, in which $V_{Ed} = V_{Ed,G} + V_{Ed,M}$), where $V_{Ed,G}$ and $V_{Ed,M}$ are the shear forces due to the gravity and moment components on the beam, respectively.

According to Section 6.6.3 of EN 1998-1, columns should be verified for the most unfavourable combination of bending moments $M_{Ed}$ and axial forces $N_{Ed}$, based on:

$$M_{Ed} = M_{Ed,G} + 1.1\gamma_{ov}\Omega M_{Ed,E} \tag{6.1}$$

$$N_{Ed} = N_{Ed,G} + 1.1\gamma_{ov}\Omega N_{Ed,E} \tag{6.2}$$

where $\Omega$ is the minimum overstrength in the connected beams ($\Omega_i = M_{pl,Rd} / M_{Ed,i}$). The parameters $M_{Ed,G}$ and $M_{Ed,E}$ are the bending moments in the seismic design situation due to the gravity loads and lateral earthquake forces, respectively, as shown in Figure 6.3 (Elghazouli, 2007); the same subscripts also apply for axial and shear actions. Additionally, the most unfavourable

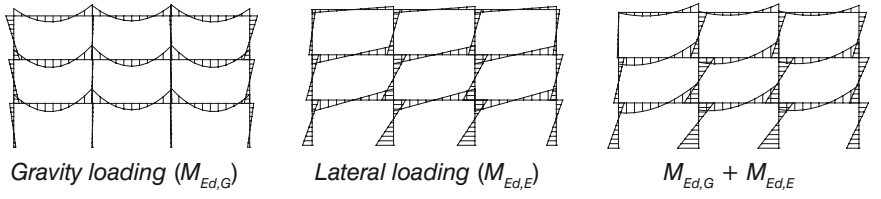



*Figure 6.3* Moments due to gravity and lateral loading components in the seismic situation

shear force ($V_{Ed}$) of the column due to seismic combination actions must be less than 50 per cent of the ultimate shear resistance of the section.

The beam overstrength parameter ($\Omega = M_{pl,Rd}/M_{Ed}$) as adopted in EC8 involves an approximation as it does not account accurately for the influence of gravity loads on the behaviour (Elghazouli, 2007). This issue becomes particularly pronounced in gravity-dominated frames (i.e. with large beam spans) or in low-rise configurations (since the initial column sizes are relatively small), in which the beam overstrength may be significantly underestimated. The extent of the problem depends on the interpretation of the code and whether $\Omega$ is used in isolation or in combination with an additional capacity design criterion based on a limiting ratio of 1.3 on the column-to-beam capacity (i.e. Equation 4.4 of Chapter 4). It is also important to note that whilst codes aim to achieve a 'weak beam/strong column' behaviour, some column hinging is often unavoidable. In the inelastic range, points of contra-flexure in members change and consequently the distribution of moments vary considerably from idealised elastic conditions assumed in design. The benefit of meeting code requirements is to obtain relatively strong columns such that beam rather than column yielding dominates over several storeys, hence achieving adequate overall performance.

### *6.4.3 Stability and drift considerations*

Deformation-related criteria are stipulated for all building types in EC8 but, as expected, they are particularly important in steel moment frames due to their inherent flexibility, which often governs the design. Two deformation-related requirements, namely 'second-order effects' and 'inter-storey drifts', are stipulated in Sections (4.4.2.2) and (4.4.3.2) of EN 1998-1. The former is associated with ultimate state whilst the latter is included as a damage-limitation (serviceability) condition.

Second-order ($P$-$\Delta$) effects are specified through an inter-storey drift sensitivity coefficient ($\theta$) given as:

$$\theta = \frac{P_{tot} d_r}{V_{tot} h} \tag{6.3}$$

where $P_{tot}$ and $V_{tot}$ are the total cumulative gravity load and seismic shear, respectively, at the storey under consideration; $h$ is the storey height and $d_r$ is the design inter-storey drift (product of elastic inter-storey drift from analysis and $q$, i.e. $d_e \times q$). Instability is assumed beyond $\theta = 0.3$ and is hence considered as an upper limit. If $\theta \leq 0.1$, second-order effects could be ignored, whilst for $0.1 < \theta \leq 0.2$, $P$-$\Delta$ may be approximately accounted for in seismic action effects through the multiplier $1/(1-\theta)$.

For serviceability, '$d_r$' is limited in proportion to '$h$' such that:

$$d_r \nu \leq \psi h \tag{6.4}$$

where $\psi$ is suggested as 0.5 per cent, 0.75 per cent and 1.0 per cent for brittle, ductile or non-interfering non-structural components, respectively; $\nu$ is a reduction factor that accounts for the smaller more frequent earthquakes associated with serviceability, recommended as 0.4–0.5 depending on the importance class.

Assessment of other codes, including US provisions, suggests that drift-related requirements in EC8 can be relatively more stringent, depending on the limit selected and the importance category under consideration. As a result, direct application of the specific rules for moment frames in EC8, followed by inter-storey drift and second-order stability checks, often results in an overall lateral capacity that is notably different from that assumed in design (Elghazouli, 2007; Sanchez-Ricart and Plumier, 2008). Significant levels of lateral frame overstrength can be present particularly when large $q$ factors are used and/or when the spectral design accelerations are not high. This overstrength is also a function of spectral acceleration and gravity design. Whereas the presence of overstrength reduces the ductility demand in dissipative zones, it also affects forces imposed on frame and foundation elements. A rational application of capacity design necessitates a realistic assessment of lateral capacity after the satisfaction of all provisions, followed by a re-evaluation of global overstrength and the required '$q$'. Although high '$q$' factors are allowed for moment frames, in recognition of their ductility and energy dissipation capabilities, such a choice may in some cases be unnecessary and undesirable (Elghazouli, 2009).

### *6.4.4 Beam-to-column connections*

Steel moment frames have traditionally been designed with rigid full-strength connections, usually of fully welded or hybrid welded/bolted configuration. Typical design provisions ensured that connections are provided with sufficient overstrength such that dissipative zones occur mainly in the beams. However, the reliability of these commonly used forms of full-strength beam-to-column connection has come under question following poor performance in large events in the mid 1990s, particularly in the Northridge earthquake of 1994 (Bertero *et al*, 1994) and the Hyogo-ken Nanbu (Kobe) earthquake of 1995 (EERI, 1995), as illustrated in Figure 6.4. The extent and repetitive nature of damage observed in several types of welded and hybrid connections have directed considerable research effort not only to repair methods for existing structures but also to alternative connection configurations to be incorporated in new designs.

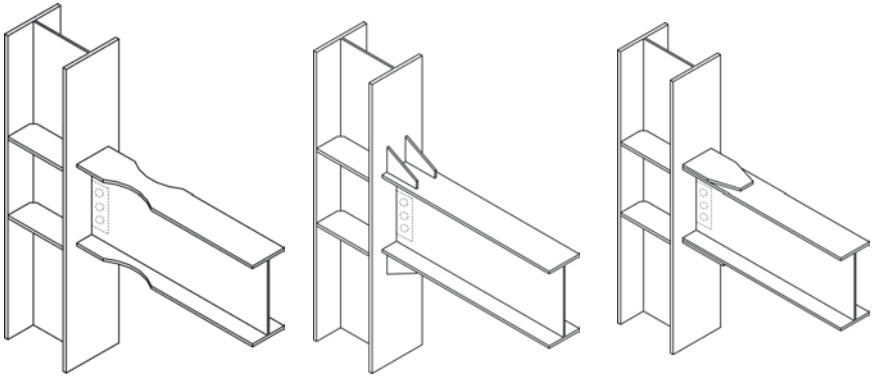
The above-mentioned problems prompted the industry in the US, in liaison with government agencies, professional institutions and academic establishments to create a joint venture (SAC) to respond to the questions raised by the extensive damage observed. Laboratory tests confirmed that connections designed and manufactured strictly to code requirements and conventional shop practice failed to provide the necessary levels of ductility



(a) *Weld fracture at bottom flange* (b) *Fracture extending into structural section*

*Figure 6.4* Examples of typical damage in connections of moment frames

(SAC, 1995, 1996a, 1996b). Observed damage was attributed to several factors including defects associated with weld and steel materials, welding procedures, stress concentration, high rotational demands and scale effects, as well as the possible influence of strain levels and rates. In addition to the concerted effort dedicated to improving seismic design regulations for new construction, several proposals have been forwarded for the upgrading of existing connections (FEMA, 1995, 1997, 2000; PEER, 2000). As shown schematically in Figure 6.5, this may be carried out by strengthening of the connection through haunches, cover or side plates, or other means. Alternatively, it can be achieved by weakening of the beam by trimming the flanges (i.e. reduced beam section 'RBS' or 'dog-bone' connections), perforating the flanges, or by reducing stress concentrations through slots in beam webs, enlarged access holes, etc. In general, the design can be based on either prequalified connections or on prototype tests. Prequalified



(a) *Reduced beam section (RBS or dog-bone)* (b) *Connection with haunches* (c) *Connection with cover plates*

*Figure 6.5* Schematic examples of modified connection configurations for moment frames

connections have been proposed in the US (AISC/ANSI, 2005), and a similar European activity is underway.

Another important aspect of connection behaviour is related to the influence of the column panel zone. This has direct implications on the ductility of dissipative zones as well as on the overall frame performance. Recent research studies (Castro *et al*, 2005; Castro *et al*, 2008), involved the development of realistic modelling approaches for panel zones within moment frames as well as assessment of current design procedures. One important issue is related to the treatment of the two yield points corresponding to the onset of plasticity in the column web and surrounding components, respectively. Another key design consideration is concerned with balancing the extent of plasticity between the panel zone and the connected beams, an issue that can be significantly affected by the level of gravity load applied on the beams. On the one hand, allowing a degree of yielding in the panel reduces the plastic hinge rotations in the beams yet, on the other hand, relatively weak panel zone designs can result in excessive distortional demands that can cause unreliable behaviour of other connection components, particularly in the welds.

Section 6.6.3 of EN 1998-1 requires the web panel to be designed to ensure adequate shear and buckling resistance. The design shear assuming plasticity in the beams ($V_{wp,Ed}$) should not exceed the web panel plastic shear resistance ($V_{wp,Rd}$) (i.e. $V_{wp,Ed} \leq V_{wp,Rd}$). The design shear should also not exceed the buckling resistance of the web panel ($V_{wb,Rd}$) (i.e. $V_{wp,Ed} \leq V_{wb,Rd}$). If strengthening is required to the web panel, additional plates can be welded to the column panel zone.

Many of the drawbacks of fully rigid welded frames can be alleviated by bolted forms. To this end, the feasibility of using partial-strength bolted connections, which are usually semi-rigid as well, for seismic resistance has been the subject of a number of investigations (e.g. Nader and Astaneh, 1992; Elnashai and Elghazouli, 1994; Astaneh, 1995; Elghazouli, 1996; Mazzolani and Piluso, 1996; Elghazouli, 1999; Faella *et al*, 2000). Despite the economic advantages in fabrication and construction, this type of connection has not been traditionally employed for earthquake resistance due to two main reasons. The first is related to the semi-rigidity, which may lead to excessive deformation under static loads. It was shown in several investigations, however, that due to the relatively longer natural period of semi-rigid frames, the deflections may often not be higher under dynamic loads as compared to rigid frames. The second reason is that insufficient information has been available on the hysteretic behaviour and local ductility of partial strength connections. In general, semi-rigid, partial-strength connections can be a viable alternative particularly in moderate seismicity areas.

Revisions have also been introduced in the current version of EC8 to reflect recent research findings. If the structure is designed to dissipate energy in the beams, connections should be designed for the required degree

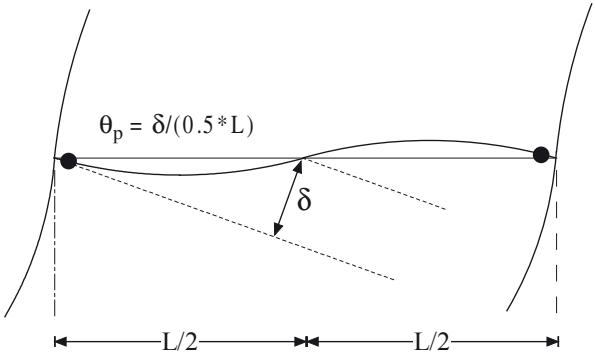



*Figure 6.6* Estimation of plastic hinge rotation θ*p*

of overstrength taking into account the plastic moment resistance of the attached beams. On the other hand, semi-rigid, partial-strength connections are now permitted provided several conditions are satisfied (according to Section 6.6.4 of the code) including: (i) all connections have rotation capacity consistent with global deformations, (ii) members framing into connections are stable at the ultimate limit state, and (iii) connection deformation is accounted for through non-linear analysis.

For all connections, whether full or partial strength, design to EC8 should ensure sufficient plastic rotation ($\theta_p$) of the plastic hinge region, such that $\theta_p \geq 35$ mrad for DCH and $\theta_p \geq 25$ mrad for DCM (with $q > 2$). The plastic rotation $\theta_p$ can be determined as $\delta/0.5L$, where $\delta$ is the beam deflection at mid-span and $L$ is the beam span, as illustrated in Figure 6.6. In tests, it should be ensured that $\theta_p$ is achieved under cyclic loading with less than 20 per cent degradation in stiffness and strength, and that the column web panel distortion does not contribute more than 30 per cent to $\theta_p$, noting that the column elastic deformation is not included in $\theta_p$. It is also important to note that if partial-strength connections are adopted, column capacity design checks need only to be verified based on the plastic capacity of the connections rather than that of the beams.

## 6.5 Concentrically braced frames

### *6.5.1 Frame characteristics*

Because of their geometry, concentrically braced frames (CBFs) such as those shown in Figure 6.7, provide truss action with members subjected largely to axial forces in the elastic range. However, during a moderate to severe earthquake, the bracing members and their connections undergo significant inelastic deformations into the post-buckling range, which has led to reported cases of damage in previous earthquakes (EERI, 1995).

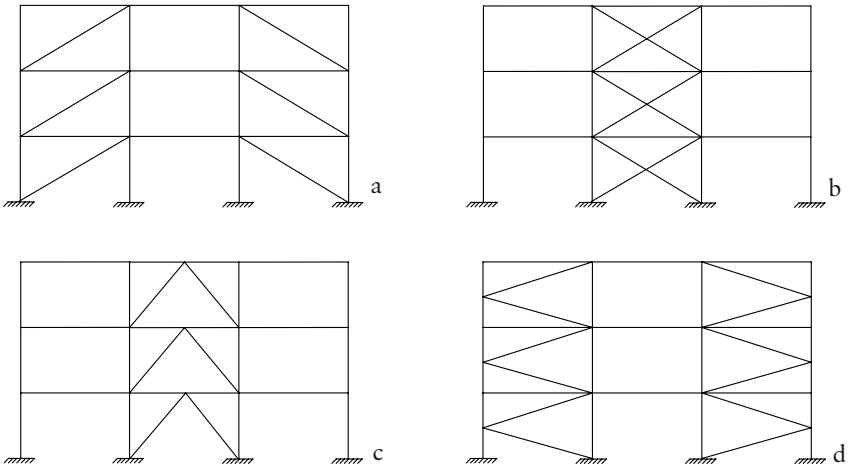



*Figure 6.7* Typical idealised configurations of concentrically braced frames

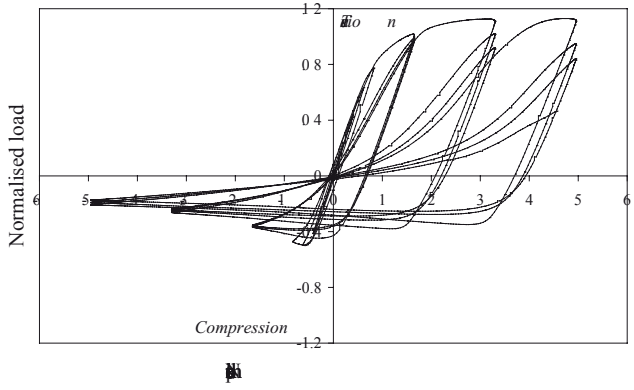



*Figure 6.8* Typical response of a bracing member under cyclic axial loading

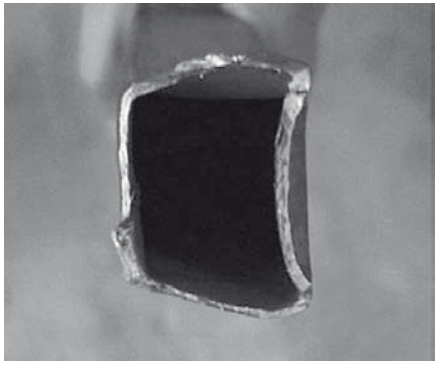
The response of concentrically braced frames is typically dominated by the behaviour of its bracing members. This behaviour has been investigated previously by several researchers (e.g. Maison and Popov, 1980; Popov and Black, 1981; Ikeda and Mahin, 1986; Goel and El-Tayem, 1986), focusing mainly on the response under idealised cyclic loading conditions. A recent collaborative European project (Elghazouli, 2003; Broderick *et al*, 2005; Elghazouli *et al*, 2005) also examined the performance of bracing members through analytical studies, which were supported by monotonic and cyclic quasi-static axial tests as well as dynamic shake table tests.

An example of the hysteretic axial load-deformation response of a bracing member is shown in Figure 6.8, in which the displacement and axial loads are normalised by the yield values. In compression, member buckling is followed

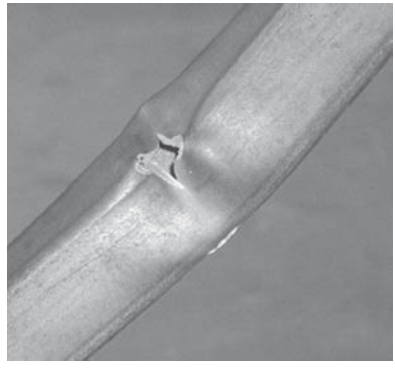
by lateral deflection and the formation of a plastic hinge at mid-length that leads to a gradual reduction in capacity. On reversing the load, elastic recovery occurs followed by loading in tension until yielding takes place. Subsequent loading in compression results in buckling at loads lower than the initial strength due to the residual deflections, the increase in length as well as the Bauschinger effect. Moreover, due to the accumulated permanent elongation, tensile yielding occurs at axial deformations that increase with each cycle of loading. Cyclic testing of diagonal bracing systems indicates that energy can be dissipated after the onset of global buckling if failures due to local buckling or at the connection are prevented.

Under the cyclic axial loading conditions applied on bracing members in seismic situations, failure can occur due to fracture of the cross section following local buckling, provided that bracing connections are adequately designed and detailed. This was clearly illustrated in recent shake-table tests on tubular bracing members (Elghazouli *et al*, 2005). High strains typically develop upon local buckling in the corner regions of the cross section. Cracks eventually form in these regions, as shown in Figure 6.9 (Elghazouli *et al*, 2005), and gradually propagate through the cross section under repeated cyclic loading.

The initiation of local buckling and fracture is influenced by the width-to-thickness ratio of the elements of the cross section, as well as the applied loading history. Seismic codes rely on the limits imposed on the width-to-thickness ratios of the cross section in order to delay or prevent local buckling and hence reduce the susceptibility to low cycle fatigue and fracture. There is also a dependence on the overall member slenderness of the brace (Elghazouli, 2003). Seismic codes also normally impose an upper limit on the member slenderness to limit sudden dynamic loading effects as well as the extent of post-buckling deformations.



(a) *Failure of tubular bracing member*



(b) *View of fracture during testing*

*Figure* 6.9 Fracture of tubular steel bracing member during shake table testing (Elghazouli *et al*, 2005).

### *6.5.2 Design requirements*

The provisions of EC8 for concentrically braced frames (provided mainly in Section 6.7 of EN 1998-1) typically consider that the horizontal seismic forces are mainly resisted by the axially loaded members. Design should allow yielding of the diagonals in tension before yielding or buckling of the beams or columns and before failure of the connections. Due to buckling of the compression braces, tension braces are considered to be the main ductile members, except in V and inverted-V configurations.

In diagonal bracings of the types shown in Figures 6.7 (a) and (b), the cyclic horizontal forces can be assumed in EC8 to be resisted by the corresponding tension members only, with the contribution of the compression bracing members neglected. To avoid significant asymmetric response effects, the value of $A\cos\alpha$ must not vary significantly between two opposite braces in the same storey, as shown in Figure 6.10 such that:

$$(A^{+}-A^{-})/(A^{+}+A^{-})\leq 0.05 \tag{6.5}$$

where $A$ is the area of the cross section of the tension diagonal and $\alpha$ is the slope of the diagonal to the horizontal.

In V-bracing, both tension and compression bracing members are needed to resist horizontal seismic forces effectively, hence both should be included in the elastic analysis of the frame. Also, the beams should be designed for gravity loading without considering the intermediate support of the diagonals, as well as account for the possibility of an unbalanced vertical action after brace buckling. In other frames, only the tension diagonals are considered. However, accounting for both braces is allowed in EC8 provided a non-linear static or time-history analysis is used, both pre-buckling and post-buckling situations are considered and background

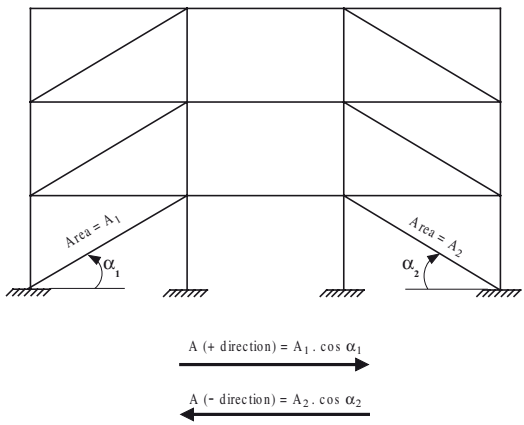



*Figure 6.10* Symmetry of lateral resistance in concentrically braced frames

studies justifying the models adopted are provided. It should be noted that ignoring the compression brace can have favourable or detrimental effects on the actual response, depending on the frame configuration and design situation (Elghazouli, 2003). On the other hand, K-bracing, such as that shown in Figure 6.7(d) where the diagonals meet at an intermediate point in the column, do not offer ductile behaviour due to the potential demand for a column yielding mechanism. Consequently, it is not appropriate for dissipative design and its use is not recommended in EC8.

In the design of the diagonal members, the non-dimensional slenderness $\bar{\lambda}$ used in EC3 plays an important role in the behaviour. This is discussed in detail elsewhere (Elghazouli, 2003). In earlier versions of EC8, an upper limit of 1.5 was proposed to prevent elastic buckling. However, further modifications have been made in the current version of EC8 and the upper limit has been revised to a value of 2.0, which results in a more efficient design. Moreover, no upper limit is needed for structures up to two storeys high. On the other hand, in frames with X-diagonal braces $\bar{\lambda}$ should be between 1.3 and 2.0. The lower limit is specified to avoid overloading columns in the pre-buckling stage of diagonals. Satisfying this lower limit can, however, result in some difficulties in practical design. It should also be noted that in frames with non-intersecting diagonal bracings (e.g. Figure 6.7a), the code stipulates that the design should account for forces that may develop in the columns due to loads from both the tension diagonals and pre-buckling forces in the compression diagonals.

All columns and beams should be capacity designed for the seismic combination actions. In summary, the following relationship applies for the capacity design of non-diagonal members, where the design resistance of the beam or column under consideration, $N_{Ed,}(M_{Ed})$, with due account of the interaction with the bending moment $M_{Ed,}$ is determined as:

$$N_{Ed}(M_{Ed}) \geq N_{Ed,G} + 1.1\, \gamma_{ov}\, \Omega\, N_{Ed,E} \tag{6.6}$$

where $N_{Ed,G}$ and $N_{Ed,E,}$ are the axial load due to gravity and lateral actions, respectively, in the seismic design situation, as illustrated in Figure 6.11; $\Omega$ is the minimum value of axial brace overstrength over all the diagonals of the frame and $\gamma_{ov}$ is the material overstrength. However, $\Omega$ of each diagonal

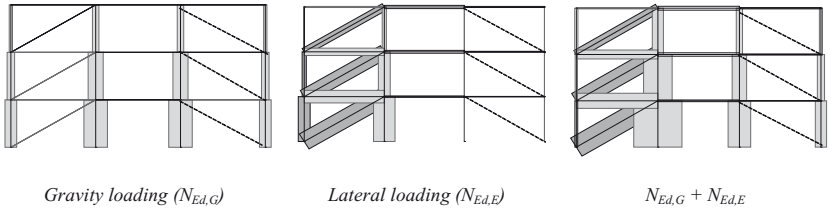



*Figure 6.11* Axial forces due to gravity and lateral loading in the seismic design situation

should not differ from the minimum value by more than 25 per cent in order to ensure reasonable distribution of ductility. It is worth noting that unlike in moment frames, gravity loading does not normally have an influence on the accuracy of Ω. It should also be noted that the 25 per cent limit can result in difficulties in practical design; it can be shown (Elghazouli, 2007) that this limit can be relaxed or even removed if measures related to column continuity and stiffness are incorporated in design.

US provisions (ASCE/SEI, 2005; AISC, 2005) differ from those in EC8 in terms of the R factors recommended as well as cross-section limits for some section types. However, the most significant difference is related to the treatment of the brace buckling in compression, which may lead to notably different seismic behaviour depending mainly on the slenderness of the braces. This is discussed in more detail elsewhere (Elghazouli, 2003), and has significant implications on the frame overstrength as well as on the applied forces and ductility demand imposed on various frame components.

### *6.5.3 Bracing connections*

Many of the failures reported in concentrically braced frames due to strong ground motion have been in the connections. In principle, bracing connections can be designed as rotationally restrained or unrestrained, provided that they can transfer the axial cyclic tension and compression effectively. The in- and out-of-plane behaviour of the connection, and their influence on the beam and column performance, should be carefully considered in all cases. For example, considering gusset plate connections (see Figure 6.12), satisfactory

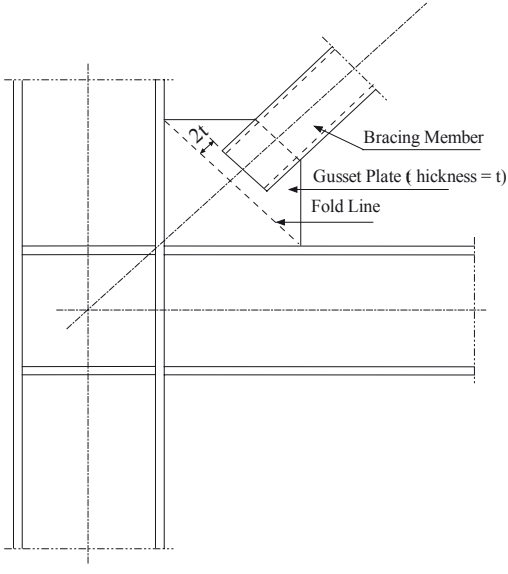



*Figure 6.12* Brace-to-gusset plate connection in concentrically braced frames

performance can be ensured by allowing the gusset plate to develop plastic rotations. This requires that the free length between the end of the brace and the assumed line of restraint (fold line) for the gusset can be sufficiently long to permit plastic rotations, yet short enough to preclude the occurrence of plate buckling prior to member buckling (Astaneh *et al*, 1986). Alternatively, connections with stiffness in two directions, such as crossed gusset plates, can be detailed.

As in the case of moment frames, the design of connections between bracing members and the beams/columns in concentrically braced frames is only dealt with in a conceptual manner in EC8. Accordingly, designers can adopt details available from existing literature, or based on prototype testing. The performance of bracing connections, such as those involving gusset plate components, has attracted significant research interest in recent years (e.g. Yoo *et al*, 2008; Lehman *et al*, 2008). Supplementary European guidance, through complementary manuals, on the design and detailing of recommended bracing connections for seismic resistance is also underway.

## 6.6 Eccentrically braced frames

### *6.6.1 Frame characteristics*

In this type of structural system, as shown in Figure 6.13, the bracing members intersect the girder at an eccentricity '*e*', and hence transmit forces by shear and bending. The length of the girder defined by *e* is termed a 'link beam', which may behave predominantly in either shear or bending. While retaining the advantages of CBFs in terms of drift control, eccentrically braced frames (EBFs) also represent an ideal configuration for failure mode control. Another important advantage is that by providing an eccentricity, a higher degree of flexibility in locating doors and windows in the structure is achieved. By careful design of the link beam, significant energy dissipation capacity can be obtained. Moreover, zones of excessive plastic deformations

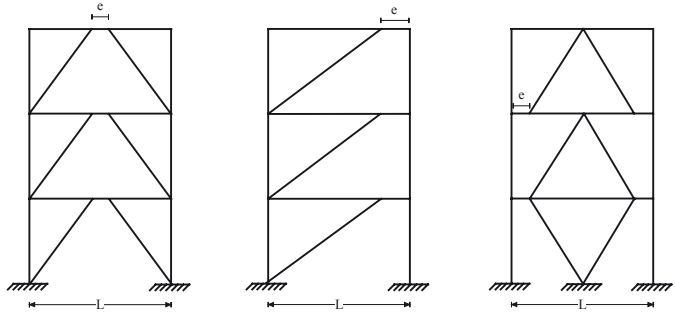



*Figure 6.13* Possible configurations of eccentrically braced frames

can be shifted away from beam–column connections, thus improving the overall integrity of the frame.

The length of the link zone has a direct influence on the frame stiffness. The relation between eccentricity ratio ($e/L$) and the lateral stiffness ($K$) is illustrated in Figure 6.14. As $e/L$ tends to unity, the stiffness of the MRF is obtained, while the zero eccentricity ratio corresponds to the CBF stiffness. There is also a direct relationship between the frame drift angle ($\theta$) and the rotational demand in the link ($\gamma$). Simple analysis of plastic collapse mechanisms of a single link in EBF gives a relationship between frame and link deformations (see Figure 6.15) as:

$$\theta L = \gamma e \tag{6.7}$$

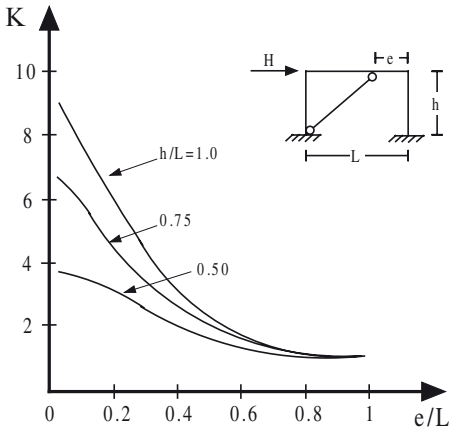



*Figure 6.14* Relationship between link length and lateral stiffness of eccentrically braced frames

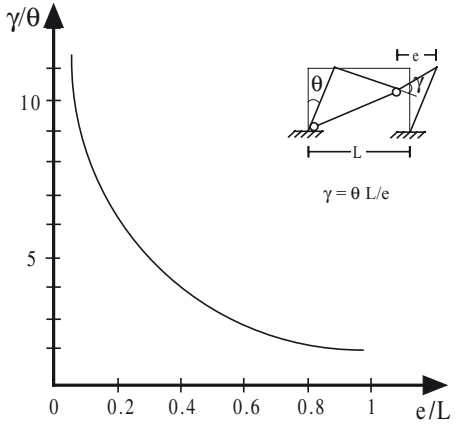



*Figure 6.15* Relationship between link length and ductility demand in eccentrically braced frames

Since the span of the frame is significantly larger than the eccentricity, $e$, it follows that the ductility demand in the link is considerably higher than that for the frame. It is also evident that shorter links would have higher demand for the same level of frame drift.

As in other codes, eccentrically braced frames are designed in EC8 so that beams are able to dissipate energy by formation of plastic bending or plastic shear mechanisms in the links. Specific rules are given to ensure that yielding in the bending/shear links of the beams will take place prior to yielding or failure in other members, which would therefore be capacity designed. The most recent version of EC8 incorporates detailed provisions (mainly in Section 6.8 of EN 1998-1) that are largely in accordance with North American design procedures.

### *6.6.2 Link beams*

Whereas short links suffer from high ductility demands, they yield primarily in shear. Experimental evidence (e.g. Hjelmstad and Popov, 1983; Kasai and Popov, 1986; Engelhardt and Popov, 1989) showed that shear link behaviour in steel is superior to that of flexural plastic hinges. However, other considerations such as architectural requirements may necessitate the use of long links. Assuming no strain-hardening or moment-shear interaction, the theoretical dividing length ($e_c$) between shear and flexural yielding is:

$$e_c = 2M_{p,link}/V_{p,link} \tag{6.8}$$

where $M_{p,link}$ and $V_{p,link}$ are the plastic moment and plastic shear capacities of the cross section, respectively.

Experimental evidence, however, shows that strain-hardening is significant in link behaviour. The ultimate shear and bending strengths may be significantly higher than $V_{p,link}$ and $M_{p,link}$, with different ratios. Accordingly, in EN 1998-1, for I-section links where equal moments occur at both ends, the links are defined as:

$$\text{short links} \quad e < e_s = 1.6\, M_{p,link}/V_{p,link} \tag{6.9}$$

$$\text{long links} \quad e > e_L = 3.0\, M_{p,link}/V_{p,link} \tag{6.10}$$

$$\text{intermediate links} \quad e_s < e < e_L \tag{6.11}$$

On the other hand, in designs in which only one plastic hinge forms at one end in I-sections:

$$\text{short links} \quad e < e_s = 0.8\,(1+\alpha)\; M_{p,link}/V_{p,link} \tag{6.12}$$

$$\text{long links} \quad e > e_L = 1.5\,(1+\alpha)\; M_{p,link}/V_{p,link} \tag{6.13}$$

intermediate links $e_s < e < e_L$ (6.14)

where $\alpha$ is the ratio of the absolute value of the smaller-to-larger bending moments at the two ends of the link.

If the applied axial force exceeds 15 per cent of the plastic axial capacity, reduced expressions for the moment and shear plastic capacities are provided in EC8 to account for the corresponding reductions in their values.

EC8 also provides limits on the rotation '$\theta_p$' in accordance with the expected rotation capacity. This is given as 0.08 radians for short links and 0.02 radians for long links, whilst the limit for intermediate links can be determined by linear interpolation. The code also gives a number of rules for the provision of stiffeners in short, long and intermediate link zones.

### *6.6.3 Other frame members*

Other members not containing seismic links, such as the columns and diagonals, should be capacity designed. These members should be verified considering the most unfavourable combination of axial force and bending moment with due account for shear forces, such that:

$$N_{Ed}(M_{Ed}, V_{Ed}) \geq N_{Ed,G} + 1.1\,\gamma_{ov}\,\Omega\,N_{Ed,E} \tag{6.15}$$

where the actions are similar to those previously defined for concentrically braced frames. However, in this case $\Omega$ is the minimum of the following: (i) min of $\Omega_i = 1.5V_{p,link,i}/V_{Ed,i}$ among all short links, and (ii) min of $\Omega_i = 1.5M_{p,link,i}/M_{Ed,i}$ among all intermediate and long links where $V_{Ed,i}$ and $M_{Ed,i}$ are the design values of the shear force and bending moment in link '*i*' in the seismic design situation, whilst $V_{p,link,i}$ and $M_{p,link,i}$ are the shear and bending plastic design capacities, respectively, of link *i*. It should also be checked that the individual values of $\Omega_i$ do not differ from the minimum value by more than 25 per cent in order to ensure reasonable distribution of ductility.

If the structure is designed to dissipate energy in the links, the connections of the links or of the elements containing the links should also be capacity designed with due account of the overstrength of the material and the links, as before. Semi-rigid and/or partial-strength connections are permitted with some conditions similar to those described previously for MRFs.

Specific guidance is given for link stiffeners in EN 1998-1. Full-depth stiffeners are required on both sides of the link web at the diagonal brace ends of the link as indicated in Figure 6.16. These stiffeners should have a combined width not less than $b_f - 2t_w$ and a thickness not less than $0.75t_w$ or 10 mm whichever is larger, where $b_f$ and $t_w$ are the link flange width and link web thickness, respectively.

Intermediate web stiffeners in shear links should be provided at intervals not exceeding $(30t_w - d/5)$ for a link rotation angle of 0.08 radians, or $(52t_w - d/5)$ for link rotation angles of 0.02 radians or less, with linear interpolation used

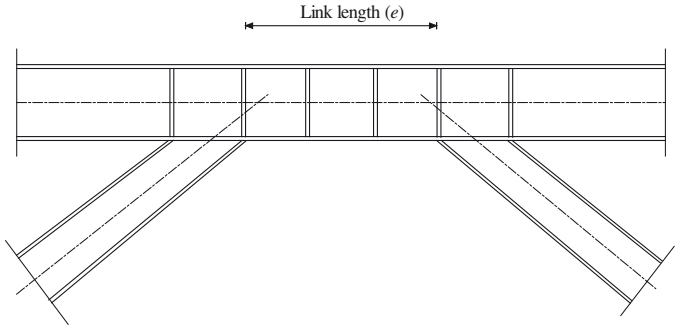



*Figure 6.16* Full-depth web stiffeners in link zones of eccentrically braced frames

in between, where $d$ is the section depth. Links of length greater than 2.6 $M_{p,link}/V_{p,link}$ and less than 5 $M_{p,link}/V_{p,link}$ should be provided with intermediate web stiffeners placed at a distance of 1.5 times $b_f$ for each end of the link. Both requirements apply for links of length between 1.6 and 2.6 $M_{p,link}/V_{p,link}$, and no intermediate web stiffeners are required in links of lengths greater than $5M_{p,link}/V_{p,link}$. Intermediate link web stiffeners are required to be full depth. For links that are less than 600 mm in depth, stiffeners are required on only one side of the link web. Lateral supports are also required at both the top and bottom link flanges at the end of the link. End lateral supports of links should have design strength of 6 per cent of the expected nominal strength of the link flange.

Design of link-to-column connections should be based upon cyclic test results that demonstrate inelastic rotation capability 20 per cent greater than that calculated at the design storey drift. On the other hand, beam-to-column connections away from links are permitted to be designed as pinned in the plane of the web.

## 6.7 Material and construction considerations

In addition to conforming to the requirements of EN 1993-1 (2005, EC3), EC8 incorporates specific rules dealing with the use of a realistic value of material strength in dissipative zones. In this respect, according to Section 6.2 of EN 1998-1, the design should conform to one of the following conditions:

- The actual maximum yield strength $f_{y,max}$ of the steel of the dissipative zones satisfies the relationship: $f_{y,max} \leq 1.1\ \gamma_{ov}\ f_y$, where $f_y$ is the nominal yield strength and the recommended value of $\gamma_{ov}$ is 1.25.
- The design of the structure is made on the basis of a single grade and nominal yield strength '$f_y$' for the steels both in dissipative and non-dissipative zones, with an upper limit '$f_{y,max}$' specified for steel in

dissipative zones, which is below the nominal value $f_y$ specified for non-dissipative zones and connections.

- The actual yield strength '$f_{y,act}$' of the steel of each dissipative zone is determined from measurements and the overstrength factor is assessed for each dissipative zone as $\gamma_{ov,act} = f_{y,act}/f_y$.

In addition to the above, steel sections, welds and bolts should satisfy other requirements in dissipative zones. In bolted connections, high strength bolts (8.8 and 10.9) should be used in order to comply with the requirements of capacity design.

In terms of detailed design and construction requirements, in addition to the rules of EN 1993-1, several specific provisions are given in Section 6.11 of EN 1998-1. The details of connections, sizes and qualities of bolts and welds as well as the steel grades of the members and the maximum permissible yield strength $f_{y,max}$ in dissipative zones should be indicated on the fabrication and construction drawings.

Checks should be carried out to ensure that the specified maximum yield strength of steel is not exceeded by more than 10 per cent. It should also be ensured that the distribution of yield strength throughout the structure does not substantially differ from that assumed in design. If any of these conditions are not satisfied, new analysis of the structure and its details should be carried out to demonstrate compliance with the code.

## 6.8 Design example – moment frame

### *6.8.1 Introduction*

The same eight-storey building considered in previous chapters is utilised in this example. The layout of the structure is reproduced in Figure 6.17. The main seismic design checks are carried out for a preliminary design according to EN 1998-1. For the purpose of illustrating the main seismic checks in a simple manner, consideration is only given to the lateral system in the X-direction of the plan, in which resistance is assumed to be provided by MRFs spaced at 4 m. It is also assumed that an independent bracing system is provided in the transverse (Y) direction of the plan. Grade S275 is assumed for the structural steel used in the example.

### *6.8.2 Design loads*

The gravity loads are adapted from those described in Chapter 3, and are summarised in Table 6.3. On the other hand, the seismic loads are evaluated based on the design response spectrum and on the fundamental period of the structure, which is estimated to be 1.06 s from the simplified expression in EC8 (Cl. 4.3.3.2.2). The total seismic mass, obtained from the self weight as well as an allowance of 30 per cent of the imposed load, is found to be 8208 t.

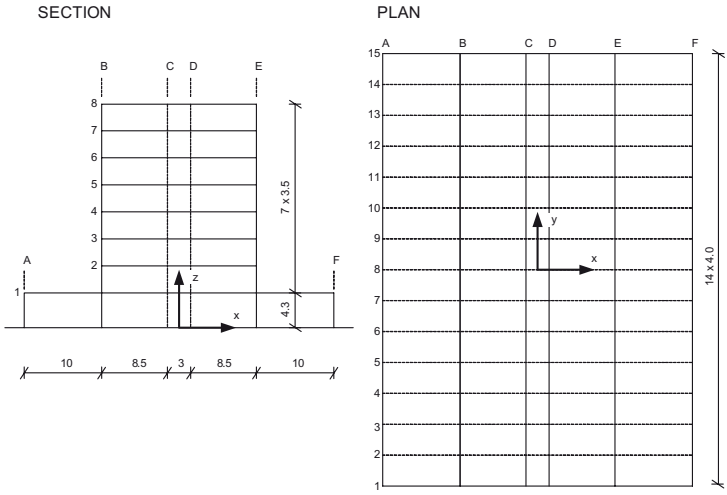



*Figure 6.17* Frame layout

*Table 6.3* Summary of gravity loads

| *Type of Load* | *Description* | *Value (kN/m²)* |
|---|---|---|
| Dead Load | 150 mm thick solid slab | 3.6 |
| | Finishing | 1.0 |
| | External walls | 3.25 |
| | Internal walls | 1.7 |
| Imposed Load | Roof | 2.0 |
| | Corridors | 4.0 |
| | Bedrooms | 2.0 |
| | Roof terrace | 4.0 |

A behaviour factor of 4 is adopted assuming ductility class medium (DCM). The total design base shear for the whole structure is therefore estimated as 8372 kN. The design base shear per frame is therefore considered as 558 kN.

The moment frame located on GL 2 is selected for illustration in this example. Although the structure is symmetric in plan, an account should be made for torsional effects resulting from the accidental eccentricity. Using the simplified approach suggested in Cl. 4.3.3.2.4(1) of EC8, the design base shear for this frame is increased by a factor of about 1.26 to approximately 703 kN.

According to Cl. 4.3.3.2.3 of EC8, the design base shear should be applied in the form of equivalent lateral loads at the floor levels. These loads are obtained by distributing the base shear in proportion to the fundamental

*Table 6.4* Floor seismic loads (GL 2 frame)

| *Floor* | *Seismic force (kN)* |
|---|---|
| 8 | 98.2 |
| 7 | 143.5 |
| 6 | 123.7 |
| 5 | 103.9 |
| 4 | 84.0 |
| 3 | 64.1 |
| 2 | 44.2 |
| 1 | 41.6 |

mode shape of the frame or, in cases where the mode follows a linear variation with height, by distributing the base shear in proportion to the mass and height of each floor. This simplified approach is adopted in this example for the floor loads, and the values are given in Table 6.4.

The frame on GL 2 was firstly designed for the non-seismic/gravity loading combinations corresponding to both ultimate and serviceability limit states according to the provisions of EC3 (EN 1993-1). On this basis, the initial column sections adopted were HEB450 for the four lower stories and HEB300 for the upper five storeys, whilst IPE550 was selected for the beams.

### *6.8.3 Seismic design checks*

#### *6.8.3.1 General considerations*

A preliminary elastic analysis was firstly carried out using the estimated seismic loads for the frame incorporating the initial member sizes. These initial member sizes were, however, found to be inadequate to fulfil both strength and damage limitation requirements. Accordingly, the columns were increased to HEA550 in the lower four storeys and to HEA500 in the upper five storeys. On the other hand, the initial size of the external (8.5 m) beams was retained, but the size of the internal (3 m) beams was reduced to IPE500 as this provided a more optimum solution in terms of the column sizes required to satisfy capacity criteria. It is also worth noting that controlling the lateral stiffness through the column sizes is often more optimal with respect to capacity design requirements.

The seismic design combination prescribed in Cl. 6.4.3.4 of EC0 (EN 1990, 2002) is:

$$\sum_{j\geq 1} G_{k,j} + \sum_{i\geq 1} \psi_{2,i} \cdot Q_{k,i} + A_{Ed}$$

*Table 6.5* Calculation of inter-storey drift sensitivity coefficient

| *Level* | $d_e$ *(mm)* | $d_s$ *(mm)* | $d_r$ *(mm)* | $P_{tot}$ *(kN)* | $V_{tot}$ *(kN)* | *h (mm)* | θ |
|---|---|---|---|---|---|---|---|
| 8 | 78.4 | 313.6 | 18.0 | 453 | 98.2 | 3500 | 0.024 |
| 7 | 73.9 | 295.6 | 28.0 | 1170 | 241.7 | 3500 | 0.039 |
| 6 | 66.9 | 267.6 | 38.4 | 1888 | 365.4 | 3500 | 0.057 |
| 5 | 57.3 | 229.2 | 46.8 | 2606 | 469.3 | 3500 | 0.074 |
| 4 | 45.6 | 182.4 | 50.4 | 3323 | 553.3 | 3500 | 0.086 |
| 3 | 33.0 | 132.0 | 53.6 | 4041 | 617.4 | 3500 | 0.100 |
| 2 | 19.6 | 78.4 | 47.6 | 4758 | 661.6 | 3500 | 0.098 |
| 1 | 7.7 | 30.8 | 30.8 | 5476 | 703.2 | 4300 | 0.056 |

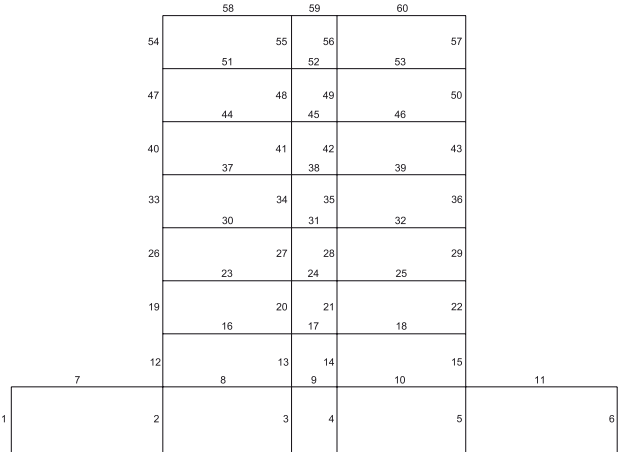



*Figure 6.18* Frame model with element numbers

where $G_K$, $Q_K$ are the action effects due to the characteristic dead and imposed loads, respectively. The parameter $\Psi_2$ is the quasi-permanent combination factor that, in this example, is taken as 0.3. In the same combination, $A_{Ed}$ refers to the action effects due to the seismic loads.

A view of the frame model showing the element numbering is given in Figure 6.18. The results of elastic analysis for the seismic loading combination are initially used in the evaluation of the inter-storey drift sensitivity coefficient, θ, as listed in Table 6.5. As shown in the table, θ does not exceed the limit of 0.1 according to Cl. 4.4.2.2(2) of EC8, and hence second-order effects do not have to be considered in the analysis.

The design checks for the beams and columns require the knowledge of the internal actions. As an example, the bending moment diagrams due to the vertical (i.e. $M_{Ed,G}$ due to $G_k + 0.3Q_k$) and earthquake (i.e. $M_{Ed,E}$ due to $E$) loads are presented in Figure 6.19. The final bending moment diagram for the seismic combination (i.e. $M_{Ed}$) is shown in Figure 6.20.

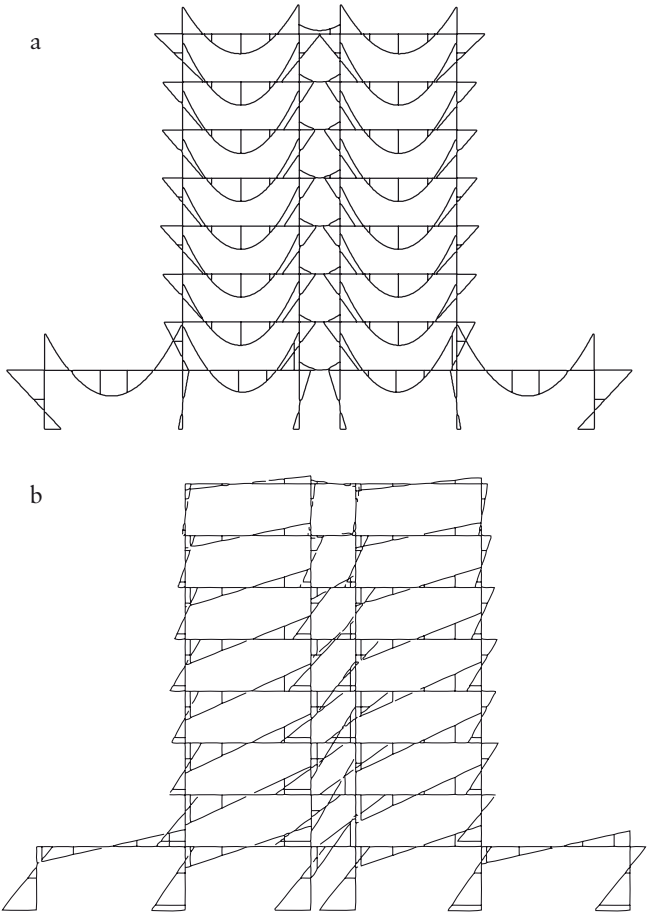



*Figure 6.19* Bending moment diagrams due to (a) gravity and (b) earthquake loads

*6.8.3.2 Beam design checks*

For illustration, the beam design checks are performed for a critical member, which is the 3 m internal beam located at the second floor (Element 17 in Figure 6.18). The internal forces at both ends of the member are listed in Table 6.6.

Based on the values from the table, the seismic demands on the beam are:

$M_{Ed} = -33.2 + (-392.6) = -425.8$ kN.m

$N_{Ed} = 24.7 + 0 = 24.7$ kN

$V_{Ed} = V_{Ed,G} + V_{Ed,M} = 45.1 + (603+603)/3.0 = 447.1$ kN

According to Cl. 6.6.2(2) of EC8 and considering the properties of the beam section (which is Class 1 according to EC3):

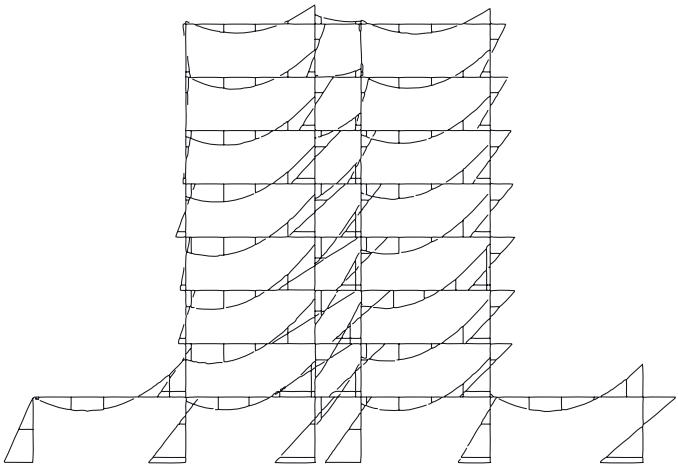


*Figure 6.20* Bending moment diagrams for the seismic combination

*Table 6.6* Internal forces in Element 17

| | *Left end* | | *Right end* | |
|---|---|---|---|---|
| | $G_k+0.3Q_k$ | $E$ | $G_k+0.3Q_k$ | $E$ |
| *M (kN.m)* | –33.2 | 392.6 | –33.2 | –392.6 |
| *V (kN)* | 45.1 | –261.7 | –45.1 | –261.7 |
| *N (kN)* | 24.7 | 0 | 24.7 | 0 |

$$M_{Ed} \le M_{pl,Rd} \rightarrow 425.8 \le 2194.10^{-6} \times 275.10^{3} \rightarrow 425.8 \text{ kN.m} \le 603 \text{ kN.m}$$

$$N_{Ed} \le 0.15.N_{pl,Rd} \rightarrow 24.7 \le 0.15 \times 116.4.10^{-4} \times 275.10^{3}$$
$$\rightarrow 24.7 \text{ kN} \le 480 \text{ kN}$$

$$V_{Ed} \le 0.5.V_{pl,Rd} \rightarrow 447.1 \le 0.5 \times 59.87.10^{-4} \times 275.10^{3}/\sqrt{3}$$
$$\rightarrow 447.1 \text{ kN} \le 476 \text{ kN}$$

*6.8.3.3 Column design checks*

The columns should be capacity designed based on the weak beam/strong column approach. According to Cl. 6.6.3 of EC8, the design forces are obtained using the following combination:

$$E_d = E_{d,Gk+0.3Qk} + 1.1\gamma_{ov}\Omega E_{d,E}$$

where $\gamma_{ov}$ is the overstrength factor assumed as 1.25

$$\Omega = \min (M_{pl,Rd,i}/M_{Ed,i}) = 603/425.8 = 1.42$$

*Table 6.7* Internal forces in Element 4

| | *Bottom end* | | *Top end* | |
|---|---|---|---|---|
| | $G_k+0.3Q_k$ | $E$ | $G_k+0.3Q_k$ | $E$ |
| *M (kN.m)* | 23.3 | –383.4 | –44.6 | 148.7 |
| *V (kN)* | –15.8 | 123.7 | –15.8 | 123.7 |
| *N (kN)* | –1502.4 | –755.0 | –1495.4 | –755.0 |

The column design combination is therefore:

$$E_d = E_{d,Gk+0.3Qk} + 1.95E_{d,E}$$

The design forces for a critical column (Element 4 in Figure 6.18) are presented in Table 6.7 for illustration.

Based on these values, the seismic demands at the bottom end of the column are:

$M_{Ed}$ = 23.3 + 1.95× (–383.4) = –724.3 kN.m

$N_{Ed}$ = –1502.4 + 1.95× (–755.0) = –2974.7 kN

$V_{Ed}$ = –15.8 + 1.95×123.7 = 225.4 kN

The combination of $M_{Ed}$ and $N_{Ed}$ given above should be used to perform all resistance checks for the member under consideration, including those for element stability, according to the provisions of EC3. The checks should consider the properties of HEA450 section (which is Class 1 according to EC3). The shear should be checked such that:

$V_{Ed} \leq 0.5V_{pl,Rd}$

225.4 ≤ 0.5×83.72×10-4×275×103/√3

225.4 kN ≤ 665 kN

In addition to the member checks, Cl. 4.4.2.3(4) of EC8 also requires that at every joint the following condition is satisfied:

$$\frac{\sum M_{Rc}}{\sum M_{Rb}} \geq 1.3$$

where $\Sigma M_{Rc}$ and $\Sigma M_{Rb}$ are the sum of the design moments of resistance of the columns and of the beams framing the joint, respectively. For illustration,

this check is performed for an internal joint located at the first floor of the frame:

$$\Sigma M_{Rc} = 2 \times 5591.10^{-6} \times 275.10^{3} = 3075 \text{ kN.m}$$

$$\Sigma M_{Rb} = 2787.10^{-6} \times 275.10^{3} + 2194.10^{-6} \times 275.10^{3} = 1370 \text{ kN.m}$$

$$\Sigma M_{Rc} / \Sigma M_{Rb} \geq 1.3$$

*6.8.3.4 Joint design checks*

According to Cl. 6.6.3(6) of EC8, the web panel zones at beam-to-column connections should be designed to resist the forces developed in the adjacent dissipative elements, which are the connected beams. For each panel zone, the following condition should be verified:

$$\frac{V_{wp,Ed}}{V_{wp,Rd}} \leq 1.0$$

where $V_{wp,Ed}$ is the design shear force in the web panel accounting for the plastic resistance of the adjacent beams/connections and $V_{wp,Rd}$ is the shear resistance of the panel zone according to EC3. For illustration, these checks are performed for an internal and an external panel.

EXTERNAL PANEL ZONE (HEA 550 + IPE 550)

$$V_{wp,Ed} = M_{pl,Rd} / (d_b - t_{bf}) = 766 / (0.550 - 0.0172) = 1438 \text{ kN}$$

$$V_{wp,Rd} = 0.9 \times f_{y,wc} \times A_{vc} / (\sqrt{3} \times \gamma_{M0}) + 4 \times M_{pl,fc.Rd} / (d_b - t_{bf})$$

(Cl. 6.2.6 of EC3 Part 1.8)

$$= 0.9 \times 275.10^{3} \times 83.72.10^{-4} / (\sqrt{3} \times 1) + 4 \times 0.300 \times 0.024^{2} \times 275.10^{3} / (4 \times (0.550 - 0.0172))$$

$$= 1196 + 89 = 1285 \text{ kN}$$

$$V_{wp,Ed} / V_{wp,Rd} = 1438 / 1285 \geq 1.0 \quad \rightarrow \text{ doubler plate required}$$

INTERNAL PANEL ZONE (IPE 550 + HEA 550 + IPE 500)

$$V_{wp,Ed} = \Sigma M_{Rb} / (d_b - t_{bf}) = 1370 / (0.500 - 0.016) = 2831 \text{ kN}$$

$$V_{wp,Ed} / V_{wp,Rd} = 2831 / 1285 \geq 1.0 \quad \rightarrow \text{ doubler plates required}$$

Design of a single supplementary doubler plate with a width of 300 mm:

$$V_{wp,Rd} \geq 1438 \text{ kN}$$

$$1285 + t_{dp} \times 0.300 \times 0.9 \times 275.10^{3} / \sqrt{3} \geq 1438$$

$t_{dp} \geq 3.6 \text{ mm} \rightarrow t_{dp} = 4 \text{ mm}$

### *6.8.4 Damage limitation*

According to Cl. 4.4.3.2(1) for the damage limitation (serviceability) limit state:

$$d_r \nu \leq 0.01h$$

here $d_r$ is the design inter-storey drift, $\nu$ is a reduction factor that takes into account the lower return period of the frequent earthquake and is assumed as 0.5, and $h$ is the storey height. The limit of 1 per cent is applicable to cases where the non-structural components are fixed to the structure in a way that does not interfere with structural deformation. For cases with non-ductile or brittle non-structural elements this limit is reduced to 0.75 per cent and 0.5 per cent, respectively.

Based on the results provided in Table 6.5, the maximum inter-storey drift occurs at the third floor:

$d_r = 53.6 \text{ mm}$

$d_r \nu \leq 0.01h$

$53.6 \times 0.5 \leq 0.01 \times 3500$

$26.8 \text{ mm} < 35 \text{mm}$

## 6.9 Design example – concentrically braced frame

### *6.9.1 Introduction*

The same eight-storey building considered previously is utilised in this example. The main seismic design checks are carried out for a preliminary design according to EN 1998-1. For the purpose of illustrating the checks in a simple manner, consideration is only given to the lateral system in the X-direction of the plan, in which resistance is assumed to be provided by concentrically braced frames spaced at 8 m. With reference to the plan shown before in Figure 6.17, eight braced frames are considered at Grid lines 1, 3, 5, 7, 9, 11, 13 and 15. It is also assumed that an independent bracing system is provided in the transverse (Y) direction of the plan. Grade S275 is considered for the structural steel used in the example.

### *6.9.2 Design loads*

The gravity loads per unit area are the same as those adopted in the moment frame example as indicated in Table 6.3. The equivalent lateral seismic loads are evaluated based on an estimated fundamental period of 0.62 s using

the simplified expression proposed in EC8 (Cl. 4.3.3.2.2). The behaviour factor considered is 4 and the total seismic mass is 8208 t. Accordingly, the resulting base shear is estimated as 14,302 kN. The design base shear per frame is therefore considered as 1,788 kN.

The braced frame located on GL 1 is selected for illustration in this example. Although the structure is symmetric in plan, an account should be made for torsional effects resulting from the accidental eccentricity. Using the simplified approach suggested in Cl. 4.3.3.2.4(1) of EC8, and for the purpose of preliminary design, the design base shear for this frame is increased by a factor of about 1.3 to approximately 2,324 kN. The base shear is applied to the frame in the form of floor loads distributed in proportion to the mass and height of each floor, as given in Table 6.8.

The frame on GL 1 was firstly designed for the non-seismic/gravity loading combinations corresponding to both ultimate and serviceability limit states according to the provisions of EC3 (EN 1993-1). On this basis, the initial column sections adopted were HEB300 for the four lower storeys and HEB220 for the upper five storeys. For the beams, IPE450 was selected for the 3 m and 8.5 m beams, whilst IPE 550 was necessary for the 10 m beams located on the first floor.

### *6.9.3 Seismic design checks*

#### *6.9.3.1 General considerations*

A preliminary elastic analysis was firstly carried out using the estimated seismic loads for the frame incorporating the initial member sizes. Preliminary considerations indicated that a suitable arrangement consists of X-bracing over each two consecutive storeys on the 8.5 m bays. Due to the different height of the first storey, there is a change of brace angle at this level, which requires particular attention when examining the actions on the first floor beams. The initial column sizes were increased to HEM360 in the lower four

*Table 6.8* Floor seismic loads (frame on GL1)

| *Floor* | *Seismic force (kN)* |
|---|---|
| 8 | 324.4 |
| 7 | 474.3 |
| 6 | 408.8 |
| 5 | 343.3 |
| 4 | 277.5 |
| 3 | 212.0 |
| 2 | 146.2 |
| 1 | 137.6 |

storeys and to HEB320 in the upper five stories, in order to satisfy strength and damage limitation requirements. The drifts and lateral shears related to the modified frame are given in Table 6.9, whilst the four different sizes selected for the braces are indicated in Table 6.10.

In the elastic analysis, the columns were assumed to be continuous along the height and pinned at the base. Beams and bracing members were also considered pinned at both ends. A view of the frame model indicating the element numbering is provided in Figure 6.21. The results of the elastic analysis for the seismic loading combination are initially used in the evaluation of the inter-storey drift sensitivity coefficients, which are listed in Table 6.9. As shown in the table, θ does not exceed the limit of 0.1 and hence second-order effects do not have to be considered in the analysis.

### *6.9.3.2 Brace design checks*

The design checks for the braces are conducted based on the axial forces, given in Table 6.10, from the structural analysis for the seismic design combination. Applying Cl. 6.7.3(5) (i.e. $N_{Ed} \le N_{pl,Rd}$) and Cl. 6.7.3 (1) (i.e. $1.3 \le \bar{\lambda} \le 2.0$) of EC8 for a critical brace in the frame (Element 71 in Figure 6.21) as an illustration:

$$N_{Ed} \le N_{pl,Rd} \rightarrow 1957 \le 72.1 \times 10^{-4} \times 275 \times 10^{3} \rightarrow 1957 \text{ kN} \le 1983 \text{ kN}$$

$$\bar{\lambda} = L_{cr}/i \times 1/\lambda_1$$

$$L_{cr} = 5.51 \text{ m}$$

$$i = 0.0466 \text{ m}$$

$$\lambda_1 = 93.9 \times \sqrt{(235/275)} = 86.8$$

$$\bar{\lambda} = (5.51 / 0.0466) \times (1 / 86.8) = 1.36 \le 2.0$$

The design checks for the remaining braces are summarised in Table 6.11.

*Table 6.9* Calculation of the inter-storey drift sensitivity coefficient

| *Level* | $d_e$ *(mm)* | $d_s$ *(mm)* | $d_r$ *(mm)* | $P_{tot}$ *(kN)* | $V_{tot}$ *(kN)* | *h (mm)* | *q* |
|---|---|---|---|---|---|---|---|
| 8 | 91.0 | 364.0 | -5.6 | 453 | 324.4 | 3500 | 0.004 |
| 7 | 92.4 | 369.6 | 68.4 | 1171 | 798.7 | 3500 | 0.043 |
| 6 | 75.3 | 301.2 | 37.6 | 1888 | 1207.5 | 3500 | 0.021 |
| 5 | 65.9 | 263.6 | 56.8 | 2606 | 1550.8 | 3500 | 0.029 |
| 4 | 51.7 | 206.8 | 57.6 | 3323 | 1828.3 | 3500 | 0.028 |
| 3 | 37.3 | 149.2 | 36.8 | 4041 | 2040.3 | 3500 | 0.016 |
| 2 | 28.1 | 112.4 | 30.4 | 4758 | 2186.5 | 3500 | 0.013 |
| 1 | 20.5 | 82.0 | 82.0 | 5476 | 2324.1 | 4300 | 0.027 |

In addition to the checks presented above, EC8 stipulates in Cl. 6.7.3 (8) that the maximum brace overstrength (Ω) does not differ from the minimum value by more than 25 per cent. As shown in Table 6.11, for this preliminary design, the overstrength in the braces exceeds this limit in several cases, with

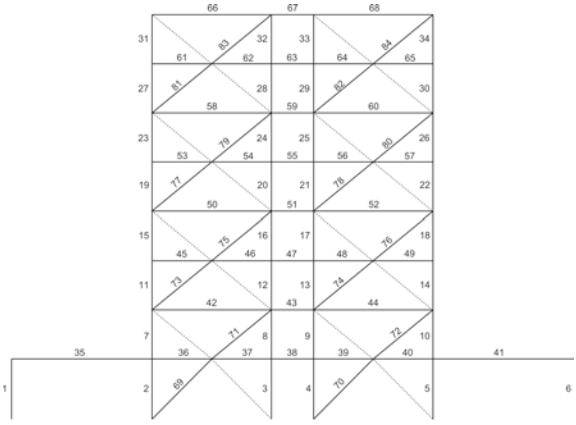



*Figure 6.21* Frame model with element numbers

*Table 6.10* Axial forces in the braces for the seismic combination

| *Storey* | *Element No.* | *Section* | $N_{Ed}$ *(kN)* |
|---|---|---|---|
| 1 | 69 | 200×120×12.5 | 1601 |
| | 70 | 200×120×12.5 | 1336 |
| 2 | 71 | 200×120×12.5 | 1957 |
| | 72 | 200×120×12.5 | 1653 |
| 3 | 73 | 200×100×10.0 | 1056 |
| | 74 | 200×100×10.0 | 1125 |
| 4 | 75 | 200×100×10.0 | 1291 |
| | 76 | 200×100×10.0 | 1339 |
| 5 | 77 | 200×100×8.0 | 788 |
| | 78 | 200×100×8.0 | 868 |
| 6 | 79 | 200×100×8.0 | 975 |
| | 80 | 200×100×8.0 | 1038 |
| 7 | 81 | 200×100×5.0 | 212 |
| | 82 | 200×100×5.0 | 274 |
| 8 | 83 | 200×100×5.0 | 335 |
| | 84 | 200×100×5.0 | 387 |

notable differences at the two upper storeys. As discussed before in Section 6.5, enforcing this limit can lead to impractical and inefficient design and may not be necessary if continuous and relatively stiff columns are adopted, as is the case in this example. By increasing the brace sizes significantly throughout the frame, the code limit may be satisfied, yet this will be at the expense of the efficiency of the design; difficulties will also be encountered in satisfying the lower slenderness limit of 1.3, which is another limit that can be replaced by appropriate consideration of the post-buckling residual compressive capacity of the braces in the design of the frame.

### *6.9.3.3 Other frame members*

Beams and columns, as well as connections, should be capacity designed to ensure that dissipative behaviour is provided primarily by the braces. According to Cl. 6.7.4 of EC8, the design forces are obtained using the following combination:

$$E_d = E_{d,G_k+0.3Q_k} + 1.1\gamma_{ov}\Omega E_{d,E}$$

where $\gamma_{ov}$ is the overstrength factor assumed as 1.25.

*Table 6.11* Summary of design checks for the braces

| *Storey* | *Element No.* | *Section* | *Slenderness* $\bar{\lambda}$ | $N_{pl,Rd} / N_{Ed} = \Omega$ |
|---|---|---|---|---|
| 1 | 69 | 200×120×12.5 | 1.50 | 1.24 |
| | 70 | 200×120×12.5 | 1.50 | 1.48 |
| 2 | 71 | 200×120×12.5 | 1.36 | 1.01 |
| | 72 | 200×120×12.5 | 1.36 | 1.20 |
| 3 | 73 | 200×100×10.0 | 1.59 | 1.43 |
| | 74 | 200×100×10.0 | 1.59 | 1.34 |
| 4 | 75 | 200×100×10.0 | 1.59 | 1.17 |
| | 76 | 200×100×10.0 | 1.59 | 1.13 |
| 5 | 77 | 200×100×8.0 | 1.56 | 1.56 |
| | 78 | 200×100×8.0 | 1.56 | 1.42 |
| 6 | 79 | 200×100×8.0 | 1.56 | 1.26 |
| | 80 | 200×100×8.0 | 1.56 | 1.19 |
| 7 | 81 | 200×100×5.0 | 1.52 | 3.72 |
| | 82 | 200×100×5.0 | 1.52 | 2.88 |
| 8 | 83 | 200×100×5.0 | 1.52 | 2.36 |
| | 84 | 200×100×5.0 | 1.52 | 2.04 |

$\Omega = \min (N_{pl,Rd,i}/N_{Ed,i}) = 1983/1957 = 1.01$

The design combination is therefore:

$$E_d = E_{d,Gk+0.3Qk} + 1.39E_{d,E}$$

The design forces for a critical beam (Element 42 in Figure 6.21) and a critical column (Element 10 in Figure 6.21) are presented in Tables 6.12 and 6.13, respectively, for illustration.

Based on these values, the seismic demands at mid-span of the beam are:

$M_{Ed} = 248.5 + 1.39\times 0 = 248.5$ kN.m

$N_{Ed} = -3.6 + 1.39\times (-908.1) = -1266$ kN

$V_{Ed} = 0$ kN

The combination of $M_{Ed}$ and $N_{Ed}$ given above should be used to perform all resistance checks for the member under consideration, including those for element stability, according to the provisions of EC3. The checks should consider the properties of IPE450 section (which is Class 1 according to EC3). On the other hand, the seismic demands on the top end of the selected column are:

$M_{Ed} = -95.6 + 1.39\times (-321.9) = -543$ kN.m

$N_{Ed} = -2555.3 + 1.39\times (-1600.5) = -4780$ kN

$V_{Ed} = 47.1 + 1.39\times 139.4 = 241$ kN

The combination of $M_{Ed}$ and $N_{Ed}$ given above should be used to perform all resistance checks for the member under consideration, including those for element stability, according to the provisions of EC3. The checks should consider the properties of HEM360 section (which is Class 1 according to EC3). The shear should be checked such that:

$V_{Ed} \le 0.5V_{pl,Rd}$

$241 \le 0.5\times 102.4\times 10^{-4}\times 275\times 10^{3}/\sqrt{3}$

$241$ kN $\le 813$ kN

### *6.9.4 Damage limitation*

According to Cl. 4.4.3.2(1), for the damage limitation (serviceability) limit state:

*Table 6.12* Internal actions in a critical beam (Element 42)

| | *Mid-span* | |
|---|---|---|
| | $G_k+0.3Q_k$ | *E* |
| *M (kN.m)* | 248.5 | 0.0 |
| *V (kN)* | 0.0 | 0.0 |
| *N (kN)* | –3.6 | –908.1 |

*Table 6.13* Internal actions in a critical column (Element 10)

| | *Bottom end* | | *Top end* | |
|---|---|---|---|---|
| | $G_k+0.3Q_k$ | *E* | $G_k+0.3Q_k$ | *E* |
| *M (kN.m)* | 106.7 | 259.1 | –95.6 | –321.9 |
| *V (kN)* | –57.8 | –166.0 | 47.1 | 139.4 |
| *N (kN)* | –1432.3 | –2555.3 | –1200.3 | –1600.5 |

$$d_r \nu \leq 0.01h$$

where $d_r$ is the design inter-storey drift, ν is a reduction factor that takes into account the lower return period of the frequent earthquake and is assumed 0.5, and $h$ is the storey height. The limit of 1 per cent is applicable to cases where the non-structural components are fixed to the structure in a way that does not interfere with structural deformation. For cases with non-ductile or brittle non-structural elements this limit is reduced to 0.75 per cent and 0.5 per cent, respectively.

Based on the results provided in Table 6.9, the maximum inter-storey drift occurs at the seventh storey:

$$d_r = 68.4 \text{ mm}$$

$$d_r \nu \leq 0.01\, h$$

$$68.4 \times 0.5 \leq 0.01 \times 3500$$

$$34.2 \text{ mm} < 35 \text{ mm}$$

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# 7 Design of composite steel/concrete structures

*A.Y. Elghazouli and J.M. Castro*

## 7.1 Introduction

The design of composite steel/concrete buildings in EC8, covered in Section 7 of EN 1998-1 (2004), largely follows the general methodology adopted for steel structures (Section 6 of EN 1998-1). Accordingly, most of the approaches and procedures discussed in the previous chapter also apply to composite steel/concrete structures, with some differences related mainly to ductility requirements and capacity design considerations. This chapter highlights these differences, discusses a number of key behavioural and design aspects, and concludes with an illustrative design example.

Three general '*design concepts*' are stipulated in Section 7 of EN 1998-1, namely:

1 *Concept a*: low-dissipative structural behaviour – which refers to DCL in the same manner as in steel structures. In this case, a behaviour factor of 1.5–2 (recommended as 1.5) can be adopted based largely on the provisions of EC3 (EN 1993-1, 2005) and EC4 (EN 1994-1, 2004) for steel and composite components, respectively.
2 *Concept b*: dissipative structural behaviour with composite dissipative zones. In this case, DCM and DCH design can be adopted with additional rules to satisfy ductility and capacity design requirements as discussed in subsequent sections of this chapter.
3 *Concept c*: dissipative structural behaviour with steel dissipative zones. In this case, critical zones are designed as steel to Section 6 of EN 1998-1 in the seismic situation, although other '*non-seismic*' design situations may consider composite action to EC4 (EN 1994). Therefore, specific measures are stipulated to prevent the contribution of concrete under seismic conditions.

This chapter deals primarily with *Concept b* in which composite dissipative zones are expected, but some discussion of *Concept c*, which implies steel-only dissipation, is also included. After outlining the structural types and associated behaviour factors, as stipulated in Section 7 of EN 1998-1, the

main ductility and capacity design requirements are summarised. Emphasis is then given to discussing design procedures related to composite beam and column members within moment frames and other lateral-resisting structural configurations.

## 7.2 Structural types and behaviour factors

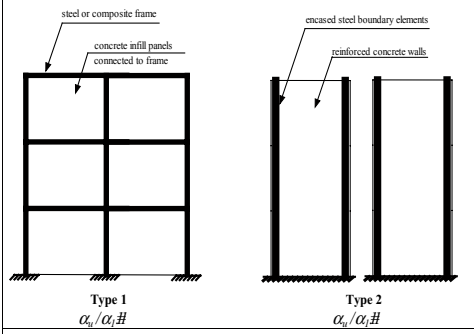
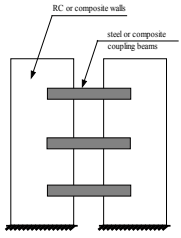
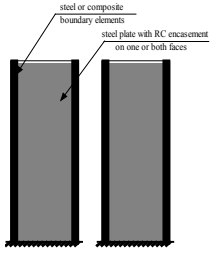
The same upper limits of the reference behaviour factors specified for steel framed structures (Section 6 of EN 1998-1) are also employed in Section 7 of EN 1998-1 for composite structures. This applies to composite moment resisting frames, composite concentrically braced frames and composite eccentrically braced frames. However, whilst in composite moment frames the dissipative beam and/or column zones may be steel or composite, the dissipative zones in braced frames are in most cases only allowed to be in steel. In other words, the diagonal braces in concentrically braced frames, and the bending/shear links in eccentrically braced frames, should in most cases be designed and detailed such that they behave as steel dissipative zones. This limitation is adopted in the code as a consequence of the uncertainty associated with determining the actual capacity and ductility properties of composite steel/concrete elements in these configurations. As a result, the design of composite braced frames follows very closely those specified for steel, and are therefore not discussed in detail herein. On the other hand, several specific criteria related to the dissipative behaviour of composite moment frames are addressed in subsequent sections of this chapter.

A number of additional composite structural systems are also referred to in Section 7 of EN 1998-1, as indicated in Table 7.1, including:

- Steel or composite frame with connected infill concrete panels (*Type 1*), or reinforced concrete walls with embedded vertical steel members acting as boundary/edge elements (*Type 2*).
- Steel or composite coupling beams in conjunction with reinforced concrete or composite steel/concrete walls (*Type 3*).
- Composite steel plate shear walls consisting of vertical continuous steel plates with concrete encasement on one or both sides of the plates and steel/composite boundary elements.

The upper limits of reference q for the above-listed systems are shown in Table 7.1 for DCM and DCH. As noted in previous chapters, these reference values should be reduced by 20 per cent if the building is irregular in elevation. Also, an estimate for the multiplier $\alpha_u/\alpha_1$ may be determined from conventional nonlinear 'pushover' analysis, but should not exceed 1.6. In the absence of detailed calculations, the default value of $\alpha_u/\alpha_1$ may be assumed as 1.1 for *Types 1–3*. For composite steel plate shear walls, the default value may be assumed as 1.2. It should be noted that for buildings that are irregular in plan, the default values of $\alpha_u/\alpha_1$ should be assumed as 1.05

*Table 7.1* Structural types and behaviour factors (additional to those in Section 6 of EN 1998-1)

| Structural Type | q-factor DCM | q-factor DCH |
|---|---|---|
|  **Type 1** $\alpha_u/\alpha_1$ # **Type 2** $\alpha_u/\alpha_1$ # | 3 $\alpha_u/\alpha_1$ | 4 $\alpha_u/\alpha_1$ |
|  **Type 3** $\alpha_u/\alpha_1$ # | 3 $\alpha_u/\alpha_1$ | 4 $\alpha_u/\alpha_1$ |
|  **Composite steel plate shear walls** $\alpha_u/\alpha_1$ # | 3 $\alpha_u/\alpha_1$ | 4 $\alpha_u/\alpha_1$ |

and 1.1 for *Types 1–3* and composite steel plate shear walls, respectively. In terms of dissipative zones, these can be located in the vertical steel sections and in the vertical reinforcement of the walls. The coupling beams in the case of *Type 3* can also be considered as dissipative elements.

## 7.3 Ductility classes and rules for cross sections

As in the case of dissipative steel zones, there is a direct relationship between the ductility of dissipative composite zones, consisting of concrete-encased or concrete-infilled steel members, and the cross-section slenderness. However, as expected, additional rules relating to the reinforcement detailing also

*Table 7.2* Cross-section requirements based on ductility classes and reference $q$ factors

| *Ductility classes and reference q factors* | *Partially or fully encased H/I sections* | *Concrete filled rectangular sections* | *Concrete filled circular sections* |
|---|---|---|---|
| DCM $(1.5 < q \leq 2.0)$ | $c/t_f \leq 20\sqrt{235/f_y}$ | $h/t \leq 52\sqrt{235/f_y}$ | $d/t \leq 90(235/f_y)$ |
| DCM $(2.0 < q \leq 4.0)$ | $c/t_f \leq 14\sqrt{235/f_y}$ | $h/t \leq 38\sqrt{235/f_y}$ | $\leq 85(235/f_y)$ |
| DCH $(q > 4.0)$ | $c/t_f \leq 9\sqrt{235/f_y}$ | $h/t \leq 24\sqrt{235/f_y}$ | $d/t \leq 80(235/f_y)$ |

apply in the case of composite members, as discussed in subsequent parts of this chapter.

If dissipative steel zones are ensured, the cross-section rules described in the previous chapter and in Section 6 of EN 1998-1 should be applied. For dissipative composite sections, the beneficial presence of the concrete parts in delaying local buckling of the steel components is accounted for by relaxing the width-to-thickness ratio as indicated in Table 7.2.

In Table 7.2 (which is adapted from Table 7.3 of EN 1998-1), partially encased elements refer to sections in which concrete is placed between the flanges of I or H sections, whilst fully encased elements are those in which all the steel section is covered with concrete. The cross-section limit $c/t_f$ refers to the slenderness of the flange outstand of length $c$ and thickness $t_f$. The limits in hollow rectangular steel sections filled with concrete are represented in terms of $h/t$, which is the ratio between the maximum external dimension $h$ and the tube thickness $t$. Similarly, for filled circular sections, $d/t$ is the ratio between the external diameter $d$ and the tube thickness $t$. The limits for partially encased sections may be relaxed even further if special additional details are provided to delay or inhibit local buckling. These aspects are discussed in subsequent sections of this chapter within the provisions related to the ductility and capacity design requirements in composite members and components.

## 7.4 Requirements for critical composite elements

### *7.4.1 Beams acting compositely with slabs*

For beams attached with shear connectors to reinforced concrete or composite profiled slabs, a number of requirements are stipulated in Section

7.6.2 of EN 1998-1 in order to ensure satisfactory performance as dissipative composite elements (*Concept b*). These requirements comprise several criteria including those related to the degree of shear connection, ductility of the cross section and effective width assumed for the slab.

Dissipative composite beams may be designed for full or partial shear connection according to EC4 (EN 1994-1, 2004). However, the minimum degree of connection should not be lower than 80 per cent. This is based on recent research studies (e.g. Bursi and Caldara, 2005; Bursi *et al*, 2005), which indicate that, at reduced connection levels, the connectors may be susceptible to low cycle fatigue under seismic loading. The total resistance of the shear connectors within hogging moment regions should also not be less than the plastic resistance of the reinforcement. In addition, EC8 requires the resistance of connectors (as determined from EC4) to be reduced by a factor of 75 per cent. These two factors of 0.8 and 0.75 therefore have the combined effect of imposing more than 100 per cent in terms of degree of shear connection.

EC8 requirements also aim to ensure ductile behaviour in composite sections by limiting the maximum strain that can be imposed on concrete in the sagging moment regions of the dissipative zones. This is achieved by limiting the ratio $x/d$, as shown in Figure 7.1, where $x$ is the distance from the neutral axis to the top concrete compression fibre and $d$ is the overall depth of the composite section, such that:

$$\frac{x}{d} < \frac{\varepsilon_{cu2}}{\varepsilon_{cu2} + \varepsilon_a} \tag{7.1}$$

in which $\varepsilon_{cu2}$ is the ultimate compressive strain of concrete and $\varepsilon_a$ is the total strain in steel at the ultimate limit state.

The code includes a table (Table 7.4 in EN 1998-1) that proposes minimum values of $x/d$, which are deemed to satisfy the ductility requirement depicted in Equation (7.1) above. The values are provided as a function of the ductility class (DCM or DCH) and yield strength of steel ($f_y$). Close observation of the limits stipulated in the table suggests that they are derived based on assumed values for $\varepsilon_{cu2}$ of 0.25 per cent and $\varepsilon_a$ of $q \times \varepsilon_y$, where $\varepsilon_y$ is the yield strain of steel.

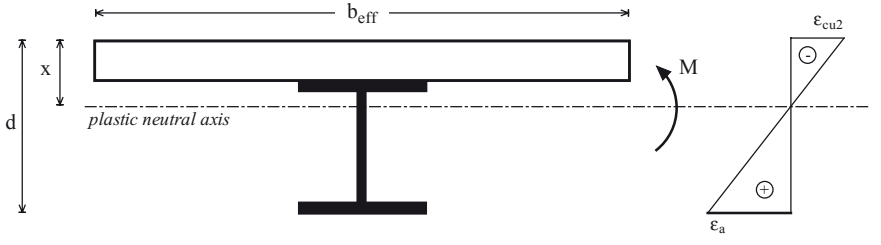



*Figure 7.1* Ductility of dissipative composite beam section under sagging moment

For dissipative zones of composite beams within moment frames, EN 1998-1 requires the inclusion of '*seismic bars*' in the slab at the beam-to-column connection region. The objective is to incorporate ductile reinforcement detailing to ensure favourable dissipative behaviour in the composite beams. The detailed rules are given in Annex C of EN 1998-1 and include reference to possible mechanisms of force transfer in the beam-to-column connection region of the slab. The provisions are largely based on background European research involving analytical and experimental studies (Plumier *et al*, 1998; Bowkamp *et al*, 1998; Doneux and Plumier, 1999). It should be noted that Annex C of the code only applies to frames with rigid connections in which the plastic hinges form in the beams; the provisions in the annex are not intended, and have not been validated, for cases with partial strength beam-to-column connections.

Another important consideration related to composite beams is the extent of the effective width $b_{eff}$ assumed for the slab, as indicated in Figure 7.1. EN 1998-1 includes two tables (Tables 7.5 I and 7.5 II in the code) for determining the effective width. These values are based on the condition that the slab reinforcement is detailed according to the provisions of Annex C since the same background studies (Plumier *et al*, 1998; Bowkamp *et al*, 1998; Doneux and Plumier, 1999) were used for this purpose. The first table (7.5 I) gives values for negative (hogging) and positive (sagging) moments for use in establishing the second moment of area for elastic analysis. These values vary from zero to 10 per cent of the beam span depending on the location (interior or exterior column), the direction of moment (negative or positive) and existence of transverse beams (present or not present). On the other hand, Table 7.5 II of the code provides values for use in the evaluation of the plastic moment resistance. The values in this case are as high as twice those suggested for elastic analysis. They vary from zero to 20 per cent of the beam span depending on the location (interior or exterior column), the sign of moment (negative or positive), existence of transverse beams (present or not present), condition of seismic reinforcement, and in some cases on the width and depth of the column cross section.

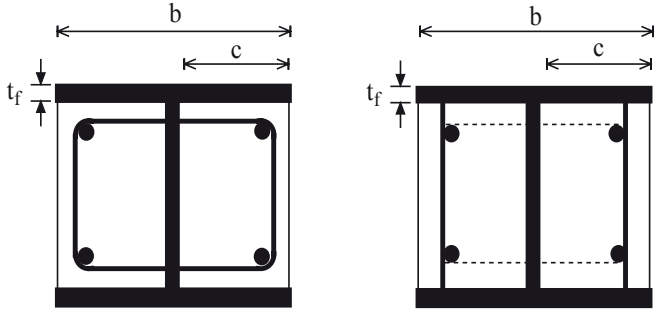
Clearly, design cases other than the seismic situation would require the adoption of the effective width values stipulated in EC4 (EN 1994-1, 2004). Therefore, the designer may be faced with a number of values to consider for various scenarios. Nevertheless, since the sensitivity of the results to these variations may not be significant (depending on the design check at hand), some pragmatism in using these provisions appears to be warranted. Recent research studies (Castro and Elghazouli, 2002; Amadio *et al*, 2004; Castro *et al*, 2007) indicate that the effective width is mostly related to the full slab width, although it also depends on a number of other parameters such as the slab thickness, beam span and boundary conditions.

### *7.4.2 Partially encased members*

Partially encased members, in which concrete is placed between the flanges as shown in Figure 7.2a, are often used in beams and columns. This configuration offers several advantages in comparison with bare steel members, particularly in terms of enhanced fire resistance (Schleich, 1988) as well as improved ductility due to the delay in local flange buckling (Ballio *et al*, 1987). In comparison with fully encased alternatives, this type of member enables the use of conventional steel connections to the flanges and reduces or eliminates the need for formwork. Several background studies on the inelastic behaviour of this type of member can be found elsewhere (Elghazouli and Dowling, 1992; Elghazouli and Elnashai, 1993; Broderick and Elnashai, 1994; Plumier *et al*, 1994; Elghazouli and Treadway, 2008).

Specific provisions for partially encased members are mainly included in Sections 7.6.1 and 7.6.5 of EN 1998-1. In dissipative zones, the slenderness of the flange outstand should satisfy the limits given in Table 7.2. However, if straight links welded to the inside of the flanges (as shown in Figure 7.2b) are provided in the dissipative zones at a spacing $s_1$ (along the length of the member), which is less than the width of the flange outstand (i.e. $s_1/c < 1.0$), then the flange slenderness limits can be relaxed. For $s_1/c < 0.5$, the limits in Table 7.2 can be increased by 50 per cent, and for $0.5 < s_1/c < 1.0$ linear interpolation can be employed. The weld of the straight bars should have a capacity of at least that of the tensile resistance of the bars. Also, a concrete cover of between 20 and 40 mm should be present, with the upper limit ensuring the effectiveness of the bar in delaying local flange buckling. The diameter $d_{bw}$ of the straight welded bars should not be less than the larger of 6 mm or the value of:

$$d_{bw} \geq \sqrt{\frac{t_f b}{8} \frac{f_{ydf}}{f_{ydw}}} \tag{7.2}$$




(a) Cross-section configuration (b) Straight bars welded to flange

*Figure 7.2* Partially encased composite sections

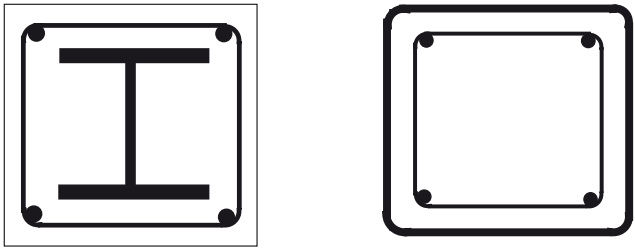
in which $b$ is the overall width of the flange and $t_f$ is the flange thickness, whilst $f_{ydf}$ and $f_{ydw}$ are the design yield strengths of the flange and straight welded bars, respectively.

Irrespective of whether straight welded bars are employed or not, the longitudinal spacing of confining reinforcement within dissipative zones of partially encased members should be limited in order to ensure an adequate level of concrete integrity. This provision becomes particularly important if local buckling cannot be prevented at large inelastic deformation levels. The length $l_{cr}$ of the critical dissipative zones, and the minimum longitudinal spacing $s$, need to be established for DCM and DCH; these requirements, which are also stipulated for fully encased members as noted in the following section, are largely based on the provisions for reinforced concrete members (Section 5 in EN 1998-1) as discussed earlier in Chapter 5 of this book.

### *7.4.3 Fully encased columns*

Composite members in which steel members are fully encased with concrete, as shown for example in Figure 7.3a, are often used as column elements in multi-storey buildings. These members clearly have inherent fire resistance properties and can provide relatively high axial and lateral loading capacity as well as significant ductility if properly designed and detailed.

A number of detailing requirements for fully encased composite columns are stipulated in Section 7.6.4 of EN 1998-1. Although, in principle, the intended plastic mechanisms in frame systems may only imply the formation of column dissipative zones at the base and perhaps at the top storey, it is important that ductile detailing is provided in other critical column regions due to the adverse consequences of overstressing non-ductile concrete. This treatment is similar to that employed in the detailing of reinforced concrete columns (Chapter 5 of this book and Section 5 of EN 1998-1) since the possibility of yielding in regions other than the intended dissipative zones exists due to factors such as higher dynamic modes, inelastic contra-flexure and bidirectional effects, amongst others.



(a) Fully encased section (b) Infilled section

*Figure 7.3* Concrete encased open sections and concrete infilled tubular sections

As noted above, the detailing rules for critical regions are largely based on those for reinforced concrete columns. The length $l_{cr}$ of the critical regions at the two ends of columns in moment frames depends on the length and depth of the column as well as on the ductility class (DCM or DCH). The code gives an expression (Equation 7.5 in EN 1998-1) for the minimum volumetric ratio of hoop reinforcement. The spacing $s$ of the confining hoops in the critical regions should also satisfy minimum values (Equations 7.7–7.9 in EN 1998-1), which depend on dimensions of the concrete core, diameter of the longitudinal bars and the ductility class. The diameter $d_{bw}$ of the hoops should satisfy minimum values (Equations 7.7–7.9 in EN 1998-1) as a function of the ductility class and the maximum diameter of the longitudinal bars, as well as the yield strength of both the hoop and longitudinal reinforcement. Also, the minimum cross-section dimensions should not be less than 250 mm.

As indicated in Table 7.2, the code suggests the same flange slenderness limits for fully encased members as those for partially encased sections on the basis that the concrete cover is ineffective in providing additional restraint against local buckling. However, the presence of closely spaced confining hoops can clearly have a beneficial effect in delaying local flange buckling. Accordingly, this is treated in the code in the same manner as that of welded straight bars in partially encased members as discussed in the previous section. Therefore, if hoops are provided with spacing $s$, which is less than the width of the flange outstand (i.e. $s/c < 1.0$), then the flange slenderness limits can be relaxed. For $s/c < 0.5$, the limits in Table 7.2 can be increased by up to 50 per cent, and for $0.5 < s/c < 1.0$ linear interpolation can be employed. Again, the diameter $d_{bw}$ of the confining hoops used to delay local buckling should satisfy the minimum value resulting from Equation (7.2) above.

### *7.4.4 Filled composite columns*

Tubular steel members of rectangular or circular/oval cross sections can be filled with concrete to provide a highly effective solution for columns in buildings. Figure 7.3b shows an example of a rectangular hollow section filled with concrete. This type of member combines aesthetic appearance with favourable structural properties including stiffness, capacity and ductility, as well as enhanced fire resistance in comparison with bare steel configurations.

As for other types of composite member, the design should conform to the requirements of EC4 (EN 1994, 2004). Additional specific criteria for filled columns are briefly described in EC8 and are given mainly in Section 7.6.6 of EN 1998-1. For dissipative zones, the cross-section slenderness, represented by $d/t$ or $h/t$, should satisfy the limits given in Table 7.2. Also, as for other types of composite member, the shear resistance in dissipative zones should be determined on the basis of the structural steel section only.

However, it can also be based on the reinforced concrete section with the steel hollow section considered only as shear reinforcement.

In general, whether the member is encased or infilled, if the concrete is assumed to contribute to the axial and/or flexural resistance, complete shear transfer between the steel and reinforced concrete parts should be ensured. Due to the expected deterioration in shear strength under cyclic loading conditions, the design shear strength given in EC4 (EN 1994-1, 2004) should be reduced by 50 per cent. If shear transfer cannot be achieved through bond and friction, shear connectors should be provided to ensure full composite action. Also, in composite columns that are subjected to predominantly axial loads, sufficient shear transfer should be provided to ensure that the steel and concrete parts share the loads applied to the column at connections to beams and bracing members.

## 7.5 Design of structural systems

### *7.5.1 Composite moment frames*

Composite moment frames, consisting of steel (or composite) columns and steel (or encased/filled) beams acting compositely with reinforced concrete (or composite) slabs, can offer several behavioural and practical advantages over bare steel and other alternatives. The seismic behaviour of composite moment frames has been examined experimentally and analytically by several researchers (e.g. Plumier *et al*, 1998; Leon, 1998; Leon *et al*, 1998; Hajjar *et al*, 1998; Thermou *et al*, 2004; Spacone and El-Tawil, 2004; Bursi *et al*, 2005; Elghazouli *et al*, 2008). Several of these studies, amongst others, have dealt with modelling and design considerations including behaviour factors, slab effects, shear interaction, connections and capacity design, and have contributed to the development of design codes such as AISC (2005) and EC8 (EN 1998-1, 2004).

Rules for the design and detailing of composite moment resisting frames are given in Section 7.7 of EN 1998-1. With the exception of a number of specific criteria, this section of the code refers directly to the ductility and capacity design rules for steel moment frames (in Section 6 of EN 1998-1 and Chapter 6 of this book), as well as the requirements for critical composite elements (Section 7.6 in EN 1998-1) discussed above in Sections 7.3 and 7.4 of this chapter.

An important consideration is related to the flexural stiffness assumed in analysis. For composite beams, the code specifies two values $EI_1$ and $EI_2$ for positive bending (uncracked section) and negative bending (cracked section) regions, respectively. However, the code also allows the alternative use of an equivalent second moment of area $EI_{eq,}$ which can be kept constant over the entire length of the beam, such that:

$$EI_{eq} = 0.6\, EI_1 + 0.4 EI_2 \tag{7.3}$$

The above equation clearly provides a more convenient representation of the composite beam for the purpose of analysis. On the other hand, if composite columns are used, the composite flexural stiffness of the column ($EI_{comp}$) can be represented as:

$$EI_{comp} = 0.9\,(E_s I_a + rE_{cm} I_c + E_s I_s) \tag{7.4}$$

where $E_s$ and $E_{cm}$ are the moduli of elasticity for steel and concrete, respectively, while $I_a$, $I_c$ and $I_s$ are the second moments of area for the steel section, concrete and reinforcement, respectively. The recommended value of $r$, which accounts for the influence of concrete cracking, is 0.5.

For composite columns, the code limits the applied axial load $N_{Ed}$ to 30 per cent of the plastic axial plastic capacity of the cross section $N_{pl,Rd}$ to ensure that ductility is not significantly reduced. The use of composite trusses as dissipative beams is also not permitted due to the uncertainty related to their performance under inelastic cyclic loading.

For steel panel zones in composite moment frames, as illustrated in Figure 7.4, the code refers to the rules for steel moment frames in Section 6 of EN 1998-1. Recent studies (Castro *et al*, 2005) have, however, shown that the behaviour of panel zones in composite moment frames differs from that in steel moment frames due to the variation in stress distribution and distortional demand imposed on the panel. Accordingly, expressions used for the modelling and assessment for panel zones in steel frames may not be realistic for composite frames, and would need to be modified in order to account for the influence of beam/slab interaction.

For situations in which partially encased beams are utilised, the concrete encasement of the column web may be accounted for in determining the

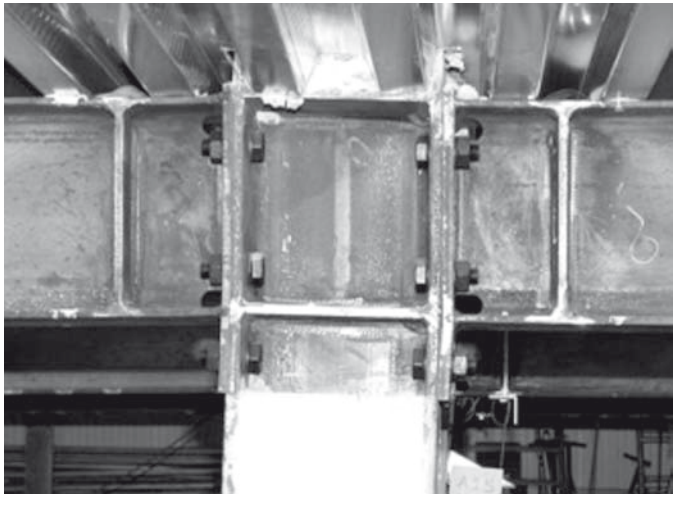


*Figure 7.4* View of column panel zone in a composite moment frame during testing

resistance of the panel zone. According to Section 7.5.4 of EN 1998-1, the resistance can be evaluated as the sum of the contributions from the concrete and steel panels. However, the aspect ratio of the panel zone $h_b/h_c$ has to be between 0.6 and 1.4, and the design shear force $V_{wp,Ed}$ derived from the plastic capacity of adjacent dissipative zones should be less than 80 per cent of the shear resistance $V_{wp,Rd}$ of the composite steel/concrete web panel according to EN 1994-1.

An important consideration is stipulated in Section 7.7.5 of EN 1998-1 whereby the dissipative zones at the beam ends of composite moment frames can be considered as steel-only sections (i.e. following *Concept c*). To achieve this, the slab needs to be *totally disconnected* from the steel members in a circular zone with a diameter of at least $2b_{eff}$ around the columns, with $b_{eff}$ determined on the basis of the larger effective width of the connected beams. This '*total disconnection*' also implies that there is no contact between the slab and the sides of any vertical element such as the columns, shear connectors, connecting plates or corrugated flange, etc.

The above consideration, of disregarding the composite action and designing for steel-only dissipative zones, can be convenient in practical design. Clearly, two *EI* values for the beams need to be accounted for in the analysis: composite in the middle and steel at the ends. The beams are composite in the middle, hence providing enhanced stiffness and capacity under gravity loading conditions. On the other hand, in the seismic situation, the use of steel dissipative zones avoids the need for detailed considerations in the slab, including those related to seismic rebars, effective width and ductility criteria associated with composite dissipative sections. This consideration also implies that the connections would be designed on the plastic capacity of the steel beams only. Also, the columns need to be capacity designed for the plastic resistance of steel instead of composite beam sections, which avoids over-sizing of the column members.

### *7.5.2 Composite braced frames*

As discussed before, in concentrically braced frames, the diagonal members, which are the main dissipative zones, should be in steel only according to the provisions of EC8. On the other hand, the beam and column members can be either steel or composite. The seismic design rules are therefore directly based on those for steel concentrically braced frames in Section 6 of EN 1998-1, since the ductility and capacity design requirements are largely related to the capacity of the diagonal braces. It should be noted, however, that buckling restrained braces (or unbonded braces) are not covered by the current version of EC8.

For composite eccentrically braced frames, the design rules again follow closely those stipulated for steel frames in Section 6 of EN 1998-1. The code recommends that the link zones are steel sections, which should not be encased, although it is noted that they can be connected to the slab. The

code stipulates that the links should be of short or intermediate length, and provides a number of additional requirements. Most importantly, if the link beam is connected to the slab, the concrete contribution should be ignored in determining the resistance of the link except when performing capacity design checks for members and components other than the dissipative zones.

### *7.5.3 Composite wall configurations*

As discussed in Section 7.2 above, a number of composite wall systems are referred to in EN 1998-1. Specific criteria related to the design and detailing of these systems are given Section 7.10 of the code. These include several useful figures outlining detailing requirements for partially encased and fully encased boundary elements for DCM and DCH, as well as details for coupling beams framing into walls. This part of the code offers guidance for the design and detailing of wall configurations including boundary elements, coupling beams and steel plates. For most aspects, it refers to the provisions of reinforced concrete design (Section 6 of EN 1998-1 and Chapter 5 of this book) as well as other parts of Section 7 of the code that are related to rules for critical members.

## 7.6 Other design considerations

In terms of material properties, apart from the requirements in the concrete and steel parts in EC8 (Sections 5 and 6 of EN 1998-1), additional criteria are specified in Section 7.2 of the composite part (Sections 7 of EN 1998-1). In dissipative zones, the concrete class should not be less than C20/25 and not higher than C40/50; the upper limit is imposed since the use of typical plastic capacity calculations for composite cross sections may become unreliable when concrete of relatively high strength is employed. For dissipative composite zones, the reinforcement should be of Class B or C for DCM, and should be Class C for DCH. In addition, Class B or C reinforcement should be used in highly stressed regions of non-dissipative zones. Except for closed stirrups or cross ties, only ribbed bars are allowed as reinforcing steel in highly stressed regions. It is also important to note that non-ductile welded meshes are not recommended in composite dissipative zones. If they are used, ductile reinforcement duplicating the mesh should be placed and their resistance should be accounted for.

A number of general requirements related to the design and detailing of dissipative zones are also included in Sections 7.5 and 7.6 of EN 1998-1. As in steel frames, it is stipulated that dissipative zones may be located in the structural members or in the connections; accordingly, capacity design checks of non-dissipative elements should be based on the plastic resistance of either the dissipative members or connections, respectively. In general, two plastic resistances for composite dissipative zones, reflecting the lower and upper bound estimates, should be determined. The former considers only

the steel and reliably ductile concrete portions (for assessing the dependable resistance), whilst the latter accounts for the steel and concrete portions (for determining the overstrength necessary for capacity design checks).

## 7.7 Design example – composite moment frame

### *7.7.1 Introduction*

The same eight-storey building considered in previous chapters is utilised in this example. The layout of the structure is reproduced in Figure 7.5. The main seismic checks are carried out for a preliminary design according to EN 1998-1. Consideration is only given to the lateral system in the X-direction of the plan, in which resistance is assumed to be provided by moment resisting frames spaced at 4 m. It is also assumed that an independent bracing system is provided in the transverse direction (Y) of the plan. Grades S275, S500 and C30/37 are assumed for structural steel, reinforcement and concrete, respectively.

The gravity and seismic loads are assumed to be the same as those adopted in the steel moment frame example presented in the previous chapter (Tables 6.3 and 6.4 of Chapter 6). As in the steel moment frame case, the example focuses on the design of the moment frame located on GL 2, as indicated in Figure 7.5.

The frame on GL 2 was firstly designed for the non-seismic/gravity loading combinations corresponding to both ultimate and serviceability limit states, according to the provisions of EC3 (EN 1993-1) and EC4 (EN 1994-1). On this basis, the initial sections adopted were partially encased HEA340 for all

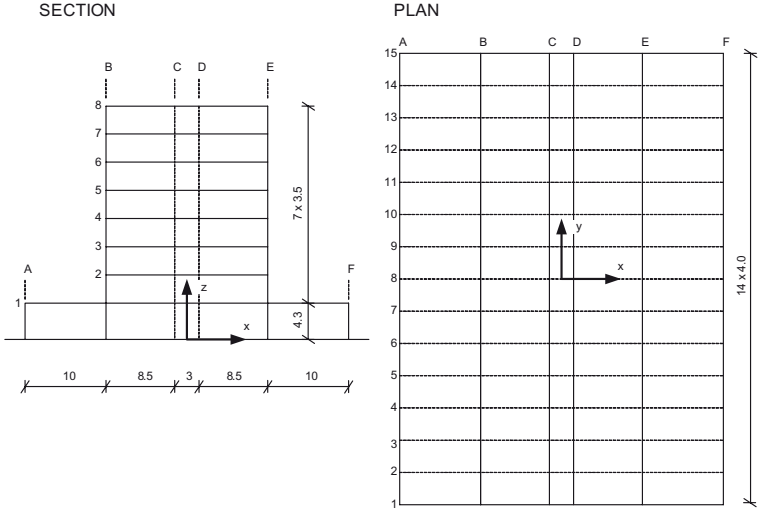



*Figure 7.5* Frame layout

columns and IPE450 steel profiles (in conjunction with a 150 mm solid slab) for the beams. Composite action is achieved through the incorporation of shear studs in order to attain full interaction according to the provisions of EC4.

### *7.7.2 Seismic design checks*

#### *7.7.2.1 Initial considerations*

A preliminary elastic analysis was firstly carried out using the estimated seismic loads for the frame incorporating the initial member sizes. These initial member sizes were, however, found to be inadequate to fulfil both strength and damage limitation requirements. Accordingly, the partially encased columns were increased to HEA500 in the lower four storeys and to HEA450 in the upper storeys.

*Concept b*, in which the contribution of concrete is accounted for in dissipative zones, is considered in this example. According to Cl. 7.6.3 of EN 1998-1, the effective widths assumed in the seismic analysis and design of the frame, are presented in Table 7.3. It is assumed that seismic rebars can be anchored to a concrete cantilever edge strip or to a transverse beam.

According to Cl. 7.7.2(3) of EN 1998-1, the structural analysis can be performed using equivalent properties for the entire beam instead of considering two flexural stiffnesses (for cracked and uncracked section). Therefore, $EI_{eq}$ (as presented in Equation (7.3) above) was used within a linear elastic analysis of the frame. A view of the frame model showing the element numbers is shown in Figure 7.6. The results from the seismic loading combination are initially used in the evaluation of the inter-storey drift sensitivity coefficients θ as listed in Table 7.4. As shown in the table, θ does not exceed the lower limit of 0.1 and hence second-order effects do not have to be considered in the analysis.

#### *7.7.2.2 Beam design checks*

For illustration, the beam design checks are performed for one of the critical members, which is the 3 m composite beam located on the third floor

*Table 7.3* Effective widths according to EC8

| | *Analysis* | *Resistance* |
|---|---|---|
| Positive moment | $2 \times 0.0375L = 0.075L$ | $2 \times 0.075L = 0.15L$ |
| Negative moment | $2 \times 0.05L = 0.1L$ | $2 \times 0.1L = 0.2L$ |

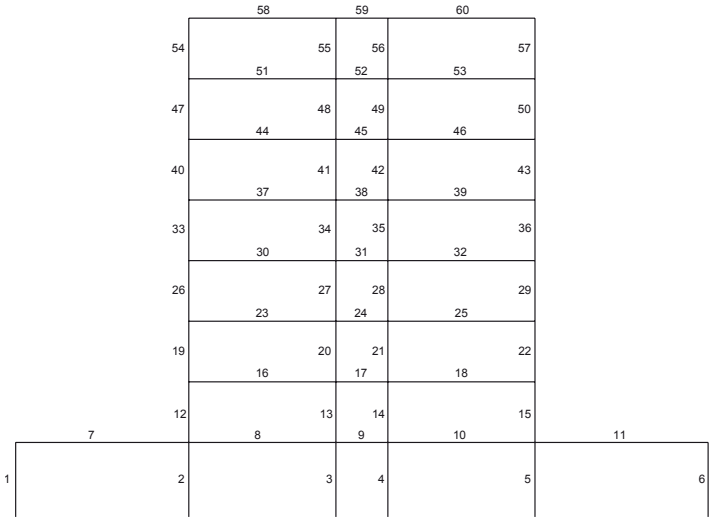



*Figure 7.6* Frame model with element numbers

*Table 7.4* Calculation of the inter-storey drift sensitivity coefficients

| *Level* | $d_e$ *(mm)* | $d_s$ *(mm)* | $d_r$ *(mm)* | $P_{tot}$ *(kN)* | $V_{tot}$ *(kN)* | *h (mm)* | θ |
|---|---|---|---|---|---|---|---|
| 8 | 77.7 | 310.8 | 16.4 | 453 | 98.2 | 3500 | 0.022 |
| 7 | 73.6 | 294.4 | 27.2 | 1170 | 241.7 | 3500 | 0.038 |
| 6 | 66.8 | 267.2 | 38.0 | 1888 | 365.4 | 3500 | 0.056 |
| 5 | 57.3 | 229.2 | 46.8 | 2606 | 469.3 | 3500 | 0.074 |
| 4 | 45.6 | 182.4 | 49.2 | 3323 | 553.3 | 3500 | 0.084 |
| 3 | 33.3 | 133.2 | 52.4 | 4041 | 617.4 | 3500 | 0.098 |
| 2 | 20.2 | 80.8 | 48.8 | 4758 | 661.6 | 3500 | 0.100 |
| 1 | 8.0 | 32.0 | 32.0 | 5476 | 703.2 | 4300 | 0.058 |

(Element 17 in Figure 7.6). The internal forces at both ends of the member are listed in Table 7.5.

Based on the values from the table, the seismic demands on the beam are:

$M_{Ed,left} = -36.9 + 414.2 = -377.3$ kN.m

$M_{Ed,right} = -36.9 + (-414.2) = -451.1$ kN.m

$N_{Ed} = 27.8 + 0.0 = 27.8$ kN

According to Cl. 7.7.3(3) of EC8, which refers to Cl. 6.6.2(2), and considering the properties of the composite beam:

*Table 7.5* Internal forces in Element 17

| | *Left end* | | *Right end* | |
|---|---|---|---|---|
| | $G_k+0.3Q_k$ | $E$ | $G_k+0.3Q_k$ | $E$ |
| *M (kN.m)* | –36.9 | 414.2 | –36.9 | –414.2 |
| *V (kN)* | 45.2 | –276.2 | –45.2 | –276.2 |
| *N (kN)* | 27.8 | 0.0 | 27.8 | 0.0 |

$M_{Ed,left} \le M_{pl,Rd} \rightarrow 377.3 \le 704.0$ kN.m
(considering 3 bars of 12 mm diameter within the 0.45 m of effective width)

$M_{Ed,right} \le M_{pl,Rd} \rightarrow 451.1 \le 529.8$ kN.m
(considering 4 bars of 12 mm diameter within the 0.60 m of effective width)

$$N_{Ed} \le 0.15N_{pl,Rd} \rightarrow 27.8 \le 0.15 \times 98.8.10^{-4} \times 275.10^{3}$$
$$\rightarrow 27.5 \text{ kN} \le 407.6 \text{ kN}$$

$$V_{Ed} \le 0.5V_{pl,Rd} \rightarrow V_{Ed} = V_{Ed,G} + V_{Ed,M}$$
$$= 45.2 + (529.8+704.0)/3.0 = 456.5 \text{ kN}$$
$$\rightarrow 456.5 \le 0.5 \times 50.85 \,.\, 10^{-4} \times 275.10^{3} / \sqrt{3}$$
$$\rightarrow 456.5 \text{ kN} \le 403.7 \text{ kN not satisfied!}$$

From the above calculations it is clear that the shear design check cannot be satisfied, which is largely a consequence of the short length of the beam. Increasing the beam strength would lead to higher moment capacity, hence higher shears. Also, as a result of capacity design requirements, this would necessitate over-sizing of the columns. It is therefore suggested that specific measures are taken in order to increase the shear capacity through supplementary web plates within the critical short beams.

Cl. 7.6.2(7) of EC8 stipulates upper limits for the ratio $x/d$ of the dissipative composite cross section as indicated in Equation (7.1) and Figure 7.1 of this chapter. For the critical beam considered above, this ratio is found to be around 0.27 (i.e. 0.16/0.60), which is lower than the limit of 0.32 derived from the equation.

The adoption of *Concept b* also requires that specific detailing rules are verified in order to ensure reliable dissipative behaviour in the concrete parts. According to Clauses C.3.2.2(2) and C.3.3.1(2) in Annex C of EC8, *transverse reinforcement* (or *seismic rebars*) should be positioned in the joint region in order to allow the mobilisation of Mechanism 2 (defined in Figure C.2 of EN 1998-1, consisting of concrete diagonal compressive struts resisted by the internal region of the column). The area of reinforcement ($A_T$) is given by:

$$A_T \ge F_{Rd2} / f_{yd,T} = 0.7 \times h_c \times d_{eff} \times f_{cd} / f_{yd,T}$$

where $h_c$ is the depth of the column section, $d_{eff}$ is the overall depth of slab, $f_{cd}$ is the concrete design compressive strength and $f_{yd,T}$ is the design yield strength of the reinforcement.

The area of transverse reinforcement required at the joints within the lower four storeys is:

$$A_T \geq 0.7 \times 0.490 \times 0.150 \times (30000 / 1.5) / (500000 / 1.15)$$

$A_T \geq 23.67$ cm$^2$ → 5 bars of 25 mm of diameter positioned over a beam length of 0.49 m.

The area of seismic rebars to use at joints in the upper five storeys is given by:

$$A_T \geq 0.7 \times 0.440 \times 0.150 \times (30000 / 1.5) / (500000 / 1.15)$$

$A_T \geq 21.25$ cm$^2$ → 5 bars of 25 mm of diameter positioned over a beam length of 0.44 m.

*7.7.2.3 Column design checks*

Except at the base, the columns should be capacity designed according to the weak beam/strong column dissipative mechanism. According to Cl. 7.7.3(5) of EC8, which refers directly to Cl. 6.6.3, the design forces are obtained using the following combination:

$$E_d = E_{d,Gk+0.3Qk} + 1.1\gamma_{ov}\Omega E_{d,E}$$

where $\gamma_{ov}$ is the overstrength factor assumed as 1.25

$$\Omega = \min (M_{pl,Rd,i}/M_{Ed,i}) = 529.8/476.5 = 1.11$$

(note that –476.5 kNm is obtained from Element 25).

The design combination for consideration in column checks is therefore given by:

$$E_d = E_{d,Gk+0.3Qk} + 1.53E_{d,E}$$

The design forces for a critical column (Element 4 in Figure 7.6) are presented in Table 7.6 for illustration.

Based on these values, the seismic demands at the bottom end of the column are:

$$M_{Ed} = 21.7 + 1.53\times(-358.0) = -526.0 \text{ kN.m}$$

$$N_{Ed} = -1529.5 + 1.53\times(-828.3) = -2796.8 \text{ kN}$$

*Table 7.6* Internal forces in Element 4

| | *Bottom end* | | *Top end* | |
|---|---|---|---|---|
| | $G_k+0.3Q_k$ | $E$ | $G_k+0.3Q_k$ | $E$ |
| *M (kN.m)* | 21.7 | –358.0 | –41.8 | 175.7 |
| *V (kN)* | –14.8 | 124.1 | –14.8 | 124.1 |
| *N (kN)* | –1529.5 | –828.3 | –1520.6 | –828.3 |

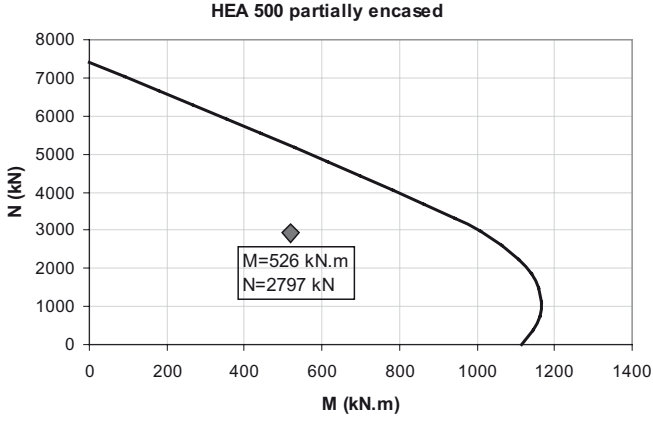



*Figure 7.7* Interaction curve for the first storey composite column

$$V_{Ed} = -14.8 + 1.53 \times 124.1 = 175.1 \text{ kN}$$

The design checks are performed according to EC4. For brevity, only cross-section checks are presented, but clearly all EC4 resistance checks including those for member stability should also be satisfied. Considering a partially encased HEA500 cross section, for which the axial/bending interaction curve is depicted in Figure 7.7, it is evident that the composite column cross section is able to satisfy the seismic demands. It is also necessary to check the level of shear applied on the cross section:

$$V_{Ed} \le 0.5 V_{pl,Rd}$$

$$175.1 \le 0.5 \times 74.72 \times 10^{-4} \times 275 \times 10^{3}/\sqrt{3}$$

$$175.1 \text{ kN} \le 593.2 \text{ kN}$$

In addition to the member checks, Cl. 4.4.2.3(4) of EC8 also requires that at every joint the following condition is satisfied:

$$\frac{\sum M_{Rc}}{\sum M_{Rb}} \ge 1.3$$

where $\sum M_{Rc}$ and $\sum M_{Rb}$ are the sum of the design moments of resistance of the columns and of the beams framing the joint, respectively. For illustration, this check is carried out for an internal joint located at the first floor of the frame, as follows:

$\sum M_{Rc} = 2 \times 1030 = 2060$ kN.m
(for a level of axial force of around 2800 kN)

$\sum M_{Rb} = 704.0 + 529.8 = 1233.8$ kN

$\sum M_{Rc} / \sum M_{Rb} \geq 1.3$

*7.7.2.4 Joint design checks*

According to Cl. 6.6.3(6) of EC8, the web panel zones at beam-to-column connections should be designed to resist the forces developed in the adjacent dissipative elements, which are the connected beams. For each panel zone the following expression should be verified:

$$\frac{V_{wp,Ed}}{V_{wp,Rd}} \leq 1.0$$

where $V_{wp,Ed}$ is the design shear force in the web panel accounting for the plastic resistance of the adjacent beams/connections and $V_{wp,Rd}$ is the shear resistance of the panel zone according to EC3. For illustration, these checks are performed for internal and external joint panel zones.

EXTERNAL PANEL ZONE (HEA 500 + IPE 450)

$V_{wp,Ed} = M_{pl,Rd} / (d_b - t_{bf}) = 704.0 / (0.450 - 0.0146) = 1617$ kN

$V_{wp,Rd} = 0.9 f_{y,wc} A_{vc} / (\sqrt{3}\, \gamma_{M0}) + 4 M_{pl,fc.Rd} / (d_b - t_{bf})$ (Cl. 6.2.6 of EC3 Part 1.8)

$= 0.9 \times 275 \times 10^3 \times 74.72 \times 10^{-4} / \sqrt{3} + 4 \times 0.490 \times 0.023^2 \times 275 \times 10^3 / (4 \times (0.450 - 0.0146)$

$= 1068 + 164 = 1232$ kN

$V_{wp,Ed} / V_{wp,Rd} = 1617 / 1232 \leq 1.0$ not satisfied $\rightarrow$ a doubler plate is required

Design of a single supplementary doubler plate with a width of 300 mm:

$1617 = 1232 + t_{dp} \times 0.300 \times 0.9 \times 275 \times 10^3 / \sqrt{3}$

$t_{dp} = 8.98$ mm $\rightarrow$ $t_{dp} = 9$ mm

INTERNAL PANEL ZONE (IPE 450 + HEA 500 + IPE 450)

$V_{wp,Ed} = \sum M_{pl,Rd} / (d_b - t_{bf}) = (704.0 + 529.8) / (0.450 - 0.0146)$
$= 2834$ kN

$V_{wp,Ed} / V_{wp,Rd} = 2834 / 1232 \leq 1.0$ not satisfied → a doubler plate is required

Design of a single supplementary doubler plate with a width of 300 mm:

$2834 = 1232 + t_{dp} \times 0.300 \times 0.9 \times 275 \times 10^3 / \sqrt{3}$

$t_{dp} = 37.3$ mm → $t_{dp} = 38$ mm

### *7.7.3 Damage limitation*

According to Cl. 4.4.3.2(1) for the damage limitation (serviceability) limit state:

$$d_r \nu \leq 0.01h$$

where $d_r$ is the design inter-storey drift, $\nu$ is a reduction factor that takes into account the lower return period of the frequent earthquake and is assumed as 0.5, and $h$ is the storey height. The limit of 1 per cent is applicable to cases where the non-structural components are fixed to the structure in a way that does not interfere with structural deformation. For cases with non-ductile or brittle non-structural elements this limit is reduced to 0.75 per cent and 0.5 per cent, respectively.

Based on the results provided in Table 7.4, the maximum inter-storey drift occurs at the third storey:

$d_r = 52.4$ mm

$d_r \nu \leq 0.01h$

$52.4 \times 0.5 \leq 0.01 \times 3500$

26.2 mm < 35 mm (satisfies limit, provided that non-interfering non-structural elements are used).

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# 8 Shallow foundations

*S.P.G. Madabhushi, I. Thusyanthan, Z. Lubkowski and A. Pecker*

## 8.1 Introduction

### *8.1.1 Dynamic properties of structures*

Practical and comprehensive seismic design methods for both shallow and deep foundations are currently some way from being fully established. The 1990s have seen the publication of a series of important papers on the seismic design of shallow foundations. However, several important gaps between theory and observation are yet to be bridged. In 1987 Dowrick could write that an authoritative rationale for the design of deep foundations did not exist. Progress has been made in the intervening years but substantial further work is required fully to establish seismic design methods both for shallow and deep foundations (e.g. Pappin, 1991; Pender, 1996). It is therefore pleasing to state that EN 1998-5:2004 provides one of the most comprehensive codes of practice for addressing seismic foundation design.

In order to develop robust and reliable design methods good observations of field performance are indispensable. Despite the problems of making such observations on foundations, evidence has accumulated from major earthquakes over the past few decades. Many of these, from Alaska in 1964 to the Hyogoken-Nambu (Kobe) earthquake of 1995 have provided evidence of the highly damaging effects of liquefaction and lateral spreading for both shallow and deep foundations (Ross et al, 1969; Tokimatsu and Asaka, 1998). The unusual soft clay conditions of Mexico City (1985) gave rise to a range of foundation failures rarely reported elsewhere (Mendoza and Auvinet, 1988; Zeevaert, 1991). In general, spread footings properly designed for static loadings have been observed to perform adequately under seismic loading although cases of significant settlement have been reported (Richards et al, 1993). In contrast, the poor performance of raked or battered piles in bridge abutments and jetties has been highlighted in several earthquakes including Loma Prieta in 1989 (Seed et al, 1991).

Current trends suggest that displacement based design methodologies will come in time to play a major role in seismic foundation design, but their full development is yet to come.

### *8.1.2 Overview of soil behaviour*

The ground presents the earthquake engineer with major challenges. Its behaviour often falls far short of that which would be desirable for the support of structures in earthquakes. However, the engineer is obliged to work with the ground, modifying its performance where necessary, to produce economic and safe foundations. The ground properties of particular interest when considering the seismic design of foundations are its stiffness, damping and strength. In addition its density and degree of over-consolidation provide important indicators of its likely behaviour under seismic excitation.

The stiffness of both granular and cohesive soils is highly non-linear with stiffness reducing with increasing strain once a threshold strain has been exceeded. While the stiffness of soils reduces with increasing strain, the hysteretic or material damping increases. It should also be noted that the stiffness and the hysteretic damping of soils are generally not dependent on the frequency of the loading.

The non-linear behaviour of soils poses significant problems to the designer for assessing the foundation response to both static and dynamic loadings. Close to the foundation structural element (e.g. the footing or pile), strains may be high and the ground response soft. However, at more remote locations the strains will be small and the ground behaviour will be stiff. The response of the foundation to loading will represent an integration of both the near field and the far field strains (see Jardine et al, 1986). This makes the use of linear elastic solutions difficult except at very small load levels where the strains may be largely below the threshold values at all locations. At higher loadings the choice of a representative stiffness and damping values requires considerable care.

The stiffness and strength of soils are dependent on the effective stresses in the ground. The effective stresses are directly affected by the pore water pressures. These can vary both as a result of fluctuations in groundwater levels and as a result of stress changes in the ground. For saturated loose sands or normally to lightly over-consolidated clays, cyclic shear stresses produce increases in pore water pressure and progressive losses of strength and stiffness.

The designer must not forget that natural ground is a heterogeneous material created by complex and often non-uniform processes. The information provided by a normal ground investigation will strictly relate only to a tiny fraction of the volume of ground that will affect the behaviour of structural foundations. The influences of foundation construction imposed loadings and of the seismic excitation itself will alter the ground's behaviour. Hence the designer requires a good understanding of the processes involved and a suitably cautious attitude.

### 8.1.3 Soil stiffness

The response of the foundation as a system under earthquake loading is strongly affected by the stiffness profile and depth of the foundation stratum. These affect the fundamental frequency of the stratum, which in turn affects the amplification of bedrock motions to the surface and the foundation system damping characteristics. Three stiffness profiles are shown in Figure 8.1. The fundamental frequencies of the strata are given by:

$$\text{Constant stiffness with depth: } f = \frac{0.25 v_s}{H} \tag{8.1}$$

$$\text{Parabolic stiffness with depth: } f = \frac{0.22 v_s}{H} \tag{8.2}$$

$$\text{Linearly increasing stiffness: } f = \frac{0.19 v_s}{H} \tag{8.3}$$

where $v_s$ = the shear wave velocity at the base of the stratum (depth = $H$).

Additional results for an arbitrary increase with depth stiffness profile are given by Pecker (2005).

The stiffness, damping and strength characteristics of the soil column affect the transmission of seismic motions from the bedrock to the ground surface. Various methods are used to assess the seismic motions at foundation level. Direct measurements may be available from the site or another site with similar characteristics. A word of caution is necessary here because measurements are only valid if the range of induced strains in the soil profile is representative of the design situation. Code requirements may be used for the class of site (usually based on depth of deposit and shear wave velocity) and site location. Alternatively the ground motions may be calculated using one-dimensional wave propagation codes such as SHAKE (Schnabel et al,

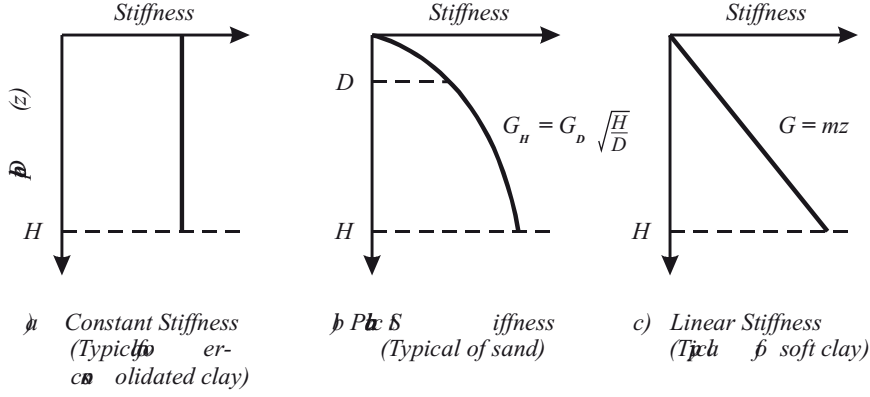



*Figure 8.1* Typical stiffness profiles for foundation strata

*Table 8.1* Average soil damping ratios and reduction factors

| *Ground acceleration ratio,* $\alpha S$ *(g)* | *Damping ratio* | $\frac{\nu_s}{\nu_{s,\max}}$ | $\frac{G}{G_{\max}}$ |
|---|---|---|---|
| 0.10 | 0.03 | 0.90 (±0.07) | 0.80 (±0.10) |
| 0.20 | 0.06 | 0.70 (±0.15) | 0.50 (±0.20) |
| 0.30 | 0.10 | 0.60 (±0.15) | 0.36 (±0.20) |

$\nu_{s,\max}$ is the average $\nu_s$ value at small strain ($< 10^{-5}$), not exceeding 360 m/s.
$G_{\max}$ is the average shear modulus at small strain.
Note: Through the ± one standard deviation ranges the designer can introduce different amounts of conservatism, depending on such factors as stiffness and layering of the soil profile. Values of $\nu_s/\nu_{s,\max}$ and $G/G_{\max}$ above the average could, for example, be used for stiffer profiles, and values of $\nu_s/\nu_{s,\max}$ and $G/G_{\max}$ below the average could be used for softer profiles.

1972), SIREN (Pappin, 1991) or DESRA-2 (Lee and Finn, 1978) and where the geometry is more complex, 2D wave propagation codes such as FLUSH (Lysmer et al, 1975) or DYNA3D.

Section 4.2.3 of EN 1998-5 provides a table that gives average soil damping and reduction factors (Table 8.1) for shear wave velocity and shear modulus, which can be used in the absence of specific measurements.

The maximum acceleration that can be transmitted through a soil stratum in a seismic event is limited by the shear strength of the soil. For dry non-cohesive soils the peak acceleration is given by:

$$k_{h\lim} = \tan\varphi \tag{8.4}$$

Application of this relationship indicates limiting horizontal accelerations of between 0.5 g and 0.8 g for dry granular soils. In the case of loose saturated deposits, the effects of cyclic loading are likely to lead to liquefaction at acceleration levels significantly below the limit given in Equation (8.4). Once liquefaction occurs the liquefied horizon substantially reduces the transmitted peak acceleration. However, it may be noted that several cycles of earthquake loading are likely to occur prior to the general onset of liquefaction. The appropriate strength to use for saturated granular materials (loose or dense) is discussed to some extent in Pecker (2005). For dense saturated sands, values in excess of those given by Equation (8.4) may be achieved.

For cohesive soils the equivalent relationship for limiting horizontal acceleration is:

$$k_{h\lim} = \frac{s_u}{\gamma h} \tag{8.5}$$

where $s_u$ = the undrained shear strength at depth $h$
$\gamma$ = the average bulk unit weight of the soil

The non-linear stress-strain response of soils result in different amplifications of the bedrock motion through the soil column, depending on the magnitude of the earthquake motions (Idriss, 1990). The greatest amplification of bedrock accelerations occurs at low peak acceleration levels. As the peak acceleration level increases for larger earthquakes, so the amplification of the soil column decreases and becomes less than unity at high bedrock acceleration levels. Observations from soft soil sites suggest that the crossover from amplification to de-amplification occurs at bedrock accelerations of about 0.3 to 0.5 g (Mohammadioun and Pecker, 1984; Idriss, 1990; Suetomi and Yoshida, 1998; Kokusho and Matsumoto, 1998).

### *8.1.4 Soil strength*

The value of the soil strength parameters applicable under static undrained conditions may generally be used. As in EN 1997 these are characteristic strength parameters, which are defined as a cautious estimate of the value affecting the occurrence of the limit state. Further discussion on selection of characteristic soil parameters can be found in Frank et al (2004).

For cohesive soils the appropriate strength parameter is the undrained shear strength $c_s$, adjusted for the rapid rate of loading and cyclic degradation effects under the earthquake loads when such an adjustment is needed and justified by adequate experimental evidence. For cohesionless soil the appropriate strength parameter is the cyclic undrained shear strength $\tau_{cy,u}$, which should take the possible pore pressure build-up into account.

Alternatively, effective strength parameters with appropriate pore water pressure generated during cyclic loading may be used. For rocks the unconfined compressive strength, $q_u$, may be used.

EN 1998-5 requires that a partial factor ($\gamma_M$) is applied to the material properties $c_s$, $\tau_{cy,u}$ and $q_u$. These are denoted as $\gamma_s$, $\gamma_{\tau cy}$ and $\gamma_{qu}$, and those for tan $\phi'$ are denoted as $\gamma_{\phi'}$. The recommended values are $\gamma_s = 1.4$, $\gamma_{\tau cy} = 1.25$, $\gamma_{qu} = 1.4$ and $\gamma_{\phi'} = 1.25$ (EN 1998-5:2004, p13).

## 8.2 Siting requirements

### *8.2.1 General*

The primary cause of building damage has been identified as ground shaking; however, in most earthquakes the overall damage to buildings is caused by more than one hazard. The principal secondary cause of building damage is ground failure, which can be divided into five elements, namely fault rupture, topographic amplification, slope instability, liquefaction and shakedown settlement (Bird and Bommer, 2004).

Section 4 of EN 1998-5:2004 requires that these earthquake phenomena are identified and hence they can be minimised. Figure 8.2 shows just a few examples where a failure to assess these phenomena has impinged



*(a)* Uplift of a building due to fault rupture – Ji-Ji earthquake 1999



*(b)* Punching failure of shallow foundations due to soil liquefaction – Kocaeli earthquake 1999



*(c)* Building damage close to a steep slope – Northridge earthquake 1994

*Figure 8.2* Examples of poorly sited structures

on the performance of structures during a major earthquake. By ensuring these potential hazards at a site are identified, the designer can then take appropriate actions to minimise those hazards. Further discussion of general siting issues is provided in Chapter 4.

### *8.2.2 Active faults*

Section 4.1.2 of EN 1998-5 states that buildings of importance classes II, III and IV (i.e. all buildings except agricultural buildings) should not be sited in the immediate vicinity of active tectonic faults.

No minimum distance requirement between a building and an active fault is quoted. Requirements in countries such as New Zealand, Russia and the USA range from about 15 m to 200 m.

In areas of high seismicity the code requires geological investigations to be carried out for important structures near active tectonic faults, in order to determine the hazard in terms of ground rupture and severity of ground shaking.

For structures that are not critical to public safety the absence of movement in the Late Quaternary (last 10,000 years) may be used to define non-active faults.

### *8.2.3 Slopes*

Structures adjacent to slopes may be subject to two different phenomena, firstly slope instability and secondly topographic amplification.

### *8.2.4 Slope instability*

As part of a natural process, slopes undergo a process of landsliding in order to reduce their slope angle and to re-establish equilibrium. This process takes place in a variety of forms such as soil creep, cambering and rotational slips. These have different effects on structures, but the degree to which they affect a structure will also depend on the foundations of the structure itself.

Where instability is shallow, for example where there is soil creep or flow sliding, the foundations of the structure may displace unless constructed beneath the plane along which slipping is occurring, as shown in Figure 8.3(a).

Where the foundations are at depth, the structure should remain stable and may help stabilise the local area, as shown in Figure 8.3(b). However, material may eventually build up behind the structure and apply a horizontal pressure to it. The possible effects of this should be taken into consideration.

Where deep landslides occur, there is a greater risk of severe structural damage, with structures being translated downhill or undermined by the slip, as shown in Figure 8.3(c). While previous deep landslides may be stabilised to some extent, for example by incorporating drainage measures and/or regrading, construction in these areas should be avoided where possible.

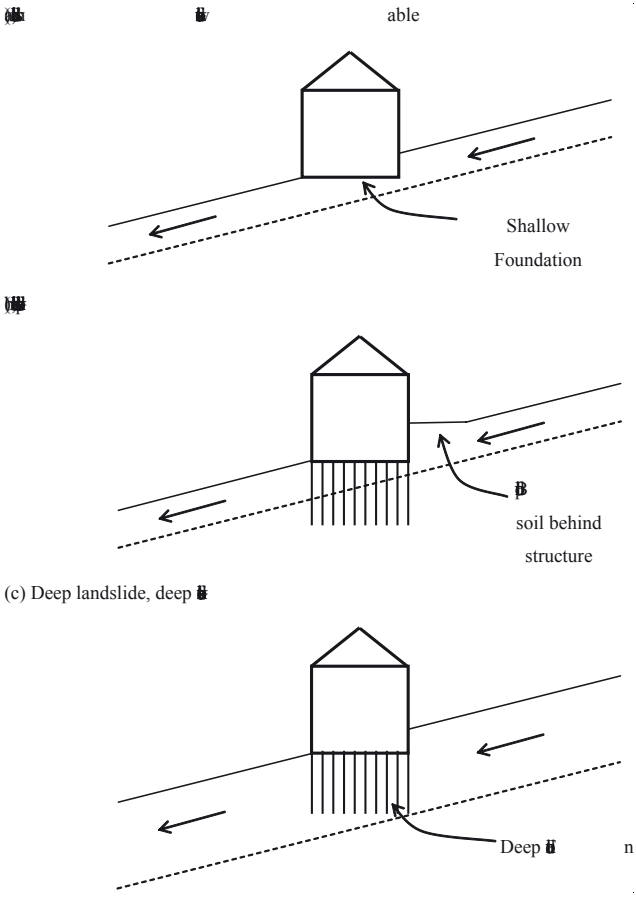



*Figure 8.3* Vulnerability of structures to landslide hazard

The code recommends the pseudo-static method of analysis to determine the degree of slope instability.

### *8.2.5 Topographic amplification*

Earthquake ground motion experienced near the top of a slope or ridge is often greater than the ground motion felt on level ground, assuming similar soil conditions. This phenomenon is referred to as the topographical effect and is understood to be a function of the height and inclination of the topography and the wavelength of the ground motion. Topographical effect studies were first conducted after the 1971 San Fernando earthquake in California. The amplitude of ground motions at the crest of hills were 2 to 3 times greater than amplitudes measured at the base of the hills (Finn

et al, 1995). Similar results have been reported from Matsuzaki in Japan, Northridge in California and many other parts of the world.

Section 4.1.3.2 of EN 1998-5:2004 requires that, for structures erected near slopes, the amplification factor ($S_T$) should be determined and applied to the seismic action derived in Section 3.2.2 of EN 1998-1 (2004). Simple guidelines on determining $S_T$ are given in Annex A of the code. The following limits on this effect should be noted:

- The amplification factor should, in general, only be applied to long ridges and cliffs of height greater than about 30 m.
- For average slope angles of less than about 15° the topography effects may be neglected.

It should be noted that ground motions are also strongly influenced by subsurface topography, though this is not considered explicitly in EN 1998. A major factor contributing to the amplification of ground motion and increased damage in alluvium filled valleys and basins is interpreted to be the generation of surface waves at the valley edges and the reflection of these waves back and forth through the alluvium infilling the valleys. This phenomenon is referred to as the basin effect (Borcherdt and Glassmoyer, 1992; Finn et al, 1995). Basin effects were interpreted to have significantly influenced the characteristics of the earthquake ground motion and location of major damage centres in the Los Angeles and San Fernando Basins during recent earthquakes (Borcherdt and Glassmoyer, 1992). The most common manifestations of the basin effect are an increase in the duration and a shift to lower frequencies that are more damaging for taller structures during an earthquake's strong ground shaking.

## 8.3 Liquefaction

### *8.3.1 Effect of soil liquefaction on structures*

Liquefaction is a process by which non-cohesive or granular sediments below the water table temporarily lose strength and behave as a viscous liquid rather than a solid when subjected to strong ground shaking during an earthquake. Typically, saturated, poorly graded, loose, granular deposits with a low fines content are most susceptible to liquefaction.

Liquefaction does not occur at random, but is restricted to certain geological and hydrological environments, primarily recently deposited sands and silts in areas with high ground water levels. Dense and more clayey soils, including well compacted fills, and older deposits (Pleistocene deposits; Youd and Perkins, 1978) have low susceptibility to liquefaction.

The liquefaction process itself may not necessarily be particularly damaging or hazardous. For engineering purposes, it is not the occurrence of liquefaction that is of importance, but the capability of the process and

associated hazards to cause damage to structures. The adverse effects of liquefaction can be summarised as follows:

- Flow failures – completely liquefied soil or blocks of intact material ride on a layer of liquefied soil. Flows can be large and develop on moderate to steep slopes.
- Lateral spreads – involve lateral displacement of superficial blocks of soil as a result of liquefaction of a subsurface layer. Spreads generally develop on gentle slopes and move toward a free face such as an incised river channel or coastline.
- Ground oscillation – where the ground is flat or the slope too gentle to allow lateral displacement, liquefaction at depth may disconnect overlying soils from the underlying ground, allowing the upper soil to oscillate back and forth in the form of ground waves. These oscillations are usually accompanied by ground fissures and fracture of rigid structures such as pavements and pipelines.
- Loss or reduction in bearing capacity – liquefaction is induced when earthquake shaking increases pore water pressures, which in turn causes the soil to lose its strength and hence bearing capacity.
- Settlement – soil settlement may occur as the pore-water pressures dissipate and the soil densifies after liquefaction. Settlement of structures may occur due to the reduction in bearing capacity or due to the ground displacements noted above.
- Increased lateral pressure on retaining walls – occurs when the soil behind a wall liquefies and so behaves as a 'heavy' fluid with no internal friction.
- Flotation of buried structures – occurs when buried structures such as tanks and pipes become buoyant in the liquefied soil.

Other manifestations of liquefaction, such as sand boils, can also occur and may pose a risk to structures, particularly through loss or reduction in bearing capacity and settlement.

### *8.3.2 Liquefaction potential*

Section 4.1.4 of EN 1998-5 describes the requirements for assessing liquefaction potential. Furthermore it provides a normative methodology in Annex B. It should, however, be noted that there have been numerous developments in liquefaction assessment methodologies in recent years (e.g. Seed et al, 2003; Boulanger and Idriss, 2004 etc.) and the methods described in the code may be potentially unconservative, especially for materials with high fines content. It is therefore recommended that an expert should be employed to carry out liquefaction assessment.

A liquefaction susceptibility evaluation should be made when the soil includes extended layers of thick lenses of loose sand (with or without

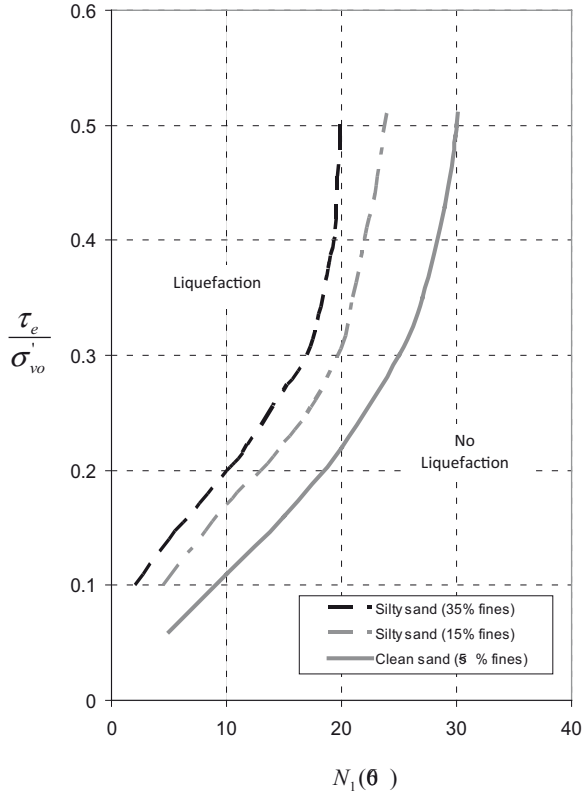



*Figure 8.4* Liquefaction assessment using corrected SPT values

silt/clay fines), beneath the water table and when the water table level is close to the ground surface. EN 1998-5 recommends that the shear stress approach is applied. In this method, the horizontal shear stresses generated by the earthquake are compared with the resistance available to prevent liquefaction. In Annex B of EN 1998-5 a set of liquefaction potential charts can be found for a magnitude $M_s$ = 7.5 earthquake. The shear stresses 'demand' are expressed in terms of a cyclic stress ratio (CSR), and the 'capacity' in terms of a cyclic resistance ratio (CRR).

The CRR is assessed based on corrected SPT blow count using the empirically derived liquefaction charts, which are shown schematically for silty sand in Figure 8.4. These charts compare CRR ($\tau/\sigma'_{v0}$), with corrected SPT blow count ($N_{1(60)}$). In Figure 8.4 the dependence of liquefaction potential on the percentage fines content in the silty soil is also seen by comparing the three lines. For a given corrected SPT blow count, clean sands with fines content of <5 per cent liquefy more easily compared to silty sands with a greater percentage of fines content. The procedure for correcting

the field $N$ values to obtain the corrected $N_{1(60)}$ is explained later in Section 8.3.3.

The CSR is assessed by first calculating the cyclic shear stress ($\tau_e$) using Equation (8.6).

$$\tau_e = 0.65 \cdot \alpha \cdot S \cdot \sigma_{v0} \tag{8.6}$$

where $\alpha$ is the ratio of the design ground acceleration on type A ground, $a_g$, to the acceleration of gravity, $g$, $S$ is the soil factor and $\sigma_{v0}$ is the overburden pressure.

It must be pointed out that Equation (8.6) is conservative because it neglects the stress reduction factor with depth ($r_d$).

This expression may not be applied for depths larger than 20 m. A soil shall be considered susceptible to liquefaction whenever $CRR > \lambda \times CSR$, where $\lambda$ is recommended to be 0.8, which corresponds to a factor of safety of 1.25.

If soils are found to be susceptible to liquefaction, mitigation measures such as ground improvement and piling (to transfer loads to layers not susceptible to liquefaction), should be considered to ensure foundation stability.

The use of pile foundations alone should be considered with caution due to the large forces induced in the piles by the loss of soil support in the liquefiable layers, and to the inevitable uncertainties in determining the location and thickness of such layers.

For buildings on shallow foundations, liquefaction evaluation may be omitted when the saturated sandy soils are found at depths greater than 15 m.

### *8.3.3 Design example on determination of liquefaction potential*

In this section we shall outline the liquefaction assessment for Site A, as described in Chapter 4. The foundations for the hotel building can take the form of shallow foundation provided that the chosen site does not pose a major risk of liquefaction. In other words, liquefaction potential of the chosen site should be low. The design of shallow foundations will be considered in this chapter. However, in certain sites where there is significant liquefaction risk, pile foundation may be preferred. The design of pile foundation will be considered in Chapter 9. In either case, it is important to carry out an assessment of liquefaction potential for any building site. The method for carrying out such an assessment on Site 'A' is shown in this section. As explained in Chapter 4, Site 'A' has loose sand layers below the water table. The borehole data obtained from site investigation is presented in Figure 8.5 along with the strength parameters and the water table.

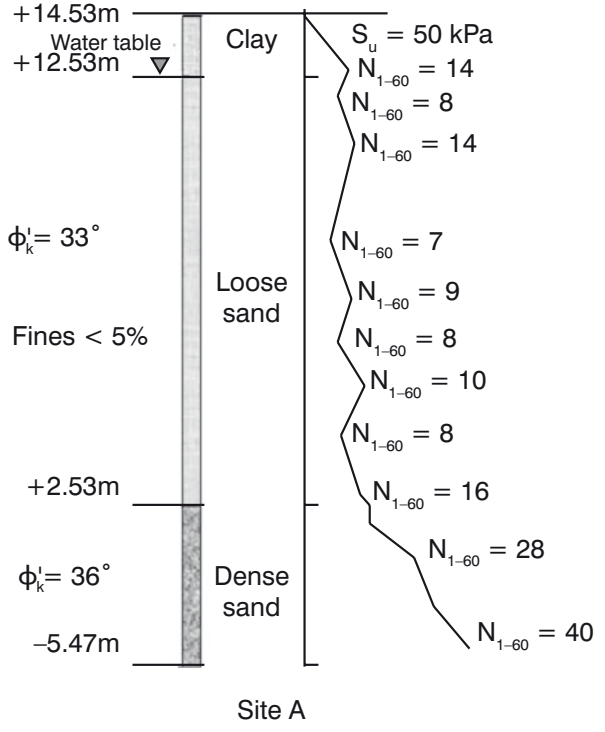



*Figure 8.5* Borehole data from Site A

*8.3.3.1 Check for liquefaction*

$N_{SPT}$ from a field SPT test are to be normalised as given below to obtain $N_1(60)$.

For the present site, this has already been done and values of $N_1(60)$ are given.

$$N_1(60) = N_{SPT}\sqrt{\frac{100}{\sigma'_{vo}}}\frac{ER}{60}$$

$\sigma'_{vo}$ in kPa (8.7)

*ER* = the ratio of the actual impact energy to the theoretical free-fall energy.

In Europe a value of *ER* = 70 per cent is commonly used. However, it is recommended that as much as possible, measurements of *ER* should be made at the start of the site investigation as the values for *ER* vary significantly from one equipment to another and even from one operator to another.

Liquefaction hazard may be neglected when $\alpha$ S < 0.15 and at least one of the following is satisfied (see pp 16–17 of EN 1998–5 (2004)).

1 the sands have clay content > 20 per cent with PI >10;
2 sands have silt content > 35 per cent and $N_1(60) > 20$;
3 sands are clean and $N_1(60) > 30$.

Seismic shear stress $\tau_e$

$$\tau_e = 0.65 \cdot \alpha \cdot S \cdot \sigma_{v0} \text{ for depths} < 20 \text{ m} \tag{8.8}$$

For the present case, $\alpha = 0.3$, $S = 1.15$. Also take the saturated unit weight of the clay and sand to be 20 kN/m$^3$.

$M_s$ is surface wave magnitude. For $M_s = 6$, from Table B.1 => CM is 2.2 (p 34, EC 8 Part 5).

In Table 8.2 the calculations for the seismic shear stress and its normalisation with the effective vertical stress and CM factor are presented for various depths.

*Table 8.2* Calculation of seismic shear stress with depth

| *Depth (m)* | *N1(60)* | *Total stress kPa* | *Effective stress kPa* | *Seismic shear stress kPa* | *Seismic shear stress/ effective stress* | *$(\tau_e/\sigma'_{vo})$/ CM* |
|---|---|---|---|---|---|---|
| 1 | 5 | 20 | 20 | 4.49 | 0.2243 | 0.10 |
| 2 | 5 | 40 | 40 | 8.97 | 0.2243 | 0.10 |
| 3 | 14 | 60 | 50 | 13.46 | 0.2691 | 0.12 |
| 4 | 8 | 80 | 60 | 17.94 | 0.2990 | 0.14 |
| 5 | 10 | 100 | 70 | 22.43 | 0.3204 | 0.15 |
| 6 | 14 | 120 | 80 | 26.91 | 0.3364 | 0.15 |
| 7 | 10 | 140 | 90 | 31.40 | 0.3488 | 0.16 |
| 8 | 7 | 160 | 100 | 35.88 | 0.3588 | 0.16 |
| 9 | 9 | 180 | 110 | 40.37 | 0.3670 | 0.17 |
| 10 | 8 | 200 | 120 | 44.85 | 0.3738 | 0.17 |
| 11 | 10 | 220 | 130 | 49.34 | 0.3795 | 0.17 |
| 12 | 8 | 240 | 140 | 53.82 | 0.3844 | 0.17 |
| 13 | 12 | 260 | 150 | 58.31 | 0.3887 | 0.18 |
| 14 | 16 | 280 | 160 | 62.79 | 0.3924 | 0.18 |
| 15 | 20 | 300 | 170 | 67.28 | 0.3957 | 0.18 |
| 16 | 28 | 320 | 180 | 71.76 | 0.3987 | 0.18 |
| 17 | 35 | 340 | 190 | 76.25 | 0.4013 | 0.18 |
| 18 | 40 | 360 | 200 | 80.73 | 0.4037 | 0.18 |

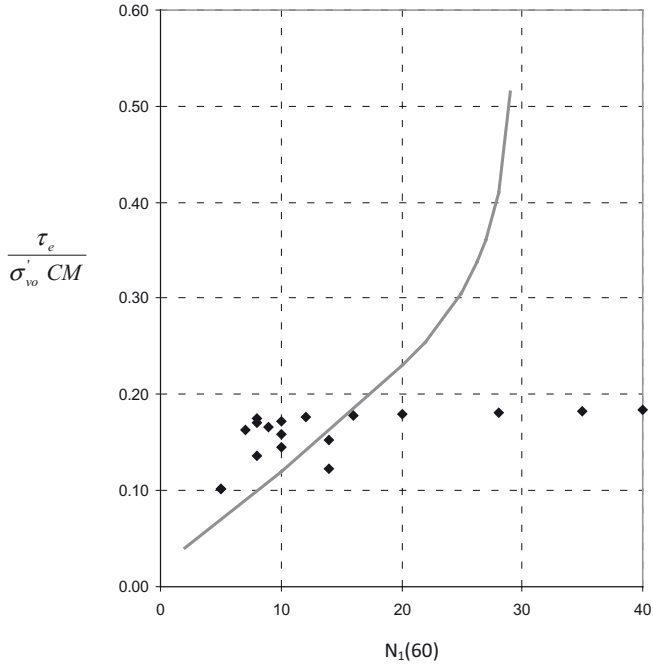



*Figure 8.6* Determination of liquefaction potential

Note that EC8 Part 5 Clause 5.1.4 (11) requires liquefaction factor of safety check.

Normalised data is put into Figure B.1 in Annex B of EC 8 Part 5.

From the plot in Figure 8.6 we can determine that *liquefaction is possible in the loose sand layer*.

*Conclusion*: the loose sand layer from elevation +12.53 to +2.53 (i.e. 10 m of sand layer) is susceptible to liquefaction.

## 8.4 Shallow foundations

### *8.4.1 Overview of behaviour*

The performance of shallow or spread foundations subject to seismic loading can be considered as consisting of several modes (see Figure 8.7). The long-term static loading will have produced some foundation displacement (1). For relatively small seismic loadings most foundations will respond in an essentially linear elastic manner (2). As the loading increases towards the ultimate dynamic capacity, non-linear soil responses become significant and the foundation response may be affected by partial uplift (3). The ultimate capacity of the foundation will be significantly influenced by the dynamic loadings imposed, with transient horizontal loads and moments acting to reduce the ultimate vertical capacity. For transient loadings that exceed

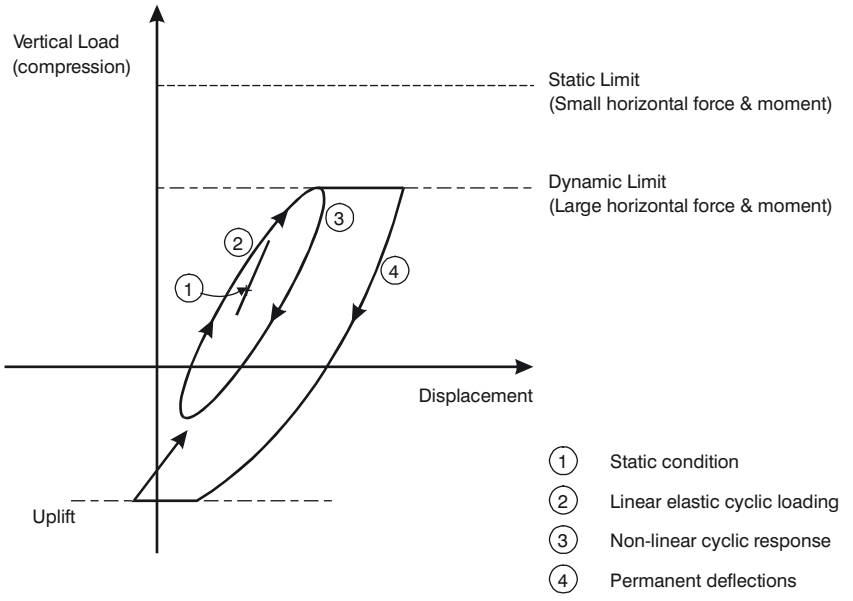



*Figure 8.7* Conceptual response of spread foundation to seismic loading

yield, permanent displacements may occur (4). Lateral loading may generate sliding with larger sliding displacements accumulating if the transient horizontal loading is biased in one direction. Uplift and rocking behaviour may result in permanent rotations while bearing capacity failure will lead to settlement, translation and tilt.

In addition to the transient and permanent deformations that arise from loads transmitted through the structure into the foundation, additional displacements may arise from ground movements imposed on the foundation. In this class of behaviour are settlements arising from densification of the soil, the effects of liquefaction and lateral spreading.

Historically, seismic foundation design has aimed to avoid yield of the foundation material. Fixed base assumptions have often been made for the structural analysis and the foundation design has attempted to produce this behaviour. However, the recent trend has been to recognise that limited foundation displacements (both transient and permanent) may absorb substantial energy and allow significant economies in construction. Practical design methodologies have been developed to enable implementation of this approach particularly in American and New Zealand practice. EN 1998:2004 does not explicitly discuss displacement based geotechnical design.

### 8.4.2 Ultimate capacity of shallow foundations

EN 1998-5 requires the ultimate seismic capacity of footings to be assessed for the onset of sliding and bearing capacity 'failure'. These modes of behaviour are considered in the following sections.

### 8.4.3 Sliding

The friction resistance for footings on cohesionless deposits above the water table, $F_{Rd}$, may be calculated from the following expression:

$$F_{Rd} = N_{Ed} \frac{\tan \delta}{\gamma_M} \tag{8.9}$$

where $N_{Ed}$ = the design normal force on the horizontal base
$\delta$ = the interface friction angle
$\gamma_M$ is the partial factor (1.25 for tan $\delta$).

For cohesive soils the equivalent relationship is:

$$F_{Rd} = \frac{s_u A}{\gamma_M} \tag{8.10}$$

where $A$ = plan area of foundation
$s_u$ = undrained strength
$\gamma_M$ is the partial factor (1.4 for $s_u$).

Most foundations are embedded and derive additional resistance to sliding by mobilising passive resistance on their vertical faces. For some classes of foundation (e.g. bridge abutments) this resistance provides a major contribution to their performance. However, the mobilisation of full passive resistance requires significant displacements, which may amount to between 2 per cent and 6 per cent of the foundation's depth of burial (see for example Martin and Yan (1995)). Such displacements may exceed the maximum allowable values for the structure and hence the foundation design may incorporate only a proportion of the full passive resistance.

EN 1998-5 requires that to ensure no failure by sliding on a horizontal base, the following expression must be satisfied:

$$V_{Ed} \leq F_{Rd} + E_{Rd} \tag{8.11}$$

where $E_{Rd}$ = the design lateral resistance from earth pressure, not exceeding 30 per cent of the full passive resistance.

### *8.4.4 Bearing capacity*

#### *8.4.4.1 Static bearing capacity*

Bearing capacity formulae for seismic loading are generally related to their static counterparts. For the static case:

$$q = cN_c s_c + 0.5\gamma BN_\gamma s_\gamma + p_0 N_q s_q \tag{8.12}$$

where $q$ = ultimate vertical bearing pressure
$c$ = cohesion
$\gamma$ = soil density
$B$ = foundation width
$p_0$ = surcharge at foundation level
$N_c$, $N_\gamma$, $N_q$ = bearing capacity factors
$s_c$, $s_\gamma$, $s_q$ = shape factors

Closed form solutions exist for $N_c$ and $N_q$ but not for $N_\gamma$. Thus while the factors $N_c$ and $N_q$ have widely accepted definitions, a considerable range of solutions have been proposed for $N_\gamma$ based on approximate numerical studies or on experimental results. A selection of the suggested values are presented in Table 8.3 and plotted in Figure 8.8.

Inclined loading is incorporated into the bearing capacity equation either by incorporating inclination factors into each term of Equation (8.13) or by direct modification of the bearing capacity factors. Thus:

$$q = cN_c s_c i_c + 0.5\gamma BN_\gamma s_\gamma i_\gamma + p_0 N_q s_q i_q \tag{8.13}$$

where $i_c$, $i_\gamma$, $i_q$ = inclination factors

*Table 8.3* Formulations for bearing capacity factors

| | |
|---|---|
| $N_q = e^{\pi \tan\phi} \tan^2\left(45 = \frac{\phi}{2}\right)$ | Terzaghi and Peck (1967) |
| $N_c = \left(N_q - 1\right)\cot\phi$ | Terzaghi and Peck (1948) |
| $N_\gamma = 2\left(N_q + 1\right)\tan\phi$ | Caquot and Kerisel (1953), API (1984) |
| $N_\gamma = 2\left(N_q - 1\right)\tan\phi$ | EN 1997-1:2004 |
| $N_\gamma = \exp\left(-1.646 + 0.173\phi\right)$ | Strip footing – Ingra and Baecher (1983) |
| $N_\gamma = 0.657\exp\left(0.141\phi\right)$ | Strip footing - Zadroga (1994) |

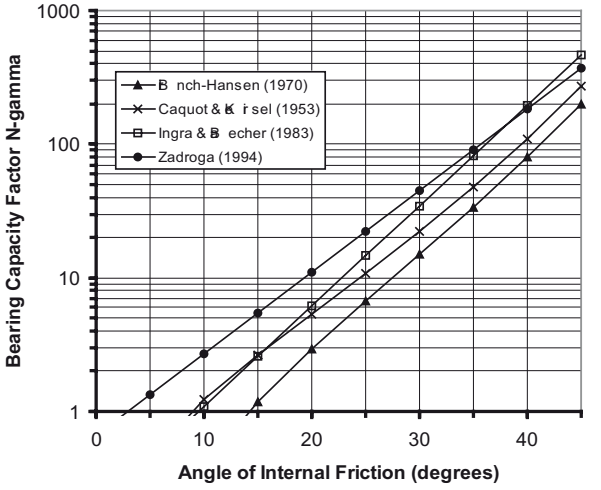



*Figure 8.8* Published relationships between $N_\gamma$ and $\phi$ for static loading

Various proposals for the inclination factors are shown in Table 8.4. While there may not be unanimity on the precise formulation of the various inclination factors, the key issue they all indicate is that the ultimate vertical capacity of a foundation is severely reduced by relatively modest horizontal loading.

A moment acting on the foundation is treated by defining an effective foundation width $B'$. The horizontal and vertical loads are applied to the effective foundation. $B'$ is defined as follows:

$$e = \frac{M}{V} \text{ and } B' = B - 2e \tag{8.14}$$

where M = applied moment
$H$ = horizontal loading (parallel to $B$)
$V$ = vertical loading
$A$ = plan area of foundation, $BL$
$s_u$ = undrained shear strength

$$m = \frac{2 + B/L}{1 + B/L}$$

*8.4.4.2 Seismic bearing capacity*

Significant earthquake events substantially reduce the ultimate bearing capacity of spread footings due principally to the following effects:

*Table 8.4* Published relationships for inclination factors

| *Inclination factor* | *EN 1997-1:2004* | *Vesic (1975)* |
|---|---|---|
| $i_c$ undrained | $0.5\left(1-\sqrt{\frac{H}{As_u}}\right)$ | $1-\frac{mH}{As_uN_c}$ |
| $i_c$ drained | $\frac{\left(i_qN_q-1\right)}{\left(N_q-1\right)}$ | $\frac{\left(i_qN_q-1\right)}{\left(N_q-1\right)}$ |
| $i_q$ | $\left(1-\frac{0.7H}{V+Ac\cot\varphi}\right)^3$ | $\left(1-\frac{H}{V+Ac\cot\varphi}\right)^m$ |

- The imposition of transient horizontal loads and moments arising from the inertia of the supported structure.
- Inertial loading of the foundation material.
- Changes in the strength of foundation materials due to rapid cyclic loading.

In addition the soil strata that comprise the foundation may act to limit the maximum seismic accelerations that can be transmitted to the foundation level.

Several solutions have recently been published for bearing capacity that take account of inertia effects in the foundation material. The methods due to Sarma and Iossifelis (1990), Budhu and Al-Karni (1993) and Shi and Richards (1995) are all based on Equation (8.11) with modified bearing capacity factors that incorporate the effects of load inclination and inertia in the foundation. Thus the seismic bearing capacity may be expressed as:

$$q = cN_{cE}s_c + 0.5\gamma BN_{\gamma E}s_\gamma + p_0N_{qE}s_q \tag{8.15}$$

where $q$ = vertical component of the ultimate bearing pressure
$N_{cE}$, $N_{\gamma E}$, $N_{qE}$ = seismic bearing capacity factors.

While this expression appears suitable for the evaluation of shallow foundation behaviour on either granular or cohesive soil, some caution is required. The rate of loading applied by seismic events is sufficiently high to cause the response of a saturated granular stratum to be essentially undrained beneath the footing. The undrained strength of sand under such loadings is not well understood.

Returning to Equation (8.15), Sarma and Iossifelis (1990) and Budhu and Al-Karni (1993) assume that the horizontal loading applied to the foundation by the structure is given by:

$$H = k_hV \tag{8.16}$$

For many real foundations subjected to seismic loading this condition will not be satisfied. For foundations of base isolated structures or bridge piers with sliding bearings, the applied horizontal loadings may be greatly reduced. Conversely, many structures will amplify the applied base accelerations leading to horizontal loadings much higher than those suggested by Equation (8.16). Phase differences between the ground accelerations and those of the structure complicate the assessment of the appropriate horizontal load. For a more comprehensive discussion on these issues and other limitations refer to Pecker (1994).

Shi and Richards (1995) have assessed the effects of a range of horizontal loadings on the seismic bearing capacity. They define the horizontal load as:

$$H = fk_h V \tag{8.17}$$

where $f$ = a shear transfer factor.

Solutions have been presented as the ratios of the static to the dynamic bearing capacity factors for cases where the shear transfer factor is 0, 1 or 2 (see Figure 8.9). The solution obtained by Shi and Richards (1995) for a shear transfer factor of unity agrees closely with those obtained by Sarma and Iossifelis (1990). It may be noted from Figure 8.9 that inertia effects within the foundation material have negligible effect on $N_{cE}$ ($\varphi = 0$), while $N_{\gamma E}$ and $N_{qE}$ are substantially affected even in cases where the horizontal loading imposed by the foundation remains at its static value.

Annex F of EN 1998-5 presents an alternative method for assessing bearing capacity of strip, shallow foundations. The result is based on a long-term European research programme, including field evidence, analytical and numerical solutions and a few experimental results (Pecker and Salençon, 1991; Dormieux and Pecker, 1995; Salençon and Pecker, 1995a, 1995b; Auvinet et al, 1996; Paolucci and Pecker 1997a, 1997b; Pecker 1997).The stability against seismic bearing failure of a shallow foundation may be checked with the following inequality:

$$\frac{\left(1-e\bar{F}\right)^{c_T}}{\bar{N}^a\left[\left(1-m\bar{F}^k\right)^k-\bar{N}\right]^b}+\frac{\left(1-f\bar{F}\right)^{c_M}\left(\gamma\bar{M}\right)^{c_M}}{\bar{N}^c\left[\left(1-m\bar{F}^k\right)^k-\bar{N}\right]^d}-1\le 0 \tag{8.18}$$

where for a footing of dimensions width $B$ and length $L$:

$$\bar{N}=\frac{\gamma_{Rd}N_{Ed}}{N_{\max,tot}} \tag{8.19}$$

$$\bar{V}=\frac{\gamma_{Rd}V_{Ed}}{N_{\max,tot}} \tag{8.20}$$

$$\bar{M}=\frac{\gamma_{Rd}M_{Ed}}{BN_{\max,tot}} \tag{8.21}$$

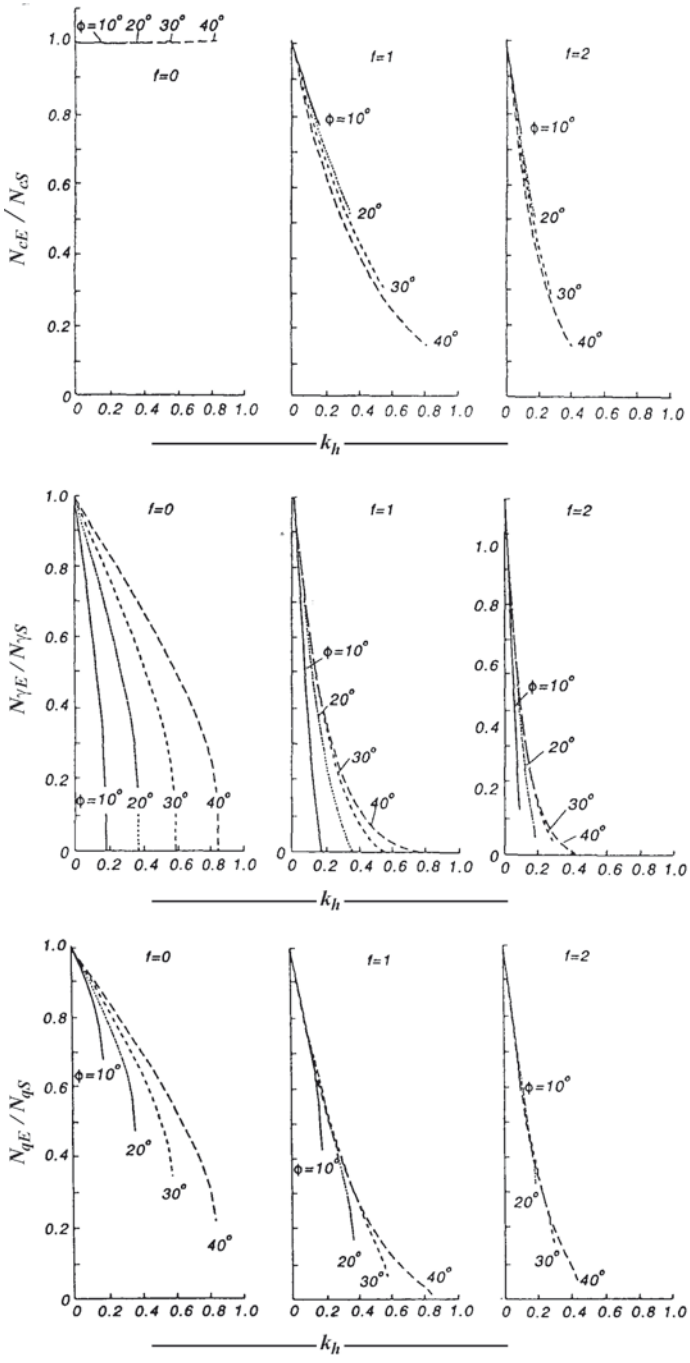



*Figure 8.9* Seismic bearing capacity factors with horizontal acceleration and angle of internal friction

*Table 8.5* Values of parameters used in Equation (8.18)

| *Purely cohesive soils* | | | | *Purely cohesionless soils* | | | |
|---|---|---|---|---|---|---|---|
| *Parameter* | *Value* | *Parameter* | *Value* | *Parameter* | *Value* | *Parameter* | *Value* |
| *A* | 0.70 | *k* | 1.22 | *a* | 0.92 | *k* | 1.00 |
| *B* | 1.29 | *k'* | 1.00 | *b* | 1.25 | *k'* | 0.39 |
| *C* | 2.14 | *ct* | 2.00 | *c* | 0.92 | $c_t$ | 1.14 |
| *D* | 1.81 | *cm* | 2.00 | *d* | 1.25 | $c_m$ | 1.01 |
| *E* | 0.21 | *c'm* | 1.00 | *e* | 0.41 | $c'_m$ | 1.01 |
| *F* | 0.44 | β | 2.57 | *f* | 0.32 | β | 2.90 |
| M | 0.21 | γ | 1.85 | *m* | 0.96 | γ | 2.80 |

*Table 8.6* Values of partial factors

| *Soil type* | $\gamma_{Rd}$ |
|---|---|
| Medium-dense to dense sand | 1.00 |
| Loose dry sand | 1.15 |
| Loose saturated sand | 1.50 |
| Non-sensitive clay | 1.00 |
| Sensitive clay | 1.15 |

where $N_{Ed}$, $V_{Ed}$ and $M_{Ed}$ are the design action effects at the foundation level, and the rest of the numerical parameters in Equations (8.18) to (8.21) depend on the type of soil and are given in Tables 8.5 and 8.6.

PURELY COHESIVE SOILS

The ultimate bearing capacity of the foundation under a vertical centred load $N_{\max}$ is given by Equation (8.22):

$$N_{\max} = (\pi + 2)\frac{s_u}{\gamma_s} B \tag{8.22}$$

where $s_u$ = the undrained shear strength of the soil
$\gamma_s$ = the partial factor for the undrained shear strength.
The dimensionless soil inertia $\bar{F}$ is given by Equation (8.23).

$$\bar{F} = \frac{\rho a_g S B}{s_u} \tag{8.23}$$

where ρ = unit mass of the soil
$a_g$ = design ground acceleration on type A ground, given by $a_g = \gamma_I a_{gR}$
$a_{gR}$ = reference peak ground acceleration
$\gamma_I$ = importance factor, depending on the building importance
$S$ = soil factor.

The following constraints apply to the general bearing capacity expression in Equation (8.18).

$$0 < \bar{N} \le 1\,, \quad \left|\bar{V}\right| \le 1 \tag{8.24}$$

PURELY COHESIONLESS SOILS

The ultimate bearing capacity of the foundation under a vertical centred load $N_{max}$ is given by Equation (8.25):

$$N_{max} = 0.5\rho g\left(1 \pm \frac{a_v}{g}\right)B^2 N_\gamma \tag{8.25}$$

where $a_v$ = vertical ground acceleration, given by $a_v = 0.5\ a_g\ S$

$N_\gamma$ = bearing capacity factor, given by Equation (8.26):

$$N_\gamma = 2\left[\tan^2\left(45° + \frac{\varphi'_d}{2}\right)e^{\pi \tan \varphi'_d} + 1\right]\tan\varphi'_d \tag{8.26}$$

where $\varphi'_d$ = design shearing resistance angle given by Equation (8.27):

$$\varphi'_d = \tan^{-1}\left(\frac{\tan\varphi'_k}{\gamma_\varphi}\right) \tag{8.27}$$

where $\varphi'$ is the shearing resistance angle.

The dimensionless soil inertia $\bar{F}$ is given by Equation (8.28):

$$\bar{F} = \frac{a_g\ S}{g\tan\varphi'_d} \tag{8.28}$$

The following constraints apply to the general bearing capacity expression in Equation (8.18):

$$0 < \bar{N} \le \left(1 - m\bar{F}\right)^k \tag{8.29}$$

where $k$ = a coefficient from Table 8.5.

The previous formulation has been recently extended to circular foundations on homogeneous and heterogeneous foundations by Chatzigogos et al (2007).

## 8.5 Seismic displacements

In cases where the transient seismic loadings exceed the available foundation resistance, permanent displacements will occur. The accelerations at which displacement commences are termed threshold accelerations. In many cases the peak earthquake accelerations can exceed the threshold values by a substantial margin with minimal foundation displacement occurring. Though EN 1998-5 generally requires that foundations remain elastic, for foundations above the water table, where the soil properties remain unaltered

and the sliding will not affect the performance of any lifelines connected to the structure, a limited amount of sliding may be tolerated.

Designing on the basis of allowable deflections can result in significant economies by comparison to alternative 'elastic' design approaches. However, a cautious approach is required to the assessment of seismic displacements because modest variations in design parameters can result in substantial variations in displacements.

### *8.5.1 Sliding displacements*

The principles whereby permanent seismic displacements can be calculated were set out by Newmark (1965) in his Rankine Lecture. These are illustrated in Figure 8.10 for a block subjected to a rectangular acceleration pulse. The method considers that the block accelerates with the ground until threshold acceleration ($N_g$) is reached. The ground acceleration continues to rise to peak acceleration ($A_g$) but the acceleration of the block is limited by the shear capacity of the base to a value of $N_g$. The equations of motion give the velocity of the block and the ground and their relative displacement.

The Newmark analysis may be used directly to calculate the sliding displacement of a foundation provided that design acceleration time-histories are available and the threshold acceleration for sliding has been established.

In many instances, design acceleration time-histories will not be available for routine foundation design. Several authors have used the Newmark approach combined with earthquake acceleration records to derive 'design lines' relating sliding displacements to the ratio of threshold to peak accelerations ($N/A$). Notable examples are those of Franklin and Chang (1977), Richards and Elms (1979), Whitman and Liao (1985) and Ambraseys and Menu (1988).

The Ambraseys and Menu relationships are shown in Figure 8.11 for various probabilities of exceedance. It may be noted that both unsymmetrical (one-way) sliding and symmetrical (two-way) sliding have been considered. Significant differences between the two cases only arise when the peak acceleration is more than twice the threshold (i.e. $N/A < 0.5$). The Ambraseys and Menu database included earthquakes of $M_S$ 6.4 to 7.7.

### *8.5.2 Shakedown settlement*

Settlements under cyclic loads should be assessed when extended layers or thick lenses of loose, unsaturated cohesionless materials exist at a shallow depth. Excessive settlements may also occur in very soft clays because of cyclic degradation of their shear strength under ground shaking of long duration. If the settlements caused by densification or cyclic degradation appear capable of affecting the stability of the foundations, consideration should be given to ground improvement methods. Dynamic settlement can be estimated using empirical relationships between volumetric strain,

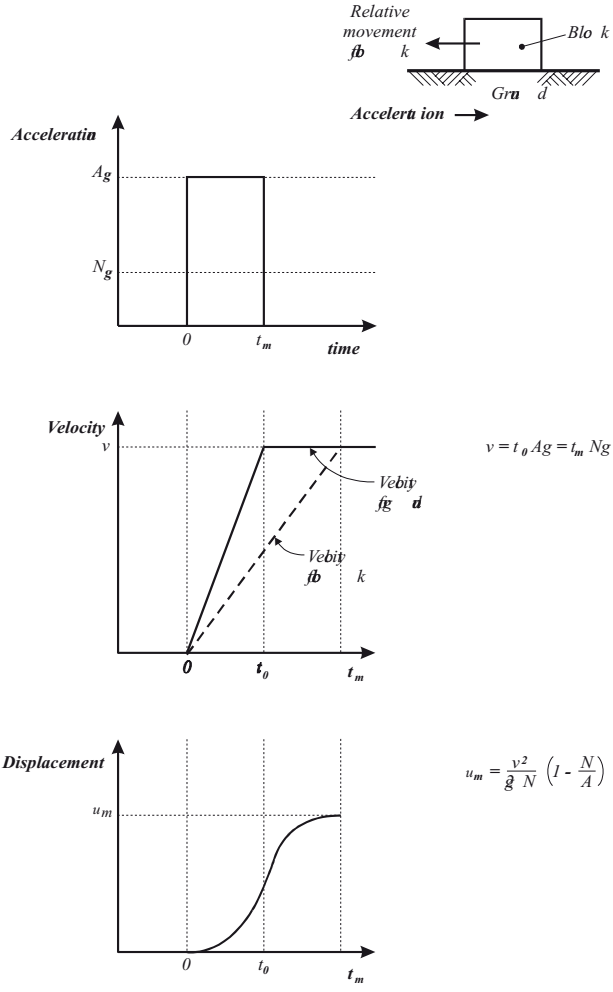



*Figure 8.10* Sliding displacement for a block with a rectangular base acceleration pulse (After Newmark (1965))

SPT *N*-values (corrected for overburden), and the cyclic shear strain. For example the approach developed by Tokimatsu and Seed (1987) is based on relationships between the volumetric strain, the cyclic shear strain and SPT *N*-values. The peak shear strain computed from the one-dimensional response analysis and the SPT corrected *N*-value at that point are entered into the Tokimatsu and Seed chart (see Figure 8.12) to yield the volumetric strain. The total settlement can then be obtained by integrating these volumetric strains as a function of depth.

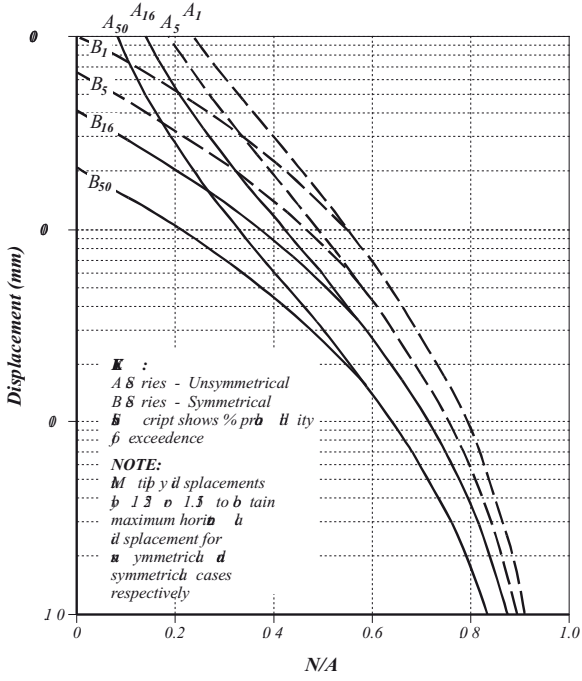



*Figure 8.11* Displacement vs N/A for various probabilities of exceedance

### *8.5.3 Foundation horizontal connections*

Tie beams should be provided between all foundations, except for ground type A (rock), or on ground type A and B (stiff soil) in areas of low seismicity.

The tie beams should be designed to withstand an axial force, considered in both tension and compression, equal to:

$$\pm 0.3\alpha S N_{Ed} \text{ for ground type B} \tag{8.30}$$

$$\pm 0.4\alpha S N_{Ed} \text{ for ground type C} \tag{8.31}$$

$$\pm 0.6\alpha S N_{Ed} \text{ for ground type D} \tag{8.32}$$

where $N_{Ed}$ = mean value of the design axial forces of the connected vertical elements

## 8.6 Design example on a shallow foundation – pad foundation

### *8.6.1 Sites*

There are four sites (A, B, C and D) that are available for construction of the hotel as stated in Chapter 4 (Section 4.8).

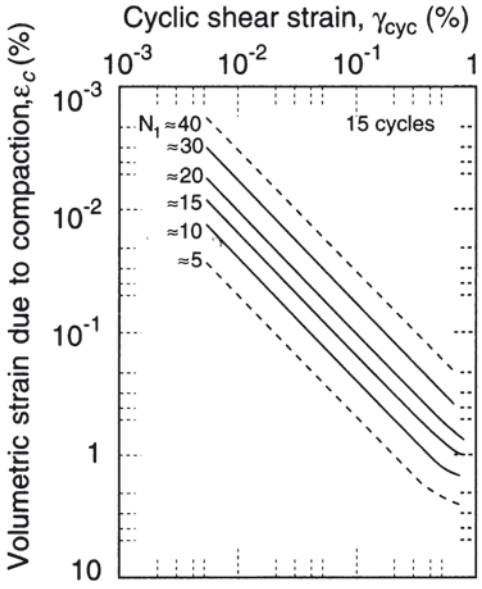



*Figure 8.12* Assessment of volumetric strain (Tokimatsu & Seed, 1987)

The design ground acceleration is taken as $a_{gR} = 0.3$g.

The building importance factor for the hotel is taken as $\gamma_I = 1$ in this example.

Preliminary site investigation was carried out at all the sites. Borehole data and SPT and field vane shear tests were carried out at each site. This information is assimilated in Figure 8.13.

Site C is selected for the hotel for the reasons stated in Chapter 4, Section 4.8.3.

In a practical design situation, Site D may also be considered, at least as an initial candidate, and the design calculations may be carried out using the undrained shear strength of the stiff clay with appropriate partial factors as outlined in Section 8.4.4. However, in this example only Site C will be considered.

Ultimate limit state (ULS) design:

1 failure by sliding
2 bearing capacity failure.

(See Section 5.4 'Verification and dimensioning criteria' in EC8.)

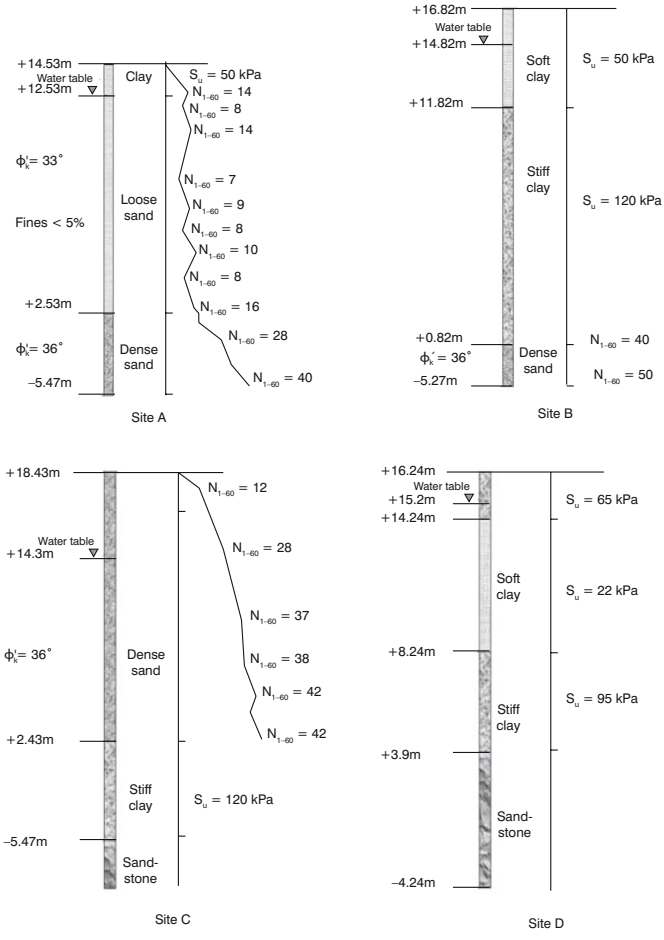



*Figure 8.13* Borehole data at the four sites from site investigation

### *8.6.2 Design of pad foundation*

The most heavily loaded columns in the hotel are along C and D lines (see Figure 8.14) separated by only 3 m spacing, so let us consider a combined PAD foundation for these two column bases.

As an initial guess, choose a 10 m × 4 m pad foundation, located 1 m below ground level to support the two columns along C and D lines shown in the plan view.

The worst loading occurs on two columns in a 4 m bay along C and D on the plan of the building. These loads are obtained in the structural design example (Chapters 3 and 5) with due consideration to the capacity design aspects and are shown Table 8.7.

Using the data from Table 8.7 we can obtain the following design loads:

*Table 8.7* Loads on the foundation from the columns

| | *Column C* | *Column D* | *Total design loads on the pad foundation* |
|---|---|---|---|
| Axial load | 5978 kN | 862 kN | 6840 kN |
| Shear load | 826 kN | 826 kN | 1652 kN |
| Moment load | 2405 kNm | 2088 kNm | 4493 kNm |

Design vertical load $N_{Ed} = 6840$ kN

Design horizontal, shear load $V_{Ed} = 1652$ kN

Design moment load $M_{Ed} = 4493$ kNm

Consider Equation F.6 (EC8 p 43):

$$N_{\max} = \frac{1}{2}\rho g\left[1 \pm \frac{a_v}{g}\right]B^2 N_\gamma$$

The mass density of the sand = 1650 kg/m$^3$ (unit weight of 16.19 kN/m$^3$).

$$a_v = 0.5 \cdot S \cdot a_g$$

Referring to Tables 3.1 and 3.2 in EC8, choose soil factor $S = 1.15$. Also,

$$a_g = \gamma_I \cdot a_{gR}$$

where $\gamma_I$ is the building importance factor, which is taken as unity for this hotel building.

From design brief, $a_{gR} = 0.3g$ :

$$a_v = 0.5 \times 1.15 \times 0.3g = 0.1725g$$

$$N_{\max} = \frac{1}{2}1650 \times 9.81[1 \pm 0.225]B^2 N_\gamma$$

The friction angle for the sand needs to be reduced using the $\gamma_m$ factor obtained before (see EC8 p 23). $\gamma_m = 1.25$

$$\varphi'_d = \tan^{-1}\left(\frac{\tan\varphi_k}{\gamma_m}\right)$$

$$\varphi'_d = \tan^{-1}\left(\frac{\tan 36^O}{1.25}\right) = 30^O$$

The bearing capacity factor can be calculated using:

$$N_\gamma = 2\left[\tan^2\left(45 + \frac{\varphi_d}{2}\right)e^{\pi\tan\varphi_d} + 1\right]\tan\varphi_d$$

$$N_{\gamma} = 38.8 \times \tan 30^{o} = 22.4$$

$$N_{\max} = \frac{1}{2} 1650 \times 9.81 \times 0.8275 \times 22.4 \times B^{2}$$

$$N_{\max} = 150 B^{2} \text{ kN/m}$$

$$N_{\max,tot} = 150 \times B^{2} \times L \text{ kN}$$

Substituting the dimensions of the footing (10 m × 4 m), we get

$$N_{\max} = 15000 \text{ kN/m and}$$
$$N_{\max,tot} = 60{,}000 \text{ kN}$$

### *8.6.3 Failure against sliding*

Design friction resistance for footing above water table:

$$F_{Rd} = N_{Ed} \frac{\tan \delta_{d}}{\gamma_{m}}$$

Choose $\gamma_{m}$ value from Equation 5.1 (EC8 p23)

$\gamma_{m} = 1.25$ and $\delta_{d} = \varphi_{k}$.

Use Equation 5.2 (EC8 p 23)

$$V_{Ed} \leq F_{Rd} + E_{Rd}$$

$V_{Ed}$ is design horizontal shear force

$E_{Rd}$ is design lateral resistance. It can be up to 30 per cent of passive resistance according to EC8.

#### *8.6.3.1 Sliding resistance*

Angle of internal friction for this sand:

$$\varphi'_{k} = 360$$

$$F_{Rd} = 6840 \times \frac{\tan 36^{o}}{1.25}$$

$$F_{Rd} = 3975.6 \text{ kN}$$

Therefore,

$$V_{Ed} \leq F_{Rd}$$

(1652 kN $\leq$ 3975.6 kN)

So, satisfies sliding check. Note that in this example we did not have to use 30 per cent of passive resistance clause in EC8 here, but if needed we could estimate 30 per cent of passive resistance for the footing, once we established the depth of the foundation below ground level.

### *8.6.4 Verification of bearing capacity*

Now calculate $\bar{N}$, $\bar{V}$ and $\bar{M}$ using Equation F.2 (EC8 Part 5 p42).
Choose $\gamma_{Rd} = 1$, using Table F.2 (see EC8 p44).

$$\bar{N} = \frac{\gamma_{Rd} N_{Ed}}{N_{\max,tot}} = \frac{1 \times 6840 \text{ kN}}{60000 \text{ kN}} = 0.114$$

$$\bar{V} = \frac{\gamma_{Rd} V_{Ed}}{N_{\max,tot}} = \frac{1 \times 1652 \text{kN}}{60000 \text{ kN}} = 0.0275$$

$$\bar{M} = \frac{\gamma_{Rd} M_{Ed}}{B N_{\max,tot}} = \frac{1 \times 4493 \text{kNm}}{10 \times 60000 \text{ kNm}} = 0.0075$$

Calculate $\bar{F}$ using Equation F.7 (EC8 p43)

$$\bar{F} = \frac{a_g S}{g \tan \varphi'_d}$$

$$\bar{F} = \frac{0.3g \times 1.15}{g \tan 30^o} = 0.598$$

Check using Equation F.8 (EC8 p43)

$$0 \leq \bar{N} \leq \left(1 - m\bar{F}\right)^{k'}$$

Constants $m$ and $k'$ are to be chosen appropriately from Table F.1 (EC8 Part 5 p44).
Values of $m = 0.96$ and $k' = 0.39$.

$$0 \leq 0.114 \leq \left(1 - 0.96 \times 0.598\right)^{0.39}$$

$$0 \leq 0.114 \leq 0.7169$$

So check is satisfied.
Check for bearing capacity failure (EC8 Part 5 Equation F.1):

*Table 8.8* Parameters for dense sand

| *Parameter* | *Value* | *Parameter* | *Value* |
|---|---|---|---|
| a | 0.92 | k | 1.00 |
| b | 1.25 | k' | 0.39 |
| c | 0.92 | $C_T$ | 1.14 |
| d | 1.25 | $C_M$ | 1.01 |
| e | 0.41 | $C'_M$ | 1.01 |
| f | 0.32 | β | 2.90 |
| m | 0.96 | γ | 2.80 |

$$\underbrace{\frac{(1-e\bar{F})^{C_T}(\beta\bar{V})^{C_T}}{(\bar{N})^{a}\left[(1-m\bar{F}^{k})^{k'}-\bar{N}\right]^{b}}}_{I}+\underbrace{\frac{(1-f\bar{F})^{C'_m}(\gamma\bar{M})^{C_m}}{(\bar{N})^{c}\left[(1-m\bar{F}^{k})^{k'}-\bar{N}\right]^{d}}}_{II}-1\le 0$$

The constants for dense sand may be chosen from EC8 Part 5 as shown in Table 8.8.

Substitute for the values and check the inequality is satisfied.

$$\frac{(1-0.41\times 0.598)^{1.14}(2.9\times 0.0275)^{1.14}}{0.114^{0.92}\left[(1-0.96\times 0.598)^{0.39}-0.114\right]^{1.25}}=0.56375 \quad \ldots \text{(I)}$$

$$\frac{(1-0.32\times 0.598)^{1.01}(2.8\times 0.0075)^{1.01}}{0.114^{0.92}\left[(1-0.96\times 0.598)^{0.39}-0.144\right]^{1.25}}=0.22628 \quad \ldots \text{II)}$$

Inequality is therefore (I+II–1):

$0.564+0.226-1\le 0$

$-0.2099\le 0$

So, check is satisfied. The *pad* foundation is *safe* against bearing failure. Recall that the design in EC8 is based on partial safety factors and therefore no other global safety factor needs to be applied.

Check the plan view of the building (see Figure 8.14). Columns along C and D lines are separated by 3 m spacing. Columns along B and E are separated by 20 m spacing. The required dimensions for the pad foundation designed above are 10 m × 4m.

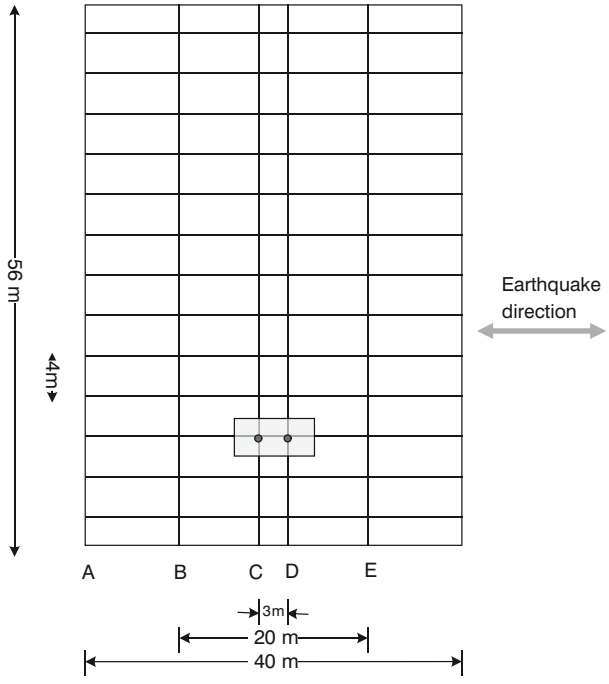



*Figure 8.14* Plan view of the hotel showing the location of the pad foundation

## 8.7 Design example on a shallow foundation – raft foundation

### *8.7.1 Design of raft foundation*

A pad foundation was designed for individual columns along lines C and D of the hotel plan in the previous section. This section demonstrates the design of a raft foundation for the hotel. The design loads for the raft foundation are obtained from structural analysis. For this design example, we will use the structural loads for a steel frame design outlined earlier. (Note: the total mass of the structure will be somewhat larger when the structural design is based on concrete, but the following calculations can be easily repeated taking into account the increased mass.) For a raft foundation design, we need to consider loads from all columns. Also we need to multiply the loads by 7, as there are seven bays in the building. The loads are summed up and shown below:

Design vertical load $N_{Ed}$ = 80.522 MN

Design moment load $M_{Ed}$ = 414.4 MNm

Design horizontal, shear load $V_{Ed}$ = 21.5 MN

### 8.7.2 Failure against sliding

#### 8.7.2.1 Sliding resistance

Angle of internal friction for this sand: $\varphi_k = 36^o$

$$F_{Rd} = 80.5 \times 10^3 \times \frac{\tan 36^o}{1.25}$$

$$F_{Rd} = 46{,}802\ kN$$

Therefore,

$$V_{Ed} \leq F_{Rd}\ (21.5\ \text{MN} \leq 46.8\ \text{MN})$$

So, satisfies sliding check.

(Note: as in the case of pad foundation, we did not have to use 30 per cent of passive resistance clause in EC8 here, but if needed we could estimate 30 per cent passive resistance for the footing, once we established depth of the foundation below ground level.)

### 8.7.3 Verification of bearing capacity

Site C has purely cohesionless soil at the depth of the pad foundation.

Consider Equation F.6 (EC8 p43)

$$N_{\max} = \frac{1}{2}\rho g\left[1 \pm \frac{a_v}{g}\right]B^2 N_\gamma$$

The mass density of the sand = 1650 kg/m$^3$ (16.19 kN/m$^3$ )

$$a_v = 0.5 \cdot S \cdot a_g$$

Referring to Tables 3.1 and 3.2 in EC7, choose soil factor $S = 1.15$. Also,

$$a_g = \gamma_I \cdot a_{gR}$$

where $\gamma_I$ is the building importance factor, which is taken as unity for this hotel building.

From design brief, $a_{gR} = 0.3g$ :

$$a_v = 0.5 \times 1.15 \times 0.3g = 0.1725g$$

$$N_{\max} = \frac{1}{2}1650 \times 9.81[1 \pm 0.1725]B^2 N_\gamma$$

The friction angle for the sand needs to be reduced using the $\gamma_m$ factor obtained before (see EC8 p23). $\gamma_m = 1.25$

$$\varphi'_d = \tan^{-1}\left(\frac{\tan\varphi_k}{\gamma_m}\right)$$

$$\varphi'_d = \tan^{-1}\left(\frac{\tan 36^O}{1.25}\right) = 30^O$$

The bearing capacity factor can be calculated using:

$$N_\gamma = 2\left[\tan^2\left(45 + \frac{\varphi_d}{2}\right)e^{\pi\tan\varphi_d} + 1\right]\tan\varphi_d$$

$$N_\gamma = 38.8 \times \tan 30^o = 22.4$$

$$N_{\max} = \frac{1}{2} \times \frac{1650}{1000} \times 9.81 \times 0.8275 \times 22.4 \times B^2$$

$$N_{\max} = 150B^2 \text{ kN/m}$$

If we assume that the raft foundation is going to be a '$B \times L$' foundation we can use, as before, the following equation:

$$N_{\max,tot} = 150 \times B^2 \times L \text{ kN}$$

Choose a 42 m × 58 m raft foundation, located 1 m below ground level to support the whole building shown in the plan view with a 1 m extension beyond the plan area.

$$N_{\max} = 264600 \text{ kN/m}$$
$$N_{\max,tot} = 15346800 \text{ kN}$$

Now calculate $\bar{N}$, $\bar{V}$ and $\bar{M}$ using Equation F.2 (see EC8 Part 5 p42). Choose $\gamma_{Rd} = 1$, using Table F.2 (see EC8 p44).

$$\bar{N} = \frac{\gamma_{Rd}N_{Ed}}{N_{\max,tot}} = \frac{1 \times 80522\,\text{kN}}{15346800\,\text{kN}} = 0.0052$$

$$\bar{V} = \frac{\gamma_{Rd}V_{Ed}}{N_{\max,tot}} = \frac{1 \times 21500\,\text{kN}}{15346800\text{ kN}} = 0.0014$$

$$\bar{M} = \frac{\gamma_{Rd}M_{Ed}}{BN_{\max,tot}} = \frac{1 \times 414500\text{ kNm}}{42 \times 15346800\text{ kNm}} = 0.00064$$

Use $\bar{F}$ value calculated before as this will not change with the loading:

$$\bar{F} = 0.598$$

Use the table of constants for dense sand in Table 8.8, which was used for the bearing capacity verification calculations for the pad foundation.

Check using Equation F.8 (EC8 p43):

$$0 \le \bar{N} \le \left(1 - m\bar{F}\right)^{k'}$$

Constants $m$ and $k'$ are taken from the table for 'dense sand' from before, as these do not change.

Values of $m = 0.96$ and $k' = 0.39$.

$$0 \le 0.0052 \le \left(1 - 0.96 \times 0.598\right)^{0.39}$$

$$0 \le 0.0052 \le 0.7169$$

So, check is satisfied.

Check for bearing capacity failure (EC8 Part 5 Equation F.1):

$$\underbrace{\frac{(1 - e\bar{F})^{C_T}(\beta\bar{V})^{C_T}}{(\bar{N})^{a}\left[(1 - m\bar{F}^{k})^{k'} - \bar{N}\right]^{b}}}_{I} + \underbrace{\frac{(1 - f\bar{F})^{C'_m}(\gamma\bar{M})^{C_m}}{(\bar{N})^{c}\left[(1 - m\bar{F}^{k})^{k'} - \bar{N}\right]^{d}}}_{II} - 1 \le 0$$

Substitute for the values and check the inequality is satisfied.

$$\frac{(1 - 0.41 \times 0.598)^{1.14}(2.9 \times 0.0014)^{1.14}}{0.0052^{0.92}\left[(1 - 0.96 \times 0.598)^{0.39} - 0.0052\right]^{1.25}} = 0.263264844 \qquad \ldots \text{(I)}$$

$$\frac{(1 - 0.32 \times 0.598)^{1.01}(2.8 \times 0.00064)^{1.01}}{0.0052^{0.92}\left[(1 - 0.96 \times 0.598)^{0.39} - 0.0052\right]^{1.25}} = 0.262201923 \qquad \ldots \text{(II)}$$

Inequality is therefore (I+II–1):

$$0.263 + 0.262 - 1 \le 0$$

$$-0.475 \le 0$$

So, check is satisfied. The raft foundation is safe against bearing failure.

Hence a *raft foundation* with dimensions of 42 m × 58 m located 1 m below the ground level is suitable for the hotel. A plan view of the raft foundation that extends the plinth area of the hotel by 1 m is shown in Figure 8.15.

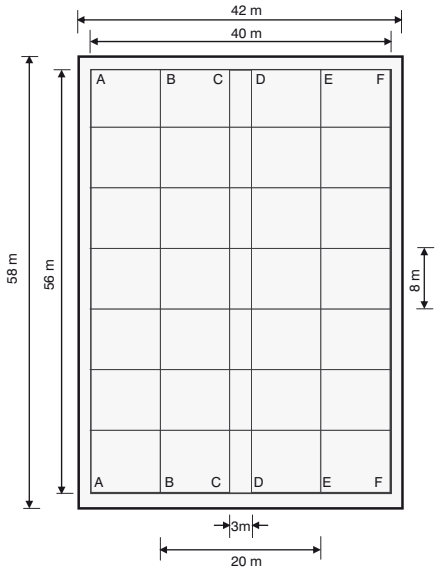



*Figure 8.15* Plan view of the hotel showing the raft foundation

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# 9 Pile foundations

*S.P.G. Madabhushi and R. May*

## 9.1 Introduction

Pile foundations are widely used both onshore and offshore to transfer heavy structural loads to competent load bearing soil strata or bedrock. Geotechnical engineers are called upon to design deep foundations when the shallow layers of soils beneath the building are either unable to support the loads imposed by the superstructure on the shallow foundations or if the shallow layers may become unstable due to the cyclic shear stresses induced by the earthquake loading. Under such circumstances it is imperative to look for pile foundations that transfer the load from the superstructure to more firm and stable soil strata at deeper levels or onto bedrock. In this chapter the seismic design of pile foundations is considered in the light of the EC8 Part 5 (2003) provisions as well as some of the current research findings. It is perhaps helpful if some of the well-known examples of failures of pile foundations during or following an earthquake loading are considered first.

### *9.1.1 Examples of pile foundation failures following earthquake loading*

Although pile foundations are widely used in the regions of high seismicity around the world, there are a number of examples where the pile foundations have failed during strong earthquake events. Such failures can cause either collapse of the superstructure or excessive settlements and rotations.

During the 1964 Niigata earthquake the Showa Bridge collapsed as shown in Figure 9a. Figure 9b shows one of the piles that was extracted during the post earthquake investigation, while Figure 9c shows the schematic diagram of the collapsed spans.

The Showa bridge collapse was attributed to many causes. For example, Hamada (1992) proposed that lateral spreading of the soil following liquefaction (see Figure 9c) caused large displacements at the pile heads and resulted in the dislodging of the spans. Bhattacharya et al (2005a) have proposed that buckling of the piles in liquefied sands could have caused the collapse of the Showa Bridge. The Showa Bridge collapse is not a unique event. There have been many other failures involving pile foundations.

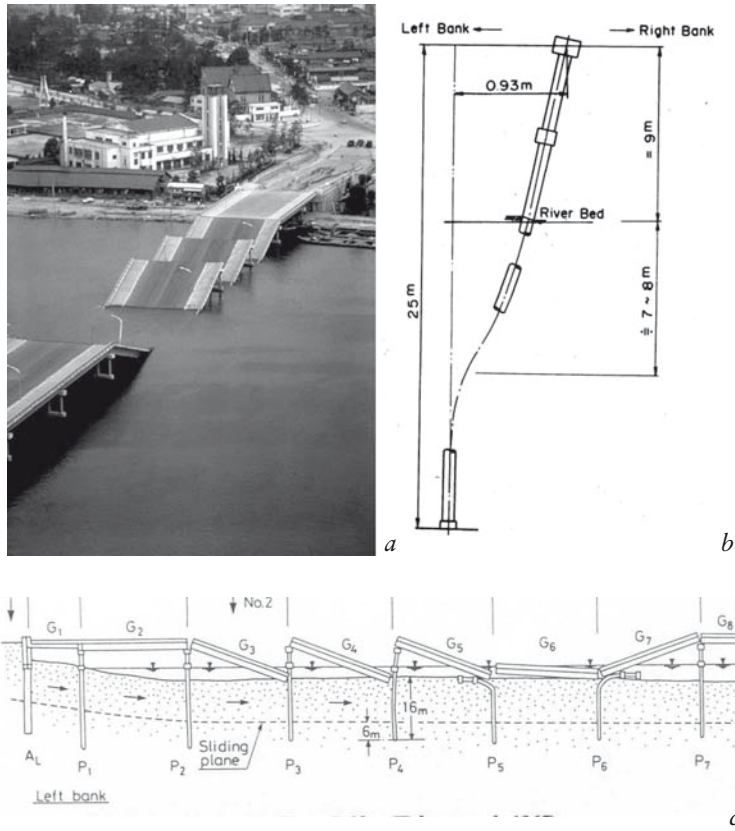



*Figure 9.1 a)* Collapse of the Showa bridge *b)* Excavated pile, after Hamada, 1992 *c)* Collapsed spans of the Showa bridge, after Takata et al., 1965

More recently the harbour master's building at Kandla Port suffered a rotation of about 11° from the vertical following the Bhuj earthquake of 2001 as shown in Figure 9.2. The pile foundations supporting this building have suffered differential settlement. Similarly Tokimatsu et al (1997) describe the failure of a three-storey building supported on pile foundations during the 1995 Kobe earthquake as shown in Figure 9.3. They suggest that the failure of the quay wall allowed the seaward movement of the soil that caused the pile foundations to fail. Lateral spreading of soil subjects the pile to additional loading. The piles need to be adequately designed to sustain these additional lateral loads.



*Figure 9.2* Rotation of a tall masonry building on pile foundations during the Bhuj earthquake (Madabhushi et al. 2005)

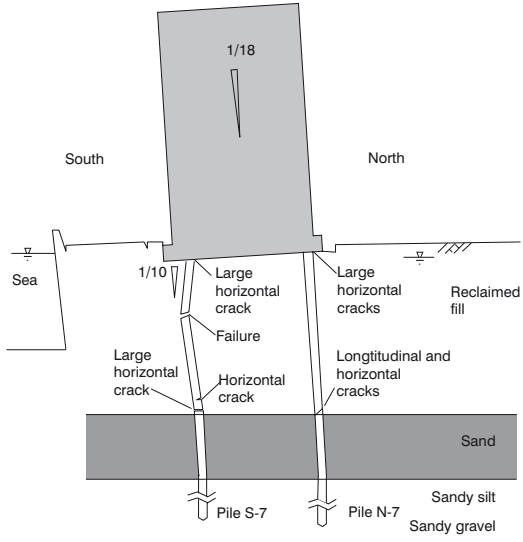



*Figure 9.3* Failure of piles in a three-storeyed building in 1995 Kobe earthquake (Tokimatsu et al. 1997)

### *9.1.2 Lessons learnt from pile foundation failures*

Pile foundations seem to suffer from earthquake loading for a variety of reasons. A comprehensive list of pile foundations of various structures that have performed poorly was compiled by Bhattacharya, Madabhushi and Bolton (2004). The load bearing soil strata into which the piles are transferring the load may change their character under strong cyclic loading. In addition the piles have to bear the inertial and kinematic loading described in detail later in Section 9.4.1. In many cases the pile foundation failures appear to be associated with 'liquefaction' of the ground to some depth around the piles. Similarly presence of a non-liquefied layer such as stiff clay overlying a liquefiable layer appears to cause additional loading particularly if the ground is on a slope. Research has shown that slopes as gentle as 1° to 3° can result in lateral spreading of liquefied soil and that of any non-liquefied soil crust overlying the liquefied soil (Haigh et al, 2000).

### *9.1.3 EC8 provisions*

The normal static design of pile foundations must be carried out under the provisions of EC7 Part 1 (1995). In addition, EC8 recommends that the liquefaction potential of all the soil layers at a given site be carefully determined based on the SPT tests conducted at the site. It is also suggested that careful consideration of any additional loading on the piles and pile caps that may arise due to the lateral spreading of the soil, particularly in the presence of a non-liquefiable soil strata overlying a liquefiable layer. In addition, where liquefaction is anticipated it is suggested that the strength of the liquefied soil must be ignored.

## 9.2 Pile foundation design under static loading

The static design of the pile foundations has to be carried out in accordance with EC7. A procedure is outlined below for cohesionless soils. A similar approach can be used for cohesive soils with suitable modification.

The pile capacity can be determined as a combination of the base capacity and the shaft capacity:

$$\text{Pile capacity } Q = \underbrace{Q_b}_{\text{Base capacity}} + \underbrace{Q_s}_{\text{Shaft capacity}} \tag{9.1}$$

### *9.2.1 Base capacity*

The base capacity depends on the bearing capacity of the soil at the pile tip level. It can be calculated using:

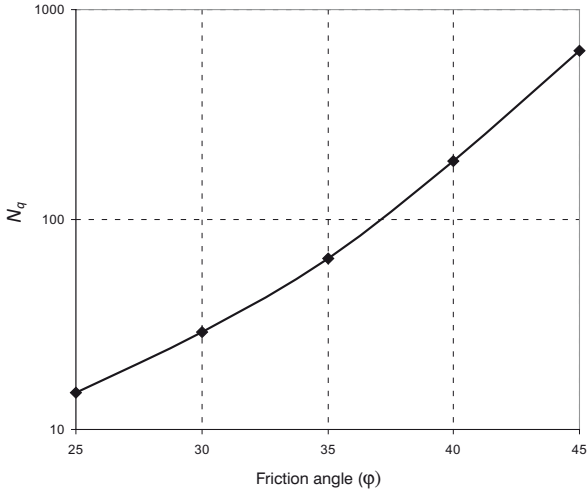



*Figure 9.4* Bearing capacity factor $N_q$ for deep foundations

$$Q_b = q_b \cdot A_b \tag{9. 2}$$

where $A_b$ = base area of pile shaft

$$q_b = \sigma'_v \cdot N_q \tag{9. 3}$$

The bearing capacity factor, $N_q$, for a deep foundation can be obtained using the chart shown in Figure 9.4. These types of charts were originally proposed by Berezantzev et al (1961) and subsequently modified by several researchers.

### *9.2.2 Shaft capacity*

The shaft capacity is obtained by estimating the shear stress generated along the shaft, which can be calculated as:

$$\tau_s = K_s \cdot \sigma'_v \cdot \tan\delta \tag{9. 4}$$

where $K_s$ depends on type of pile and installation (driven or cast *in situ* piles), $\sigma'_v$ is the effective stress at the elevation where shear stress is being calculated and $\delta$ is the friction angle between the pile and the soil.

For driven piles, $K_\sigma \leq 1$, so choose $K_s = 1$ conservatively.

In order to obtain the shaft capacity due to skin friction, we need to integrate the shear stress over the surface area of the pile using the following equation:

$$Q_s = 2\pi r \times \int_0^L \tau_s \tag{9. 5}$$

where $r$ is the pile radius and $L$ is the length of the pile.

The overall pile capacity can be determined by adding the base capacity and shaft capacity. It must be noted that where multiple piles are present, the pile spacing must be nominally 2 to 3 pile diameters; the closer the spacing of the piles, the lower the efficiency of the pile group.

In addition, the piles need to be designed following the procedure outlined in EC7 taking into account appropriate National Annex. This aspect is outlined in the design example presented in Section 9.7.

## 9.3 Liquefaction effects on pile foundations

Soil liquefaction is the association of phenomena like piping, boiling, mud volcanoes etc. that lead to severe loss of strength in loose saturated soils. It is well known that loose sandy soils and sandy silts are particularly vulnerable from a liquefaction point of view. In Chapter 8, Sections 8.3.1 to 8.3.3, we have seen how to determine whether a given site is susceptible to liquefaction by using *in situ* tests such as Standard Penetration Tests (SPT) or Cone Penetration Tests (CPT). EC8 requires the assessment of a site to determine its vulnerability to liquefaction.

Pile foundations can suffer the effects of soil liquefaction around them in a number of ways. These are considered in detail next.

### *9.3.1 Buckling of piles in liquefiable soils*

Piles are slender columns that are supported by surrounding soil. Normally pile foundations do not suffer buckling except when placed in very soft soils and are carrying large axial loads. As discussed in Section 9.1, the load-carrying mechanism is via base capacity and skin friction. The horizontal stresses generated around the surface area of the pile provide the skin friction and also offer lateral support by acting like closely spaced struts.

During earthquake loading, if the soil suffers liquefaction (as determined using the EC8 Part 5 (2003) procedure outlined in Chapter 8, Section 8.3.3) then the lateral support (and the skin friction) may be lost. According to EC8 Part 5 the strength of any soil layer that liquefies must be ignored. In addition if the piles are carrying large axial loads, then they may become vulnerable to buckling failure. Recent research at Cambridge (Bhattacharya et al, 2004, 2005b) has shown that pile buckling is a possible mechanism of failure if the following conditions are satisfied:

- the pile is fully end bearing, i.e. the pile tip is socketed into the bedrock, *and*
- if the pile is carrying a relatively large axial load compared to the Euler buckling load of an equivalent column.

The Euler buckling load can be calculated quite easily using the following equation:

$$P_E = \frac{\pi^2 EI}{L_e^2} \tag{9.6}$$

where $EI$ is the flexural rigidity of the pile and $L_e$ is the equivalent length of the pile.

*9.3.1.1 Fixity condition and equivalent pile length*

Equivalent length, $L_e$, depends on the end conditions of the pile. For example the top of the pile is almost always connected to a pile cap. This provides a rotational restraint. However, the pile cap may be able to 'sway' laterally especially if there is laterally spreading non-liquefied crust present around the pile cap. So at the top of the pile, we generally expect a rotational restraint but not a translational restraint. At the base of the pile, if the pile tip is socketed into the bedrock to sufficient depth, then there will be both rotational and translational restraints. These conditions will yield the equivalent length, $L_e$, to be the length of the pile in the liquefied soil. On the other hand, if the top of the pile is free to rotate and translate and the base of the pile is fixed in both rotation and translation then the $L_e$ will be twice the length of the pile in the liquefied soil. This is demonstrated in Figure 9.5.

Similarly we can easily obtain equivalent lengths for other end conditions by considering the buckling mode shape of the pile. It must be pointed out that the Euler buckling load is very sensitive to the equivalent length.

Once the equivalent length, $L_e$, is determined and the pile's flexural rigidity, $EI$, is known, the Euler buckling load can be calculated.

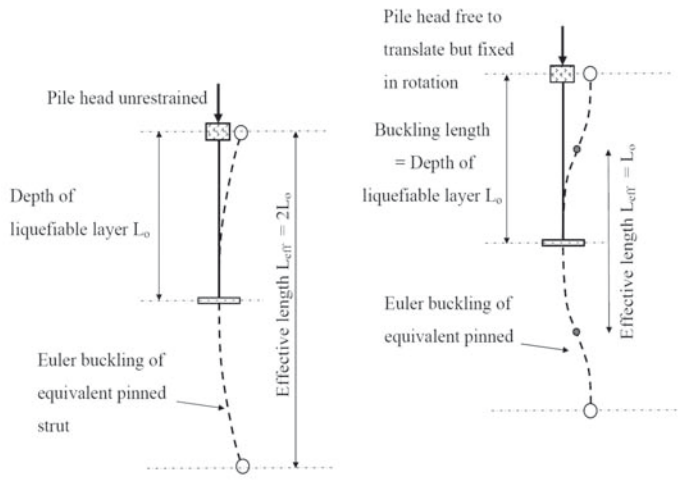



*Figure 9.5* Buckling mode shape and effective length

*9.3.1.2 Slenderness ratio*

The concept of slenderness ratio of the pile can also be used to check the pile design for any possible buckling. Slenderness ratio may be defined simply as:

$$\kappa = \frac{L_e}{r} \tag{9.7}$$

where $r$ is the minimum radius of gyration of the pile section given by $\sqrt{\frac{I}{A}}$, $I$ is the second moment of area about the weakest axis and $A$ is the cross-sectional area of the pile. For a tubular pile the minimum radius of gyration can be estimated as 0.35 times the outside diameter of the pile.

The slenderness ratio $\kappa$ can be used to quickly check the vulnerability of the pile to buckle, if liquefaction potential is high, i.e. liquefaction of soil around the pile is to be expected based on methodology explained in Chapter 8, Sections 8.3.1 to 8.3.3. If the slenderness ratio $\kappa$ is very much less than 50, the piles can be considered generally safe from buckling as the pile would behave more as a short column rather than a long column. For slenderness ratio $\kappa$ greater than 50, buckling of piles needs to be considered in view of the axial load anticipated on the pile and end conditions of the pile etc. as described earlier.

*9.3.1.3 Critical load on piles*

Normally the applied axial load must be less than the Euler buckling load by a factor of 5 or more. Euler buckling load calculation is for 'idealised' situations where the load is completely concentric to the axis of the pile and the pile does not have any imperfections. Any deviation from these conditions can result in a large drop in the estimated buckling load. Therefore it is prudent to have a large factor between the Euler buckling load and the design axial load on the pile.

Similarly, it must be noted that the above simple calculation does not account for the moment loading applied on the pile cap. Of course presence of a moment load on the pile cap again reduces the buckling load. In addition, any lateral displacement of the pile cap due to lateral spreading of the soil following liquefaction can induce additional P-$\Delta$ effects. Also any errors in the pile alignment due to pile wander during installation will reduce the buckling load. Due consideration must be given to these factors.

### *9.3.2 Lateral spreading of sloping ground*

One of the side effects of liquefaction of soil is that the sloping ground starts to move in the down slope direction. This is often termed as 'lateral spreading'. Recent earthquakes such as the 921 Ji-Ji earthquake in Taiwan and the Bhuj earthquake in India provided many examples of lateral spreading of ground. In Figures 9.6 to 9.9 some examples of lateral spreading are presented. In

Figure 9.6 the lateral spreading of a slope past a bridge pier is seen. Clearly such a lateral spread will generate large lateral forces due to the passive pressures generated in the upslope soil wedge. Unrestrained, the soil would spread down the slope as seen in Figure 9.7. Riverbanks, as seen in Figure 9.7, often exhibit tension crack parallel to the river as the whole slope tries to spread into the river following earthquake induced liquefaction.

Similar damage was also seen during the Bhuj earthquake of 2001 in India. In Figure 9.8 the lateral spreading that occurred next to a railway line that serviced the bulk material transportation port of Navlakhi in Gujarat is seen. This led to serious disruption to the port operations and a large section of the railway had to be relayed. In Figure 9.9 the lateral spreading that occurred on the downstream slope of an earth dam in Gujarat is seen.



*Figure 9.6* Lateral spreading past a bridge pier at the new Taichung bridge, Taiwan



*Figure 9.7* Lateral spreading of slopes of a river bank in Taichung, Taiwan

Again piles are often used to stabilise the upstream and downstream slopes of earth dams. Such piles need to resist the large lateral forces created by the soil passive pressures once the whole slope is subjected to lateral spreading.



*Figure 9.8* Lateral spreading next to a railway track at the Navalakhi port in Gujarat, India



*Figure 9.9* Lateral spreading of the downstream slope of an earth dam in Gujarat, India

*9.3.2.1 Slope angle*

As the liquefied soil has very little shear resistance, by definition, it is likely that even gently sloping ground is liable to suffer lateral spreading. Based on the dynamic centrifuge tests carried out at Cambridge, Haigh et al (2000) concluded that ground sloping even at 3° to 6° will suffer lateral spreading. Similar results were reported by other researchers at Rensselaer Polytechnic Institute (RPI) (Dobry et al, 2003) and University of California, Davis (Brandenberg et al, 2005). The amount of lateral spreading suffered by sloping ground is usually in the order of several metres. If the excess pore water pressures generated in the liquefied ground are retained for several tens of seconds, it is possible to estimate the amount of lateral spreading suffered by the soil using Newmarkian style sliding block analysis with suitable modification to include effective stress on the sliding plane. Haigh et al (2000) have shown that such calculations yield lateral spreading of several metres.

In the context of pile foundations that pass through laterally spreading soil, it is sufficient to recognise that the lateral spreading will be in the order of metres and therefore sufficient soil strains are mobilised to generate full passive earth pressures. This becomes more important when there are non-liquefied layers above the liquefied layers. Further if these upper layers are of clayey nature with low hydraulic conductivity then they exacerbate the problem by helping the liquefied layer to retain the excess pore water pressure for a longer duration.

*9.3.2.2 Presence of non-liquefied crust layers and their effects*

The laterally spreading ground layers can impose additional loading on pile foundations passing through them. More importantly it is possible that non-liquefied layers exist above a liquefied layer that can also start to move laterally riding on the liquefied layer as illustrated in Figure 9.10. Such non-

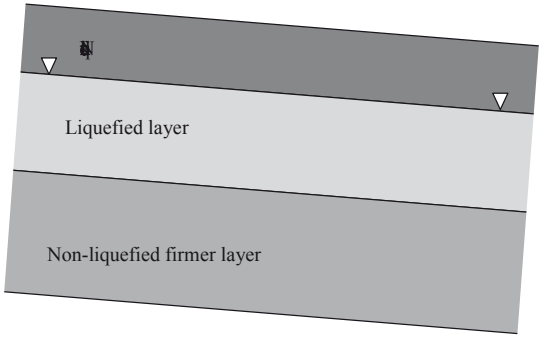



*Figure 9.10* Non-liquefiable soil crust on a liquefiable soil layer

liquefied soil crust can apply large lateral loads on pile foundations passing through them. In some cases the passive earth pressure exerted by the non-liquefied crust can dominate the lateral loading on the pile foundations making the lateral loading applied by the liquefied layer on the piles to be relatively small. Dobry et al (2003) proposed that for simplified design the lateral loading generated by the liquefied layer can be ignored provided that the passive earth pressures generated by the non-liquefied crust are accounted for.

In Section 9.7.5 of this chapter a simplified methodology is included to estimate the loading imposed by laterally spreading ground on the pile foundation. This can be used to estimate the 'upper bound' of the lateral load that can be expected to act on the pile foundation.

## 9.4 Comparison of static and dynamic performance requirements of pile foundations

As explained in Section 9.2 the static design of pile must be carried out according to the guidelines provided in EC7 and its provisions. However, it is important to compare the performance requirements of pile foundations under static and dynamic loading.

### *9.4.1 Kinematic and inertial loading*

For many classes of structure the predominant static loading on piled foundations is vertical compressive loading. Earthquake loading will impose requirements on the piles to resist significant lateral loads and moments with the further possibility of piles being required to carry tensile loads. The deformation of piles may be substantially affected by the permanent deformations of the ground in which they are embedded and in particular liquefaction induced lateral spreading can impose severe damage on piled foundations as discussed in Section 9.1.1.

The loading requirements imposed by seismic events on piles require different geotechnical and structural design of these elements compared with the static equivalent. Earthquake loading differs from other forms of environmental and machinery induced cyclic loading because the in-ground motions produce pile loadings in addition to the pile loadings derived from the motion of the supported structure. The in-ground motion generates 'kinematic interaction' between the piles and the soil while the loading imposed by the structure generates 'inertial interaction' (Figure 9.11).

EC8 Part 5 notes that bending moments due to kinematic interaction only need to be considered when all the following conditions apply:

- The ground profile is of type D, $S_1$ or $S_2$ and contains consecutive layers of sharply differing stiffness.

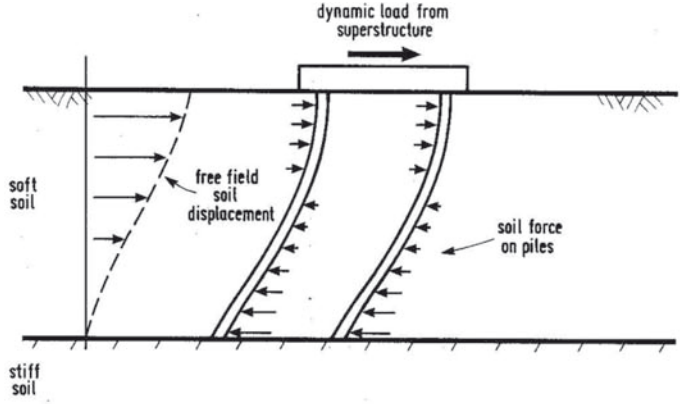



*Figure 9.11* Kinematic and inertial interaction

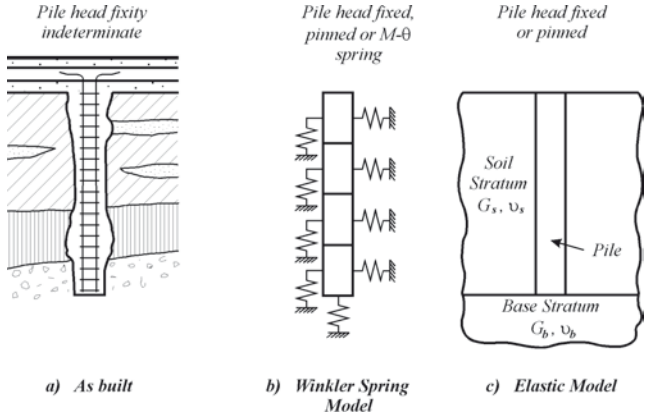



*Figure 9.12* Alternative models for pile load – deflection analyses

- The zone is of moderate or high seismicity (i.e. $\alpha S$ exceeds 0.1g), and the structure is of importance class III or IV.

### *9.4.2 Static pile load-deflection analyses*

The static load-deflection analysis of piles has developed in two principle directions, which should be seen as complementary. These methods are the Winkler spring approach in which the pile is modelled as a beam supported by a series of independent springs, and the elastic continuum approach in which an elastic pile is considered to be embedded in an elastic soil continuum (Figure 9.12).

The different approaches have different strengths and weaknesses:

- The Winkler spring method allows the non-linear loading response of the soil-to-pile deflection to be easily incorporated through the use of non-linear p-y or t-z curves. These springs can be modified to incorporate the effects of imposed ground movements around the piles. In addition, complex soil profiles can also be accommodated in a straightforward manner. However, the springs do not account for the effects of soil movement at one location on soil movements at adjacent locations. This limits the reliability of the empirical methods used to derive p-y and t-z curves and makes the analysis of pile groups difficult with this method.
- The elastic continuum approach is more satisfactory from a theoretical standpoint as the stress and strain fields in the soils around the pile are correctly analysed. This makes the technique suitable for the analysis of the interaction of piles in pile groups. However, the available solutions are predominantly linear-elastic and based on rather simple soil profiles.

Solutions for single piles under static loading are given by Poulos and Davis (1980) with further solutions by Davies and Budhu (1986), Budhu and Davies (1987 and 1988) and Gazetas (1991a and 1991b). These are summarised by Pender (1993) for the Winkler spring and elastic models for a variety of stiffness distributions. The strengths of both methodologies can be harnessed by using the Winkler spring model to refine the soil stiffnesses selected for horizontal and vertical elastic analysis of single piles. The refined parameters from the single pile analysis may then be employed in an elastic analysis of the pile group.

For static lateral loading of an elastic pile embedded in an elastic soil, the displacement, $u$, and rotation, $q$, of the pile head are given by:

*Table 9.1* Pile head flexibility coefficients for static loading

| *Flexibility coefficient* | *Soil stiffness profile* | | |
|---|---|---|---|
| | *Constant* | *Parabolic* | *Linear* |
| $f_{uH}$ | $\frac{1.3}{E_s D}\left(\frac{E_p}{E_{sD}}\right)^{-0.18}$ | $\frac{2.14}{E_{sD} D}\left(\frac{E_p}{E_{sD}}\right)^{-0.29}$ | $\frac{3.2}{mD^2}\left(\frac{E_p}{E_{sD}}\right)^{-0.333}$ |
| $f_{\theta H} = f_{uM}$ | $\frac{2.2}{E_s D^2}\left(\frac{E_p}{E_{sD}}\right)^{-0.45}$ | $\frac{3.43}{E_{sD} D^2}\left(\frac{E_p}{E_{sD}}\right)^{-0.53}$ | $\frac{5.0}{mD^3}\left(\frac{E_p}{E_{sD}}\right)^{-0.556}$ |
| $f_{\theta M}$ | $\frac{9.2}{E_s D^3}\left(\frac{E_p}{E_{sD}}\right)^{-0.73}$ | $\frac{12.16}{E_{sD} D^3}\left(\frac{E_p}{E_{sD}}\right)^{-0.77}$ | $\frac{13.6}{mD^4}\left(\frac{E_p}{E_{sD}}\right)^{-0.778}$ |

$$u = f_{uH}H + f_{uM}M \tag{9. 8}$$

$$\theta = f_{\theta H}H + f_{\theta M}M \tag{9. 9}$$

where $H$ is the horizontal load
$M$ is the moment
$f_{uH}$, $f_{uM}$, $f_{\theta H}$, $f_{\theta M}$ are flexibility coefficients
with $f_{\theta H} = f_{uM}$

The pile head flexibility coefficients for the three soil stiffness profiles given in Figure 9.13 may be expressed as shown in Table 9.1, where $m$ = rate of increase of stiffness with depth

The matrix of pile head flexibility coefficients can be inverted to obtain the matrix of pile head stiffness coefficients $K_{HH}$, $K_{HM}$, $K_{MH}$ and $K_{MM}$. These can be employed to define horizontal and rotational springs, which reproduce the pile head response, thus:

$$\begin{bmatrix} K_{HH} & K_{HM} \\ K_{MH} & K_{MM} \end{bmatrix} = \frac{1}{(f_{uH}f_{\theta M} - f_{uM}^2)} \begin{bmatrix} f_{\theta M} & -f_{uM} \\ -f_{\theta H} & f_{uH} \end{bmatrix} \tag{9. 10}$$

where $K_{HM} = K_{MH}$

$$K_h = \frac{K_{HH}K_{MM} - K_{HM}^2}{K_{MM} - eK_{HM}} \text{ (kN/mm)} \tag{9. 11}$$

$$K_\theta = \frac{K_{HH}K_{MM} - K_{HM}^2}{K_{HH} - \dfrac{K_{HM}}{e}} \text{ (kNm/mrad)} \tag{9. 12}$$

where $e = M / H$.

The variation in stiffness of the soil in the analyses considered here can be taken in an idealised fashion as shown below. These would be reasonably good approximations for the soil types indicated in Figure 9.13 if the soil layer is homogeneous. However, when the soil strata under consideration have distinct layers, suitable approximations have to be made.

### *9.4.3 Dynamic pile load deflection analyses*

#### *9.4.3.1 General behaviour*

Summaries of the methods used to assess the responses of piles and pile groups to seismic loading are provided by Gazetas (1984), Novak (1991) and Pender (1993). Numerical studies indicate that the response of a pile shaft under seismic loading can be considered in three zones:

1 The near surface zone. This zone extends to approximately eight pile diameters beneath the soil surface and is dominated by inertial loading effects.
2 An intermediate zone. This zone exists between the near surface and deep zones and is influenced by both inertial and kinematic effects.
3 The deep zone. This zone is below 12 to 15 pile diameters from the surface and is dominated by kinematic effects.

The effective length of pile, $L_{ad}$, which participates in the inertial response, may be determined for elastic soil profiles (Gazetas, 1984) as a function of the stiffnesses of the pile and the soil and the pile diameter. With reference to the idealised soil profiles shown in Figure 9.13:

$$\text{Constant stiffness with depth: } L_{ad} = 2D\left(\frac{E_p}{E_{sD}}\right)^{0.25} \tag{9.13}$$

$$\text{Parabolic stiffness with depth: } L_{ad} = 2D\left(\frac{E_p}{E_{sD}}\right)^{0.22} \tag{9.14}$$

$$\text{Linear increasing stiffness: } L_{ad} = 2D\left(\frac{E_p}{E_{sD}}\right)^{0.20} \tag{9.15}$$

where $D$ = diameter of pile
$E_p$ = Young's modulus of pile
$E_{sD}$ = Young's modulus of soil at depth.

These active lengths are somewhat greater than the equivalent lengths that can be determined for piles under static loading. Field studies such as those by Hall (1984) and Makris et al (1996) on instrumented piled structures under significant levels of seismic loading show that the stiffness of the pile group tends to decrease significantly as the number of load cycles increases.

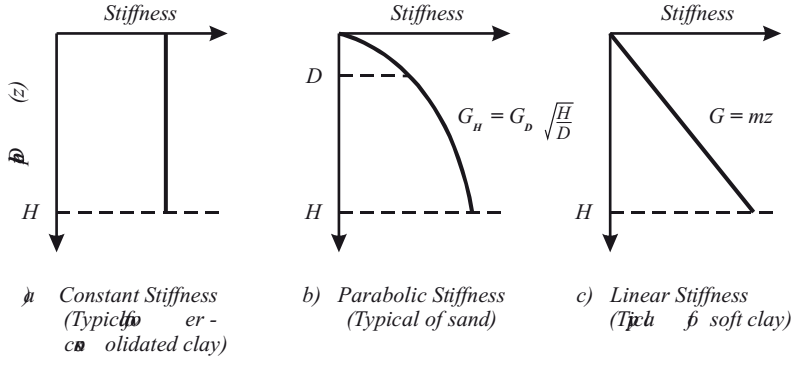



*Figure 9.13* Idealised soil stiffness profiles

This is due to effects such as a decrease in soil stiffness as shear induced pore water pressures increase in the near surface zone and the development of gapping around the top of the pile shafts. These effects will increase the effective or active length of the piles to be considered in the inertial loading response.

The effective pile length concept is useful for differentiating between 'long' and 'short' piles. For 'long' piles an increase in length does not affect the horizontal response to inertial loading. 'Short' piles, of length less than $L_{ad}$, exhibit a softer response to inertial load, which is a function of pile length.

### *9.4.3.2 Pile flexibility*

As an alternative to the method suggested above, the flexibility of the pile can be determined using the following procedure. The elastic length of pile can be determined using elastic length of pile, *T:*

$$T = \left( \frac{E_p I_p}{k} \right)^{0.2} \tag{9. 16}$$

where $E_p I_p$ is the flexural stiffness of the pile and $k$ is gradient of the soil modulus, which may vary from 200 to 2000 kN/m$^3$. $k$ takes the value of 2000 kN/m$^3$ for loose saturated conditions. Thus elastic length of the pile is determined as a function of the relative pile-soil stiffness.

Using the value of $T$ calculated above, the $Z_{\max}$ is calculated as follows:

$$Z_{\max} = \frac{L_p}{T} \tag{9. 17}$$

If $Z_{\max} > 5$, the pile is considered to be *flexible*, i.e. its behaviour is not affected by the length, and collapse is always caused by a flexural failure, with formation of a plastic hinge.

The pile is *semi-flexible* if $5 > Z_{\max} > 2.5$, and the pile is *rigid* if $Z_{\max} < 2.5$.

Piles that are classified as flexible will 'move' with the surrounding soil and therefore would attract the inertial shear load imposed by the superstructure during earthquake loading. Rigid piles, on the other hand, will attract significant soil load, as the piles stay in position and the soil would exert passive pressures on either side of the piles in alternative load cycles. This additional lateral load applied by the soil must be considered in the pile design.

## 9.5 Kinematic response

### *9.5.1 Classical approach*

It is convenient to analyse the kinematic response of the pile or pile group separately from the inertial response. The kinematic response at depth may be used to assess the structural requirement of the pile in the intermediate and deep zones. The kinematic response of the pile head is an input into the inertial response analysis.

In the deep zone the presence of piles has little effect on the ground motion or natural frequency of the stratum. The pile and soil motions are likely to be practically coincident for frequencies up to at least 1.5 times the natural frequency, $f_n$, of the stratum. This observation is of practical significance as the deflected shape of the pile can be obtained from a 1D equivalent linear shear wave propagation analysis. Having obtained the deflected shape of the pile, its bending moments and shear forces may readily be determined. Makris et al (1996) discuss field studies that make useful observations on this mode of behaviour. It should be noted that substantial bending moments may be induced in piles at the levels of interfaces between zones of appreciably different stiffness. EC8 Part 5 requires piles to remain elastic, though under certain conditions they are allowed to develop plastic hinges at their heads. The regions of plastic hinging should be designed according to EC8 Part 1, Clause 5.8.4.

In order to perform the inertial response analysis the kinematic pile head response is required. Numerical studies indicate that the kinematic response derived for single piles is applicable to pile groups and that the kinematic interaction between the soil and a free head pile is conservative if applied to a fixed head pile.

Pender (1993) describes an approximate technique based on Gazetas (1984) that may be used to evaluate the kinematic response of the pile head. Firstly the free field response at the top of the soil column is determined at a point remote from the pile group. The horizontal amplitude of the free field motion is $u_o$. Then a frequency dependent horizontal interaction factor, $I_u$, is determined such that:

$$I_u = \frac{u_p}{u_o} \tag{9.18}$$

where $u_p$ = the horizontal amplitude of the pile head motion.

For constant stiffness with depth: $$F = \left(\frac{f}{f_n}\right)\left(\frac{E_p}{E_{sD}}\right)^{0.30}\left(\frac{L}{D}\right)^{-0.5} \tag{9.19}$$

For parabolic stiffness with depth: $$F = \left(\frac{f}{f_n}\right)\left(\frac{E_p}{E_{sD}}\right)^{0.16}\left(\frac{L}{D}\right)^{-0.35} \tag{9.20}$$

*Table 9.2* Coefficients for horizontal kinematic interaction factor

| *Coefficient* | *Soil stiffness profile* | | |
|---|---|---|---|
| | *Constant* | *Parabolic* | *Linear* |
| A | 0 | $3.64 \times 10^{-6}$ | $-6.75 \times 10^{-5}$ |
| B | 0 | $-4.36 \times 10^{-4}$ | $-7.0 \times 10^{-3}$ |
| C | –0.21 | $6.0 \times 10^{-3}$ | $3.3 \times 10^{-2}$ |

For linear increasing stiffness: $$F = \left(\frac{f}{f_n}\right)\left(\frac{E_p}{E_{sD}}\right)^{0.10}\left(\frac{L}{D}\right)^{-0.4} \quad (9.21)$$

where $f$ = response spectrum frequency considered
$f_n$ = natural frequency of stratum
$E_p$ = Young's modulus of pile
$E_{sD}$ = Young's modulus of soil at depth $D$
$L$ and $D$ = length and diameter of pile.

Using the appropriate equation, values of $F$ are calculated for discrete frequencies across the frequency range of interest (e.g. 0.5 Hz to 40 Hz). Corresponding values of $I_u$ are calculated from the following expression (Gazetas, 1984):

$$I_u = aF^4 + bF^3 + cF^2 + 1.0 \quad (9.22)$$

with a minimum value of $I_u = 0.5$.

The coefficients $a$, $b$ and $c$ in above equation are given in Table 9.2.

The interaction factors produced by this procedure are strictly applicable only to a Fourier spectrum. However, approximate results can be obtained by applying the interaction factors directly to the free field spectral acceleration versus the frequency response spectrum. The horizontal spectral acceleration of the pile head is simply obtained by multiplying the free field acceleration by the value of $I_u$ for each frequency considered.

Study of spectral acceleration responses produced by the above procedure show that the piles damp the higher the frequency excitation seen in the free field. The extent of that damping depends on the ratio of pile to soil stiffness and particularly on the soil stiffness profile. The linear increasing stiffness profile produces damping at lower frequencies than the other stiffness profiles. Pender (1993) observes that the response of instrumented piles in earthquakes tends to that of the linear increasing stiffness profile even if the Site Investigation (SI) data suggest a constant or parabolic profile. This is considered to be due to softening of the soil close to the surface under seismic loading.

The studies undertaken by Gazetas (1984) show that the rotational interaction factor is sufficiently small to be neglected.

### *9.5.2 Kinematic loading induced by laterally spreading soil*

A more common problem is when pile foundations are used to transfer load from the superstructure through a shallow layer of clay and through a liquefiable soil layer into firmer soil stratum or bedrock below. If the ground at such a site is sloping (even with very small slope angles), earthquake induced liquefaction may cause the clay layer to spread laterally thereby inducing large lateral loads on the piles and pile cap. Recent research at Cambridge (Haigh and Madabhushi, 2005), at Renessealer Polytechnic Institute, New York (Dobry et al, 2003) and at University of California, Davis (Brandenberg et al, 2005) has looked at the loading applied by non-liquefied crust onto the pile cap and piles due to lateral spreading.

In such situations Dobry et al (2005) propose that in pile design, the lateral load from the non-liquefied crust plays the most important role and the contribution of the resistance offered by liquefied soil can be ignored.

Thus the lateral load applied by a clay crust can be determined by using shallow foundation bearing capacity factors. For example if the clay crust overlying the liquefied layer had an undrained strength of $S_u$ then the lateral pressure applied on the pile cap and piles in this region can simply be calculated using:

$$q = (\pi + 2) \cdot S_u \tag{9. 23}$$

For other soil types, the above expression can be suitably modified.

The lateral deflection of the pile cap and rotation of the pile cap and pile heads can be determined under the action of the lateral load, induced by $q$ above, over the resisting surface area (sides of the pile cap and the portion of the piles in this region). The above expression is based on *Upper Bound Theorem of Plasticity* and therefore should provide a safe bound.

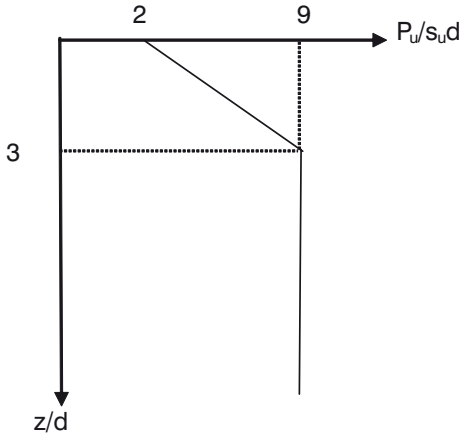



*Figure 9.14* Variation in normalised lateral load with normalised depth

Alternatively, the upper bound theory by Murff & Hamilton (1993) for lateral resistance can also be used in cohesive soil (shown in Figure 9.14 where $P_u$ is lateral resistance). This expression allows for a more gradual change of the undrained shear strength with depth from the surface of the soil crust to deeper regions and therefore should be used if the non-liquefied crust is reasonably deep.

It must be noted that the lateral loading due to inertia from the superstructure and the kinematic loading due to the lateral spreading of the soil will not generally occur at the same time. However, for design these can be superposed, which is a conservative assumption. The superposed load can be used to estimate the lateral deflection of the pile heads and their rotation.

### *9.5.3 Adoption of static pile head static stiffness concept*

EC8 Part 5 Annex C provides guidance on the pile head stiffness coefficients for the three types of idealised soil stiffness profiles presented in Figure 9.13. These are reproduced in Table 9.3 and can be used in preference to the flexibility coefficients presented earlier in Section 9.4.2, Table 9.1. As before, the key parameters are:

$E$ = Young's modulus of the soil model equal to 3G
$E_p$ =Young's modulus of the pile material
$E_s$ = Young's modulus of the soil at 1 pile diameter depth
$d$ = pile diameter
$z$ = pile depth.

Once the loads on the pile heads and the pile cap are determined as discussed in Section 9.4.3, the stiffness coefficients in Table 9.3 can be used

*Table 9.3* Static stiffness of flexible piles embedded in three soil models

| *Soil model* | $\frac{K_{HH}}{dE_s}$ | $\frac{K_{MM}}{d^3E_s}$ | $\frac{K_{HM}}{d^2E_s}$ |
|---|---|---|---|
| Linear variation $E = E_s\frac{z}{d}$ | $0.60\left(\frac{E_p}{E_s}\right)^{0.35}$ | $0.14\left(\frac{E_p}{E_s}\right)^{0.80}$ | $-0.17\left(\frac{E_p}{E_s}\right)^{0.60}$ |
| Square root variation $E = E_s\sqrt{\frac{z}{d}}$ | $0.79\left(\frac{E_p}{E_s}\right)^{0.28}$ | $0.15\left(\frac{E_p}{E_s}\right)^{0.77}$ | $-0.24\left(\frac{E_p}{E_s}\right)^{0.53}$ |
| Constant $E = E_s$ | $1.08\left(\frac{E_p}{E_s}\right)^{0.21}$ | $0.16\left(\frac{E_p}{E_s}\right)^{0.75}$ | $-0.22\left(\frac{E_p}{E_s}\right)^{0.50}$ |

to determine the lateral and cross pile head displacements and the pile cap rotation.

## 9.6 Inertial response

The inertial response analysis uses the dynamic response obtained from the kinematic interaction study to assess the seismic displacements and rotations of the pile head or of the structure. The forces driving the pile head are derived from the mass and stiffness of the structure.

Typically the structure may be simplified to a single degree of freedom system while the piled foundation is considered to have translational and rotational degrees of freedom.

### *9.6.1 Relative stiffness of pile-soil system*

The response of the foundation to the horizontal inertial loading and moments is determined by a combination of stiffness and damping in a manner analogous to the response of a shallow foundation. While the single pile stiffness is not sensitive to frequency, the pile group interaction terms and the radiation damping are frequency dependent.

A common way of addressing the response of the single pile or group to inertial loading is by the use of the impedance concept:

$$S(\omega) = \frac{R(t)}{U(t)} \tag{9. 24}$$

with $S(\omega) = K(\omega) + i\omega C$
where $S(\omega)$ = impedance for mode of response (sliding, rocking etc.)
$R(t)$ = dynamic force or moment
$U(t)$ = dynamic displacement or rotation
$K(\omega)$ = dynamic pile stiffness (kN/m)
$\omega$ = frequency (rad/s)
$C$ = damping coefficient (kN.s/m)
$i = (-1)^{0.5}$.

The impedance function is conveniently expressed as a complex variable because the damping component, being a function of velocity, is out of phase with the elastic stiffness. The damping may also be expressed as dimensionless frequency dependent coefficients, $\zeta(\omega)$, for the various modes of response where:

$$\zeta(\omega) = \frac{\pi f C}{K} = \frac{\omega C}{2K} \tag{9. 25}$$

This enables an alternative expression for the impedance to be developed:

$$S(\omega) = K\left[k(\omega) + 2\zeta(\omega)i\right] \tag{9. 26}$$

With the impedance functions defined in the above equations, any appropriate static expression for single pile or pile group loading response can be used for the dynamic loading case, substituting the complex impedance terms for their static counterparts.

Numerical studies undertaken by Gazetas (1984) show that $k(\omega)$ is approximately unity for most practical values of pile – soil stiffness ratio over the frequencies of interest and for the horizontal, rocking and vertical modes. Hence the dynamic stiffnesses for the various modes can be taken as similar to their static counterparts.

### *9.6.2 Damping coefficients*

Values for the damping coefficients, $\zeta$, are given by Gazetas (1991a) for single piles embedded in elastic soils with the stiffness profiles shown in Figure 9.13, as shown in Table 9.4. Note that $\zeta_{HH}$ is the damping due to horizontal movement under horizontal loading, $\zeta_{HM}$ refers to horizontal movement due to applied moment and $\zeta_{MM}$ refers to rotation due to applied moment.

All of the expressions in Table 9.4 apply only when $f > f_n$ for the stratum. If the exciting frequency is below the natural frequency of the stratum then there will be no radiation damping and the damping coefficients will be the left-hand term in each case.

Calculations based on the formulae in Table 9.4 have been compared with a limited amount of field data mainly derived from experiments where vibrators have been mounted on single piles (Pender, 1993). The field data suggests that the damping coefficient values obtained from these expressions under-predict actual damping by about 30 per cent.

Using the impedance terms, the pile head behaviour may be reduced to translational and rotational springs. The inertial loading may be determined using a single degree of freedom (SDOF) structural model. The equations

*Table 9.4* Dimensionless pile head damping coefficients

| *Damping coefficient* | *Soil stiffness profile* | | |
|---|---|---|---|
| | *Constant* | *Parabolic* | *Linear* |
| $\zeta_{HH}$ | $0.8\beta = \dfrac{1.10fD}{v_s}\left(\dfrac{E_p}{E_{sD}}\right)^{0.17}$ | $0.7\beta = \dfrac{1.20fD}{v_s}\left(\dfrac{E_p}{E_{sD}}\right)^{0.08}$ | $0.6\beta = \dfrac{1.8fD}{v_s}$ |
| $\zeta_{HM}$ | $0.8\beta = \dfrac{0.85fD}{v_s}\left(\dfrac{E_p}{E_{sD}}\right)^{0.18}$ | $0.6\beta = \dfrac{0.70fD}{v_s}\left(\dfrac{E_p}{E_{sD}}\right)^{0.05}$ | $0.3\beta = \dfrac{1.0fD}{v_s}$ |
| $\zeta_{MM}$ | $0.35\beta = \dfrac{0.35fD}{v_s}\left(\dfrac{E_p}{E_{sD}}\right)^{0.2}$ | $0.22\beta = \dfrac{0.35fD}{v_s}\left(\dfrac{E_p}{E_{sD}}\right)^{0.1}$ | $0.2\beta = \dfrac{0.4fD}{v_s}$ |

required to solve the response of such an SDOF system are given by Wolf (1985). Useful worked examples are given by Pender (1993).

Because the impedance terms are complex numbers, the calculated displacements also have real and imaginary parts. The maximum (real) response is readily determined by applying the Square Root of the Sum of Squares (SRSS) technique.

Calculations on the response of pile groups require the use of dynamic pile group interaction factors. These are frequency dependent complex functions. Interaction factors for various loading directions and responses are given by Gazetas (1991a); Gazetas et al (1991); and Makris and Gazetas (1992).

### *9.6.3 Combination rules*

Since the response of the soil and structure will be at different natural frequencies, the combination rules given in Clause 4.3.3.5 of EC8 Part 1 can be used to calculate the cumulative effect of kinematic and inertial interaction.

## 9.7 Design example on a pile foundation

In this section we shall outline the design of a pile foundation for a typical column of the building for which the seismic designs were carried out in earlier chapters. Of course in reality the design of pile foundations will be carried out for individual columns with the associated reductions in the pile lengths and/or pile diameters to suit the design load on the column. Here we shall only consider one typical column along the D line on the plan of the building.

Another premise that is made here is the requirement of the pile foundations. It is assumed that the building will be located at 'Site A' for economic and operational reasons.

### *9.7.1 Configuration of the problem*

In Chapter 8, the EC8 Part 5 provisions were used to determine the liquefaction potential of 'Site A'. The soil profile at this site as determined from borehole data is presented in Figure 9.15. Based on this it was determined that this site has:

- A non-liquefiable clay crust of 2 m thickness close to ground.
- Liquefaction potential analysis confirms that a 10 m thick layer of loose sand underlying the clay layer is 'liquefiable' during the design earthquake event.

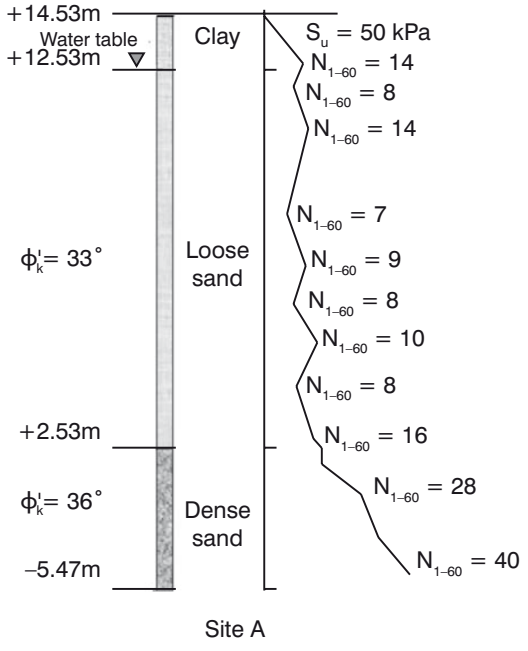



*Figure 9.15* Borehole data from Site A

*Table 9.5* Loading on the foundation from the columns

| | *Column C* | *Column D* |
|---|---|---|
| Axial load | 5978 kN | 862 kN |
| Shear load | 826 kN | 826 kN |
| Moment load | 2405 kNm | 2088 kNm |

The above ground conditions at this site would necessitate the requirement of pile foundations. The pile foundations would be required to pass through the loose sand layer and end bearing fully into the dense sand layer.

### *9.7.2 Structural loading on piles*

In Chapter 3 the structural analysis of the building frame is considered. Here we use the loading obtained from those analyses (using *q* factor of 3.9 and choosing the concrete frame building that has the more severe loading case). These loads are obtained with due consideration to the capacity design aspects and are shown in Table 9.5. Please note that the worst loading occurs

on columns along the lines C and D, each line reaching a maximum load while the other is at a minimum.

Therefore the loading on the pile group is:

Design vertical load $N_{Ed}$ = 5978 kN

Design moment load $M_{Ed}$ = 2505kNm

Design horizontal, shear load $V_{Ed}$ = 826 kN

Based on the above requirements, the following will be assumed regarding the pile foundations.

Choose:

- 2 × 2 pile group for columns along the D line
- steel tubular driven pile
- ~ 15 m pile length
- pile diameter 800 mm; pile wall thickness 20 mm
- pile spacing = 2D = 1.6 m
- pile group efficiency η = 70 per cent (conservatively).

Various other pile types can be considered for this application, such as concrete bored piles, precast concrete driven piles or steel H-piles for example.

### *9.7.3 Static pile design*

The piles are required to be designed according the provisions of EC7. Here the UK National Annex provisions are also taken into consideration.

#### *9.7.3.1 Assumptions and simplifications*

Assume pile density is equal to soil density.

Assume moment on group is carried by couple in piles.

Individual axial pile load, $Q_A$, is given by:

$$Q_A = \frac{N_{Ed}}{4} \pm \frac{M_{Ed}}{2 \times 1.6\text{m}}$$

$Q_A$ = 1495 ± 752 kN $\qquad$ $Q_{A\,\max}$ = 2247 kN

Ignore shaft friction from upper clay layer.

Assume pile is plugged and can develop full end bearing capacity.

### *9.7.4 Axial pile design*

Use BS EN 1997 Design Approach DA-1.

Two combinations must be considered. In Combination 1, partial factors are applied to the pile loading. In Combination 2, partial factors are applied to components of the pile resistance. Note: refer to the UK National Annex for appropriate partial factors for pile design.

*9.7.4.1 Combination 1*

Partial factor sets A1 + M1 + R1 apply.

From A1 adopt factor $\gamma_G = 1.35$. (Note: this is a simplification. Separate factors apply to permanent and transient loads.)

For M1 all material factors $\gamma_M = 1$.

For R1 all resistance factors $\gamma_R = 1$.

Note: a model factor, $M_F$, is also required. From the UK National Annex the model factor is 1.4 if the pile has been designed from soil test data alone. If the pile capacity has been verified using a maintained load test the model factor is 1.2.

BS EN 1997 is not prescriptive concerning the method of calculating the pile capacity, only requiring that the method should be one that is verified against pile load test data.

END BEARING

Consider the end bearing of a pile.

End bearing of pile may be calculated as:

$$Q_b = N_q \sigma_v^{'}$$

where $A_b$ = base area of shaft, assuming fully plugged base.

The friction angle for dense sand $\varphi_k = 36°$.

The bearing capacity factor, $N_q$, for a deep foundation can be obtained using the chart given by Berezantzev et al (1961) shown earlier in Figure 9.4. Reading the value of $N_q$ from this figure for the friction angle of 36°, we get:

$N_q = 65$

Assume that the pile starts from 1 m below ground level to allow for pile cap of 1 m thickness.

For a 15 m long pile, the base is at 16 m below ground level (allowing for 1 m thick pile cap). So calculate the effective stress at 16 m depth.

The pile diameter is 800 mm (assume that the pile is plugged at the base).

$$\sigma_v^{'} = 16 \times 20 - 14 \times 10 = 180 \text{kPa}$$

$$q_b = \sigma_v^{'} \pm N_q$$

$$q_b = 180 \times 65 = 11700 \text{kPa}$$

$$Q_b = \pi r^2 \times q_b = \frac{\pi \times 0.4^2 \times 11700}{1000} = 5881 \text{kN}$$

SHAFT FRICTION

We can ignore the 1 m of clay layer just below the pile cap in estimating the shaft capacity (conservative assumption):

$$\tau_s = K_s \cdot \sigma_v' \cdot \tan\delta$$

Assume the following:
For driven piles $K_s \leq 1$, choose $K_s = 1$.
For the loose sand layer around the shaft, $\phi_k = 30°$.
As we are using driven, smooth, steel tubular piles:

$$\delta_d = \frac{2}{3}\phi_{cvd} = \frac{2}{3} \times 30^\circ = 20^\circ$$

Therefore, at +12.53 m elevation:

$$\tau_s = 1 \times 40 \times \tan 20^\circ = 14.55 \cong 15\text{kPa}$$

At –1.47 m elevation (15 m long pile):

$$\tau_s = 1 \times 180 \times \tan 20^\circ = 65.5 \cong 66\text{kPa}$$

$$q_s = 2\pi r \times \int_0^L \tau_s$$

$$Q_s = 2\pi \times 0.4 \times \frac{(15+66)}{2 \times 1000} \times 15 = 1527\text{kN}$$

The applied design load for Combination 1 is 2247 × 1.35 = 3033kN

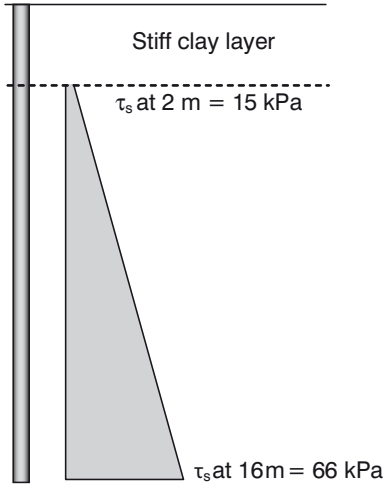



*Figure 9.16* Variation of shear stress along the length of the pile

(See the shear stress distribution along the length of the pile in Figure 9.16.)

(In practice shear stress is limited to 100 kPa, $\tau_s < 100$ kPa.)

For Combination 1 the total design resistance $R_{cd1}$ is calculated as follows:

$R_{cd1} = R_{bd1} + R_{sd1}$

where $R_{bd1} = \dfrac{Q_b}{M_F}$ and $R_{sd1} = \dfrac{Q_s}{M_F}$

Therefore, $R_{cd1} = \dfrac{5881 + 1527}{1.4} = 5291\text{kN} > 3033\text{kn}$

*9.7.4.2 Combination 2*

Partial factor sets A2 + M1 + R4 apply.

For A2 factors on actions $\gamma_G = 1$.

For M1 all material factors $\gamma_M = 1$.

For R4 various factors apply to the shaft and base resistances.

For a driven pile without explicit load test verification of the Service Limit State (SLS) the following factors apply (see UK National Annex):

$\gamma_b = 1.7$

$\gamma_s = 1.5$

$M_F = 1.4$ as for Combination 1.

As before, the unfactored end bearing and shaft resistances are given as:

$Q_b = 5881$ kN

$Q_s = 1527$ kN

For Combination 2 the total design resistance $R_{cd2}$ is calculated as follows:

$R_{cd2} = R_{bd2} + R_{sd2}$

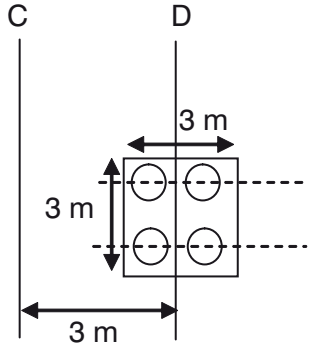



*Figure 9.17* Plan view of pile cap

where $R_{bd2} = \dfrac{Q_b}{\gamma_b \times M_F}$ and $R_{sd1} = \dfrac{Q_s}{\gamma_s \times M_F}$

Therefore,

$$R_{cd2} = \frac{5881}{1.7 \times 1.4} + \frac{1527}{1.5 \times 1.4} = 2471 + 727 = 3918\text{kN} > 2247\text{kN}$$

In this calculation Combination 2 is more critical than Combination 1 but nevertheless the required inequality is satisfied. In the case that the loose sand layer is subject to liquefaction, the shaft capacity in the liquefied layer will temporarily be lost. Raised pore pressures will also migrate into the dense sand reducing the effective stresses and therefore also reducing the shaft friction in this stratum. During subsequent reconsolidation of the loose sand, down-drag forces will be applied to the pile shaft. The designer will wish to take these issues into account when deciding whether the pile length is adequate.

Choose pile cap of 3 m × 3 m, with pile spacing of 2 diameters as shown in Figure 9.17.

### *9.7.5 Factor of safety against pile buckling*

Normally piles rely on lateral resistance offered by soil. However, when the surrounding soil liquefies, this lateral resistance is significantly reduced. Long, slender piles in liquefiable soils can suffer buckling failure based on recent research based on dynamic centrifuge modelling (Bhattacharya et al, 2004).

A simple check can be carried out to see if piles are vulnerable to buckling mode of failure. For Euler buckling, we know that:

$$P_E = \frac{\pi^2 EI}{L_e^2}$$

where $EI$ is the flexural rigidity of the pile and $L_e$ is the equivalent length of the pile, which depends on the end conditions. For our case the base of the pile is fixed with sufficient embedment length into the dense sand (5 pile diameters = 5 × 0.8 = 4 m). At the pile cap, the pile head has rotational fixity but not translational fixity. This yields a buckling mode shape that dictates the effective length of the pile to be equivalent to length of the pile. Ignoring the 1 m of clay above the liquefiable layer, and considering the embedment length into dense sand:

$$L_E = 1 + 10 + 4 = 15\text{m}$$

$E$ = 210 GPa for steel. Also the steel tubular piles are assumed with outer diameter of 0.8 m and wall thickness of 20 mm.

$$I_P = \frac{\pi}{64}\left(D_0^4 - D_i^4\right) = \frac{\pi}{64}\left(0.8^4 - 0.76^4\right) = 0.00373\text{m}^4$$

$$P_E = \frac{\pi^2 EI_P}{L_e^2} = \frac{3.14^2 \times 210 \times 10^9 \times 0.00373}{15^2} = 34355450.66\text{N}$$

$$P_E = 34.35\text{MN}$$

This is significantly more than the applied vertical load, so the pile should be safe against buckling. However, please note that the Euler buckling does not take into account any imperfections in the pile or pile wandering during driving. Similarly any lateral displacements due to lateral spreading may cause additional P-Δ effects, which you must consider.

### *9.7.6 Flexibility of the pile*

Elastic length of pile $T$

$$T = \left(\frac{E_P I_P}{k}\right)^{0.2} \text{ where}$$

$E_P I_P$ is the flexural stiffness of the pile

$$E_P = 210\text{GPa}$$

$$I_P = \frac{\pi}{64}\left(D_0^4 - D_i^4\right) = \frac{\pi}{64}\left(0.8^4 - 0.76^4\right) = 0.00373\text{m}^4$$

$k$ = gradient of the soil modulus ($k$ takes the value of 2000 kN/m$^3$ for loose saturated conditions and may vary from 200 to 2000 kN/m$^3$).

For the present case $k$ = 5000 kN/m$^3$ and length of pile, $L_p$ = 15 m.

$$T = \left[\frac{E_p I_p}{k}\right]^{0.2} = \left[\frac{210 \times 10^9 \times 0.00373}{5000 \times 10^3}\right]^{0.2} = 2.75$$

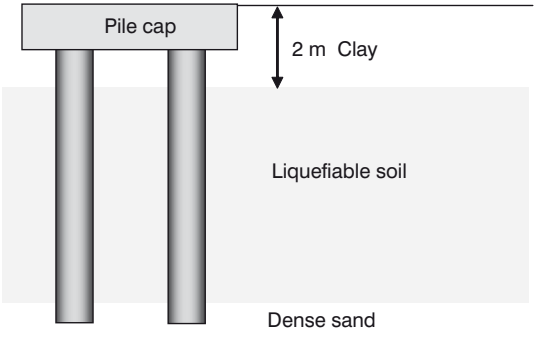



*Figure 9.18* Sectional view through the pile cap

$$Z_{\max} = \frac{L_p}{T} = \frac{15}{2.75} = 5.46$$

Since $Z_{\max} > 5$, the pile is *flexible*, i.e. its behaviour is not affected by the length, and collapse is always caused by a flexural failure, with formation of a plastic hinge.

The pile is *semi-flexible* if $5 > Z_{\max} > 2.5$, and the pile is *rigid* if $Z_{\max} < 2.5$.

Piles that are classified as flexible will 'move' with the surrounding soil and therefore would attract the inertial shear load imposed by the superstructure during earthquake loading.

### 9.7.7 *Lateral loading due to the clay layer below the surface*

Pile cap side area $= 1 \times 3 = 3$ m$^2$

Projected area of one pile for 1 m $= 1 \times 0.8 = 0.8$ m$^2$

The maximum loading on the pile cap due to the laterally spreading clay layer occurs when the clay layer fails loading the pile cap.

Lateral load (upper bound) $= (\pi + 2) \cdot s_u \cdot$Area

$= 5.14 \times 50 \times (3 + 4 \times 0.8) = 1593.4$ kN

Alternatively, the upper bound theory by Murff and Hamilton (1993) for lateral resistance can also be used in cohesive soil (as shown in Figure 9.14, in which $P_u$ is lateral resistance). Note that the calculation carried out above falls at 5.14, which is close to the average of points 2 and 9 on the x-axis of Figure 9.14.

In our design, let us assume that the liquefied soil zone does not offer any additional loading that contributes to the pile group movement. This is also suggested by Dobry et al (2003) as a design approximation. However, Japanese Road Association (JRA) 1990 suggest that a resistance of 30 per cent of total vertical stress can be used for fully liquefied soil zones. In general, non-liquefied clay crust contributes the larger proportion of the loading on the pile cap.

From Table 9.3 for the soil model with square root variation in strength with depth, choose the normalised stiffness coefficients:

$$E = E_s \sqrt{\frac{z}{d}}$$

$E_p = 210$ GPa

This elastic modulus can be used directly for a solid pile. However, our pile is a tubular pile. Therefore the elastic modulus must be reduced in proportion to the reduced second moment of area. This can be done as follows:

$$E_{p_corrected} = \frac{E_p}{\left(\frac{I_{solid}}{I_{tubular}}\right)}$$

$$E_{p_corrected} = \frac{E_p}{\left(\frac{D_0^4}{\left\{D_0^4 - D_i^4\right\}}\right)}$$

The outer diameter of the pile is 800 mm and the inner diameter is 760 mm. Substituting these, we get:

$$E_{p_corrected} = \frac{210}{\left(\frac{0.8^4}{\left\{0.8^4 - 0.76^4\right\}}\right)} = 38.95\text{GPa}$$

Take $E_s$ = 30 Mpa for clay at one depth equivalent to one pile diameter. (Note: this is quite a critical parameter that affects the estimation of lateral displacement and rotation of the pile cap. In practice, we have to determine this carefully based on laboratory data from experiments on the clay crust.)

$$\frac{K_{HH}}{dE_s} = 0.79\left(\frac{E_p}{E_s}\right)^{0.28}$$

$$K_{HH} = 0.8 \times 30 \times 10^6 \times 0.79 \times \left[\frac{38.95 \times 10^3}{30}\right]^{0.28} = 1.41 \times 10^8 \text{N/m}$$

$$\frac{K_{MM}}{d^3E_s} = 0.15\left(\frac{E_p}{E_s}\right)^{0.77}$$

$$K_{MM} = 0.8^3 \times 30 \times 10^6 \times 0.15 \times \left(\frac{38.95 \times 10^3}{30}\right)^{0.77} = 5.75 \times 10^8 \text{Nm/rad}$$

$$\frac{K_{HM}}{d^2E_s} = -0.24\left(\frac{E_p}{E_s}\right)^{0.53}$$

$$K_{HM} = 0.8^2 \times 30 \times 10^6 \times -0.24 \times \left(\frac{38.95 \times 10^3}{30}\right)^{0.53} = -2.06 \times 10^8 \text{Nm/rad}$$

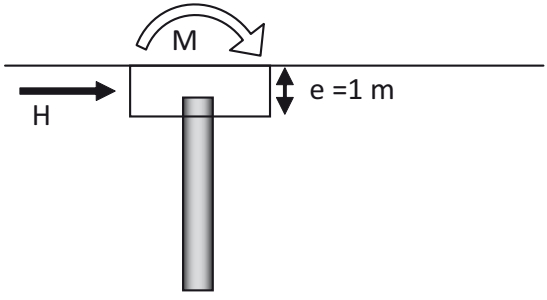



*Figure 9.19* Horizontal and moment loading on pile cap

### 9.7.8 Pile cap deflection and rotation

Design horizontal, shear load $V_{Ed} = 826$ kN
Design moment load $M_{Ed} = 2405$ kNm
Lateral load due to the clay layer = 1593.4 kN.

Note that the lateral load due to the clay layer (kinematic loading) and the design shear load (inertial load) from the superstructure do not occur at the same time. Therefore it is sufficient to consider the largest of these loads in estimating the displacement. However, let us choose to superpose both these loads as a conservative approximation.

$$\text{Horizontal force one each pile} = \frac{1593.4 + 826}{4} = 604.85\text{kN}$$

$$\text{Moment load on each pile} = \frac{2405}{4} = 601.25\text{kNm}$$

#### 9.7.8.1 Pile cap displacement

The pile cap displacement can be calculated using the following equation:

$$\delta_{HH} = \frac{H}{K_{HH}} + \frac{\frac{M}{h}}{K_{HM}}$$

where $h$ is the equivalent height at which a horizontal force will cause a moment of $M$. Assume that this equivalent height is 6 m corresponding to the first floor level of the building:

$$\delta_{HH} = \frac{604.85 \times 10^3}{1.41 \times 10^8} + \frac{601.25 \times 10^3 / 6}{2.06 \times 10^8} = 0.004776\text{m}$$

So lateral displacement of pile cap will be 4.78 mm, which is very small.

#### 9.7.8.2 Pile cap rotation

The pile cap rotation can be calculated using the following equation:

$$\theta_{HH} = \frac{M}{K_{MM}} + \frac{H}{K_{HM}}$$

$$\theta_{MM} = \frac{601.25 \times 10^3}{5.75 \times 10^8} + \frac{604.85 \times 10^3}{2.06 \times 10^8} = 3.982 \times 10^{-3}\text{rad}$$

So the pile cap rotation will be 0.228 degrees, which is very small.

(Note: although the pile cap deflection and rotation is small, the above calculations were based on the assumption of the elastic modulus of the soil at the depth of a one pile diameter. So the deflection in reality may be somewhat larger than what we calculated.)

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